



HBM3 DRAM

MT65B12G16080A00QG — 16 channels × 192 Meg × 68 I/O
MT65B18G16120A00QH — 16 channels × 288 Meg × 68 I/O

Features

- V_{DDC} , V_{DDQ} = 1.067V to 1.177V
- V_{DDQL} = 0.38V to 0.44V
- V_{PP} = 1.746V to 1.95V
- Data rate: 6.4, 7.2, 8.0, and 9.2 Gb/s
- Peak bandwidth: up to 1177 GB/s
- 16 independent channels with two pseudo channels each
 - 272 (256 + 16) bit per array read or write access
- Bank Count:
 - 8-High (24GB): 32 banks in 8 bank groups
 - 12-High (36GB): 48 banks in 12 bank groups
- Differential clock input CK_t/CK_c for command/address
- Semi-independent row and column command interfaces, allowing ACT/PRE commands to be issued in parallel with READ/WRITE commands
- Double data rate (DDR) command and address (CK)
- Differential write data strobes WDQS_t/WDQS_c and read data strobes RDQS_t/RDQS_c, each associated with four data bytes
- Double data rate (DDR) data (WDQS, RDQS)
- Programmable READ latency (RL)
- Programmable WRITE latency (WL)
- Programmable parity latency (PL)
- Burst length = 8 (fixed)
- Data bus inversion (DBIac) for writes and reads
- Parity for command/address and data monitoring
- On-die Reed Solomon (RS) ECC with 16-bit symbol error correction and error severity reporting
- Auto error check and scrub (ECS) support with error reporting function
- Auto precharge option for each burst access
- Auto refresh mode (32ms, 8k cycles) with per-bank and all-bank refresh options
- Temperature sensor controlled self refresh rate
- 'RAS lockout
- Programmable output driver strength
- Unterminated clock, command, address and data interfaces
- Temperature sensor with read-out

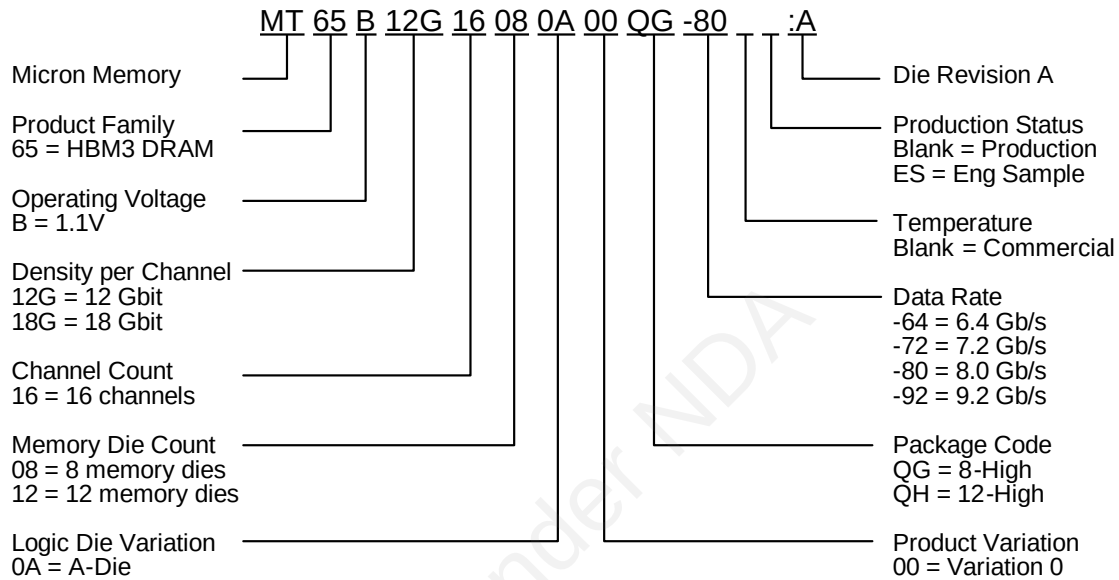
- IEEE1500 standard serial test interface

Options¹

Marking

- | | |
|---|------|
| • Per-channel density | |
| – 12Gb | 12G |
| – 18Gb | 18G |
| • Memory die count | |
| – 8 DRAM layers | 08 |
| – 12 DRAM layers | 12 |
| • Microbump package | |
| – 8-High (24GB) | QG |
| – 12-High (36GB) | QH |
| • Per-signal data rate | |
| – 6.4 Gb/s | -64 |
| – 7.2 Gb/s | -72 |
| – 8.0 Gb/s | -80 |
| – 9.2 Gb/s | -92 |
| • Operating temperature | |
| – Commercial (0°C ≤ T _{OPER} ≤ +105°C) | None |
| • Revision | :A |

Note: 1. Not all options listed can be combined to define an offered product. Use the part catalog search on <http://www.micron.com> for available offerings.


Figure 1: Part Numbering




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Important Notes and Warnings

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Signal Description

The HBM3 DRAM defines 120 signals per channel as described in Table 1 which are all located in the AWORD and DWORD regions of the micro-bump matrix. Additional 27 global signals located in the MIDSTACK region of the micro-bump matrix are described in Table 2, and remaining other signals and supplies are described in Table 3.

Table 1: Per-Channel Signals (AWORD and DWORD)

Symbol	Type	Description
CK[a:p]_t, CK[a:p]_c	Input	Clock: CK_t and CK_c are differential clock inputs. Row and column command and address inputs are latched on the rising and falling edges of CK. All latencies are referenced to the rising edge of CK.
C[a:p][7:0]	Input	Column command and address: the command code, bank and column address for Write and Read operations and the mode register address and code to be loaded with MODE REGISTER SET commands are received on the C[7:0] inputs.
R[a:p][9:0]	Input	Row command and address: the command code, bank and row address for Activate, Precharge and Refresh commands are received on the R[9:0] inputs.
APAR[a:p]	Input	Command and address parity: one command and address parity input per AWORD
ARFU[a:p]	Input	Reserved command and address: sampled along with R, C; and APAR inputs
RA[a:p]	Input	Redundant command and address microbump in AWORD, used with lane repair
AERR[a:p]	Output	Address parity error: one command and address parity error flag per AWORD
DQ[a:p][63:0]	I/O	Data input/output: 64-bit data bus. DQ[31:0] are associated with DWORD0 (PC0) and DQ[63:32] are associated with DWORD1 (PC1)
DBI[a:p][7:0]	I/O	Data bus inversion: DBI0 is associated with DQ[7:0], DBI1 is associated with DQ[15:8], ... , and DBI7 is associated with DQ[63:56].
ECC[a:p][3:0]	I/O	ECC: 4-bit additional data bus for system ECC bits. ECC[1:0] are associated with DWORD0 (PC0) and ECC[3:2] are associated with DWORD1 (PC1)
SEV[a:p][3:0]	I/O	Error severity: SEV[1:0] are associated with DWORD0 (PC0) and SEV[3:2] are associated with DWORD1 (PC1)
DPAR[a:p][1:0]	I/O	Data parity: one data parity I/O per DWORD. DPAR0 is associated with DWORD0 (PC0) and DPAR1 is associated with DWORD1 (PC1)
WDQS[a:p][1:0]_t, WDQS[a:p][1:0]_c	Input	Write Data Strobe: WDQS_t and WDQS_c are differential strobe inputs for latching write data. One WDQS pair per DWORD: WDQS0_t/_c are associated with DWORD0 (PC0) and WDQS1_t/_c are associated with DWORD1 (PC1)
RDQS[a:p][1:0]_t, RDQS[a:p][1:0]_c	Output	Read data strobe: RDQS_t and RDQS_c are differential strobe outputs and sent along with read data. One RDQS pair per DWORD: RDQS0_t/_c are associated with DWORD0 (PC0) and RDQS1_t/_c are associated with DWORD1 (PC1)
RD[a:p][3:0]	I/O	Redundant microbumps in DWORD, used with lane repair. RD[1:0] are associated with DWORD0 (PC0) and RD[3:2] are associated with DWORD1 (PC1);
DERR[a:p][1:0]	Output	Data parity error: one data parity error flag per DWORD. DERR0 is associated with DWORD0 (PC0) and DERR1 is associated with DWORD1 (PC1)

Note: 1. Index [a:p] represents the channel indicator "a" to "p" of the HBM3 DRAM; signal names including the channel indicator are used whenever more than one channel is referenced, as e.g., with the HBM3 ballout. The channel indicator is omitted whenever features and functions common to all channels are described.


Table 2: Global Signals (MIDSTACK)

Symbol	Type	Description
RESET_n	Input	Reset: RESET_n LOW asynchronously initiates a full chip reset of the HBM3 DRAM.
TEMP[1:0]	Output	DRAM temperature report
CATTRIP	Output	DRAM catastrophic temperature report
WRCK	Input	IEEE1500 test port clock
WRST_n	Input	IEEE1500 test port reset
SELECTWIR	Input	IEEE1500 test port instruction register select
SHIFTWIR	Input	IEEE1500 test port shift
CAPTUREWR	Input	IEEE1500 test port capture
UPDATEWR	Input	IEEE1500 test port update
WSI	Input	IEEE1500 test port data in
WSO[a:p]	Output	IEEE1500 test port per-channel data out

- Notes:
- Two micro bumps are associated with each signal in this table except for TEMP[1:0].
 - Unused micro bumps at locations CY-94 and DB-94 (MRFU[1:0]) are "No Connect" (NC) bumps for this device.

Table 3: Other Signals and Supply

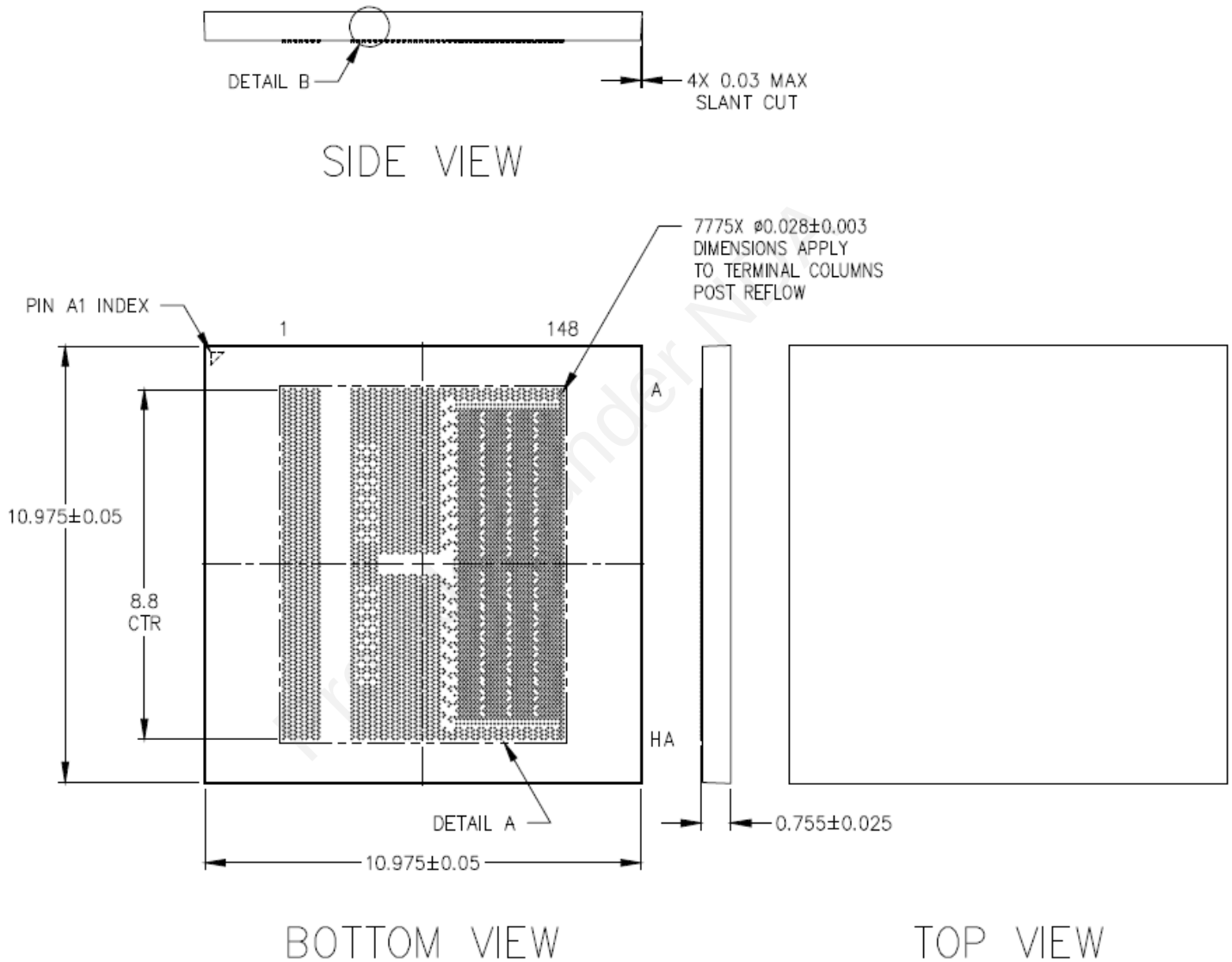
Symbol	Type	Description
DA[39:0] ²	I/O	Direct Access Input/Output: these signals are provided for direct access test mode and must be routed to an external package pin
NC		No connect pad: electrically isolated micro bump
NOBUMP		Depopulated pad: no pad, or reserved as test pad for probing
V _{SS}	Supply	Ground
V _{DDC}	Supply	Core supply voltage
V _{DDQ}	Supply	I/O supply voltage
V _{DDQL}	Supply	Output driver supply voltage
V _{PP}	Supply	Pump voltage

- Notes:
- All power supply microbumps defined in this table and in the micro bump matrix must be present and connected with their respective power nets.
 - Two micro bumps are associated with each DA signal.

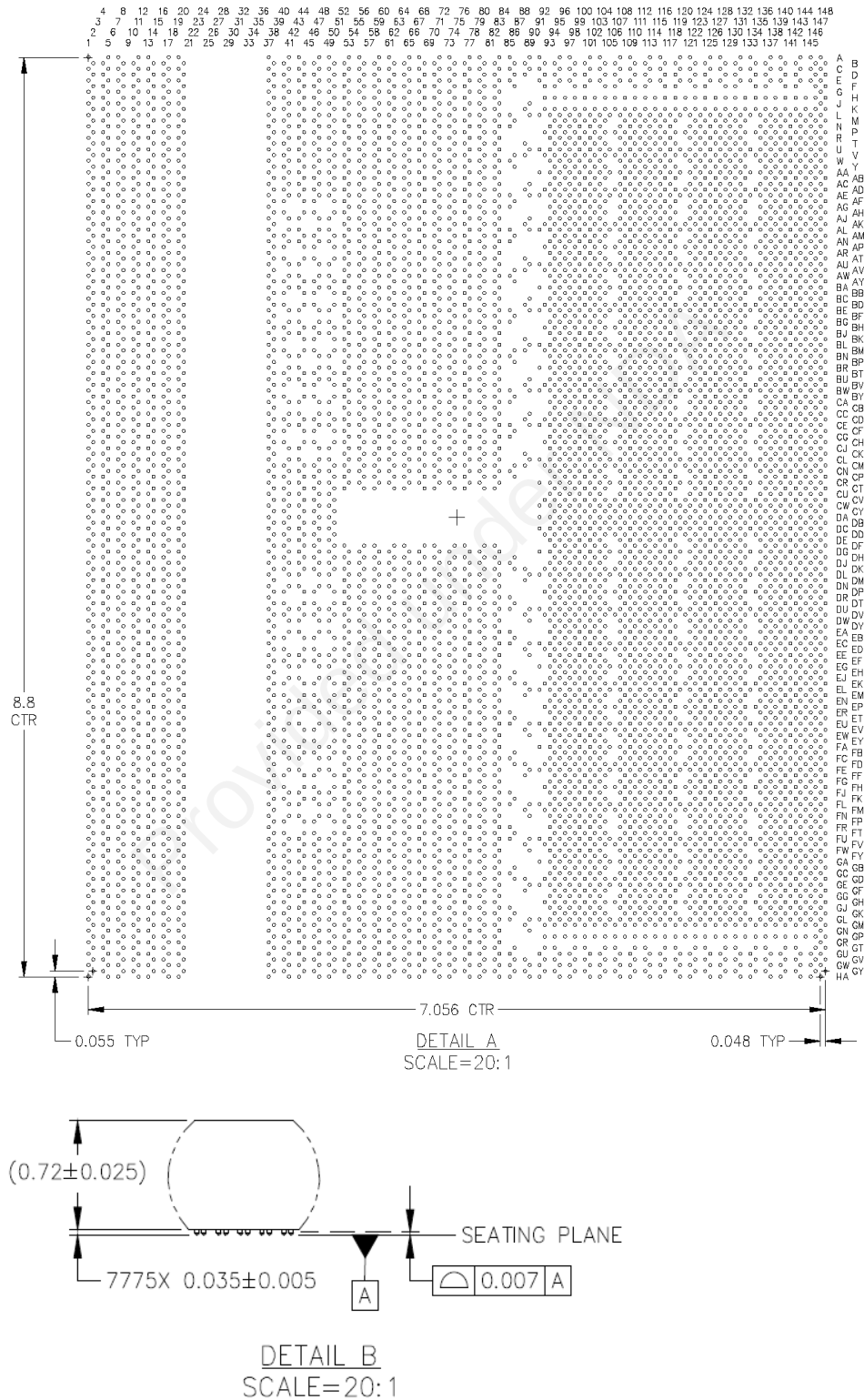


Package Dimensions

Figure 2: Mechanical Outline (Part 1 – Overview)



- Notes:
1. All dimensions in millimeter.
 2. Package X and Y dimensions are measured on the pillar side.
 3. Package height does not include room temperature warpage.


Figure 3: Mechanical Outline (Part 2 – Details A and B)


- Notes:
1. Refer to JEDEC MO-349.
 2. Bump metallurgy: Cu/Ni/Cu/SnAg 15/3/18.



Functional Description

HBM3 DRAM Organization

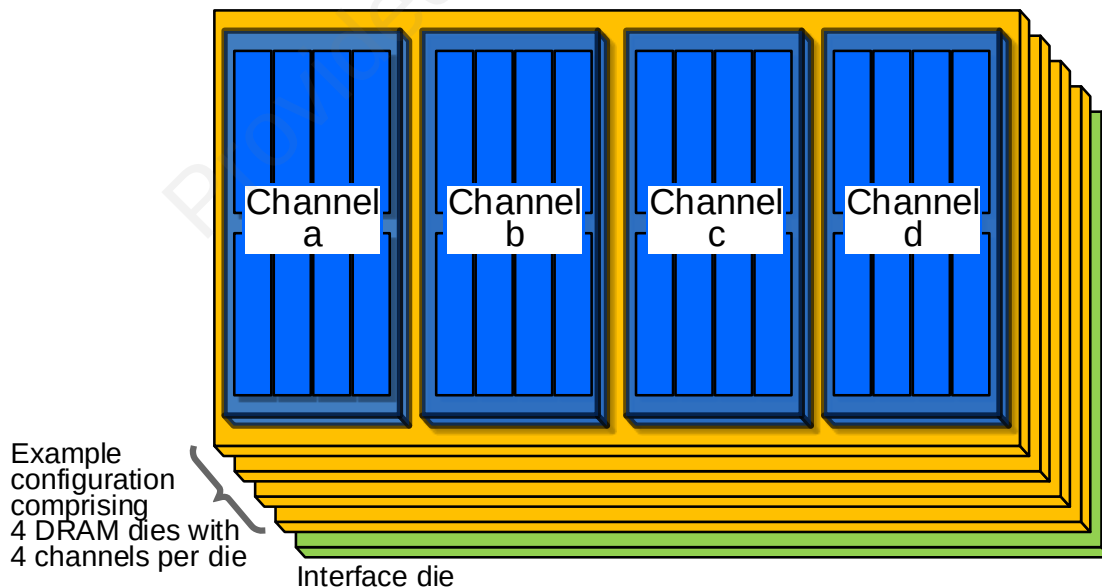
The HBM3 DRAM is optimized for high-bandwidth operation to a stack of multiple DRAM devices across a number of independent interfaces called channels. It is anticipated that each DRAM stack will support up to 16 channels.

Channel Definition

A channel provides access to an independent set of DRAM banks. Requests from one channel do not access data in a different channel. Channels are individually clocked and need not to operate synchronously. Each channel consists of an independent command and data interface comprising 120 I/O signals. Reset and temperature signals are common to all channels. Refer to Table 1 for a list of per-channel signals and to Table 2 for global signals and those associated with the IEEE1500 test access port.

The figure below shows an example stack containing four DRAM dies and one interface die. Each DRAM die comprises four channels and contributes additional capacity and additional channels to the stack, with a total of 16 channels per stack. The division of channels among the DRAM dies within a stack is irrelevant to the memory controller, and the example below is not a required organization. The actual mapping of channels to memory dies is summarized in Table 4.

Figure 4: HBM3 DRAM Stack With Channels



Because each channel is independent, much of this document will describe a single channel. Where signal names are used, families of signals belonging to a given channel will have the suffix a, b, ... , p for channels a through p; if no suffix is present, the signal(s) being described are generic instances of the various per-channel signals.

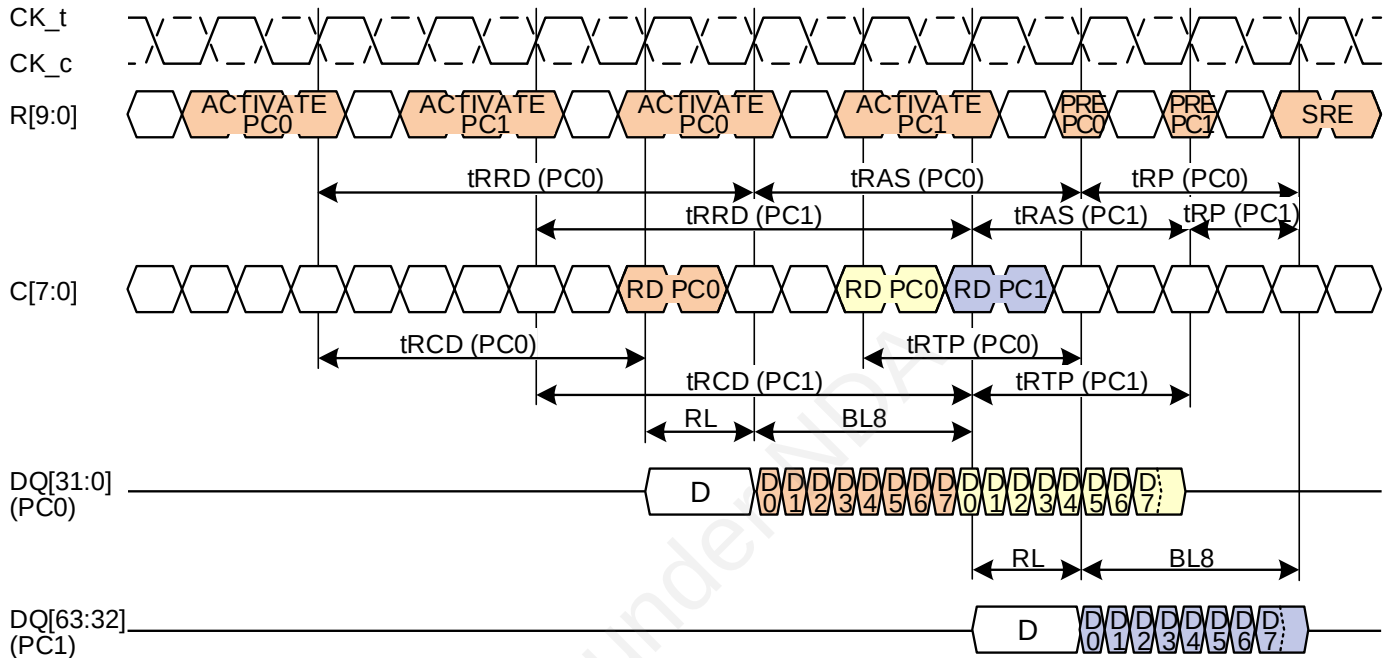

Table 4: Channel to Memory Die Physical Mapping

Die	Channels	PC	8-High	12-High
11	E, F, G, H, M, N, O, P	1		SID[1:0] = 10
10	A, B, C, D, I, J, K, L			
9	E, F, G, H, M, N, O, P	0		
8	A, B, C, D, I, J, K, L			
7	E, F, G, H, M, N, O, P	1	SID = 1	SID[1:0] = 01
6	A, B, C, D, I, J, K, L			
5	E, F, G, H, M, N, O, P	0		
4	A, B, C, D, I, J, K, L			
3	E, F, G, H, M, N, O, P	1	SID = 0	SID[1:0] = 00
2	A, B, C, D, I, J, K, L			
1	E, F, G, H, M, N, O, P	0		
0	A, B, C, D, I, J, K, L			

Pseudo Channel

Each channel is further divided into two pseudo channels (PC). Both pseudo channels operate semi-independently: they share the channel's row and column command bus as well as the CK clock but decode and execute commands individually as illustrated in the figure below: address bit PC directs commands either to pseudo channel 0 (PC0) or pseudo channel 1 (PC1). Power-down and self refresh are common to both pseudo channels. Both pseudo channels also share the channel's mode registers.

Each pseudo channel provides 272-bit prefetch (256 data and 16 meta data bits) per memory read and write access. All I/O signals of DWORD0 are associated with PC0, and all I/O signals of DWORD1 are associated with PC1.


Figure 5: Pseudo Channel Mode Operation


- Notes:
1. PC0 = pseudo channel 0; PC1 = pseudo channel 1.
 2. RL = 1 is shown as an example. Other timings parameters ('MRD, 'RRD, 'RAS, 'RP, 'RTP) are not to scale.
 3. Self refresh entry (SRE) requires that 'RP is satisfied in both pseudo channels.

Array access timings as listed in the table below apply individually for each pseudo channel. For example, an ACTIVATE command to PC0 can be followed by an ACTIVATE command to PC1 as shown in Figure 5. However, a subsequent ACTIVATE command to PC0 can only be done after 'RRD (PC0) has been met. For commands that are common to both pseudo channels (PDE, PDX, SRE, SRX and MRS) it is required that the respective timing conditions are met by both pseudo channels when issuing that command.

Table 5: Array Access Timings Counted Individually Per Pseudo Channel

Array Timing Group	Timing Parameters
Row Access Timings	'RC, 'RAS, 'RCDRD, 'RCDWR, 'RRDL, 'RRDS, 'FAW, 'RTP, 'RP, 'WR
Column Access Timings	'CCDL, 'CCDS, 'CCDR, 'WTRL, 'WTRS, 'RTW
Refresh Timings	'RFCab, 'RFCpb, 'RREFD, 'REFI, 'REFIpb, 'RTW



Addressing

Table 6: Channel Addressing

Parameter	8-High	12-High
Density per channel	12Gb	18Gb
Density per pseudo channel (PC)	6Gb	9Gb
Prefetch size per PC (bits)	256	256
Bank Address	SID, BA[3:0]	SID[1:0] ⁴ , BA[3:0]
Row Address	RA[14:0] ⁵	RA[14:0] ⁵
Column Address	CA[4:0]	CA[4:0]
Page size (per PC)	1KB	1KB
Density code ⁶	1001	1010

- Notes:
1. Prefetch size and page size reflect the effective addressing along with row and column commands. Both do not include the meta data bits.
 2. The burst order is fixed for Reads and Writes, and the HBM3 DRAM does not assign column address bits to distinguish between the eight UI of a BL8 burst. A memory controller may internally assign such column address bits, but those bits are not transmitted to the HBM3 DRAM.
 3. Page Size = $2^{\text{COLBITS}} \times (\text{Prefetch Size} / 8)$; where COLBITS is the number of column address bits.
 4. SID[1:0] = 11_B is invalid.
 5. RA[14:13] = 11_B is invalid.
 6. The density code refers to the encoding of the per-channel density in the DEVICE_ID WDR, bits [27:24].

Bank Groups

The activity within a bank group must be restricted to ensure proper operation of the HBM3 DRAM. The banks within a device are divided into four, eight or twelve bank groups as shown in the table below.

**Table 7: Bank Groups**

Bank Address ¹			Banks	8-High	12-High
SID1	SID0	BA[3:2]			
0	0	00	[3:0]	Group A	Group A
0	0	01	[7:4]	Group B	Group B
0	0	10	[11:8]	Group C	Group C
0	0	11	[15:12]	Group D	Group D
0	1	00	[19:16]	Group E	Group E
0	1	01	[23:20]	Group F	Group F
0	1	10	[27:24]	Group G	Group G
0	1	11	[31:28]	Group H	Group H
1	0	00	[35:32]	N/A	Group I
1	0	01	[39:36]		Group J
1	0	10	[43:40]		Group K
1	0	11	[47:44]		Group L

Note: 1. Address bit SID0 is SID for the 24GB device, and address bit SID1 is Don't Care.

Different timing parameters are specified depending on whether back-to-back accesses are within the same bank group or across bank groups as shown in the table below.

Table 8: Command Sequences Affected by Bank Groups

Command Sequence	Corresponding AC Timing Parameter		Notes
	Accesses To Different Bank Groups	Accesses Within The Same Bank Group	
ACTIVATE to ACTIVATE	^t RRDS	^t RRDL	
WRITE to WRITE	^t CCDS	^t CCDL	
READ to READ	^t CCDS or ^t CCDR	^t CCDL	1
Internal WRITE to READ	^t WTRS	^t WTRL	
READ to PRECHARGE	—	^t RTP	2

Notes: 1. Parameter ^tCCDR replaces parameter ^tCCDS for consecutive READs between different stack IDs (SID).
 2. Parameter ^tRTP applies when READ and PRECHARGE go to the same bank.



Mode Registers

The mode registers define the specific mode of operation for the HBM3 DRAM. Sixteen 8-bit wide mode registers (MR0 to MR15) are defined. MR10 and MR12 are special mode registers and reserved for vendor specific features. Mode registers are common to both pseudo channels (PC0 and PC1). Reprogramming the mode registers does not alter the contents of the memory array.

Mode registers are programmed via the MODE REGISTER SET (MRS) command and retain the stored information until they are reprogrammed, chip reset, or until the device loses power. Mode Register can also be programmed via the IEEE1500 instruction `MODE_REGISTER_DUMP_SET`; this instruction can also be used to retrieve the Mode Register content.

Mode registers must be loaded when all banks are idle and the time `'RDMRS` from a preceding READ command has elapsed. The controller must wait the specified time `'MOD` before initiating any subsequent operations. Violating either of these requirements will result in unspecified operation.

No default states are defined for mode registers except when otherwise noted. Users therefore must fully initialize all mode registers to the desired values upon power-up or after a subsequent chip reset.

When an entire mode register is marked as RFU (“reserved for future use”), then it is considered as not supported by the HBM3 DRAM, and its content is Don’t Care.

Reserved states should not be used, as unknown operation or incompatibility with future versions may result. RFU bits in these registers must be programmed to 0.

**Table 9: Mode Registers**

MR	M[4:0] ¹	OP7	OP6	OP5	OP4	OP3	OP2	OP1	OP0
MR0	00000 _B	Test Mode (TM)	CA Parity (CAPAR)	Write Parity (WPAR)	Read Parity (RPAR)	RFU	TCSR	Write DBI (WDBI)	Read DBI (RDBI)
MR1	00001 _B	Parity Latency (PL)			Write Latency (WL)				
MR2	00010 _B	Read Latency (RL)							
MR3	00011 _B	Write Recovery For Auto Precharge (WR)							
MR4	00100 _B	Activate to Precharge (RAS)							
MR5	00101 _B	RFU				Read to Auto Precharge (RTP)			
MR6	00110 _B	RFU		Pullup Driver Strength			Pulldown Driver Strength		
MR7	00111 _B	CATTRIP	RFU	DWORD MISR Control			RFU	DWORD Read Mux Control	DWORD Loopback
MR8	01000 _B	RFU		RFM Levels (RFML)		WDQS-to-CK Training (WDQS2CK)	ECS error log auto reset (ECSLOG)	Duty Cycle Monitor (DCM)	DA Port Lockout
MR9	01001 _B	ECS error type and address reset (ECSRES)	ECS multi-bit error correction (ECSCEM)	Auto ECS during Self Refresh (ECSSRF)	Auto ECS via REFAb (ECSREF)	ECC Engine Test Vector (ECCVEC)	ECC Engine Test Mode (ECCTM)	Severity Reporting (SEVR)	Meta Data (MD)
MR10	01010 _B	DFE	DFE Strength (DFES)				RFU		Soft Repair Undo/Lock (SPPRUL)
MR11	01011 _B	DCA code for WDQS1 (PC1) (DCAC1)				DCA code for WDQS0 (PC0) (DCAC0)			
MR12	01100 _B	Reserved for vendor specific features				Duty Cycle Training Mode	Reserved for vendor specific features		
MR13	01101 _B	RFU							
MR14	01110 _B	RFU	Reference voltage for AWORD inputs (VREFCA)						RFU
MR15	01111 _B	RFU	Reference voltage for DWORD inputs (VREFD)						RFU

Note: 1. MA4 is a valid mode register address bit that must be set to 0_B for the mode registers defined in this table.

**Table 10: Mode Register 0 (MR0)**

Field	Bits	Description	Notes
Test Mode	OP7	0 _B – Normal operation (default) 1 _B – Test Mode. No functional operation is specified with test mode enabled.	
Command / Address Bus Parity (CAPAR)	OP6	0 _B – Disabled (default) 1 _B – Enabled	1
Data Write Parity (WPAR)	OP5	0 _B – Disabled 1 _B – Enabled	2
Data Read Parity (RPAR)	OP4	0 _B – Disabled 1 _B – Enabled	2
RFU	OP3	0 _B	
Temperature Compensated Self Refresh (TCSR)	OP2	0 _B – Disabled 1 _B – Enabled (default)	
Write DBI (WDBI)	OP1	0 _B – Disabled 1 _B – Enabled	3
Read DBI (RDBI)	OP0	0 _B – Disabled 1 _B – Enabled	3

- Notes:
1. Refer to the Command/Address Parity section for details regarding CA Parity.
 2. Refer to the Data Parity section for details regarding Write Parity and Read Parity.
 3. Refer to the Data Bus Inversion (DBIac) section for details regarding WDBI and RDBI.

Table 11: Mode Register 1 (MR1)

Field	Bits	Description	Notes
Parity Latency (PL)	OP[7:5]	011 _B – 3 nCK All others – Reserved	1
Write Latency (WL)	OP[4:0]	00110 _B – 6 nCK 00111 _B – 7 nCK ... 01101 _B – 13 nCK 01110 _B – 14 nCK All others – Reserved	2

- Notes:
1. Refer to the Data Parity section for details regarding Parity Latency (PL) definition and use with write and read operations. The PL value is fixed by design and cannot be changed.
 2. Refer to the WRITE Command (WR/WRA) section for details regarding the Write Latency (WL) definitions and use.

**Table 12: Mode Register 2 (MR2)**

Field	Bits	Description	Notes
Read Latency (RL)	OP[7:0]	00001110 _B – 14 nCK 00001111 _B – 15 nCK ... 01010101 _B – 85 nCK 01010110 _B – 86 nCK All others – Reserved	1

Note: 1. Refer to the READ Command (RD/RDA) section for details regarding the Read Latency (RL) definitions and use.

Table 13: Mode Register 3 (MR3)

Field	Bits	Description	Notes
Write Recovery for Auto Precharge (WR)	OP[7:0]	00000110 _B – 6 nCK 00000111 _B – 7 nCK ... 00101111 _B – 47 nCK 00110000 _B – 48 nCK All others – Reserved	1

Note: 1. WR must be programmed with a value greater than or equal to $RU\{^tWR/^tCK\}$, where RU stands for round up, tWR is the analog value (see AC Timings) and tCK is the operating clock cycle time.

Table 14: Mode Register 4 (MR4)

Field	Bits	Description	Notes
Activate to Precharge (RAS)	OP[7:0]	00000000 _B	1

Note: 1. This HBM3 DRAM does not support the mode register definition of tRAS in clock cycles. The RAS mode register settings will be ignored.

Table 15: Mode Register 5 (MR5)

Field	Bits	Description	Notes
RFU	OP[7:4]	0000 _B	
Read to Auto Precharge (RTP)	OP[3:0]	0100 _B – 4 nCK 0101 _B – 5 nCK ... 1110 _B – 14 nCK 1111 _B – 15 nCK All others – Reserved	1

Note: 1. RTP must be programmed with a value greater than or equal to $RU\{^tRTP/^tCK\}$, where RU stands for round up, tRTP is the analog value (see AC Timings) and tCK is the operating clock cycle time.

**Table 16: Mode Register 6 (MR6)**

Field	Bits	Description	Notes
RFU	OP[7:6]	00 _B	
Pullup Driver Strength	OP[5:3]	000 _B – 8mA 001 _B – 10mA 010 _B – 12mA (default) 011 _B – 14mA All others – Reserved	1
Pulldown Driver Strength	OP[2:0]	000 _B – 8mA 001 _B – 10mA 010 _B – 12mA (default) 011 _B – 14mA All others – Reserved	1

Note: 1. Refer to the Transmit Driver Current table for the details.

Table 17: Mode Register 7 (MR7)

Field	Bits	Description	Notes
CATTRIP	OP7	0 _B – CATTRIP signal drives a LOW or HIGH depending on CATTRIP sensor output (default) 1 _B – CATTRIP signal drives a static HIGH	1
RFU	OP6	0 _B	
DWORD MISR Control	OP[5:3]	The bits are only evaluated if DWORD Loopback is enabled in OP0 000 _B – Preset: the DWORD MISR is preset to AAAAAAAAAAh, and all DWORD LFSR_COMPARE_STICKY bits are reset to 0. 001 _B – LFSR mode (READ direction) 010 _B – Register mode (WRITE and READ directions): DWORD writes are captured directly into the MISR without compression. The MISR will contain the most recent write data. 011 _B – MISR mode (WRITE direction) 100 _B – LFSR Compare mode (WRITE direction) All others – Reserved	2, 3
RFU	OP2	0 _B	
DWORD Read Mux Control	OP1	The bit is only evaluated with READ commands and if DWORD Loopback is enabled in OP0 0 _B – Return data from DWORD MISR (see OP[5:3]) 1 _B – Return LFSR_COMPARE_STICKY bits (OP[5:3] is ignored)	2, 3
DWORD Loopback	OP0	0 _B – Disabled (default) 1 _B – Enabled. Does not require any row activation. Column addresses associated with WRITE and READ commands are ignored in loopback test mode.	2

Notes: 1. CATTRIP can be asserted to HIGH from any of the channels [a:p] MR7 OP7 bit 7 (logic OR).
2. See Loopback Test Modes for DWORD MISR mode features and use.
3. Refer to Table 18 for details on DWORD MISR operation with WRITE and READ commands.


Table 18: DWORD MISR Read and Write Operations in Loopback Test Mode (MR7 OP0 = 1_B)

MR7 OP1	MR7 OP[5:3]	DWORD MISR Operation ¹		Comments
		WRITE	READ	
0 _B	000 _B (Preset)	Write data are ignored	Read Preset value (clock-like pattern)	Neither Writes nor Reads alter the MISR content
	001 _B (LFSR)		Generate read data from LFSR	Writes do not alter the MISR content
	010 _B (Register)	MISR stores the second half (UI 4 to 7) of the most recent write	Read MISR content (UI 0 to 3 and repeated for UI 4 to 7)	Reads do not alter the MISR content
	011 _B (MISR)	Write data are accumulated in the MISR		
	100 _B (LFSR Compare)	Write data are compared against data generated by the LFSR		
1 _B	XXX _B	Write data are ignored	Read sticky error bits	Neither Writes nor Reads alter the MISR content

Note: 1. See Loopback Test Modes for DWORD MISR and LFSR features and use.

Table 19: Mode Register 8 (MR8)

Field	Bits	Description	Notes
RFU	OP[7:6]	0000 _B	
RFM Levels (RFML)	OP[5:4]	00 _B – Default level (RFM may be required or not) 01 _B – Level A (RFM is required) 10 _B – Level B (RFM is required) 11 _B – Level C (RFM is required)	1
WDQS-to-CK Training (WDQS2CK)	OP3	0 _B – Disabled (default) 1 _B – Enabled	2
ECS error log auto reset (ECSLOG)	OP2	0 _B – Disabled (default) 1 _B – Enabled	
Duty Cycle Monitor (DCM)	OP1	0 _B – Disabled (default) 1 _B – Enabled	
DA Port Lockout	OP0	0 _B – Access to the DA port is enabled (default) 1 _B – Access to the DA port is locked	3

- Notes:
1. RAAIMT, RAAMMT and RAADEC values for the default RFM level and RFM levels A to C can be read via the IEEE1500 DEVICE_ID WDR
 2. Refer to the WDQS-to-CK Alignment Training section for details.
 3. The DA Port lockout bit is defined for channels **a** and **e** only. The bit is RFU for all other channels. Once enabled, the bit can only be cleared by powering off the device. The MODE_REGISTER_DUMP_SET instruction cannot be used to set or clear the bit but allows reading the bit.


Table 20: Mode Register 9 (MR9)

Field	Bits	Description	Notes
ECS error type and address reset (ECSRES)	OP7	0 _B – Maintain the ECS error type and address log (default) 1 _B – Reset the ECS error type and address log (self-clearing)	1
ECS multi-bit error correction (ECSCM)	OP6	0 _B – Correction of multi-bit errors during ECS cycles is disabled 1 _B – Correction of multi-bit errors during ECS cycles is enabled	
Auto ECS during Self Refresh (ECSSRF)	OP5	0 _B – Auto ECS during self refresh mode is disabled (default) 1 _B – Auto ECS during self refresh mode is enabled	2
Auto ECS via REFab (ECSREF)	OP4	0 _B – Auto ECS via REFab command is disabled (default) 1 _B – Auto ECS via REFab command is enabled	2, 3
ECC Engine Test Vector (ECCVEC)	OP3	The bit is only evaluated when ECC Engine Test Mode is enabled in OP2 0 _B – Code-word 0 (CW0): Data '1' means error bit and Data '0' means non-error bit 1 _B – Code-word 1 (CW1): Data '0' means error bit and Data '1' means non-error bit	
ECC Engine Test Mode (ECCTM)	OP2	0 _B – ECC Engine Test Mode is disabled (default) and the SEV signals are High-Z. 1 _B – ECC Engine Test Mode is enabled. The SEV signals drive error severity information during Reads and otherwise are High-Z.	
Severity Reporting (SEVR)	OP1	0 _B – Error severity reporting is disabled and the SEV signals are High-Z 1 _B – Error severity reporting is enabled	4
Meta Data (MD)	OP0	0 _B – ECC signals are disabled. Read and write operations do not include meta data 1 _B – ECC signals are enabled. Read and write operations include meta data transmitted via ECC signals	

- Notes:
1. The bit is self clearing meaning that it automatically returns back to 0_B after the reset function has been issued.
 2. For ECS operation either ECSSRF or ECSREF (or both) must be enabled.
 3. When ECS during REFab is enabled, the host must issue REFab commands at an average rate of 'ECSint.
 4. Input data on SEV signals during write operations will be ignored regardless of the SEVR setting.

**Table 21: Mode Register 10 (MR10)**

Field	Bits	Description	Notes
Decision Feedback Equalization for DWORD inputs (DFE)	OP7	0 _B – Disabled (default) 1 _B – Enabled	
Decision Feedback Equalization Strength for DWORD inputs (DFES)	OP[6:3]	0000 _B – lowest DFE strength ... 1111 _B – highest DFE strength	
RFU	OP[2:1]	00 _B	
Soft Repair Undo / Lock (SPPRUL)	OP0	0 _B – Disabled (default) 1 _B – Enabled	1

Note: 1. Once Soft Repair Undo / Lock has been enabled by setting the SPPRUL bit to 1_B, it must remain enabled; disabling the feature by setting the SPPRUL bit back to 0_B must be followed by a chip reset (RESET_n = LOW).

Table 22: Mode Register 11 (MR11)

Field	Bits	Description	Notes
DCA code for WDQS1 (PC1) (DCAC1)	OP[7:4]	0000 _B – 0 steps (default; no correction) 0001 _B – -1 step 0010 _B – -2 steps ... 0110 _B – -6 steps 0111 _B – -7 steps 1000 _B – Reserved 1001 _B – +1 step 1010 _B – +2 steps ... 1110 _B – +6 steps 1111 _B – +7 steps	
DCA code for WDQS0 (PC0) (DCAC0)	OP[3:0]	0000 _B – 0 steps (default; no correction) 0001 _B – -1 step 0010 _B – -2 steps ... 0110 _B – -6 steps 0111 _B – -7 steps 1000 _B – Reserved 1001 _B – +1 step 1010 _B – +2 steps ... 1110 _B – +6 steps 1111 _B – +7 steps	

Table 23: Mode Register 12 (MR12)

Field	Bits	Description	Notes
Vendor specific	OP[7:4]	0000 _B	
Duty Cycle Training Mode	OP3	0 _B – Disabled (default) 1 _B – Enabled	
Vendor specific	OP[2:0]	000 _B	

**Table 24: Mode Register 13 (MR13)**

Field	Bits	Description	Notes
RFU	OP[7:4]	00000000 _B	

Table 25: Mode Register 14 (MR14)

Field	Bits	Description	Notes
RFU	OP7	0 _B	
Reference voltage for AWORD inputs (VREFCA)	OP[6:1]	000000 _B – $0.18 \times V_{DDQL}$ 000001 _B – $0.19 \times V_{DDQL}$... 011111 _B – $0.49 \times V_{DDQL}$ 100000 _B – $0.50 \times V_{DDQL}$ (default) 100001 _B – $0.51 \times V_{DDQL}$... 111110 _B – $0.80 \times V_{DDQL}$ 111111 _B – $0.81 \times V_{DDQL}$	
RFU	OP0	0 _B	

Note: 1. Refer to the DC & AC Operating Conditions section for the AWORD input receiver voltage level specification.

Table 26: Mode Register 15 (MR15)

Field	Bits	Description	Notes
RFU	OP7	0 _B	
Reference voltage for DWORD0 and DWORD1 inputs (VREFD)	OP[6:1]	000000 _B – $0.18 \times V_{DDQL}$ 000001 _B – $0.19 \times V_{DDQL}$... 011111 _B – $0.49 \times V_{DDQL}$ 100000 _B – $0.50 \times V_{DDQL}$ (default) 100001 _B – $0.51 \times V_{DDQL}$... 111110 _B – $0.80 \times V_{DDQL}$ 111111 _B – $0.81 \times V_{DDQL}$	
RFU	OP0	0 _B	

Note: 1. Refer to the DC & AC Operating Conditions section for the DWORD input receiver voltage level specification.



Initialization

To power-up and initialize the HBM3 device into functional operation the sequence in the HBM3 Power-Up and Initialization Sequence section must be followed.

At any time after the power-up initialization, the HBM3 device may be reset using the sequence in the Initialization Sequence with Stable Power section. All IEEE 1500 port instructions may be used within the initialization sequences, as described in the Initialization Sequence For Use Of IEEE 1500 Instructions Including Lane Repairs section.

The interactions between HBM functional reset and the IEEE 1500 port reset are as follows:

- Functional reset requires that the IEEE 1500 port also be reset.
- The IEEE 1500 port can be reset at any time without impacting normal operation.
- The IEEE 1500 port may be brought out of reset and a limited set of instructions may be used after a minimum time t_{INIT3} after RESET_n has been deasserted. See section Initialization Sequence For Use Of IEEE 1500 Instructions Including Lane Repairs
- If not needed, the IEEE 1500 port may be left in reset (WRST_n = LOW) during normal operation.

HBM3 Power-Up and Initialization Sequence

The HBM3 DRAM must be powered up and initialized in a predefined manner following the sequence below. Refer also to Figure 6.

1. Apply power to the V_{DDC} , V_{DDQ} , V_{DDQL} and V_{PP} supplies following the requirements in the Power Ramp Conditions table. V_{PP} must ramp at the same time or earlier than V_{DDC} and V_{DDQ} . V_{DDC} and V_{DDQ} must ramp simultaneously and at the same time or earlier than V_{DDQL} . During power supply ramp time t_{INIT0} , RESET_n, WRST_n and all other input signals may be in an undefined state (driven LOW or HIGH, or High-Z). After T_a is reached, V_{DDC} and V_{DDQ} must be greater than $V_{DDQL} - 200\text{mV}$. V_{DDC} and V_{DDQ} must be powered within 1 minute after power is applied to V_{PP} , and the total time for which V_{PP} is powered and V_{DDC} and V_{DDQ} are unpowered should not exceed 24 cumulative hours.
2. RESET_n and WRST_n must be driven LOW (below $0.2 \times V_{DDQ}$) before or at the same time when t_{INIT0} expires. All other input signals may be in an undefined state (driven LOW or HIGH, or High-Z) at this point. RESET_n must be maintained LOW for a minimum of t_{INIT1} time with stable power. However, RESET_n must go HIGH within 1 minute after V_{DDC} and V_{DDQ} have ramped and reached stable levels. After t_{INIT6} time has elapsed, the HBM3 device drives RDQS_t and RDQS_c to LOW and HIGH static levels, respectively, and AERR, DERR and CATTRIP signals LOW.
3. A time t_{INIT2} before RESET_n is pulled HIGH, CK_t and CK_c must be driven to static LOW and HIGH levels, respectively.
4. After RESET_n is driven HIGH, R[3:0] must be driven to PDE state (HIGH, LOW, HIGH, LOW) and C[2:0] must be driven to CNOP state (HIGH, HIGH, HIGH) a time t_{INIT7} before the CK clock is started. R[9:4] and C[7:3] are allowed to remain in an undefined state. The HBM3 device resets into the precharged power-down state. During t_{INIT3} , the HBM3 device will read and apply internal fuse configuration data and perform I/O driver impedance calibration. At the same time the WRST_n signal may be optionally driven HIGH to enable a subset of the IEEE 1500 instructions (see Initialization Sequence For Use Of IEEE1500 Instructions Including Lane Repairs and Channel Disable and IEEE Standard 1500 sections). In that case, all other



IEEE1500 inputs (WRCK, SELECTWIR, SHIFTWR, CAPTUREWR, UPDATEWR, WSI) must be driven per IEEE1500 Port Input and Output Timings figure at time $t_{W\text{INIT}2}$ before WRST_n is pulled HIGH (see IEEE1500 Test Port AC Timing Parameters). The CATTRIP output is forced to LOW from the end of $t_{\text{INIT}6}$ to the end of $t_{\text{INIT}3}$ and is valid after $t_{\text{INIT}3}$.

5. While $\text{R}[3:0]$ and $\text{C}[2:0]$ remain driven to PDE and CNOP states as defined in step 4, the CK clock shall be started and stable clocks shall be maintained for minimum of $t_{\text{INIT}4}$ time before $\text{R}[3:0]$ is driven HIGH. Since $\text{R}0$ is a synchronous signal, the corresponding setup time to clock (t_{IS}) must be met. Also, RNOP and CNOP commands must be registered with $t_{\text{IS}}/t_{\text{IH}}$ satisfied. After $\text{R}[3:0]$ are registered HIGH, a minimum $t_{\text{INIT}5}$ time must be satisfied before issuing a first MRS command. At or before the time that $\text{R}[3:0]$ are driven HIGH, WDQS_t and WDQS_c must be driven to LOW and HIGH static levels, respectively. A stable CK clock shall be maintained except when a channel is in power-down or self refresh state. See Power-Down and Self Refresh sections for conditions about stopping and re-starting the CK clock.

A stable CK clock must be applied within 3 seconds after RESET_n was pulled HIGH. For debug purposes, the 1 minute and 3 second delay limits may be extended to 6 minutes each provided the device is operated in this debug mode for no more than 24 cumulative hours.

6. Issue all MRS commands to configure the HBM3 DRAM appropriately for the application setting.
7. The HBM3 DRAM is now ready for normal operation.

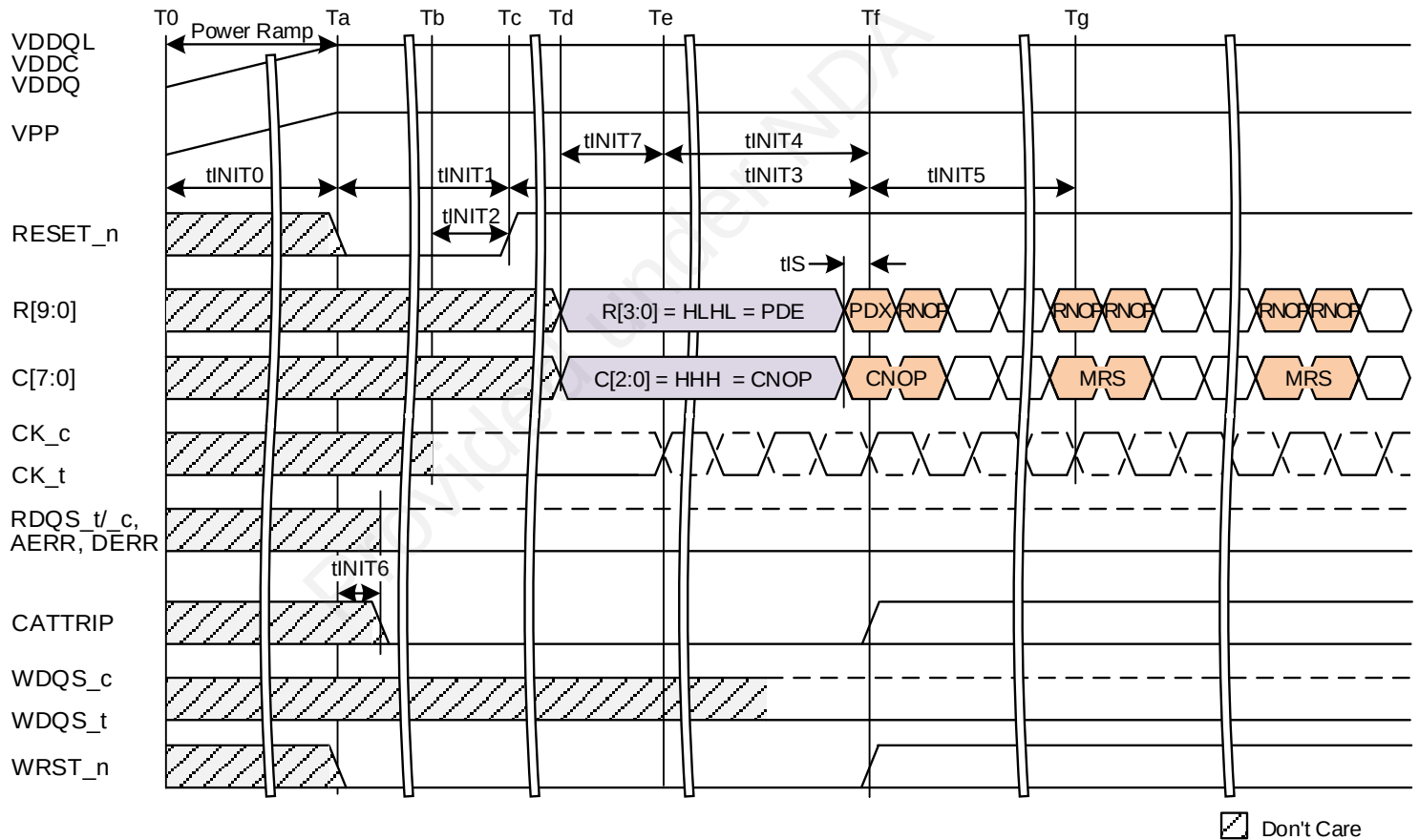
Table 27: Initialization Timing Parameters

Symbol	Parameter	Min	Max	Unit
$t_{\text{INIT}0}$	Power supply ramp time	0.01	200	ms
$t_{\text{INIT}1}$	RESET_n signal LOW time at power-up (after stable power)	200	—	μs
$t_{\text{INIT}2}$	Time for driving CK_t and CK_c to static LOW and HIGH levels before RESET_n deassertion	10	—	ns
$t_{\text{INIT}3}$	Duration of precharged power-down state and WRST_n LOW time after RESET_n deassertion	4	—	ms
$t_{\text{INIT}4}$	Stable CK clock before $\text{R}[3:0]$ going HIGH	10	—	nCK
$t_{\text{INIT}5}$	Idle time before first MRS command	200	—	ns
$t_{\text{INIT}6}$	RDQS_t , RDQS_c driven valid and AERR, DERR and CATTRIP driven LOW after RESET_n assertion	—	100	ns
$t_{\text{INIT}7}$	$\text{R}[3:0] = \text{HLHL}$ (PDE state) and $\text{C}[2:0] = \text{HHH}$ (CNOP state) before enabling the CK clock	2		nCK
$t_{\text{PW_RESET}}$	RESET_n signal LOW time with stable power	1	—	μs


Table 28: Power Ramp Conditions

After	Application Condition
T0	V _{PP} must be greater than V _{DDC} , V _{DDQ} . V _{DDC} and V _{DDQ} must be greater than V _{DDQL} -200mV

- Notes:
1. T0 is the point when any power supply first reaches 300mV.
 2. Voltage ramp conditions apply between time T0 and controlled power-off.
 3. Ta is the point at which all supply voltages are within their defined ranges.
 4. Power ramp duration t_{INIT0} (Ta - T0) must not exceed 200ms.

Figure 6: HBM3 Power-up and Initialization


Note: 1. CATTRIP is valid after t_{INIT3} .

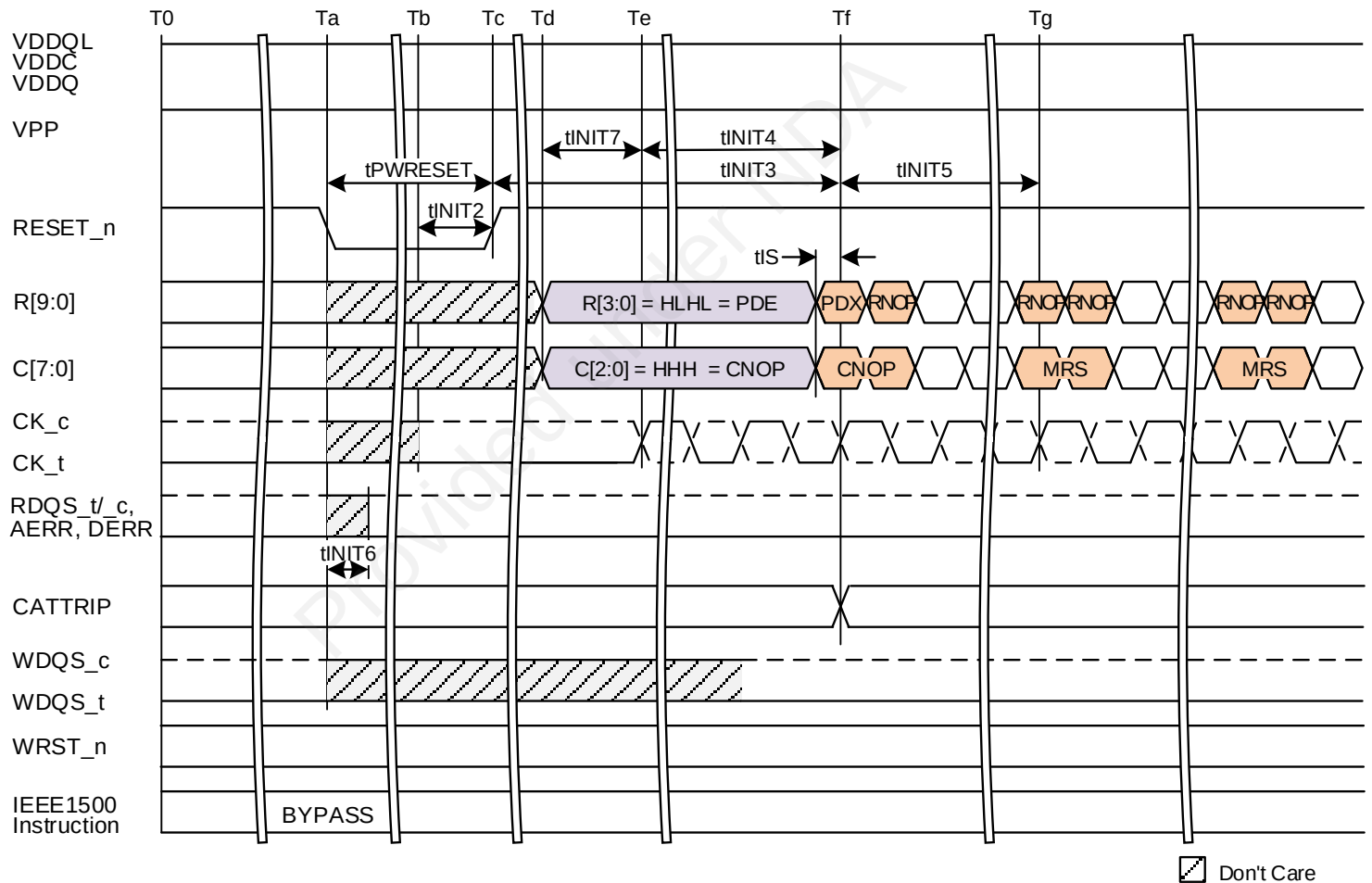
Initialization Sequence with Stable Power

The following sequence must be satisfied to perform a functional reset when power is kept stable at the HBM3 DRAM. See Figure 7.

1. RESET_n must be driven LOW anytime when a functional reset is needed. All other input signals may be in an undefined state (driven LOW or HIGH, or High-Z) at this point except WRST_n and CATTRIP as shown in Figure 7. RESET_n must be maintained LOW for a minimum of t_{PW_RESET} .
2. A time t_{INIT2} before RESET_n is pulled HIGH, CK_t and CK_c must be driven to static LOW and HIGH levels, respectively.



3. R[3:0] must be driven to PDE state (HIGH, LOW, HIGH, LOW) and C[2:0] must be driven to CNOP state (HIGH, HIGH, HIGH) a time t_{INIT7} before the CK clock is started. R[9:4] and C[7:3] are allowed to remain in an undefined state. Alternately, the IEEE1500 port HBM_RESET instruction may be used to perform a re-initialization, with RESET_n continuing to be driven HIGH. Refer to the HBM_RESET Instruction section.
4. Follow steps 5 to 7 as described in section HBM3 Power-Up and Initialization Sequence. Note that the CATTRIP output is not cleared by a functional reset.

Figure 7: HBM3 Reset and Initialization Sequence with Stable Power


Note: 1. CATTRIP will retain its value from before reset until t_{INIT3} has elapsed.



Initialization Sequence For Use Of IEEE1500 Instructions Including Lane Repairs and Channel Disable

All IEEE1500 port instructions are allowed after t_{INIT3} without completing the full initialization sequence.

Figure 8 illustrates the usage of the EXTEST and SOFT_LANE_REPAIR instructions and Figure 9 the usage of the CHANNEL_DISABLE instruction within the initialization sequence. These sequences may be applied as part of the power-up or stable-power initialization sequences to check for and correct failed connections on the row and column command buses, which must be correctly driven to RNOP and CNOP as part of the above initialization sequences. DWORD lane repairs are also allowed. The sequence may also be used to disable a faulty channel before normal operation mode is entered.

1. At time T_a , RESET_n and WRST_n must be driven LOW.
2. After a minimum time t_{INIT1} (if during an initial power-up sequence) or after 'PW_RESET (if during a stable power initialization sequence) RESET_n shall be driven HIGH. t_{INIT2} must also be met.
3. After t_{INIT3} , WRST_n is driven HIGH. IEEE1500 port instructions may now be used. (Note that the WRST_n LOW pulse width t_{WRSTL} is met since t_{WRSTL} is less than the t_{INIT1} or 'PW_RESET). Refer to IEEE1500 Port I/O Timings for timing requirements for operating the IEEE1500 port, including 'SWRST. At this point, a defective channel may be disabled; also, defective lane detection and soft lane repair may be executed. EXTEST operations may be applied to identify lanes needing repair. If lane repair is needed, SOFT_LANE_REPAIR and HARD_LANE_REPAIR instructions can be issued after another RESET_n toggle, which is required after EXTEST instruction operation. A BYPASS instruction should be applied to return all HBM3 signals to their normal functional mode after SOFT_LANE_REPAIR operations. Alternately, WRST_n may be driven LOW.
4. The initialization sequence may then continue per steps 3 to 7 of the HBM Power-Up and Initialization Sequence as needed.

During the t_{INIT3} period before WRST_n is driven HIGH, the HBM3 DRAM executes various internal configuration operations, including applying hard lane repairs based on previously fused data. Executing soft lane repair instructions after t_{INIT3} overwrites any previously programmed hard lane repair data. It is suggested that the hard lane repair data is read from the HBM3 DRAM and merged in any new lane repairs before applying the new soft lane repair operations. Any applicable IEEE1500 port instructions timings must be met before continuing to time point T_h , such as 'SLREP if a SOFT_LANE_REPAIR instruction has been applied.

The EXTEST instructions are not required before applying the soft lane repair(s). Previously determined needed lane repairs may be applied as part of each initialization event.

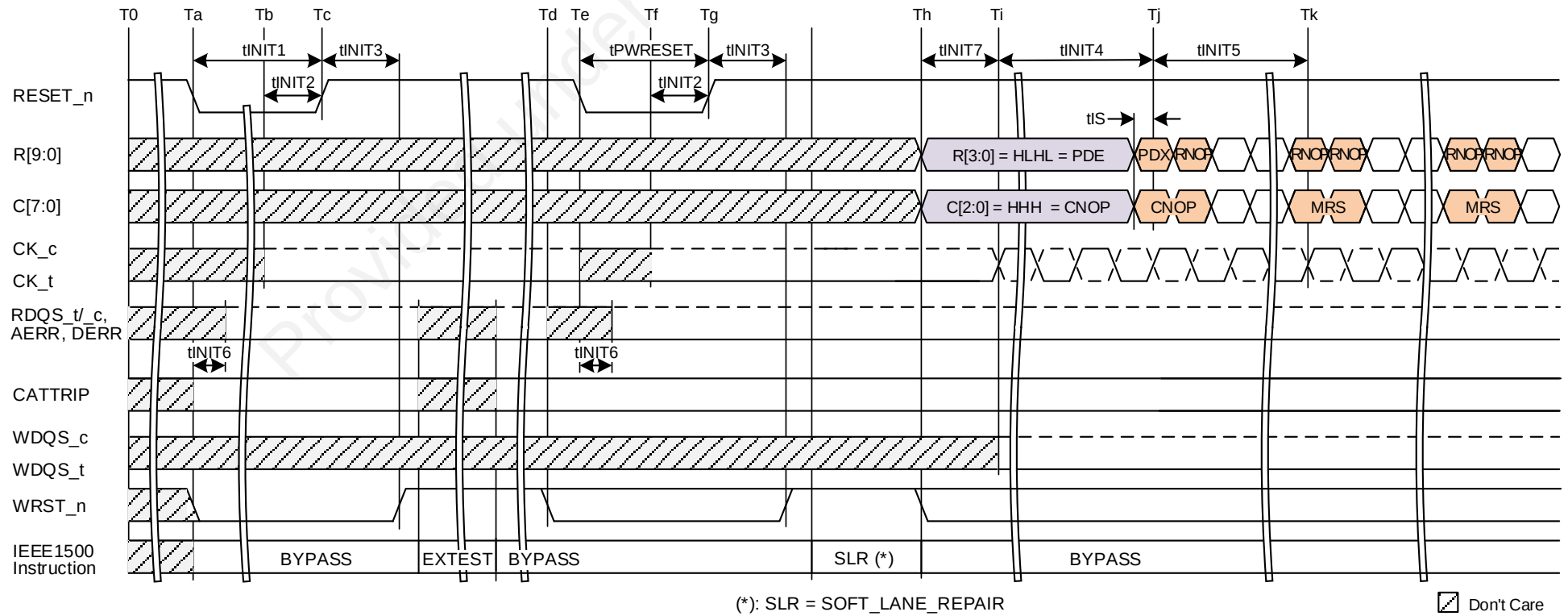
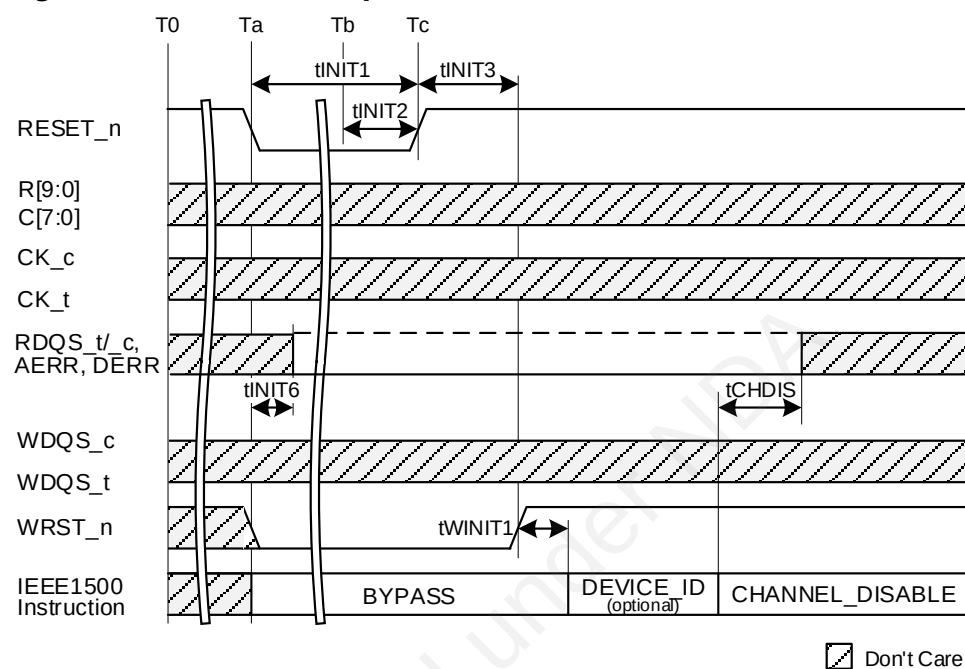

Figure 8: Initialization Sequence with Lane Repairs



Figure 9: Initialization Sequence with Channel Disable


Note: 1. A disabled channel will turn off all AWORD and DWORD input and output buffers including CK_t/CK_c and WDQS_t/WDQS_c inputs, thus allowing all external signals to float. The CK clock is allowed to be High-Z throughout this initialization sequence.

Power-Off Sequence

For a controlled power-off, the following conditions must be met: while powering off, all input levels must be between V_{SS} and V_{DDQ} or V_{DDQL} during voltage ramp to avoid latch-up.

Table 29: Power Supply Conditions

Between	Application Condition
Tx and Tz	V_{PP} must be greater than V_{DDC} , V_{DDQ} . V_{DDC} and V_{DDQ} must be greater than $V_{DDQL} - 200\text{mV}$

Notes: 1. Tx is the point when any power supply drops below the minimum value specified.
 2. Tz is the point where all power supplies are below 300mV. After Tz, the HBM3 device is powered off.

Operation

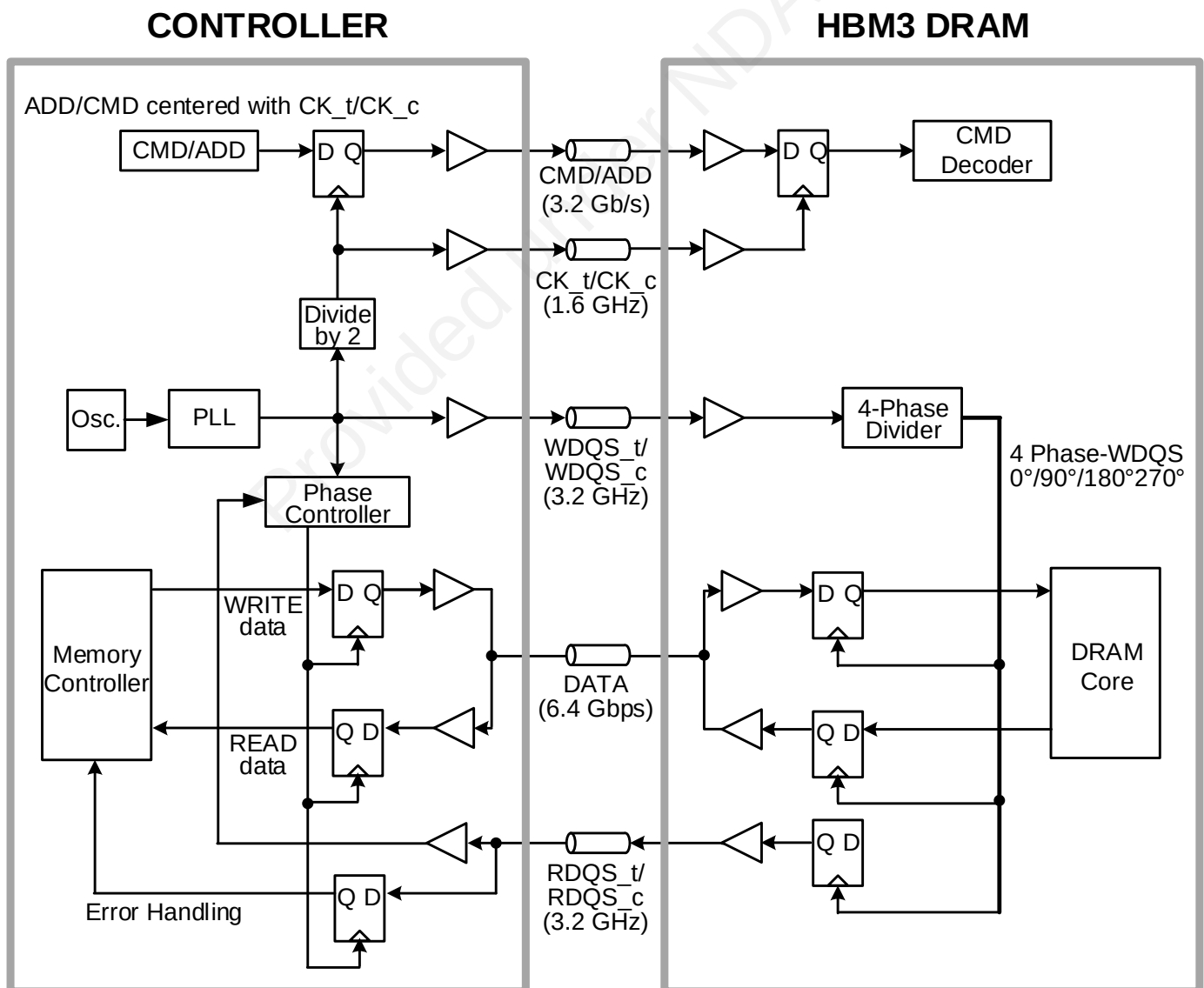
Clocking

The HBM3 DRAM captures commands and addresses on the row and column buses using a differential clock CK_t/CK_c. Both buses operate at double data rate (DDR).

The HBM DRAM has uni-directional differential write strobes (WDQS_t/WDQS_c) and read strobes (RDQS_t/RDQS_c).per 32 DQ (DWORD). The data bus operates at double data rate (DDR).

A high-level block diagram of an example clocking scheme is shown in Figure 10.

Figure 10: High Level Block Diagram of Example HBM3 Clocking Scheme



Note: 1. The frequency and data rate values shown are examples, provided to illustrate the relationship between these. Refer to the Part Numbering section for the actual supported speed bins.

The frequency of WDQS and RDQS strobes is nominally 2x the CK clock frequency. CK clock and WDQS are generated from the same clock source. The HBM3 DRAM generates the RDQS strobe from the incoming WDQS strobe.

The WDQS frequency is internally divided by 2 to match the CK clock frequency. Four internal WDQS phases (0°, 90°, 180° and 270°) are generated for the 4 UI of write or read data per CK clock cycle as shown in Figure 11. The internal WDQS divider is reset at exit from reset state, power-down exit and self refresh exit.

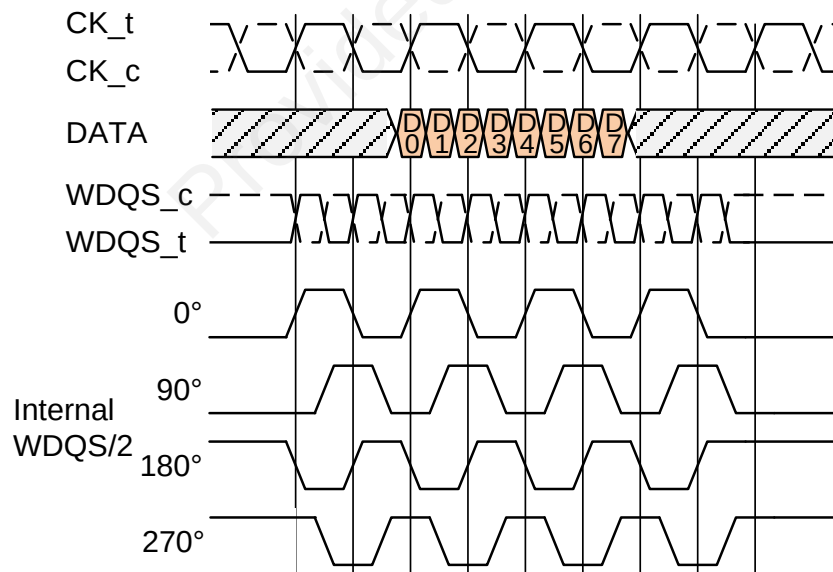
The sum of the WDQS pulses including pre- and postambles must be an even number for all write and read transactions such that the internal WDQS divider state is back at its reset state once the transaction has finished. With that, no specific synchronization between CK and WDQS is required.

During idle states, WDQS_t and WDQS_c must be driven to LOW and HIGH static levels, respectively, and the HBM3 DRAM drives RDQS_t and RDQS_c to LOW and HIGH static levels, respectively. WDQS and RDQS preamble pulses are provided to reduce inter-symbol interference (ISI) at the receiver

During WDQS-to-CK alignment training, the WDQS phase shall be shifted against the CK clock phase to meet the 'DQSS timing. See WDQS-to-CK Alignment Training for details.

During WRITE data training for the unmatched DQ-to-WDQS path, the write data shall be shifted against the WDQS strobe while the CK and WDQS phases shall be kept stable.

Figure 11: WDQS-to-CK Alignment



The following nomenclature is being used throughout this specification:

- a rising CK (or WDQS, RDQS) edge is defined as the crossing of the rising edge of CK_t (or WDQS_t, RDQS_t) and the falling edge of CK_c (or WDQS_c, RDQS_c);
- a falling CK (or WDQS, RDQS) edge is defined as the crossing of the falling edge of CK_t (or WDQS_t, RDQS_t) and the rising edge of CK_c (or WDQS_c, RDQS_c).

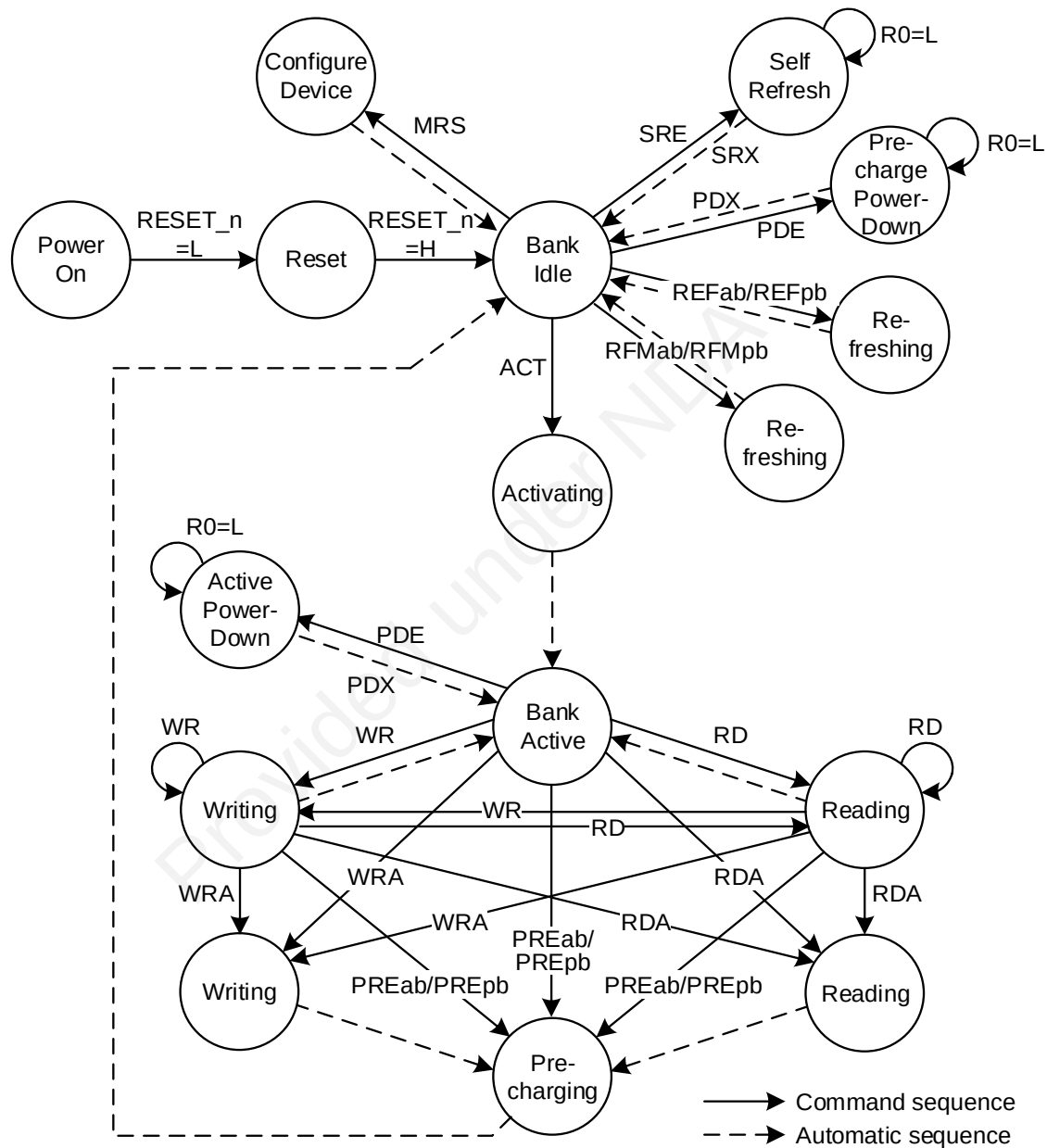


State Diagram

The state diagram provides a simplified illustration of the allowed state transitions and the related commands to control them. The following operations are either not shown or not fully shown in the diagram:

- power-up initialization and initialization with stable power;
- state transitions involving more than one bank;
- interactions from the use of IEEE1500 instructions to load mode registers or execute test functions;
- the immediate transition from any state to reset state by asserting RESET_n LOW or by loading the IEEE1500 instructions HBM_RESET;
- ECS and ECC engine test mode operation.
- Interface trainings that do not affect the bank state like DCA, DCM, WDQS-to-CK alignment training and loopback test modes.

For a complete description of the device behavior, use the information provided in the simplified state diagram along with the command truth tables and AC timing specifications.

Figure 12: Simplified State Diagram


Code	Command	Code	Command	Code	Command
ACT	ACTIVATE	WRA	WRITE w/ AP	MRS	MODE REGISTER SET
PREab	PRECHARGE ALL	REFab	REFRESH	PDE	POWER-DOWN ENTRY
PREpb	PRECHARGE	REFpb	PER-BANK REFRESH	PDX	POWER-DOWN EXIT
RD	READ	RFMab	REFRESH MANAGEMENT	SRE	SELF REFRESH ENTRY
RDA	READ w/ AP	RFMpb	PER-BANK REFRESH MANAGEMENT	SRX	SELF REFRESH EXIT
WR	WRITE				

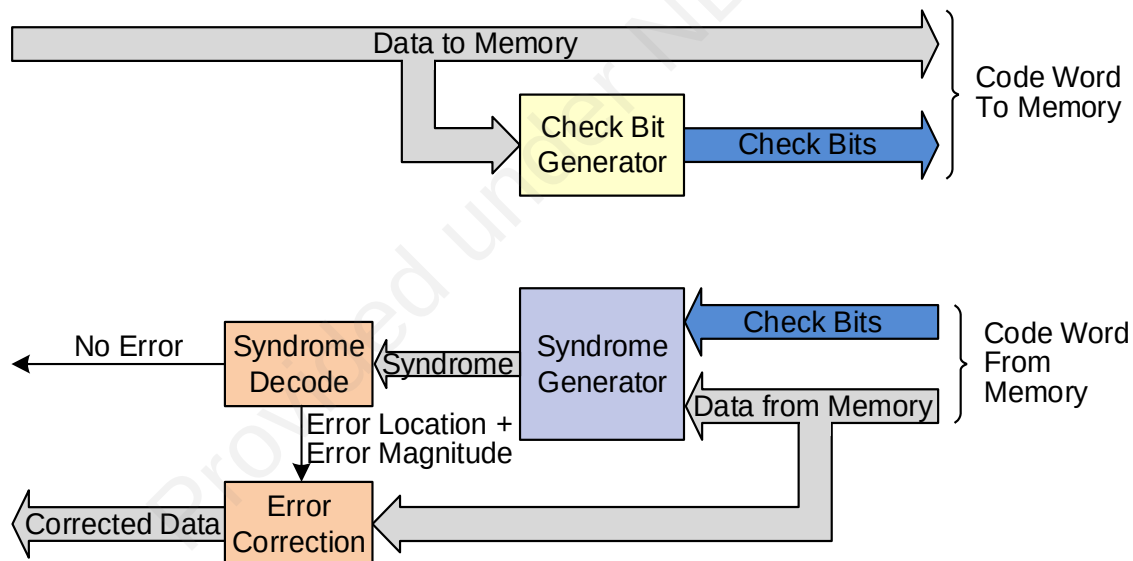
Error Correction Code (ECC)

The HBM3 DRAM uses a symbol-based on-die error correction code (ECC), read/write meta-data bits, an error scrubbing mechanism, an error transparency protocol, interface transmission parity, and fault isolation limits to achieve a high level of system RAS.

32 bits of on-die ECC bits enable a single symbol correction scheme over 272-bit user data (256 bits of data and 16 bits of meta data). These 272 bits of user data are transmitted on write and read transactions as a burst of eight via 32 DQ signals for data and 2 ECC signals for meta data. User data and on-die ECC bits are stored together in the array as a 304-bit code-word. The on-die ECC cannot be disabled.

An overview of the HBM3 on-die ECC engine is shown in Figure 13.

Figure 13: On-Die ECC Overview



On writes the HBM3 DRAM computes the check bits and writes the data and check bits to the array. On reads the HBM3 DRAM corrects all errors up to at least the single symbol size before returning the data to the memory controller. The HBM3 DRAM does not write the corrected data back to the memory array during a read cycle.

The transmission of meta data on the ECC signals can be disabled via the MD bit in MR9 OP0. When MD is off, the ECC input receivers and output buffers are disabled. The meta data written to the array are not specified, and any meta data read from the array are not transmitted. The DRAM can only guarantee valid array meta data if written while the MD function is enabled. The basic array write and read operations and the ECC engine's operation are not affected by the MD bit.

Fault Isolation

Fault isolation is the management of errors caused by various faults to be isolated within certain boundaries regardless of the on-die-ECC operation. The fault isolation boundaries have been chosen in accordance with the ECC symbol size to maximize the correction capability of most common multi-bit faults.



Error Correction Transparency

The HBM3 device provides transparency of actions by the on-die ECC engine. The severity of an error denotes the outcome of the on-die ECC processing during a read operation. The error severity information is conveyed on the SEV signals along with the read data, if this error reporting is enabled by the SEVR bit in MR9 OP1. The severity is encoded for each read transaction as shown in Table 30. The SEV signals will be High-Z when the error severity reporting is disabled.

A read transaction with a corrected single bit error (CEs) will be reported as error-free (NE) unless the error count (ERRCNT) maintained by the Error Check and Scrub (ECS) engine has exceeded the error threshold (ERRTH). Refer also to Figure 17 regarding error signaling on SEV outputs.

Table 30: Error Severity Encoding on SEV Signals for Reads

Severity	Code	SEV3 / SEV1 (Burst Positions 0 to 7)	SEV2 / SEV0 (Burst Positions 0 to 7)
No error	NE	0000 0000	0000 0000
Corrected single bit error	CEs	0000 0000	0000 1111
Corrected Multi-Bit Error	CEm	0000 1111	0000 1111
Uncorrectable error	UE	0000 1111	0000 0000

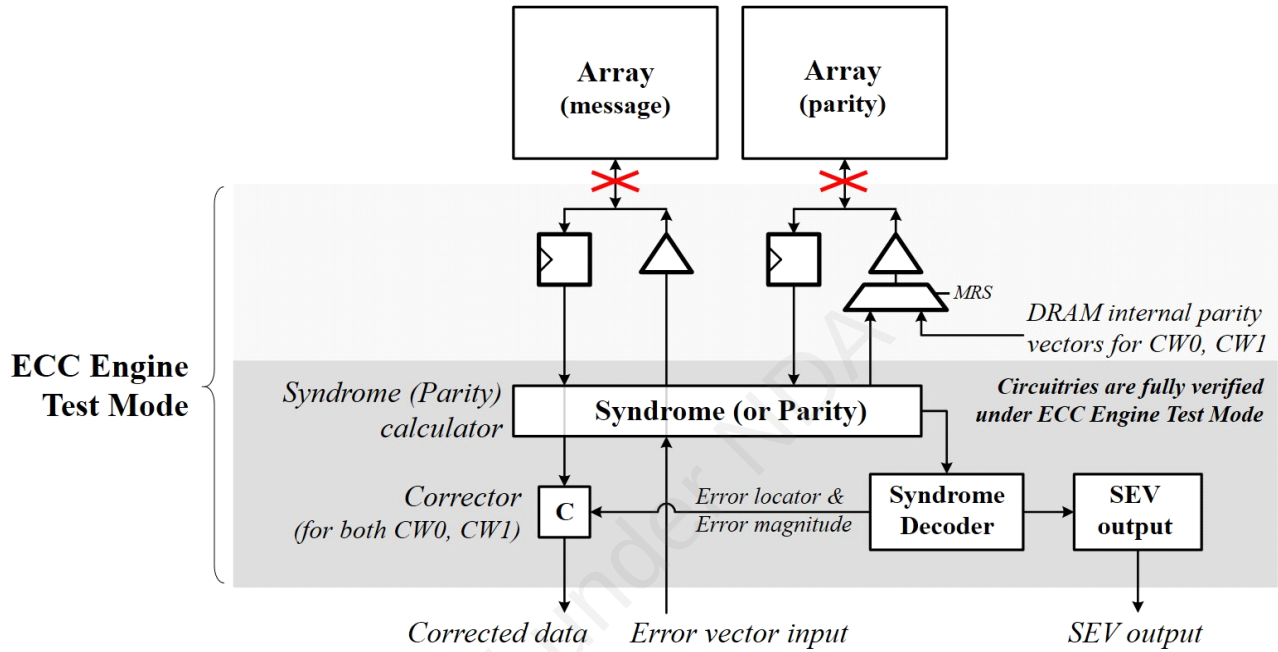
ECC Engine Test Mode

The HBM3 DRAM provides a test mode for the ECC error correction engine that enables a health check of this logic without accessing the memory array as illustrated in Figure 14. The test mode is controlled by bits in MR9 OP[3:2].

The test mode utilizes the regular data interface (DQ, ECC) and regular WRITE and READ commands to write an error vector to a DRAM internal register and read back the corrected error vector from that register and associated error severity information.

While in this test mode the ECC engine will not calculate check bits (OD-ECC bits) from the write data but instead assume one of two hard-coded check bit words CW0 (all 0s) or CW1 (all 1s) as selected by bit in MR9 OP3. On a READ the ECC engine will correct the data word based on the selected check bit word.

WRITE and READ commands can be issued without preceding ACTIVATE command, leaving the memory array in bank idle state. No commands other than RNOP, CNOP, WRITE, READ and MRS to disable this test mode are allowed while this test mode is active.

Figure 14: Block Diagram of On-Die ECC Engine and Data Paths for ECC Engine Test Mode

Table 31: Example of Error Vectors and Corresponding Severity

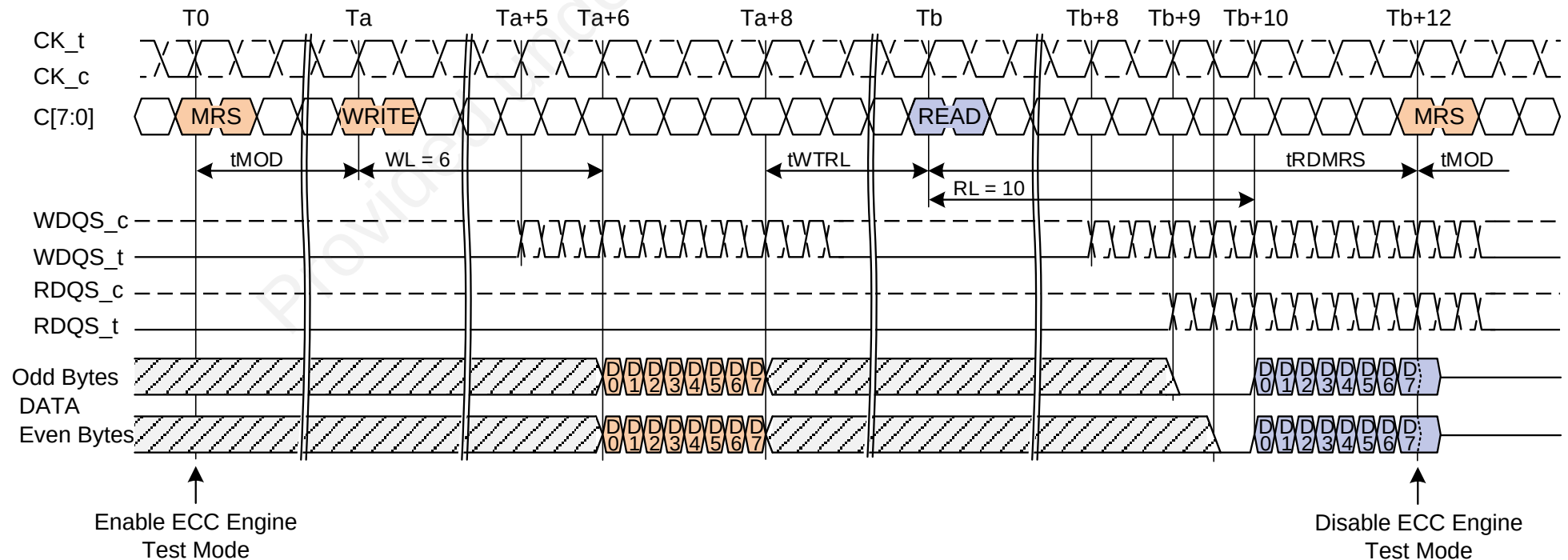
Case	Error Vector Pattern (MR9 OP3)	Error Input Vector [271:0] (WRITE data)	Error Output Vector [271:0] (READ data)	Error Severity (SEV [1:0])	Notes
No error	CW0	0000 ... 00000000 _B	0000 ... 00000000 _B	NE	1
	CW1	1111 ... 11111111 _B	1111 ... 11111111 _B	NE	2
Corrected single bit error	CW0	1000 ... 00000000 _B ... 0000 ... 00000001 _B	0000 ... 00000000 _B ... 0000 ... 00000000 _B	CEs	1, 3
	CW1	0111 ... 11111111 _B ... 1111 ... 11111110 _B	1111 ... 11111111 _B ... 1111 ... 11111111 _B	CEs	2, 3
Corrected multi-bit error	CW0	0000 ... 00001111 _B ... 1111 ... 00000000 _B	0000 ... 00000000 _B ... 0000 ... 00000000 _B	CEm	1, 4
	CW1	1111 ... 11110000 _B ... 0000 .. 11111111 _B	1111 ... 11111111 _B ... 1111 ... 11111111 _B	CEm	2, 4
Uncorrectable error	CW0 or CW1	None of the above	Not specified	UE	

- Notes:
1. Parity vector CW0 selected: 1_B = error bit, 0_B = error-free bit
 2. Parity vector CW1 selected: 0_B = error bit, 1_B = error-free bit.
 3. No error threshold for CEs is active in this test mode and all CEs will be flagged.
 4. The correction capability for multi-bit errors is limited to one symbol.



A basic command sequence of this ECC engine test is illustrated in the figure below.

Figure 15: ECC Engine Test Mode



- Notes:
1. WL = 6 and RL = 10 are shown as examples.
 2. DATA = DQ[31:0], DBI[3:0], ECC[1:0] for PC0, and DQ[63:32], DBI[7:4], ECC[3:2] for PC1.
 3. D0 ... D7 = BL8 data burst.
 4. WDQS_t/_c is WDQS0_t/_c for PC0, and WDQS1_t/_c for PC1. RDQS_t/_c is RDQS0_t/_c for PC0, and RDQS1_t/_c for PC1.
 5. 'DQSS = 0, 'WDQS2DQ_I = 0, 'DQSS = 0, 'WDQS2DQ_O = 0 and nominal 'LZ, 'HZ is shown for illustration purposes.
 6. WDBI and RDBI could be on or off and are controlled with MR0 OP[1:0].
 7. Identical SID and bank addresses are required for a pair of WRITE and READ commands.
 8. Read Parity (RPAR) off is assumed for the tRDMRS timing.
 9. Error severity reporting must be enabled via the SEVR bit in MR9 OP1.
 10. The MD bit in MR9 OP0 must be set according to the user's desire to include the ECC signals in this test or not.



1. The test mode is enabled by an MRS command that sets bit MR9 OP2 to 1, and the check bit word is selected by bit MR9 OP3 (CW0 or CW1). Error severity reporting must be enabled via the SEVR bit in MR9 OP1. The MD bit in MR9 OP0 must be set according to the user's desire to include the ECC signals in this test or not.
2. After 'MOD, a first WRITE command is issued, with the error input vector set to the desired error injection mode (NE, CEs, CEm or UE).
3. A first READ command is issued after the 'WTR timing has been met. Note that the SID and bank addresses must be the same for WRITE and READ, while the column addresses are not evaluated.
4. The sequence of alternating WRITE and READ commands can be repeated after waiting for at least 'RTW after the READ command.
5. The ECC engine test mode can be terminated by setting bit MR9 OP2 back to 0.

Error Check and Scrub (ECS)

The HBM3 DRAM implements an ECC error check and scrub (ECS) mode with an error counting scheme for transparency. Five mode register bits control ECS operation:

Table 32: Error Check and Scrub (ECS) Related Mode Registers

Mode Register	Purpose
MR9 OP4 (ECSREF)	Controls the repurposing of REFab commands for ECS during normal operation
MR9 OP5 (ECSSRF)	Controls ECS operation during self refresh
MR9 OP6 (ECSCEM)	Controls whether a code-word will be corrected and written back to the memory array in case of a correctable multi-bit error
MR9 OP7 (ECSRES)	Resets the ECS error type and address log for both PCs. This bit is self-clearing meaning that it automatically returns back to the 0 _B value after the reset function has been issued. The bit does not reset the internal ECS address counter for ECS operation
MR8 OP2 (ECSLOG)	Controls the ECS error type and address log auto reset upon a read (CaptureWR event)

Note: 1. See also Table 19 and Table 20 for MR8 and MR9 details.

ECS will operate in the background during REFab and self refresh periods when ECS is enabled by the ECSREF and ECSSRF bits, respectively. It allows the HBM3 DRAM to internally detect errors and write the corrected data back to the array (scrub errors) while providing transparency. Errors corrected during ECS cycles are logged in the transparency registers as outlined below.

ECS is disabled by default. The HBM3 DRAM must have all array bits written prior to enabling ECS operations to avoid generating false failures. It is recommended that a full error scrub of the array is performed at least once every 24 hours ('ECS).

ECS modes in MR9 OP[6:4] shall be programmed during initialization and shall not be changed once the first ECS operation occurs unless followed by a device reset or an ECS error type and address reset (MR9 OP7), otherwise an unknown operation could result during subsequent ECS operations.

ECS can be disabled by setting both bits MR9 OP[5:4] back to 00_B. Disabling ECS does not reset the ECS error type and address log, nor does it reset the internal ECS error counter (ERRCNT, see below).

During each single auto ECS operation, the HBM3 DRAM generates the following internally self-timed command sequence: ACTIVATE – READ – WRITE – PRECHARGE. This internal read-modify-write cycle will:

1. read one entire code-word from the array;
2. check the code-word for errors;
3. write the corrected code-word back to the array in case a single-bit error (CEs) was detected;
4. write the corrected code-word back to the array in case a multi-bit error (CEm) was detected and correction of multi-bit errors is enabled via the ECSCem bit;
5. not write any data back to the array in case the error is uncorrectable (UE).

After each ECS operation, the internal ECS address counter increments the column address such that the next column address (code-word and check bits) is selected. Once the column counter wraps (all columns on a row have been accessed), the row counter will increment until all rows within a bank are accessed. When the row counter wraps (all rows within the bank have been accessed), the bank counter will increment until all banks within one SID have been accessed. In that case all counter wrap to 0, equivalent to the completion of one 'ECS period.

Error scrubbing is performed simultaneously on all SID and logged separately per SID.

The ECS operation timing is shown in Figure 16. The maximum spacing between REFab commands to complete the scrub within the recommended 24 hours is 'ECSint as outlined in Table 33. Meeting this requirement allows the DRAM to perform the ECS operations without placing additional restrictions on refresh mode usage, i.e. all-bank / per-bank refresh. The ECS operation with REFab must be scheduled separately for PC0 and PC1 by issuing REFab commands evenly to PC0 and PC1. REFab commands issued in excess of required by the DRAM for ECS operations (one REFab per 'ECSint) may be used by the DRAM for normal refresh operation.

The HBM3 DRAM performs ECS operations while in self refresh mode if this function is enabled by the ECSSRF bit. These ECS operations occur approximately at 'ECSint spacing in order to generate the same scrubbing rate as recommended with REFab commands. However, some variation in the scrubbing rate may be encountered while in self refresh since the DRAM will need to synchronize the internal operations to an internal oscillator. Entering and exiting self refresh will not reset the ECS error log registers. All timings for the maximum spacing between REFab commands or another self refresh entry are allowed to restart upon self refresh exit.

Figure 16: Error Check and Scrub (ECS) Operation Timing

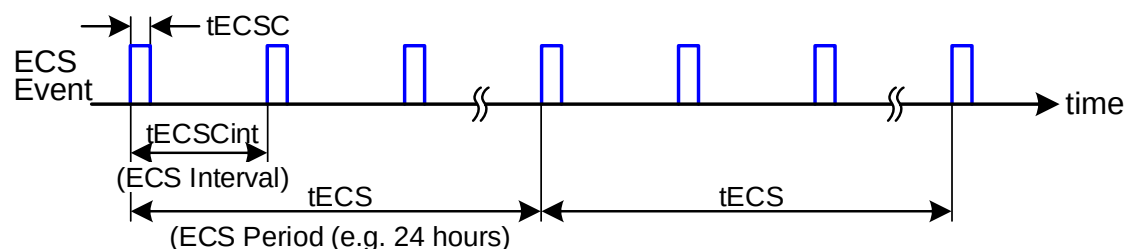



Table 33: Error Check and Scrub (ECS) Interval

Parameter	Values	Notes
Number of unique memory locations (code-words) per PC and per SID	1.5×2^{23}	
'ECSint per PC	6.866ms	1
Error threshold per Gb of memory cells	512	

Note: 1. Calculation: 'ECSint = 86,400s / (16 [banks] 24,576 [rows] x 32 [columns]). 'ECSint is the same for all stack heights.

ECS Error Logging and Counting

Errors that are detected during ECS operations are internally logged (see below for special conditions regarding logging of single bit errors (CEs)). The IEEE1500 instruction ECS_ERROR_LOG can be used to retrieve the error log information. Errors are logged within 'ECS separately per PC in the form of SID, bank, row and column address and the error severity as summarized in Table 34.

The HBM3 DRAM supports two methods to clear the error type and address log for both PCs:

- the error log can be cleared by setting the ECSRES bit in MR9 OP7 to 1B; this bit is self-clearing meaning that it automatically returns to the default state (0B) once the reset has been performed a;
- when the ECSLOG bit in MR8 OP2 is set to 1B, the error log is automatically cleared after each CaptureWR event of the ECS_ERROR_LOG WDR. Note that the logging of the newest error may be lost in case the auto reset inadvertently occurs simultaneously with an internal ECS event.

A reset of the ECS error log clears all VALID bits of the ECS_ERROR_LOG WDR to 0_B and the error type priority log to “NE” (no error).”

Table 34: Error Overwrite Priority Rules

		Current Error			
		NE	CEs	CEm	UE
Previous Errors	NE	None	Update	Update	Update
	CEs	Maintain	Update	Update	Update
	CEm	Maintain	Maintain	Update	Update
	UE	Maintain	Maintain	Maintain	Update

Notes: 1. NE = No Error; CEs = Corrected Single-Bit Error; CEm = Corrected Multi-Bit Error; UE = Uncorrectable Error.

2. Maintain = keep the current error log; Update = overwrite the error log.

The error log is updated according to these rules:

- the initial state after enabling ECS or at the beginning of a subsequent ECS period is “no error”, and all error logs are flagged as invalid (0_B);
- once a first error is detected, that error is logged with its address and severity, and the corresponding VALID bit is set to 1_B;



- if another error of the same or a higher severity is detected, the error log will be updated with that error's address and severity; otherwise the current error log will be maintained (not updated).

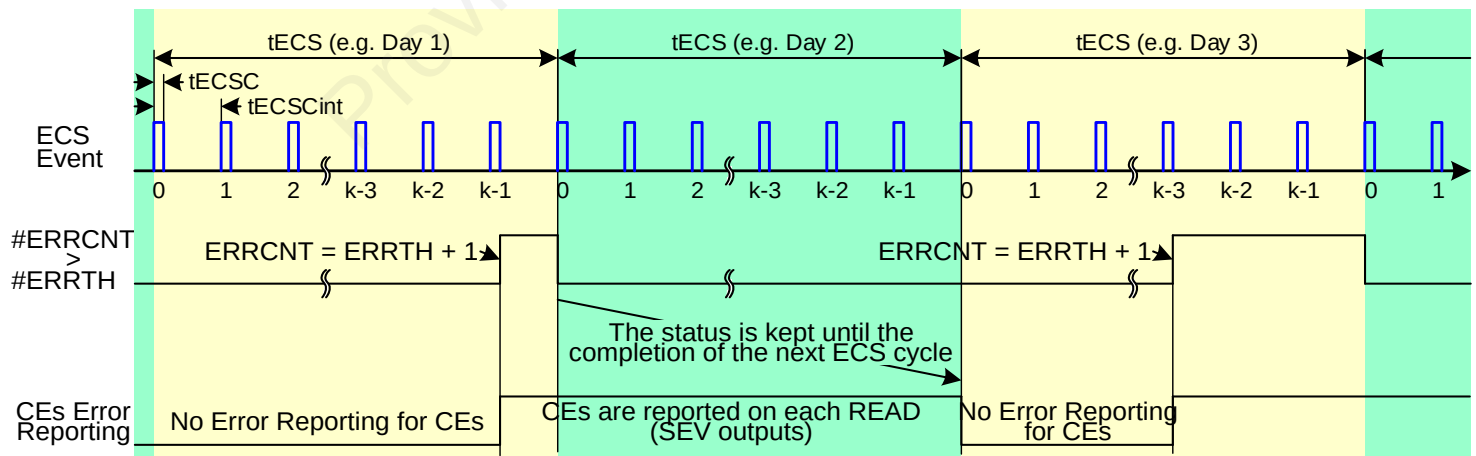
The number of corrected single bit errors (CEs) is counted during the ECS operation and compared against an error threshold value (ERRTH). CEs will not be logged and be reported as error-free (NE) on the SEV signals during READ operations until the error count has exceeded the error threshold ($ERRCNT > ERRTH$).

Rules for error counting:

- Error counting assumes one code-word covering each access;
- The error count (ERRCNT) is maintained separately per PC and per SID;
- Each CEs counts as one event towards ERRCNT;
- Multi-bit errors (CEm) and uncorrectable errors (UE) do not count towards ERRCNT and are always reported;
- a CEs in both code-words counts as a CEm;
- The ERRCNT is internally reset after each t_{ECS} period. It will keep its current value and not be reset when ECS is disabled.

The timing for enabling the CEs reporting on read transactions is illustrated in Figure 17. While the ERRCNT is reset each t_{ECS} period, the error reporting for corrected single cell fails (CEs) will remain enabled during the following t_{ECS} period(s) in case the ERRTH limit was exceeded in one t_{ECS} period. The error reporting will be disabled again when the error count has not exceeded the ERRTH limit in a subsequent full t_{ECS} period.

Figure 17: ECS Timing For Reporting Corrected Single Bit Errors (CEs) on SEV Signals During READs





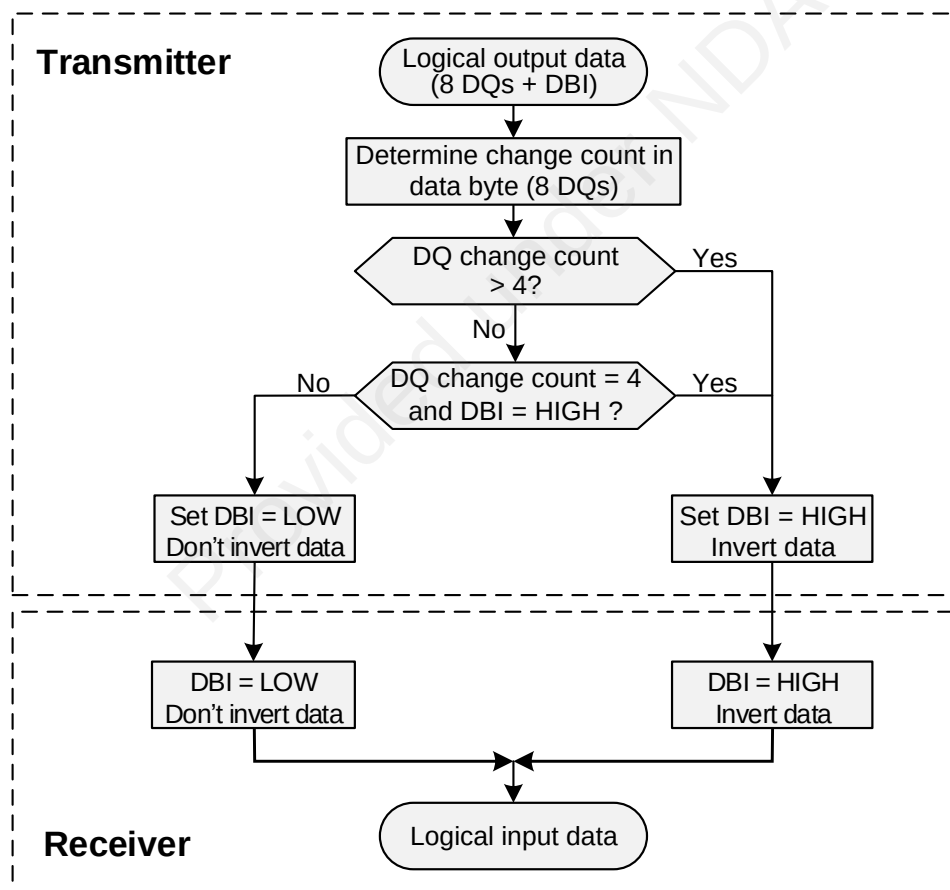
Data Bus Inversion (DBIac)

HBM3 DRAMs supports a byte granular Data Bus Inversion (DBIac). The corresponding DBI signal is a DDR I/O and driven or sampled along with the DQs for read and write operations.

The word DBI refers to the internal state of the device unless explicitly noted as DBI signal. The DBIac function can be enabled or disabled independently for writes per MR0 OP1 (WDBI) and for reads per MR0 OP0 (RDBI).

The DBI input is a don't care and the DBI input receivers are disabled when WDBI is disabled. The DBI output buffers are turned off when RDBI is disabled.

Figure 18: DBIac Algorithm

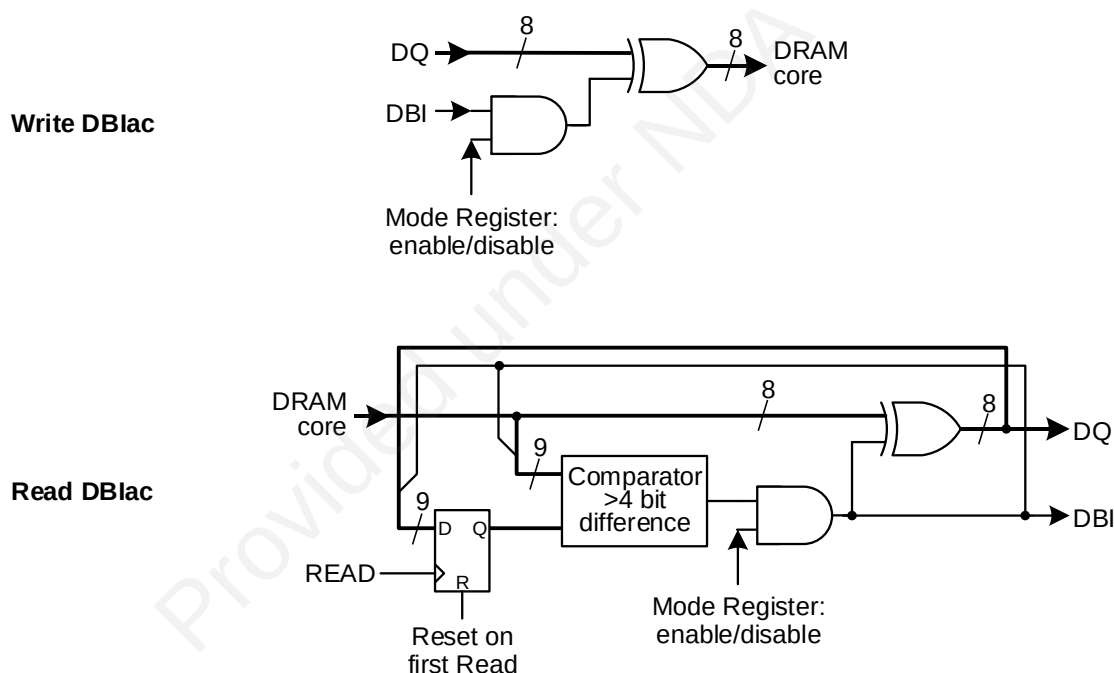


Write operation: the HBM3 DRAM inverts write data received on the DQ inputs in case DBI is sampled HIGH or leaves the write data non-inverted in case DBI is sampled LOW. Note that the ECC inputs are not affected by the DBIac function.

Read operation: the HBM3 DRAM counts the number of DQ signals that are transitioning from the previous state. Note that the ECC and SEV outputs are not affected by DBIac. See Internal DBIac States with Read for bus pre-conditioning. The HBM3 DRAM inverts read data and sets DBI HIGH when the number of changing data bits within a byte is greater than 4, or when the number of changing data bits within a byte equals 4 and DBI was HIGH; otherwise the HBM3 DRAM does not invert the read data and sets DBI LOW.

Table 35: DBI(ac) Truth Table

DQ Change Count	Previous DBI State	New DBI State	New DQ State
0 to 3	X	LOW	Not inverted
4	LOW		
	HIGH	HIGH	Inverted
5 to 8	X		

Figure 19: Example DBIac Logic for Write and Read

Command/Address Parity

The HBM3 DRAM includes a command/address parity checking function controlled by the CAPAR bit in MR0 OP6. The function is disabled by default. The APAR input and AERR output are associated with the function. APAR is enabled only when the function is enabled.

If enabled, the parity is calculated every CK clock cycle separately on both the rising and falling CK clock edges over input signals R[9:0], C[7:0], ARFU and APAR as summarized in Table 36. The AERR output indicates whether a parity error has occurred or not on either the rising or falling CK clock edge (or both edges). The HBM3 DRAM executes commands regardless of a command/address parity error.

Table 36: Command/Address Parity Function Table

Inputs	Sum of Inputs Received HIGH	AERR
R[9:0], C[7:0], ARFU, APAR	Even	LOW
	Odd	HIGH

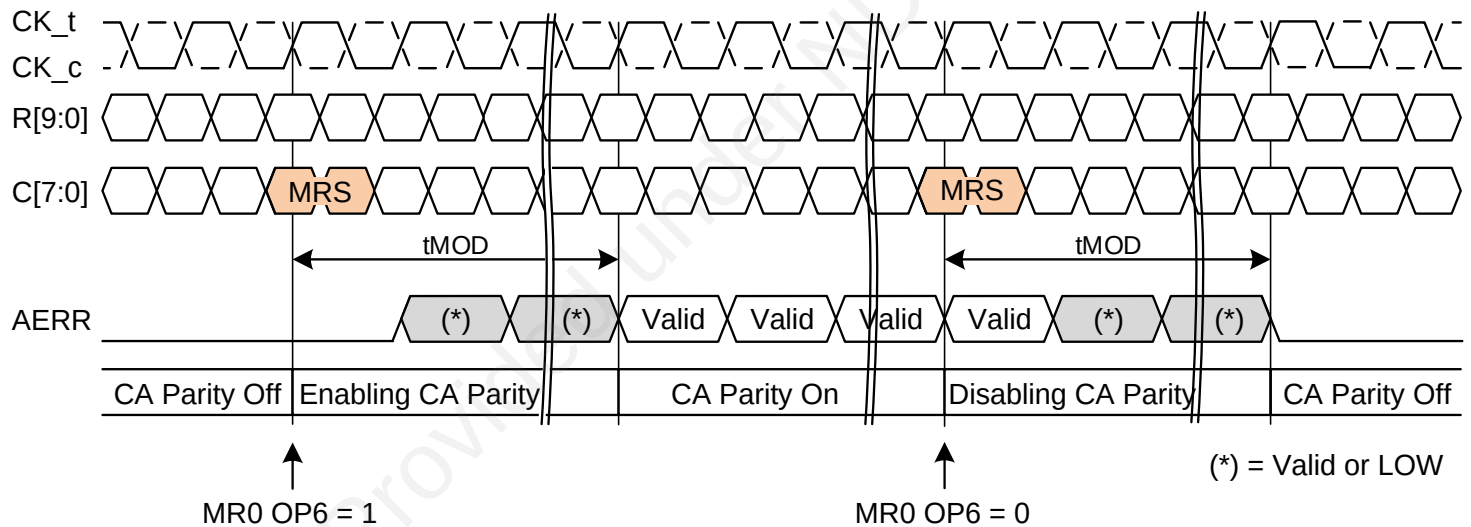
Note: 1. See Command Truth Tables for command and device state exceptions.



The HBM3 DRAM may begin to check parity on the next clock cycle following the MODE REGISTER SET command that enables the parity checking function; it will have the parity check enabled latest when t_{MOD} has expired after that MODE REGISTER SET command. The HBM3 DRAM therefore requires all subsequent commands including RNOP and CNOP to be issued with correct parity until when t_{MOD} has expired for the MODE REGISTER SET command that disables the parity calculation. See also the Power-Down and Self Refresh sections. AERR is driven LOW by the HBM3 DRAM at reset.

For every parity error, AERR is driven HIGH for 1 t_{CK} , t_{PARAC} after the corresponding cycle of the error inputs. In the case of consecutive errors, the AERR signal will stay HIGH during the next cycle. The parity function should not be disabled within t_{PARAC} after an access command.

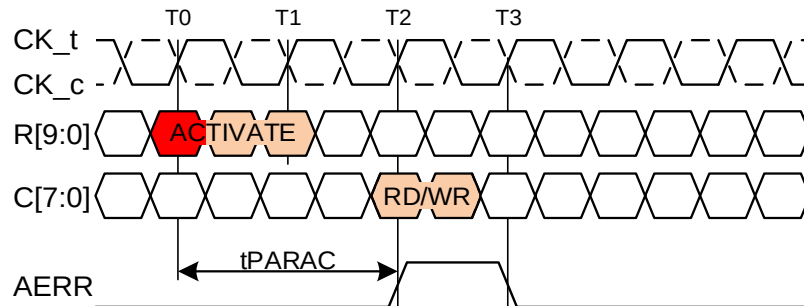
Figure 20: Enabling and Disabling Command/Address Parity



- Notes:
1. For illustration purpose, t_{PARAC} is shown with 0 t_{CK} digital and 0ns analog output delay.
 2. See Power-Down and Self Refresh sections for details on disabling and enabling parity check in conjunction with power-down and self refresh entry and exit.

Figure 21 illustrates a single parity error occurrence on the R inputs. In this case, the error occurs at the rising edge of the first cycle of the ACTIVATE command at time T_0 . After t_{PARAC} , AERR is driven HIGH for 1 t_{CK} and then LOW since no subsequent errors occur.

Figure 21: Single Command/Address Parity Error

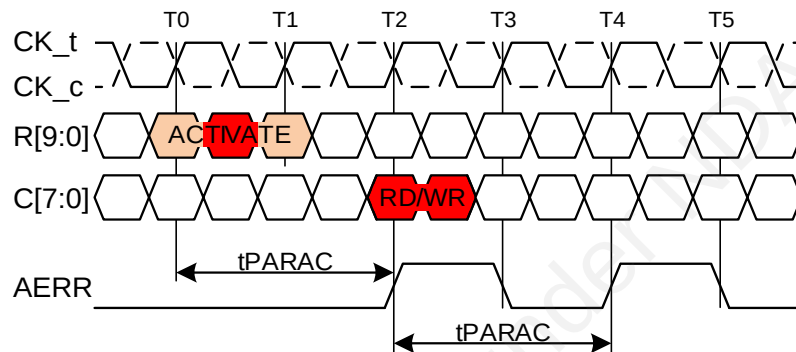


- Notes:
1. For illustration purpose, t_{PARAC} is shown with 2 t_{CK} digital and 0ns analog output delay.

- MR0 OP6 shall be maintained as 1 for at least t_{PARAC} after the access command.

Figure 22 illustrates parity error occurrences on the R and the C inputs. In this case, the error occurs at the falling edge of the first cycle of the ACTIVATE command at time T0. After t_{PARAC} , AERR is driven HIGH for 1 t_{CK} and then LOW for 1 t_{CK} . Since an error also occurs in T2 at both the rising and the falling edges of the READ or WRITE command, the AERR is again driven HIGH for 1 t_{CK} and then LOW since no subsequent errors occur.

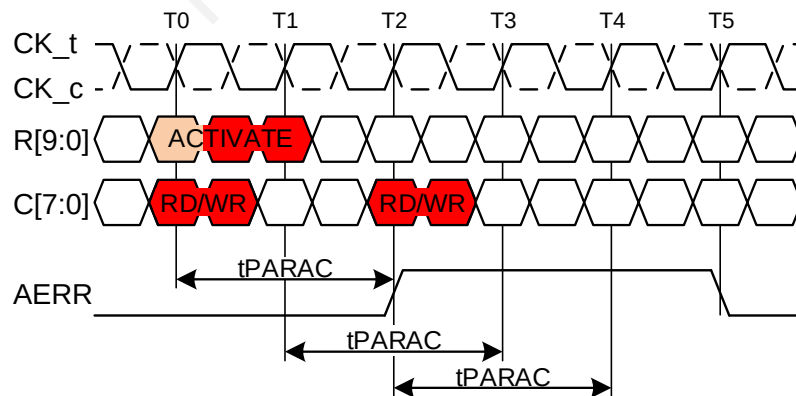
Figure 22: Separated Command/Address Parity Errors



- Notes:
- For illustration purpose, t_{PARAC} is shown with 2 t_{CK} digital and 0ns analog output delay.
 - MR0 OP6 shall be maintained as 1 for at least t_{PARAC} after the access command.

Figure 23 illustrates consecutive parity error occurrences on the R and the C inputs during the T0, T1, and T2 cycles and either the rising, the falling or both clock edges. Due to the common AERR output, parity error occurrences on both interfaces are indistinguishable.

Figure 23: Consecutive Command/Address Parity Errors



- Notes:
- For illustration purpose, t_{PARAC} is shown with 2 t_{CK} digital and 0ns analog output delay.
 - MR0 OP6 shall be maintained as 1_B for at least t_{PARAC} after the access command.



Data Parity

The HBM3 DRAM includes a data parity checking function for writes controlled by the WPAR bit in MR0 OP5, and a data parity generation function for reads controlled by the RPAR bit in MR0 OP4. Both WPAR and RPAR functions are disabled by default. There is one DPAR bidirectional DDR I/O and one DERR output signal per DWORD associated with the function. The DPAR input is enabled with WPAR during writes, and the DPAR output is enabled with RPAR during reads, otherwise DPAR is disabled.

The data parity function includes a programmable parity latency PL between the corresponding data and the DPAR signal. PL is programmed in MR1 OP[7:5] and is the same for writes and reads. The corresponding DPAR signal will be received and sent PL cycles later. The WDQS and RDQS strobes will have additional strobe cycles with the same preamble and postamble to accommodate the latching of the delayed DPAR signal at both ends. Examples of reads and writes with DQ parity enabled can be found in the Write Command and Read Command sections.

On read transactions, the HBM generates parity and transmits the parity on the DPAR signal along with the corresponding data on DQ, DBI and ECC.

On write transactions, the HBM3 DRAM compares the DPAR input with the corresponding data received on DQ, DBI and ECC inputs as summarized in Table 37. The parity calculation is performed separately for each UI of a write burst.

If an error occurs in any single or in multiple UIs within one clock cycle of a write burst (D0 ... D3 or D4 ... D7), DERR is driven HIGH for 1 'CK 'PARDQ after the corresponding cycle of error inputs. The 'PARDQ interval for errors occurring during the first clock cycle of a write burst begins (WL + PL) clock cycles after the WRITE command was issued. In case of errors within the first and the second clock cycle of a write burst, DERR will stay HIGH during the next cycle. DERR is driven LOW by the HBM3 DRAM at reset.

When an error occurs, the HBM3 DRAM does not block the write data. The HBM3 DRAM completes the write transaction to the array as normal.

WPAR should not be disabled within (WL + PL + 1 'CK + 'PARDQ) after the WRITE command in order to not create a conflict with any ongoing parity operation. For the same reason RPAR should not be disabled within 'RDMRS after the READ command.

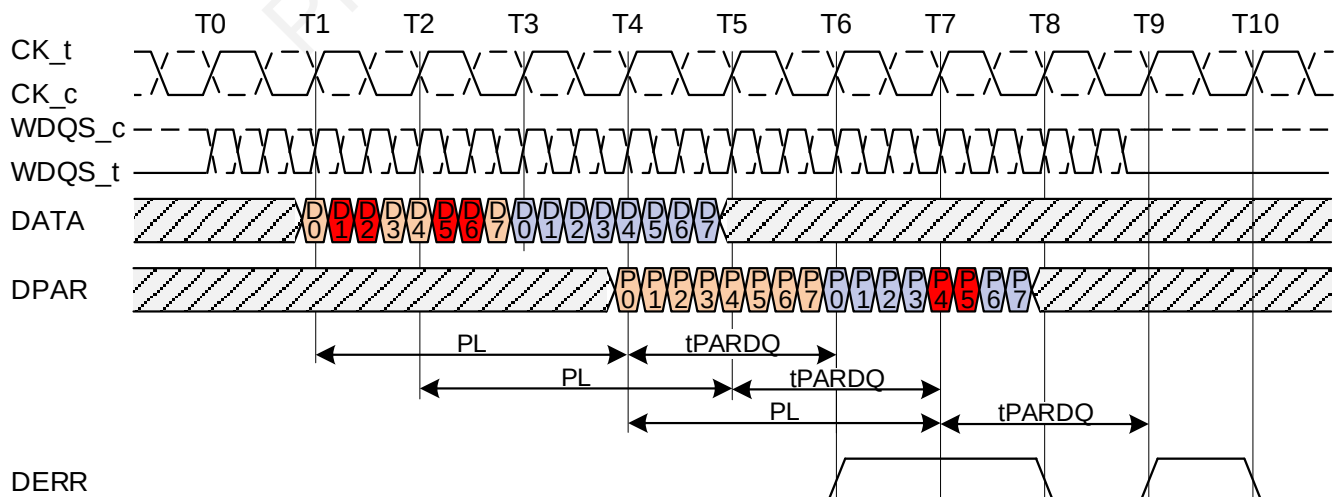
As outlined in Table 37, meta data received and sent via the ECC signals are included in the parity check and parity generation only when these signals are enabled by the MD bit in MR9 OP0. Similarly, the DBI signals are included in the parity check and parity generation only when these signals are enabled by the WDBI and RDBI bits in MR0 OP[1:0]. The SEV signals are not included in the parity check or parity generation.


Table 37: Data Parity Function Table

Configuration		Inputs		Sum of Inputs Received HIGH	DERR
MD (MR9 OP0)	WDBI or RDBI (MR0 OP[1:0])	DWORD0 (PC 0)	DWORD1 (PC 1)		
Enabled	Enabled	DQ[31:0], ECC[1:0], DBI[3:0], DPAR0	DQ[63:32], ECC[3:2], DBI[7:4], DPAR1	Even	LOW
				Odd	HIGH
	Disabled	DQ[31:0], ECC[1:0], DPAR0	DQ[63:32], ECC[3:2], DPAR1	Even	LOW
				Odd	HIGH
Disabled	Enabled	DQ[31:0], DBI[3:0], DPAR0	DQ[63:32], DBI[7:4], DPAR1	Even	LOW
				Odd	HIGH
	Disabled	DQ[31:0], DPAR0	DQ[63:32], DPAR1	Even	LOW
				Odd	HIGH

Note: 1. The DBI inputs are disabled and excluded from the parity check when WDBI is disabled in MR0 OP1. The DBI outputs are disabled and excluded from the parity generation when RDBI is disabled in MR0 OP0. The ECC I/Os are disabled and excluded from the parity check and parity generation when MD is disabled in MR9 OP0.

The figure below illustrates data parity error occurrences on two seamless write bursts. In this example errors occur in the second (D1), third (D2), sixth (D5) and seventh (D6) UI of the first write burst, and in the fifth (P4) and sixth (P5) UI of the DPAR input of the second write burst. After t_{PARDQ}, DERR is driven HIGH for 2 t_{CK} at T6 and T7, then driven LOW for 1 t_{CK} and again driven HIGH for 1 t_{CK} at T9.

Figure 24: Write Parity Errors with PL = 3


- Notes:
1. D0 ... D7 = data-in for WRITE command (BL8 burst). P0 ... P7 = parity-in for WRITE command.
 2. DATA = DQ[31:0], DBI[3:0], ECC[1:0] for PC0, and DQ[63:32], DBI[7:4], ECC[3:2] for PC1. WDQS_t/_c is WDQS0_t/_c for PC0, and WDQS1_t/_c for PC1.
 3. Two seamless bursts are shown, with parity errors in the second (D1), third (D2), sixth (D5) and seventh (D6) UI of the first write burst, and in the fifth (P4) and sixth (P5) UI of the DPAR input of the second write burst.
 4. PL = 3 is assumed.



5. The parity check is performed separately for the first clock cycle (UI = D0 ... D3) and the second clock cycle (UI = D4 ... D7) of a BL8 burst.
6. ¹PARDQ is shown with 2 tCK digital and 0ns analog output delay.
7. WDBI could be on or off and is controlled with MR0 OP1. WDBI shall be maintained enabled for at least ¹PARDQ after the access command.

Command Truth Tables

The HBM3 DRAM features DDR commands entered on both rising and falling CK clock edges. The ACTIVATE command requires one-and-a-half-cycle and other row commands require only a half-cycle. All column commands require one cycle.

The command interface includes a reserved DDR input signal ARFU which is omitted from subsequent truth tables but required to be driven to a valid signal level along with the other AWORD inputs.

Table 38: Row Command Table

Command ⁴	Symbol	R/F	R0	R1	R2	R3	R4	R5	R6	R7	R8	R9	Notes
ROW NO OPERATION	RNOP	R or F	H	H	H	H	V	V	V	V	V	V	1, 2, 3
ACTIVATE	ACT	R	L	H	H	PC	SID0/V	SID1/V	BA0	BA1	BA2	BA3	1, 2, 3, 5, 6, 9
		F	H	H	RA8	RA9	RA10	RA11	RA12	RA13	RA14	V	
		R	H	H	RA0	RA1	RA2	RA3	RA4	RA5	RA6	RA7	
PRECHARGE	PREpb	R or F	H	L	L	PC	SID0/V	SID1/V	BA0	BA1	BA2	BA3	1, 2, 3, 5, 6
PRECHARGE ALL	PREab	R or F	H	L	H	PC	V	V	V	V	V	V	1, 2, 3, 5
PER-BANK REFRESH	REFpb	R	L	L	L	PC	SID0/V	SID1/V	BA0	BA1	BA2	BA3	1, 2, 3, 5, 6
ALL-BANK REFRESH	REFab	R	H	H	L	PC	V	V	V	V	L	V	1, 2, 3, 5
PER-BANK REFRESH MANAGEMENT	RFMpb	R	L	L	H	PC	SID0/V	SID1/V	BA0	BA1	BA2	BA3	1, 2, 3, 5, 6, 7
ALL-BANK REFRESH MANAGEMENT	RFMab	R	H	H	L	PC	V	V	V	V	H	V	1, 2, 3, 5, 7
POWER-DOWN ENTRY	PDE	R	L	H	L	H	V	V	V	V	V	V	1, 2, 3
		F	L	H	L	H	V	V	V	V	V	V	
SELF REFRESH ENTRY	SRE	R	L	H	L	L	V	V	V	V	V	V	1, 2, 3
		F	L	H	L	L	V	V	V	V	V	V	
POWER-DOWN or SELF REFRESH EXIT	PDX / SRX	R	H	H	H	H	V	V	V	V	V	V	1, 2, 8

- Notes:
1. BA = Bank address; RA = Row address; SID = Stack ID; V = Valid (either H or L, but not floating).
 2. R[9:0] must be driven to a valid signal level even if a stack ID address (SID) is not defined for a specific density. APAR must be driven to a valid signal level even if CA parity is disabled in MR0 OP6.
 3. Parity is evaluated on all inputs if CA parity is enabled in MR0 OP6.
 4. All other command encodings not shown in the table are reserved for future use.



5. PC = 0_B selects pseudo channel 0 (PC0), and PC = 1_B selects pseudo channel 1 (PC1). The pseudo channel not selected by PC performs a RNOP.
6. The SID bits act as bank address bits in conjunction with ACT, PREpb, REFpb and RFMpb commands, and related timing diagrams shall be interpreted accordingly. All other row commands do not use SID. Refer to Table 6 for HBM3 configurations using SID.
7. An HBM3 DRAM not requiring refresh management (RFM) will execute an RNOP command instead of RFMab or RFMpb.
8. No parity checking at power-down exit or self refresh exit. The HBM3 DRAM requires RNOP and CNOP commands on row and column bus respectively with valid parity if CA parity is enabled during the power-down exit period ('XP) and self refresh exit period ('XS).
9. ACTIVATE is a 1.5-cycle long command that cannot be interrupted by other commands.

Table 39: Column Command Table

Command ⁴	Symbol	R/F	C0	C1	C2	C3	C4	C5	C6	C7	Notes
COLUMN NO OPERATION	CNOP	R	H	H	H	V	V	V	V	V	1, 2, 3
		F	V	V	V	V	V	V	V	V	
READ	RD	R	H	L	H	L	PC	SID0/V	SID1/V	BA0	1, 2, 3, 5, 6, 7
		F	BA1	BA2	BA3	CA0	CA1	CA2	CA3	CA4	
READ w/ AP	RDA	R	H	L	H	H	PC	SID0/V	SID1/V	BA0	1, 2, 3, 5, 6, 7
		F	BA1	BA2	BA3	CA0	CA1	CA2	CA3	CA4	
WRITE	WR	R	H	L	L	L	PC	SID0/V	SID1/V	BA0	1, 2, 3, 5, 6
		F	BA1	BA2	BA3	CA0	CA1	CA2	CA3	CA4	
WRITE w/ AP	WRA	R	H	L	L	H	PC	SID0/V	SID1/V	BA0	1, 2, 3, 5, 6
		F	BA1	BA2	BA3	CA0	CA1	CA2	CA3	CA4	
MODE REGISTER SET	MRS	R	L	L	L	MA4	OP5	OP6	OP7	MA0	1, 3, 6
		F	MA1	MA2	MA3	OP0	OP1	OP2	OP3	OP4	

- Notes:
1. BA = Bank address; CA = Column address; SID = Stack ID; V = Valid (either H or L, but not floating).
 2. C[7:0] must be driven to a valid signal level even if a stack ID address (SID) is not defined for a specific density, or if parity is disabled in the mode register. APAR must be driven to a valid signal level even if CA parity is disabled in MR0 OP6. C[7:0] are Don't Care when the device is in power-down or self refresh.
 3. Parity is evaluated on all inputs if CA parity is enabled in MR0 OP6.
 4. All other command encodings not shown in the table are reserved for future use.
 5. PC = 0_B selects pseudo channel 0 (PC0), and PC = 1_B selects pseudo channel 1 (PC1). The pseudo channel not selected by PC performs a CNOP.
 6. The SID bits act as bank address bits in conjunction with READ and WRITE commands, and related timing diagrams shall be interpreted accordingly. All other column commands do not use SID. Refer to Table 6 for HBM3 configurations using SID.
 7. HBM configurations using the SID specify a timing parameter 'CCDR for consecutive READs to different SID. Refer to the AC Timings table for details.

Table 40: Precharge Command Table

Command on Rising Clock Edge	Allowed PREab/PREpb Command(s) on Falling Clock Edge (Same Cycle)			Notes
	Same PC, Same Bank	Same PC, Different Bank	Different PC, Any Bank	
RNOP	PREab, PREpb		PREab, PREpb	1
ACT	--	PREpb	PREab, PREpb	
PREab	--	--	PREab, PREpb	
PREpb	--	PREpb	PREab, PREpb	
REFab	--	--	PREab, PREpb	
REFpb	--	PREpb	PREab, PREpb	
RFMab	--	--	PREab, PREpb	
RFMpb	--	PREpb	PREab, PREpb	
PDE, PDX	--			
SRE, SRX	--			

Note: 1. See Table 38 for row commands.

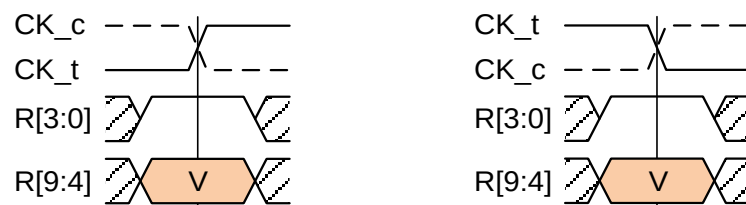
Row No Operation Command (RNOP)

The ROW NO OPERATION (RNOP) command is a half-cycle command received on the row command inputs R[9:0] and latched either with the rising or with the falling CK clock edge (or both edges) as shown in Figure 25. RNOP is used to instruct the HBM3 device to perform a NOP as row command; this prevents unwanted row commands from being registered during idle or wait states. Operations already in progress are not affected.

Row commands other than RNOP are defined either as half-cycle or as one-and-a-half-cycle commands that begin and end on a rising CK clock edge. These commands must be padded with RNOP on the falling CK clock edge of the same cycle. As an alternative, some row commands may be paired with PRECHARGE or PRECHARGE ALL commands on the falling CK clock edge instead of RNOP, with the specific conditions for these commands being explicitly described for each row command.

Parity is evaluated with the RNOP command when CA parity is enabled in MR0 OP6.

RNOP is assumed for the R[9:0] inputs on subsequent timing diagrams unless other row commands are explicitly shown.

Figure 25: RNOP Command


Note: 1. V = Valid (H or L)

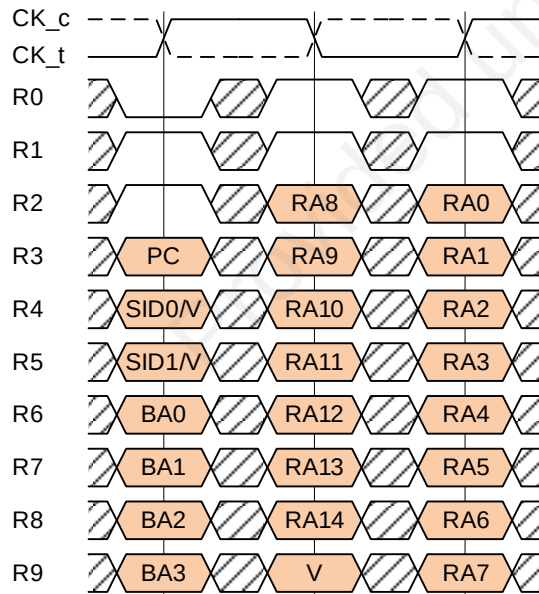
ACTIVATE Command (ACT)

Before a READ or WRITE command can be issued to a bank, a row in that bank must be opened. This is accomplished via the ACTIVATE command, which selects the SID, bank and row to be activated. Once a row is open, a READ or WRITE command could be issued to that row, subject to the t_{RCD} specification.

The ACTIVATE command is a one-and-a-half-cycle command received on the row command inputs R[9:0] and latched with the rising and falling CK clock edges as shown in Figure 26. The command must be followed either by RNOP, PRECHARGE or PRECHARGE ALL on the falling CK clock edge of the second clock cycle. Note that a PRECHARGE ALL in that case must be for the other pseudo channel. A PRECHARGE command could be to any bank in the other pseudo channel as well as to a different bank in the same pseudo channel. In all cases the timing requirements for issuing these commands must be met.

The actual bank and row activation is initiated with the second rising CK clock edge of the ACTIVATE command; therefore, all relevant timing parameters refer to this second rising CK clock edge as shown in subsequent timing diagrams.

Figure 26: ACTIVATE Command



Note: 1. PC = Pseudo Channel 0 or 1; BA = Bank address; RA = Row address; SID = Stack ID; V = Valid (H or L)

Parity is evaluated with the ACTIVATE command when CA parity is enabled in MR0 OP6.

A subsequent ACTIVATE command to another row in the same bank can only be issued after the previous row has been closed (precharged). The minimum time interval between successive ACTIVATE commands to the same bank is defined by t_{RC} , as shown in Figure 27. A minimum time t_{RAS} must have elapsed between opening and closing a row. The figure also shows two cases of t_{RAS} timings and command slots of the PRECHARGE command.

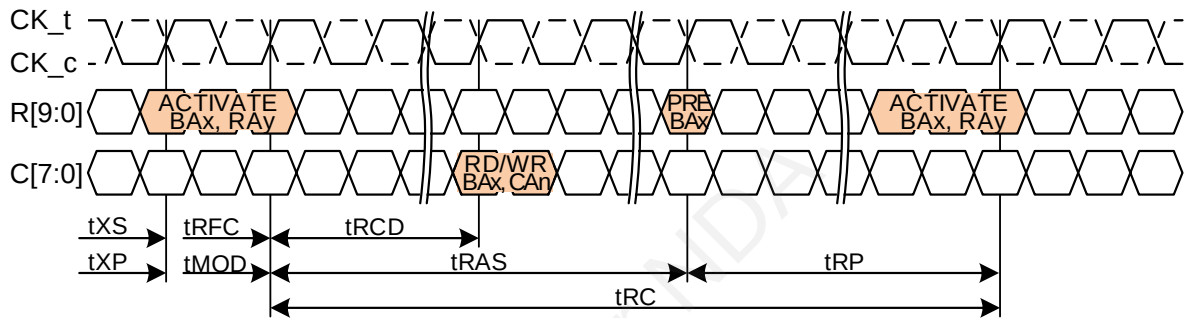
A subsequent ACTIVATE command to another bank can be issued while the first bank is being accessed, which results in a reduction of total row access overhead. The minimum time interval between successive ACTIVATE commands to different banks is defined by t_{RRD} . The row remains



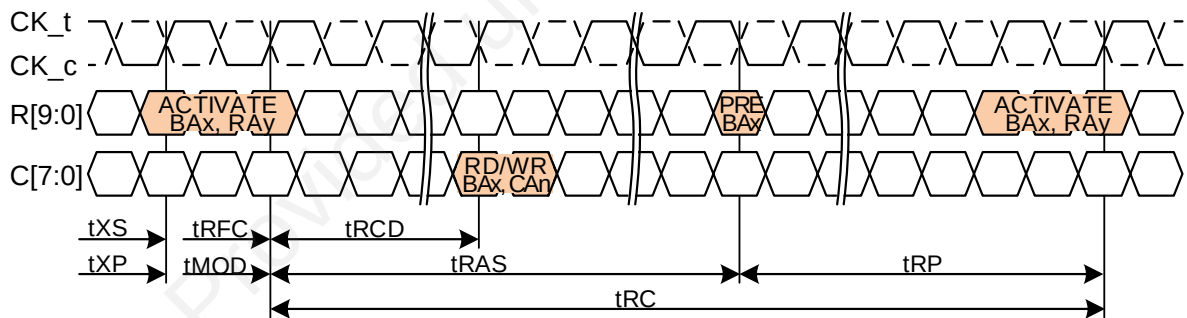
active until a PRECHARGE command (or READ or WRITE command with Auto Precharge) is issued to the bank.

Figure 27: Bank and Row Activation Command Cycle

Case 1: tRAS timing met at rising CK clock edge



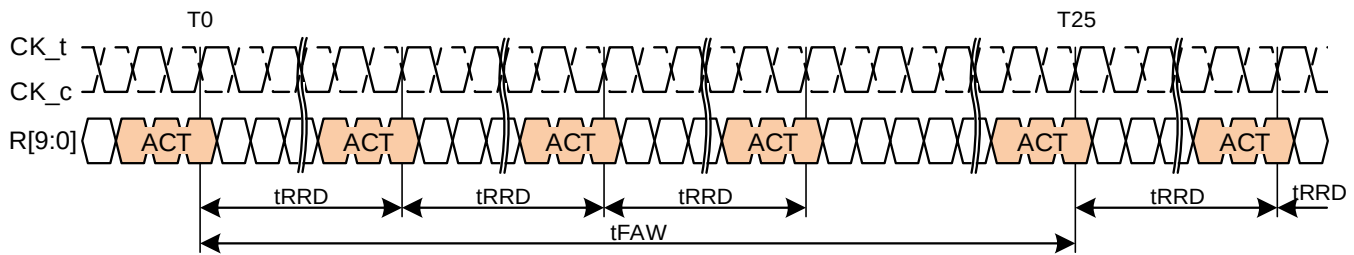
Case 2: tRAS timing met at falling CK clock edge



- Notes:
1. BAX = Bank address x; RAY,z = Row address y,z; CAn = Column address n.
 2. The PRECHARGE command shown could also be a PRECHARGE ALL command.
 3. 'RCD = 'RCDRD or 'RCDWR, depending on command; 'RFC = 'RFCab or 'RFCpb, depending on command.
 4. The reference for 'XP and 'XS timings is the first clock cycle of an ACTIVATE command, and the reference for 'RFC and 'MOD timing is the second clock cycle of an ACTIVATE command.

Bank Restrictions

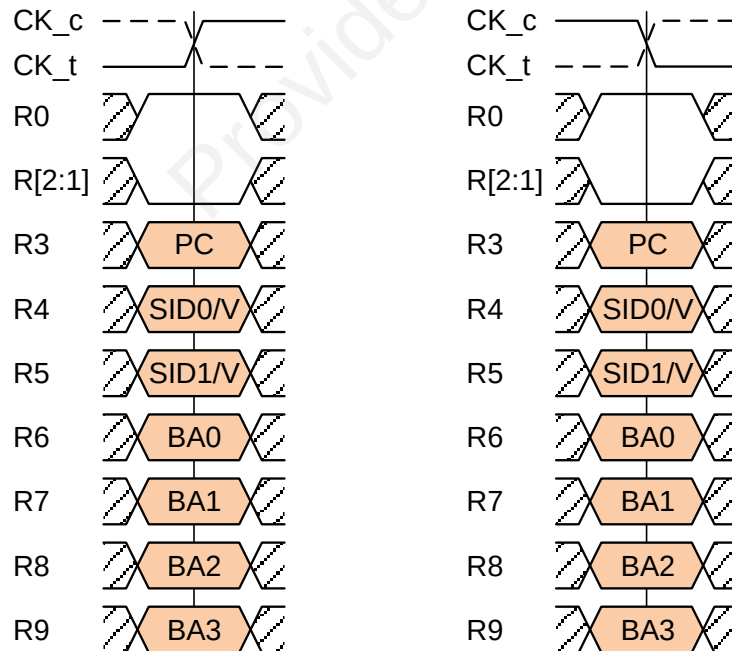
There is a need to limit the number of bank activations in a rolling window to ensure that the instantaneous current supplying capability of the device is not exceeded. To reflect the short-term current supply capability, the parameter 'FAW (four activate window) is defined: no more than 4 banks may be activated in a rolling 'FAW window. Converting to clocks is done by dividing 'FAW (ns) by 'CK (ns) and rounding up to next integer value. As an example of the rolling window, if ('FAW/'CK) rounds up to 25 clocks, and an ACTIVATE command is issued at clock T0, no more than three further ACTIVATE commands may be issued at clocks T1 through T24 as illustrated in Figure 28.

Figure 28: Multi Bank Activations

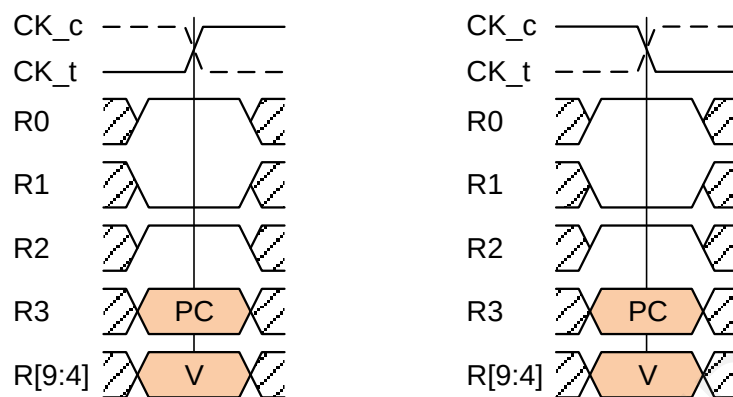
- Notes:
1. tRRD = tRRDS or tRRDL depending on accessed banks.
 2. Refer to Table 41 for timing restrictions between all combinations of ACTIVATE and PER-BANK REFRESH commands.

PRECHARGE (PREpb) and PRECHARGE ALL (PREab) Commands

The PRECHARGE (PREpb) and PRECHARGE ALL (PREab) commands are half-cycle commands received on the row command inputs R[9:0] and latched either with the rising or with the falling CK clock edge as shown in Figure 29 and Figure 30. The commands are used to deactivate the open row in a particular bank (PREpb) or the open rows in all banks (PREab). The bank(s) will be in idle state and available for a subsequent row access a specified time t_{RP} after the PRECHARGE command is issued.

Figure 29: PRECHARGE (PREpb) Command

- Note:
1. BA = Bank address; PC = Pseudo Channel 0 or 1; SID = Stack ID; V = Valid (H or L)

Figure 30: PRECHARGE ALL (PREab) Command


Note: 1. PC = Pseudo Channel 0 or 1; V = Valid (H or L)

The fact that both are half-cycle commands and defined on both the rising and the falling CK clock edges allows to issue one PREpb or PREab command on the rising CK clock edge and a second PREpb or PREab command on the falling CK clock edge of the same cycle and thus deactivate the open row in two different banks or even all banks in both pseudo channels within a single clock cycle, provided the ^tPPD timing has been met. It is pointed out that the ^tRP timing is always referenced from the CK clock edge on which the PREpb or PREab command is issued.

Parity is evaluated with the PRECHARGE and PRECHARGE ALL commands when CA parity is enabled in MR0 OP6.

Input R2 determines whether one or all banks are to be precharged. In case where only one bank is to be precharged, bank addresses {SID[1:0], BA[3:0]} select the bank. Otherwise the bank addresses are treated as “Don’t Care”.

Once a bank has been precharged, it is in the idle state and must be activated prior to any READ or WRITE command being issued to that bank. A PRECHARGE command is allowed if there is no open row in that bank (idle state), or if the previously open row is already in the process of precharging. However, the precharge period (^tRP) shall be determined by the most recent PRECHARGE command issued to the bank.

Auto Precharge

Auto precharge is a feature which performs the same individual bank precharge function described above, but without requiring an explicit PRECHARGE command. Auto precharge is nonpersistent meaning that it is enabled or disabled along for each individual READ or WRITE command.

For read bursts an auto precharge of the bank and row that is addressed with the READ command begins RTP clock cycles after the READ command was issued or after RAS has been met, with RTP as programmed in clock cycles in MR5 OP[3:0] and RAS as programmed in clock cycles in MR4 OP[7:0].

For write bursts an auto precharge of the bank and row that is addressed with the WRITE command begins (WL + 2 + WR) clock cycles after the WRITE command was issued or after RAS has been met, with WR as programmed in clock cycles in MR3 OP[7:0] and RAS as programmed in clock cycles in MR4 OP[7:0].

Auto precharge ensures that the precharge is initiated at the earliest valid stage within a burst. The user must not issue another command to the same bank until the precharge (^tRP) is completed.



This is determined as if an explicit PRECHARGE command was issued at the earliest possible time, as described for READ or WRITE commands. A precharge resulting from a READ or WRITE with auto precharge may occur in parallel with an explicit PRECHARGE (or PRECHARGE ALL) command. It is pointed out that an auto precharge is internally always issued with a rising CK clock edge, while explicit PRECHARGE (or PRECHARGE ALL) commands are supported on both clock edges.

Table 41: Precharge and Auto Precharge Timings

From Command	To Command	Min. Delay Between “From Command” to “To Command”	Unit	Notes
READ	PRECHARGE (same bank)	t_{RTP}	nCK	
	PRECHARGE (different bank)	0	nCK	4
	PRECHARGE ALL	t_{RTP}	nCK	
	WRITE or WRITE w/ AP (any bank)	t_{RTW}	ns	
	READ or READ w/ AP (any bank)	t_{CCD}	nCK	5
READ w/ AP	PRECHARGE ALL	t_{RTP}	nCK	
	PRECHARGE (different bank)	0	nCK	4
	ACTIVATE or PER-BANK REFRESH (same bank)	$RTP + RU(t_{RP}/CK)$	nCK	2
	WRITE or WRITE w/ AP (same bank)	Illegal		
	WRITE or WRITE w/ AP (different bank)	t_{RTW}	ns	
	READ or READ w/ AP (same bank)	Illegal		
	READ or READ w/ AP (different bank)	t_{CCD}	nCK	5
WRITE	PRECHARGE (same bank)	$WL + 2 + RU(t_{WR}/CK)$	nCK	2, 6
	PRECHARGE (different bank)	0	nCK	4
	PRECHARGE ALL	$WL + 2 + RU(t_{WR}/CK)$	nCK	2, 6
	WRITE or WRITE w/ AP (any bank)	t_{CCD}	nCK	5
	READ w/ AP (same bank)	$WL + 2 + \text{MAX}[RU(t_{WR}/CK) - t_{RTP}, t_{WTR}]$	nCK	2, 6, 7
	READ (same bank)	$WL + 2 + t_{WTR}$	nCK	6, 7
	READ or READ w/ AP (different bank)	$WL + 2 + t_{WTR}$	nCK	6, 7



From Command	To Command	Min. Delay Between “From Command” to “To Command”	Unit	Notes
WRITE w/ AP	PRECHARGE ALL	$WL + 2 + RU('WR/'CK)$	nCK	2, 6
	PRECHARGE (different bank)	0	nCK	4
	ACTIVATE or PER-BANK REFRESH (same bank)	$WL + 2 + WR + RU('RP/'CK)$	nCK	2, 6
	WRITE or WRITE w/ AP (same bank)	Illegal		
	WRITE or WRITE w/ AP (different bank)	$'CCD$	nCK	5
	READ or READ w/ AP (same bank)	Illegal		
	READ or READ w/ AP (different bank)	$WL + 2 + 'WTR$	nCK	6, 7
PRECHARGE	PRECHARGE (any bank)	$'PPD$	nCK	3
	PRECHARGE ALL	$'PPD$	nCK	3
PREALL	PRECHARGE or PREALL	$'PPD$	nCK	3

- Notes:
1. A command issued during the minimum delay time is illegal.
 2. RU = round up to next integer.
 3. A PRECHARGE command is allowed if there is no open row in that bank (idle state), or if the previously open row is already in the process of precharging. However, the precharge period shall be determined by the most recent PRECHARGE command issued to the bank.
 4. READ or WRITE and PRECHARGE commands may be issued simultaneously.
 5. $'CCD$ could either be $'CCDS$ or $'CCDL$; for READs, $'CCD$ could also be $'CCDR$.
 6. WL = write latency.
 7. $'WTR$ could either be $'WTRS$ or $'WTRL$.

Rounding Rules for Row Access Timings

The HBM3 DRAM allows the PRECHARGE (PREpb) and PRECHARGE ALL (PREab) commands to be issued on both rising and falling CK clock edges, as e.g. illustrated in the Bank and Row Activation Command Cycle figure. To let a system take advantage of this flexibility in command scheduling, it is required to adapt the rounding rules for related row access timings.

Traditionally, basic row access timings are converted into clock cycles using the formula $NXX = RU('XX/'CK)$, with XX representing either RAS, RTP, WR or RP parameters. This formula rounds the analog timings up to the next integer such that the subsequent command can be issued on the next rising clock edge that meets the analog value.

For HBM3 DRAM, this formula is replaced by $NXX = 0.5 \times RU(2 \times 'XX/'CK)$, which rounds analog timings to the next rising or following clock edge that meets the analog value. The result may be the same as with the traditional formula, or 0.5 nCK less. The formula may be applied to row timings $'RAS$, $'RTP$, $'WR$ and $'RP$ only. If rounding the $'RP$ timing results in a falling clock edge as the command slot for a subsequent row access command, it is required to add 0.5nCK to the result because all row commands following a row precharge can be issued on a rising clock edge only.

Examples:

1. $'RAS = 33ns$, $'CK = 0.7ns$; $NRAS = 0.5 \times RU(2 \times 'RAS/'CK) = 0.5 \times RU(2 \times 33/0.7) = 0.5 \times RU(94.29) = 47.5$.

Conclusion: when the ACTIVATE command was issued at T0, the earliest possible slot for a PRECHARGE command is at T47.5 (falling clock edge).

2. $t_{RP} = 15\text{ns}$, $t_{CK} = 0.7\text{ns}$; $\text{NRP} = 0.5 \times \text{RU}(2 \times t_{RP}/t_{CK}) = 0.5 \times \text{RU}(2 \times 15/0.7) = 0.5 \times \text{RU}(42.85) = 21.5$.

Conclusion: when the PRECHARGE command was issued at T0 (rising edge), the earliest possible slot for a subsequent ACTIVATE command is at T22, because the falling edge at T21.5 is not supported for an ACTIVATE command and $0.5n\text{CK}$ must be added to the result. However, when the PRECHARGE command was issued at T0.5 (falling edge), the earliest possible slot for a subsequent ACTIVATE command is again at T22.

REFRESH Command (REFab)

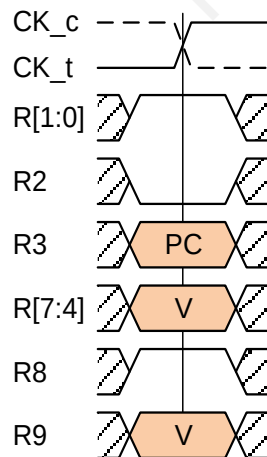
The (all-bank) REFRESH command is used during normal operation of the HBM3 device. The command is a half-cycle command received on the row command inputs R[9:0] and latched with the rising CK clock edge as shown in Figure 31. The command must be followed either by RNOP, PRECHARGE or PRECHARGE ALL on the falling CK clock edge of the same cycle. Note that PRECHARGE and PRECHARGE ALL commands in this case must be for the other pseudo channel and the timing requirements for issuing these commands must be met. The REFRESH command also requires a CNOP command on the column command inputs C[7:0], unless the column command is for the other pseudo channel.

Parity is evaluated with the REFRESH command when CA parity is enabled in MR0 OP6.

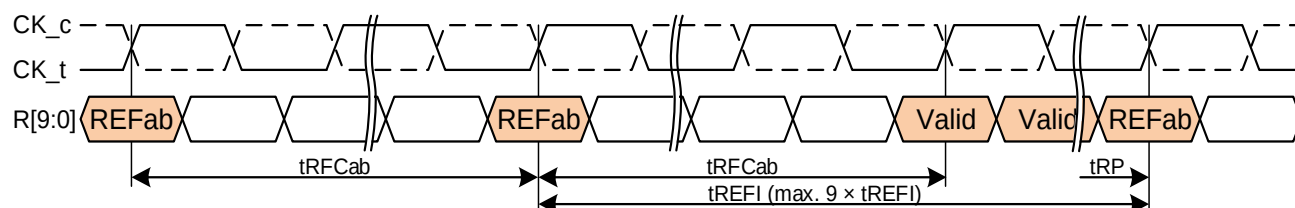
The REFRESH command is nonpersistent, so it must be issued each time a refresh is required. A minimum time t_{RFCab} is required between two REFRESH commands or a REFRESH command and any subsequent access command after the refresh operation. All banks must be precharged with t_{RP} satisfied prior to the REFRESH command. The banks are in idle state after completion of the REFRESH command.

The refresh addressing is generated by an internal refresh controller. This makes the address bits “Don’t Care” during a REFRESH command.

Figure 31: REFRESH Command



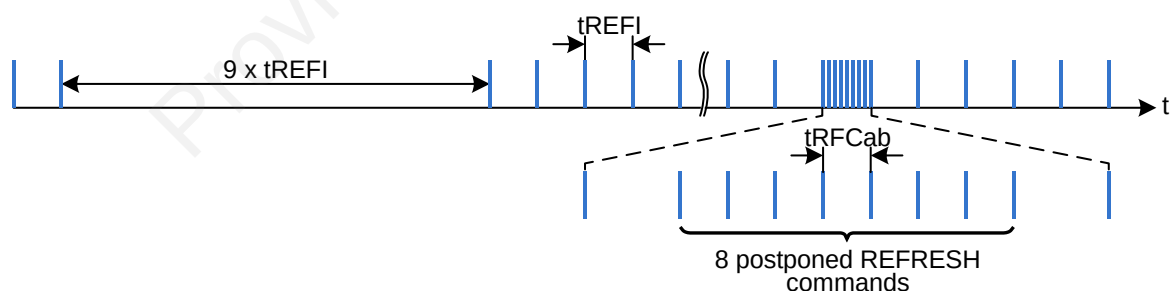
Note: 1. PC = Pseudo Channel 0 or 1; V = Valid (H or L)

Figure 32: Refresh Cycle


- Notes:
1. Only RNOP and CNOP commands allowed after REFRESH command registered until t_{RFCab} expires.
 2. The time interval between two REFRESH commands may be extended to a maximum of $9 \times t_{REFI}$.

The HBM3 DRAM requires REFab cycles at an average periodic interval of t_{REFI} (MAX). To allow for improved efficiency in scheduling and switching between tasks, some flexibility in the absolute refresh interval is provided. A maximum of eight REFRESH commands can be postponed during operation of the device, meaning that at no point in time more than a total of eight REFRESH commands are allowed to be postponed. In case that eight REFRESH commands are postponed in a row, the resulting maximum interval between the surrounding REFRESH commands is limited to $9 \times t_{REFI}$ (see Figure 33). At any given time, a maximum of 9 REFRESH commands can be issued within t_{REFI} .

This flexibility to postpone refresh commands also extends to REFpb commands (see below). The maximum interval between refreshes to a particular bank is limited to $9 \times t_{REFI}$. At any given time, a maximum of 9 REFpb commands to a particular bank can be issued within t_{REFI} .

Figure 33: Postponing Refresh Commands (Example)


Self refresh mode may be entered with a maximum of eight REFRESH commands being postponed. After exiting self refresh mode with one or more REFRESH commands postponed, additional REFRESH commands may be postponed to the extent that the total number of postponed REFRESH commands (before and after the self refresh) will never exceed eight. During self refresh mode, the number of postponed or pulled-in REFRESH commands does not change.

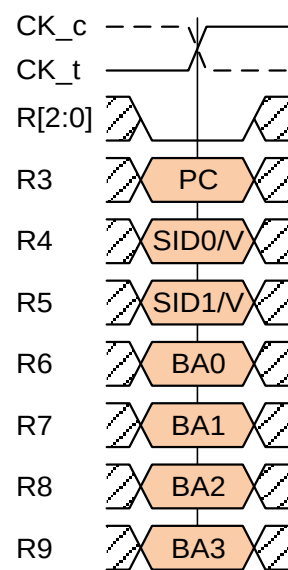
PER-BANK REFRESH Command (REFpb)

The PER-BANK REFRESH command (REFpb) provides an alternative solution for the refresh of the HBM3 device. The command initiates a refresh cycle on a single bank while accesses to other banks including writes and reads are not affected. PER-BANK REFRESH is a half-cycle command received on the row command inputs R[9:0] and latched with the rising CK clock edge as shown in Figure 34. The command must be followed either by RNOP, PRECHARGE or PRECHARGE ALL on the falling CK clock edge of the same cycle. Note that a PRECHARGE ALL must be for the other pseudo channel. A PRECHARGE command could be to any bank in the other pseudo

channel as well as to a different bank in the same pseudo channel. In all cases the timing requirements for issuing these commands must be met.

Parity is evaluated with the PER-BANK REFRESH command when CA parity is enabled in MR0 OP6.

Figure 34: PER-BANK REFRESH Command

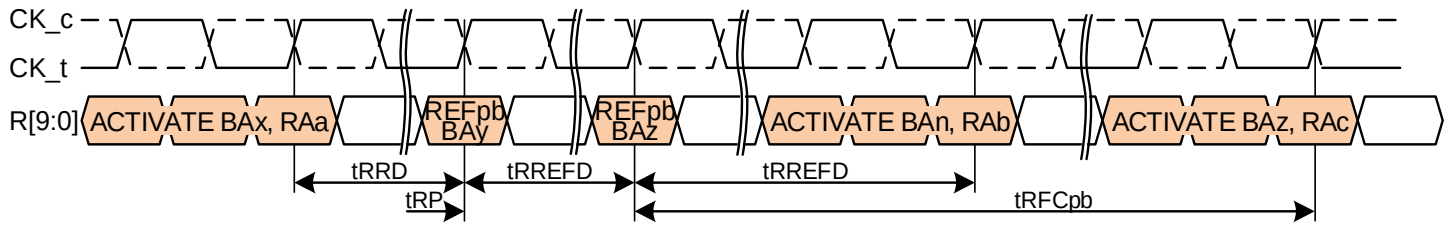


Note: 1. PC = Pseudo Channel 0 or 1; BA = Bank address; SID = Stack ID; V = Valid (H or L)

The PER-BANK REFRESH command is nonpersistent, so it must be issued each time a refresh is required. A minimum time 'RRD is required between an ACTIVATE command and a PER-BANK REFRESH command to a different bank. A minimum time 'RREFD is required between any two PER-BANK REFRESH commands (see below for an exception requiring 'RFCpb), and between a PER-BANK REFRESH command and an ACTIVATE command to a different bank as shown in Figure 35. A minimum time 'RFCpb is required between a PER-BANK REFRESH command and an access command to the same bank that follows. The bank to be refreshed must be precharged with 'RP satisfied prior to the PER-BANK REFRESH command. The bank is in idle state after completion of the PER-BANK REFRESH command.

The row address for each bank is provided by internal refresh counters. This makes the row address bits “Don't Care” during PER-BANK REFRESH commands.

Rules for issuing PER-BANK REFRESH commands to the banks apply to each SID individually. A PER-BANK REFRESH command to one of the 16 banks per SID can be issued in any order. After all banks within an SID have been refreshed using the PER-BANK REFRESH command and after waiting for at least 'RFCpb, the internal refresh counter is incremented and the controller can issue another set of PER-BANK REFRESH commands in the same or a different order. However, it is illegal to send another PER-BANK REFRESH command to a bank unless all banks within an SID have been refreshed using the PER-BANK REFRESH command. The controller must track the bank being refreshed by the PER-BANK REFRESH command.

Figure 35: Per-Bank Refresh Cycle


- Notes:
1. BAn,x,y,z = Bank address n,x,y,z; RAa,b,c = Row address a,b,c.
 2. ^tRRD timing must be met between ACTIVATE commands and REFpb command to different banks.
 3. ^tRREFD timing must be met between consecutive REFpb commands to different banks, and between a REFpb command followed by an ACTIVATE command to the different bank.
 4. ^tRFCpb timing must be met between a REFpb command followed by an ACTIVATE command to the same bank.

PER-BANK REFRESH MANAGEMENT commands can be inserted as needed between PER-BANK REFRESH commands provided the ^tRREFD timing (any bank) and ^tRFCpb timing (same bank) between the commands have been met as illustrated in Figure 36. However, neither a PER-BANK REFRESH nor a PER-BANK REFRESH MANAGEMENT command is allowed to be issued during the ^tRFCpb period following the completion of a set of PER-BANK REFRESH commands.

The bank count is synchronized between the host and the HBM3 DRAM by resetting the bank count to zero. Synchronization can occur upon exit from reset state or by issuing a REFRESH or SELF REFRESH ENTRY command. Both commands may be issued at any time even if a preceding sequence of PER-BANK REFRESH commands has not completed cycling through all banks. A REFRESH command will unconditionally reset the bank count to zero even if the command is internally re-purposed for an ECS event (see Error Check and Scrub (ECS)).

The average rate of PER-BANK REFRESH commands ^tREFIpb can be calculated by the following formula: $\text{REFIpb} = \text{REFI} / 16$.

The example in Table 42 (with 16 banks shown as an example) shows two full sets of REFpb commands with the bank counter reset to 0 and the refresh counter incremented after 16 REFpb commands each. The third set of REFpb commands is interrupted by the REFRESH command which resets the bank counter to 0 and performs refreshes to all banks indicated by the refresh counter.


Table 42: Refresh Counter Increments

Count	Sub-Count	Command	Bank Address	Refresh Bank	Bank Counter	Refresh Counter
0	0	Reset, REFab or SRE command			To 0	
1	1	REFpb	0000	0	0 to 1	n
2	2	REFpb	0001	1	1 to 2	
3	3	REFpb	0010	2	2 to 3	
4	4	REFpb	0011	3	3 to 4	
...		...		...		
15	15	REFpb	1110	14	14 to 15	
16	16	REFpb	1111	15	15 to 0	
17	1	REFpb	0100	4	0 to 1	n + 1
18	2	REFpb	0111	7	1 to 2	
19	3	REFpb	1011	11	2 to 3	
20	4	REFpb	0110	6	3 to 4	
...		...		...		
31	15	REFpb	1100	12	14 to 15	
32	16	REFpb	0001	1	15 to 0	
33	1	REFpb	0010	2	0 to 1	n + 2
34	2	REFpb	1001	9	1 to 2	
35	3	REFpb	0000	0	2 to 3	
36	0	REFab	V	all	To 0	n + 2
37	1	REFpb	1010	10	0 to 1	n + 3
38	2	REFpb	0101	5	1 to 2	
...		...		...		

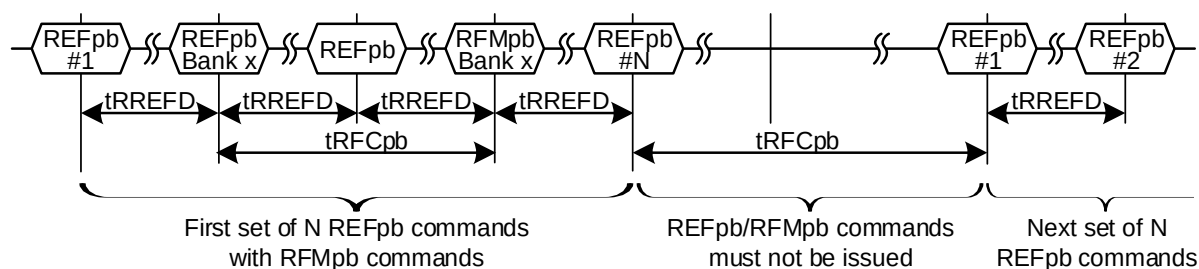
Figure 36: Sets of Per-Bank Refresh Commands with Per-Bank Refresh Management Commands



Table 43: Refresh and Refresh Management Command Scheduling Requirements

From Command	To Command	Min. Delay Between “From Command” to “To Command”	Notes
REFab	REFab	¹ RFCab	
	REFpb (any bank)	¹ RFCab	
	RFMab	¹ RFCab	
	RFMpb (any bank)	¹ RFCab	
	ACT	¹ RFCab	
REFpb	REFab	¹ RFCpb	
	REFpb (different bank)	¹ RREFD	
	REFpb (any bank)	¹ RFCpb	3
	RFMab	¹ RFCpb	
	RFMpb (different bank)	¹ RREFD	
	RFMpb (any bank)	¹ RFCpb	4
	ACT (same bank)	¹ RFCpb	
	ACT (different bank)	¹ RREFD	1
RFMab	REFab	¹ RFCab	
	REFpb (any bank)	¹ RFCab	
	RFMab	¹ RFCab	
	RFMpb (any bank)	¹ RFCab	
	ACT	¹ RFCab	
RFMpb	REFab	¹ RFCpb	
	REFpb (same bank)	¹ RFCpb	
	REFpb (different bank)	¹ RREFD	
	RFMab	¹ RFCpb	
	RFMpb (same bank)	¹ RFCpb	
	RFMpb (different bank)	¹ RREFD	
	ACT (same bank)	¹ RFCpb	
	ACT (different bank)	¹ RREFD	
ACT	REFRESH	¹ RC	2
	REFpb (same bank)	¹ RC	2
	REFpb (different bank)	¹ RRD	1
	RFMab	¹ RC	
	RFMpb (same bank)	¹ RC	
	RFMpb (different bank)	¹ RRD	

Notes: 1. ¹FAW parameter must be observed as well.
2. A bank must be in the idle state with ¹RP satisfied before it is refreshed.

3. ^tRFCpb parameter must be observed when the first REFpb command completes a set of 16 per-bank refresh operations within an SID and the second REFpb command initiates the next set of 16 per-bank refresh operations within an SID.
4. ^tRFCpb parameter must be observed when the REFpb command completes a set of 16 per-bank refresh operations within an SID, and the following RFMpb command operation.

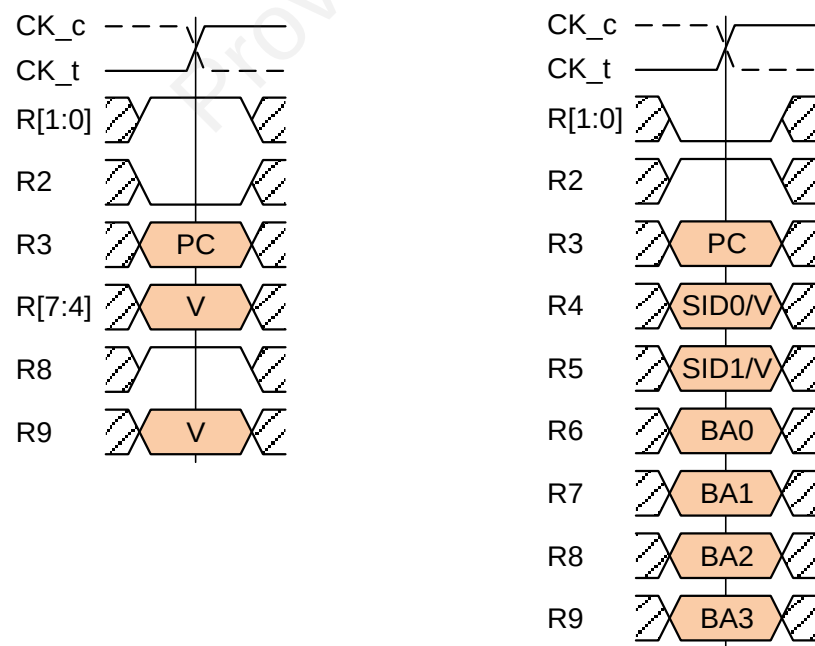
Refresh Management (RFM)

Periods of high DRAM activity may require additional refresh commands to protect the integrity of the stored data. The requirement for additional Refresh Management (RFM) is indicated in the RFM field of the DEVICE_ID WDR (see Table 90): RFM = 0_B indicates that no additional refresh is needed beyond the refreshes specified in the REFRESH section of the specification; RFM = 1_B indicates additional DRAM refresh management is required.

A suggested implementation of refresh management by the controller monitors ACTIVATE commands issued per bank to the device. This activity can be monitored as a rolling accumulated ACTIVATE (RAA) count. Each ACTIVATE command will increment the RAA count by 1 for the individual bank receiving the ACTIVATE command.

When the RAA count reaches a DRAM vendor specified Initial Management Threshold (RAAIMT), which is indicated by the HBM3 DRAM in the RAAIMT field of the DEVICE_ID WDR, additional refresh management is needed. Executing a refresh management command allows additional time for the HBM3 DRAM to manage refresh internally. The RFM operation can be initiated to all banks with the (all-bank) REFRESH MANAGEMENT (RFMab) command, or to a single bank with the PER-BANK REFRESH MANAGEMENT (RFMpb) command. The encoding of RFM related commands RFMab and RFMpb is shown in Figure 37.

Figure 37: REFRESH MANAGEMENT and PER-BANK REFRESH MANAGEMENT Commands



Note: 1. PC = Pseudo Channel 0 or 1; BA = Bank address; SID = Stack ID; V = Valid (H or L)



Both half-cycle commands are received on the R[9:0] inputs and latched with the rising CK clock edge. They must be followed either by RNOP, PRECHARGE or PRECHARGE ALL on the falling CK clock edge of the same cycle. Note that a PRECHARGE ALL must be for the other pseudo channel. In case of a PER-BANK REFRESH MANAGEMENT command a PRECHARGE command could be to any bank in the other pseudo channel as well as to a different bank in the same pseudo channel. In all cases the timing requirements for issuing these commands must be met.

Parity is evaluated with the REFRESH MANAGEMENT and PER-BANK REFRESH MANAGEMENT commands when CA parity is enabled in MR0 OP6.

An HBM3 DRAM not requiring refresh management will ignore RFMab and RFMpb commands and execute an RNOP command instead.

The REFRESH MANAGEMENT and PER-BANK REFRESH MANAGEMENT command scheduling shall meet the same minimum separation requirements as those for the REFRESH and PER-BANK REFRESH commands, respectively (see REFRESH Command (REFab) and PER-BANK REFRESH Command (REFpb) sections). The REFRESH MANAGEMENT command period is the same as the REFRESH command period ('RFCab), and the PER-BANK REFRESH MANAGEMENT command period is the same as the PER-BANK REFRESH command period ('RFCpb). The requirement for PER-BANK REFRESH commands to be issued to all banks in a rolling fashion does not apply to PER-BANK REFRESH MANAGEMENT commands.

When an RFM command is issued to the HBM3 DRAM, the RAA counter in any bank receiving the command can be decremented by the RAAIMT value, down to a minimum RAA value of 0 (no negative or "pull-in" of RFM commands is allowed). Issuing an RFMab command allows the RAA count in all banks to be decremented by the RAAIMT value. Issuing an RFMpb command allows the RAA count only in the bank selected by {SID[1:0], BA[3:0]} to be decremented by the RAAIMT value.

RFM commands are allowed to accumulate or "postpone", but the RAA counter shall never exceed a vendor specified RAA Maximum Management Threshold (RAAMMT), which is indicated by the HBM3 DRAM in the RAAMMT field of the DEVICE_ID WDR (see Table 90). If the RAA counter reaches RAAMMT, no additional ACTIVATE commands are allowed to the bank until one or more REF or RFM commands have been issued to reduce the RAA counter below the maximum value.

An RFM command does not replace the requirement for the controller to issue periodic REF commands to the HBM3 DRAM, nor does an RFM command affect internal refresh counters. The RFM commands are bonus time for the HBM3 DRAM to manage refresh internally. However, issuing a REF command also allows decrementing the RAA counter by a value indicated the RAA_CNT_DEC field of the DEVICE_ID WDR. Hence, any periodic REF command issued to the HBM3 DRAM allows the RAA counter of the banks being refreshed to be decremented by that value. Issuing an REFab command allows the RAA count in all banks to be decremented by that value. Issuing an REFpb command allows the RAA count only in the bank selected by {SID[1:0], BA[3:0]} to be decremented by that value.

The per-bank RAA count values may be reset to 0 when the HBM3 DRAM is held in self refresh for at least 'RAASRF time. No decrement to the per-bank RAA count values is allowed for entering or exiting self refresh and when the HBM3 DRAM is held in self refresh for less than 'RAASRF time.



Adaptive Refresh Management (ARFM)

This HBM3 DRAM supports a refresh management mode called Adaptive Refresh Management (ARFM). The HBM3 DRAM indicates the support of ARFM via the ARFM bit in the IEEE1500 DEVICE_ID WDR. Since RFM related parameters RAAIMT, RAAMMT and RAADEC are read-only, the ARFM mode allows the controller flexibility to choose additional (lower) RFM threshold settings called "RFM levels". The RFM levels permit alignment of the controller issued RFM commands with the DRAM internal management of these commands. MR8 OP[5:4] select the RFM level as shown in in Table 44.

Table 44: Mode Register Definition for Adaptive RFM Levels

MR8 OP[5:4]	RFM Level	RFM Requirement	RAAIMT	RAAMMT	RAA Decrement per REF Command	Notes
00	Default	Default	RAAIMT	RAAMMT	RAADEC	1
01	Level A	RFM is required	RAAIMT_A	RAAMMT_A	RAADEC_A	
10	Level B	RFM is required	RAAIMT_B	RAAMMT_B	RAADEC_B	
11	Level C	RFM is required	RAAIMT_C	RAAMMT_C	RAADEC_C	

Notes: 1. RAAIMT, RAAMMT and RAADEC values are set by DRAM vendor in the IEEE1500 DEVICE_ID WDR.

The Adaptive RFM mode inherits the RAA counting and decrement attributes of the standard RFM mode, while using the alternate RAAIMT, RAAMMT and RAADEC values for the selected RFM level. Increasing the RFM level results in increased need for RFM commands. Level C is highest RFM level. The alternate RAAIMT, RAAMMT and RAADEC values for RFM level A to C can be retrieved from the corresponding fields of the IEEE1500 DEVICE_ID WDR.

Setting the bits in MR8 OP[5:4] to something other than the default "00" will select one of the RFM levels A, B or C. The host shall decrement the Rolling Accumulated ACT (RAA) count to 0, either with RFM or pending REF commands, prior to making a change to the ARFM level.

It is required to set the same RFM level on all channels of the HBM3 DRAM.

Adaptive RFM also allows an HBM3 DRAM shipped with 'RFM not required' (RFM bit in IEEE1500 DEVICE_ID WDR = 0) to override that initial setting and enable RFM by programming a non-default ARFM level. The HBM3 DRAM internally manages the change to treat the RFM command as an RFM command in this special override case as shown in Table 45.

Table 45: RFM Commands Perceived by HBM3 DRAM

Command	Bit in DEVICE_ID WDR		RFM Level MR8 OP[5:4]	Command perceived by HBM3 DRAM	Notes
	RFM	ARFM			
RFMab / RFMpb	0 _B (RFM not required)	0 _B (ARFM not supported)	00	RNOP	1
			01, 10 or 11	Illegal	
		1 _B (ARFM supported)	00	RNOP	2
			01, 10 or 11	RFMab / RFMpb	
	1 _B (RFM required)	0 _B (ARFM not supported)	00	RFMab / RFMpb	1
			01, 10 or 11	Illegal	
		1 _B (ARFM supported)	00, 01, 10 or 11	RFMab / RFMpb	

- Notes:
1. These cases are marked as 'Illegal' because HBM3 DRAMs not supporting Adaptive RFM do not support the selection of an ARFM level via MR8 OP[5:4] and therefore define these bits as RFU which means that the only supported setting for these bits is 00_B.
 2. Adaptive RFM enables an HBM3 DRAM shipped with RFM = 0_B (RFM not required) to override the initial setting and enable Adaptive RFM by programming a non-default RFM level.

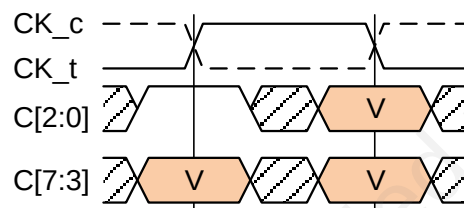
Column No Operation Command (CNOP)

The COLUMN NO OPERATION (CNOP) command is a 1-cycle command as shown in Figure 38 and is used to instruct the HBM3 DRAM to perform a NOP as the column command; this prevents unwanted column commands from being registered during idle or wait states. Operations already in progress are not affected.

Parity is evaluated with the CNOP command when CA parity is enabled in MR0 OP6.

CNOP is assumed for the C[7:0] inputs on subsequent timing diagrams unless other column commands are explicitly shown.

Figure 38: CNOP Command



Note: 1. V = Valid (H or L)

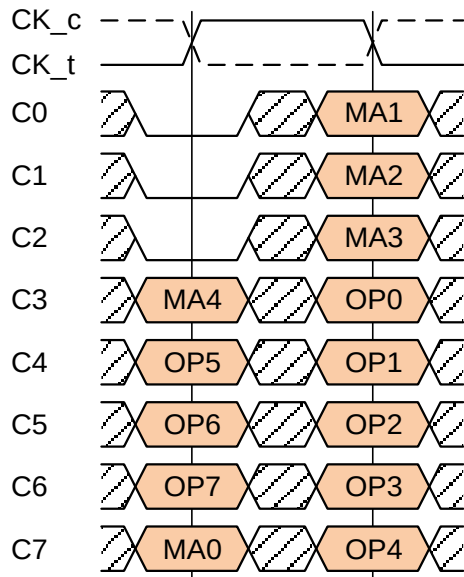
Mode Register Set Command (MRS)

MODE REGISTER SET is a one-cycle command received on the column command inputs C[7:0] and latched with the rising and falling CK clock edges. The command requires an RNOP command on the row command inputs R[9:0].

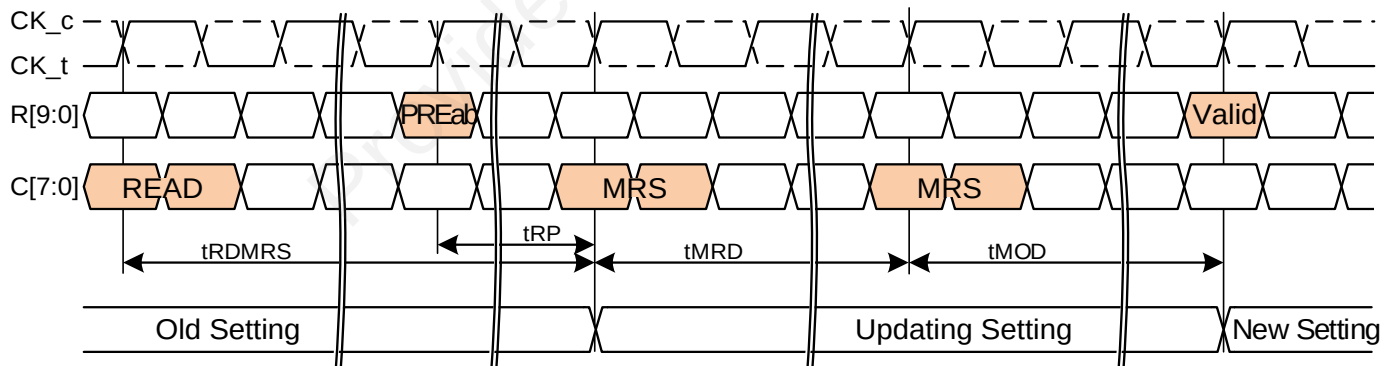
The command is used to load the mode registers of the HBM3 DRAM. Inputs MA[4:0] select one of the sixteen mode registers, and the inputs OP[7:0] determine the opcode to be loaded as shown in Figure 39. Refer to the mode register section for the register definition.

The MODE REGISTER SET (MRS) command can only be issued when all banks are idle, the time t_{RDMRS} from a preceding READ command has elapsed and the time t_{WRMRS} from a preceding WRITE command has elapsed. The MRS command cycle time t_{MRD} is required to complete the write operation to the mode register and is the minimum time required between two MRS commands. The MRS command to Non-MRS command delay, t_{MOD} , is required by the HBM3 DRAM to update the features, and is the minimum time required from an MRS command to a non-MRS command excluding RNOP and CNOP.

Parity is evaluated with the MODE REGISTER SET command when CA parity has already been enabled in MR0 OP6 prior to this MODE REGISTER SET command. When CA parity is enabled by a MODE REGISTER SET command, the HBM3 DRAM requires all subsequent commands including RNOP and CNOP to be issued with correct parity until t_{MOD} has expired for the MODE REGISTER SET command that disables CA parity. See Figure 20 for enabling and disabling CA parity.

Figure 39: MODE REGISTER SET Command


Note: 1. MA = Mode register address; OP = mode register opcode; V = Valid (H or L)

Figure 40: Mode Register Set Timings


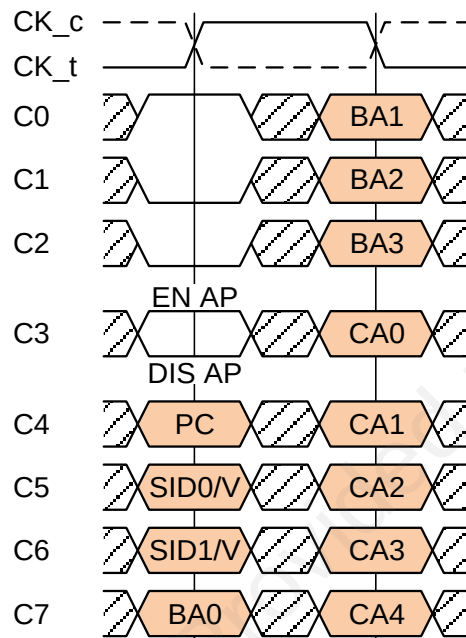
Valid = Any row command allowed in bank idle state

WRITE Command (WR/WRA)

A write burst is initiated with a WRITE command. WRITE is a one-cycle command received on the column command inputs C[7:0] and latched with the rising and falling CK clock edges as shown in Figure 41. The SID, bank and column addresses are provided with the WRITE command, and auto precharge is either enabled or disabled for that access.

Parity is evaluated with the WRITE command when CA parity is enabled in MR0 OP6.

Figure 41: WRITE Command



Note: 1. PC = Pseudo Channel 0 or 1; SID = Stack ID; BA = Bank address; CA = Column address; V = Valid (H or L); EN AP = Enable auto precharge; DIS AP = Disable auto precharge

The length of the burst initiated with a WRITE command is eight, and the column address is unique for this burst of eight. There is no interruption nor truncation of write bursts.

The write latency (WL) is defined from the rising CK edge on which the WRITE command is issued to the rising CK edge from which the tDQSS delay is measured, with the number of clock cycles as programmed in the WL field of MR1.

The write data strobe for a write burst must include two WDQS preamble pulses and two WDQS postamble pulses. The first WDQS edge must be driven $(WL-1) \times tCK + tDQSS$ after the rising CK edge when the WRITE command was issued.

The HBM3 DRAM uses an un-matched WDQS-to-data path to achieve lower power consumption. The write data (DQ, DBI, ECC) must arrive center-aligned to the internal latching clock a time tWDQS2DQ_I after the corresponding WDQS edges. tWDQS2DQ_I is the WDQS-to-DQ/DBI/ECC offset; the offset varies with PVT and must be trained prior to any write operation.

The first valid write data must arrive $(WL \times tCK + tDQSS + tWDQS2DQ_I)$ after the rising CK edge when the WRITE command was issued. Each subsequent write data must be applied nominally tCK/4 later and will be latched with subsequent WDQS edges.



The internal WDQS clock tree is optimized for lowest signal skew among signals within a group as outlined in Table 46. The grouping is aligned with the physical location of signals in a DWORD (see HBM3 microbump matrix) with no lane being repaired.

Each group contains 5 to 8 signals. The internal WDQS clock tree however compensates the different loading by e.g. adding dummy loads. The per-group de-skew is also deterministic and not frequency dependent. A larger signal skew should be expected between different groups. A per-group de-skew is recommended to achieve the largest signaling margin for write data.

Table 46: Signal Groups for Write Data De-Skew

Group	Signal List (DWORD0)	Signal List (DWORD1)
T0	DQ0, DQ1, DQ2, DQ8, DQ9, DQ10, ECC0, ECC1	DQ32, DQ33, DQ34, DQ40, DQ41, DQ42, ECC2, ECC3
T1	DQ3, DQ4, DQ11, DQ12, RD0, DPAR0	DQ35, DQ36, DQ43, DQ44, RD2, DPAR1
T2	DQ5, DQ6, DQ7, DQ13, DQ14, DQ15, DBI0, DBI1	DQ37, DQ38, DQ39, DQ45, DQ46, DQ47, DBI4, DBI5
T3	DQ16, DQ17, DQ18, DQ24, DQ25, DQ26	DQ48, DQ49, DQ50, DQ56, DQ57, DQ58
T4	DQ19, DQ20, DQ27, DQ28, RD1	DQ51, DQ52, DQ59, DQ60, RD3
T5	DQ21, DQ22, DQ23, DQ29, DQ30, DQ31, DBI2, DBI3	DQ53, DQ54, DQ55, DQ61, DQ62, DQ63, DBI6, DBI7

Clock to Write Data Strobe Timings: the CK clock to write data strobe (WDQS) relationship is shown in the figure below. Related parameters:

- $t_{DQSS(min/max)}$ describes the allowed range for a rising or falling WDQS edge relative to the rising or falling CK clock edge;
- t_{DQSS} is the actual position of a WDQS edge relative to CK;
- t_{WQSH} describes the WDQS HIGH pulse width;
- t_{WQSL} describes the WDQS LOW pulse width;
- t_{WPRE1} describes the WDQS preamble for writes;
- t_{WPST1} describes the WDQS postamble for writes.

Write Data Strobe and Data In Timings: the write data strobe (WDQS) to data in relationship is shown in the figure below. Related parameter:

- $t_{WDQS2DQ_I}$ describes the minimum-to-maximum WDQS-to-DQ/DBI/ECC offset for writes
- t_{DIVW} describes allowed range for receiver minimum setup/hold time for sampling at DQ
- V_{DIVW} describes allowed range for receiver voltage peak to peak size

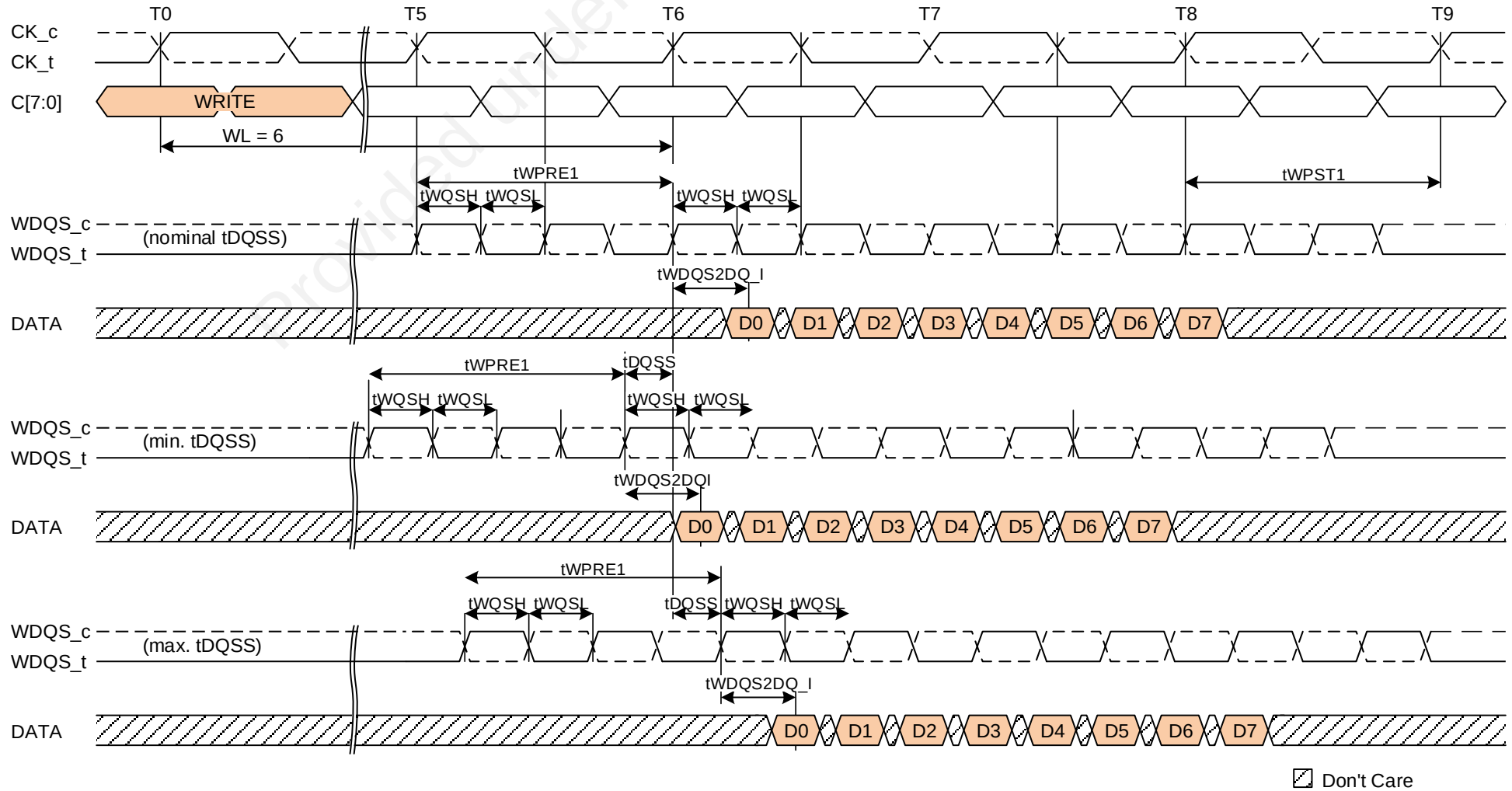
Common notes: the following notes apply to all subsequent timing diagrams related to the WRITE command. See also each figure for additional notes.

- WL = 6, RL = 10 and PL = 3 (if applicable) are shown as examples.
- DATA = DQ[31:0], DBI[3:0], ECC[1:0] for PC0, and DQ[63:32], DBI[7:4], ECC[3:2] for PC1.
DPAR = DPAR0 for PC0 and DPAR 1 for PC1 (if applicable).



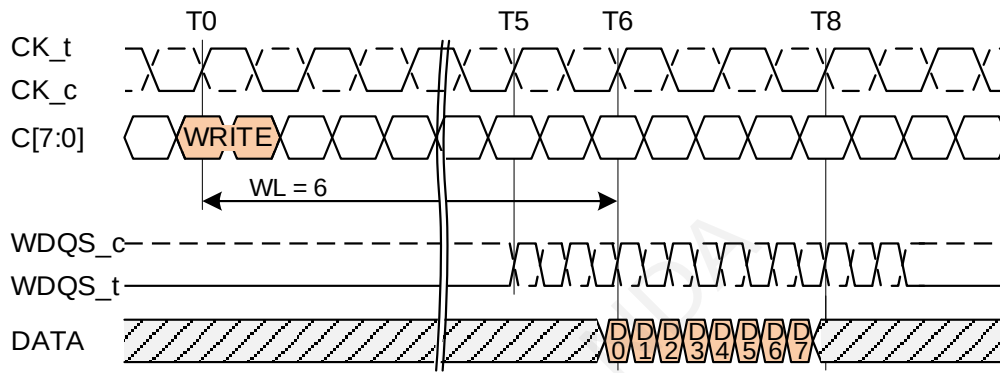
- D0 ... D7 = BL8 data burst. P0 ... P7 = BL8 parity data burst.
- WDQS_t/_c is WDQS0_t/_c for PC0, and WDQS1_t/_c for PC1. RDQS_t/_c is RDQS0_t/_c for PC0, and RDQS1_t/_c for PC1 (if applicable).
- Write data timings are shown with $t_{DQSS} = 0\text{ns}$ and $t_{WDQS2DQ_I} = 0\text{ns}$. Write training is required to determine the actual $t_{WDQS2DQ_I}$ value.
- WDBI could be on or off and is controlled with MR0 OP1.

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Figure 42: Clock to WDQS and Data Input Timings


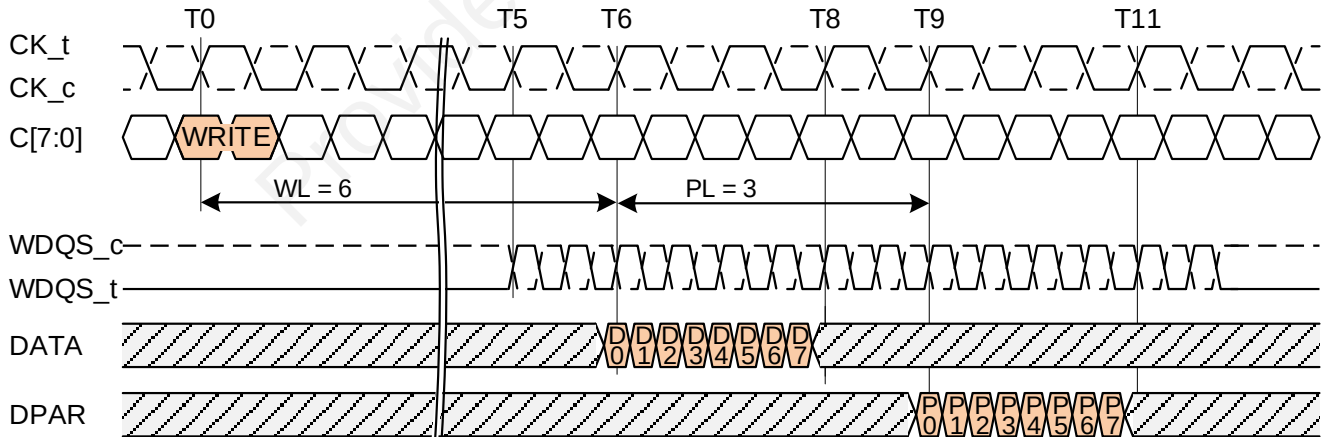
A basic single write burst is shown in Figure 43. Two WDQS preamble pulses are received at T5, and two WDQS postamble pulses at T8.

Figure 43: Single Write Burst

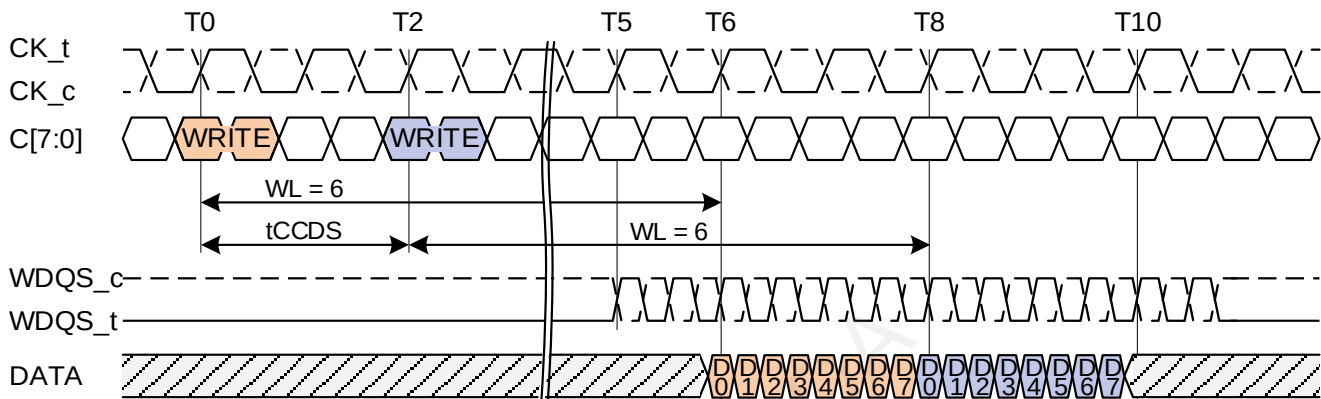


An example of a basic single write burst with write data parity enabled is shown in Figure 44. With PL = 3 four additional WDQS pulses are received at cycles T9 and T10 to latch the DPAR input, followed by the WDQS postamble pulses at T11.

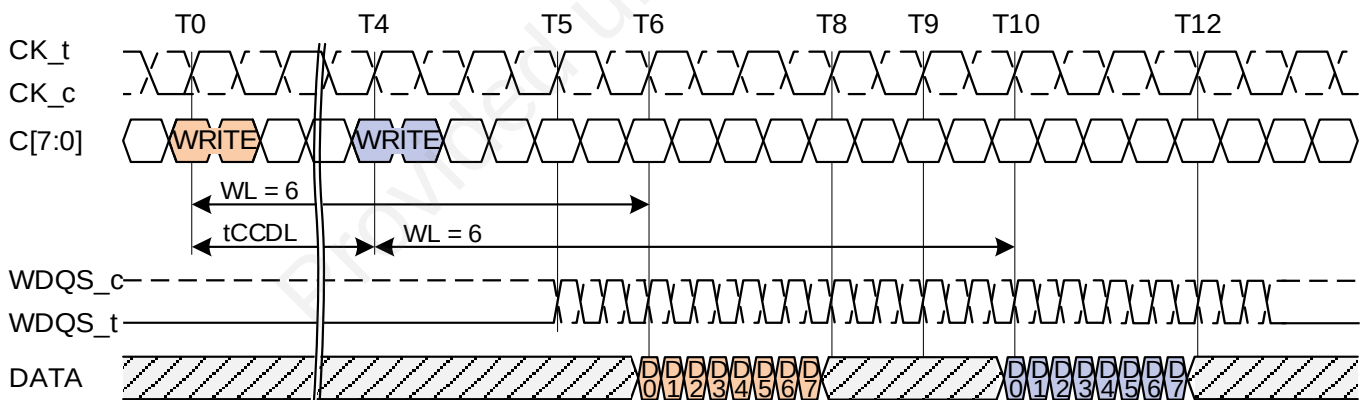
Figure 44: Single Write Burst with PL = 3



Data from any write burst may be concatenated with data from a subsequent WRITE command. A continuous flow of data can be maintained as shown in Figure 45. The first data element from the new burst follows the last element of a completed burst. The new WRITE command should be issued after the previous WRITE command according to the t_{CCDS} timing. If that WRITE command is to another idle bank then an ACTIVATE command must precede the WRITE command and t_{RCDWR} also must be met.

Figure 45: Seamless Write Bursts ($t_{CCDS} = 2$)


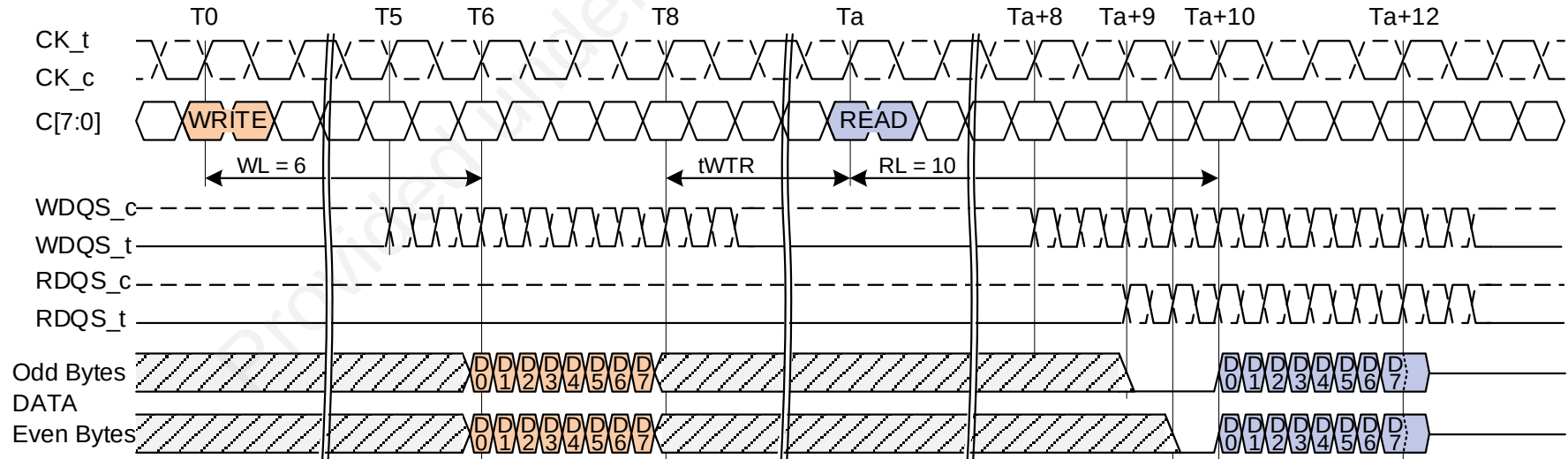
An example of non-seamless write bursts is shown in Figure 46. With $t_{CCDL} = 4$ the timing of each of the two write bursts is identical to a single write burst.

Figure 46: Non-Seamless Write Bursts ($t_{CCDL} = 4$)


A READ can be issued any time after a WRITE command as long as the bus turnaround time t_{WTR} is met as shown in Figure 47. If that READ command is to another idle bank, then an ACTIVATE command must precede the READ command and t_{RCDRD} also must be met.

The data bus is preconditioned for the first read burst by being driven LOW one clock cycles (even bytes) and half a clock cycle (odd bytes) prior to the first valid data element of the read burst regardless of whether RDBI is enabled in MR0 or not.

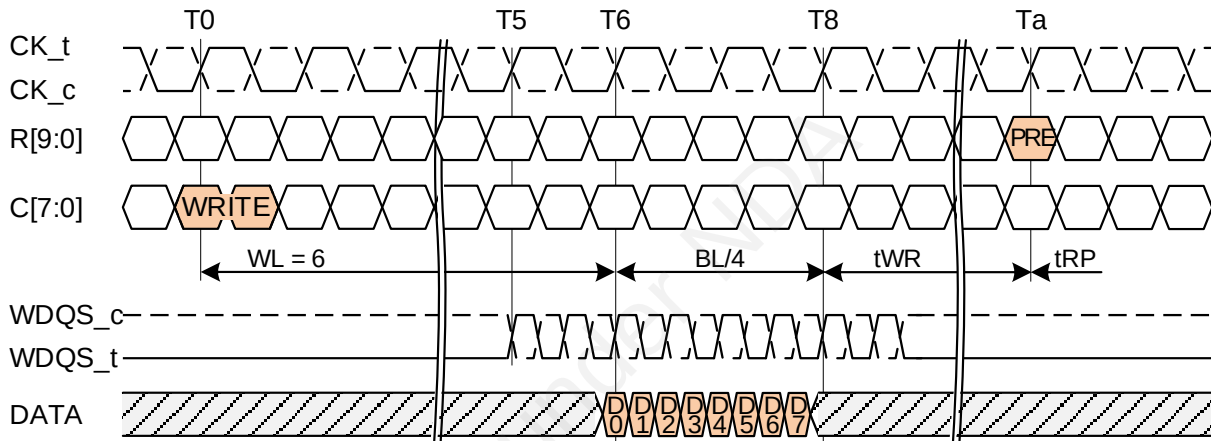
Four WDQS preamble pulses are received at $Ta+8$ and $Ta+9$, and two RDQS preamble pulses are returned at $Ta+9$. Two WDQS postamble pulses are received and two RDQS postamble pulses are returned at $Ta+12$.


Figure 47: Write to Read


- Notes:
1. $t_{DQSS} = 0$, $t_{WDQS2DQ_I} = 0$, $t_{WDQS2DQ_O} = 0$ and nominal t_{LZ} , t_{HZ} is shown for illustration purposes.
 2. $t_{WTR} = t_{WTRL}$ when both WRITE and READ access banks in the same bank group, $t_{WTR} = t_{WTRS}$ otherwise.
 3. RDBI could be on or off and is controlled with MR0 OP0.

The write recovery time t_{WR} must have elapsed before a PRECHARGE command can be issued to that bank as shown in the figure below. The t_{WR} interval begins with the completion of the write burst at $(WL + BL/4 = WL + 2)$ clock cycles after the WRITE command was issued. Also, t_{RAS} must be met when the PRECHARGE is issued. After the PRECHARGE command, a subsequent ACTIVATE command to the same bank cannot be issued until t_{RP} is met.

Figure 48: Write To Precharge



READ Command (RD/RDA)

A read burst is initiated with a READ command. READ is a one-cycle command received on the column command inputs C[7:0] and latched with the rising and falling CK clock edges as shown in Figure 49. The SID, bank and column addresses are provided with the READ command, and auto precharge is either enabled or disabled for that access.

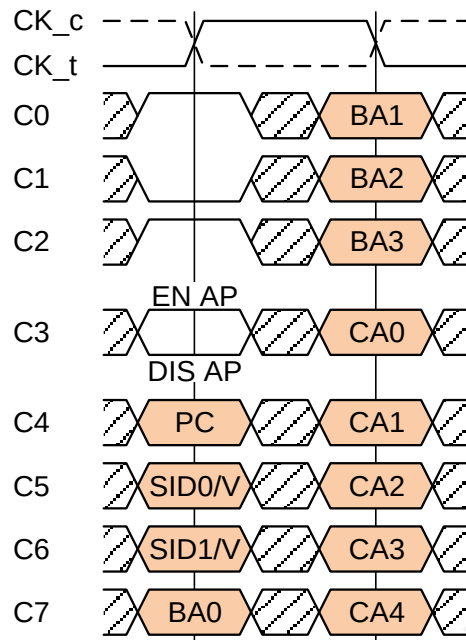
Parity is evaluated with the READ command when CA parity is enabled in MR0 OP6.

The length of the burst initiated with a READ command is eight, and the column address is unique for this burst of eight. There is no interruption nor truncation of read bursts.

The read latency (RL) is defined from the rising CK edge on which the READ command is issued to the rising CK edge from which the t_{DQSS} delay is measured, with the number of clock cycles as programmed in the RL field of MR2. The first valid data is available $(RL \times t_{CK} + t_{DQSS} + t_{WDQS2DQ_O})$ after the rising CK edge when the READ command was issued.

The write data strobe (WDQS) is the clock source for read data (DQ, DBI, ECC, SEV) and the read data strobe (RDQS). WDQS must meet the same t_{DQSS} timing relative to the CK clock as for writes.

The write data strobe for a read burst must include four WDQS preamble pulses and two WDQS postamble pulses; the first WDQS edge must be driven $((RL-2) \times t_{CK} + t_{DQSS})$ after the rising CK edge when the READ command was issued.

Figure 49: READ Command


Note: 1. PC = Pseudo Channel 0 or 1; SID = Stack ID; BA = Bank address; CA = Column address; V = Valid (H or L); EN AP = Enable auto precharge; DIS AP = Disable auto precharge

The read data strobe derived from the received write data strobe includes two RDQS preamble pulses and two RDQS postamble pulses; the first RDQS edge occurs $((RL-1) \times 'CK + 'DQSS + 'WDQSDQ_O + 'DQSQ)$ after the rising CK edge when the READ command was issued.

The first data bit of the read burst is driven at $(RL \times 'CK + 'DQSS + 'WDQSDQ_O + 'DQSQ)$ after the rising CK edge when the READ command was issued. Each subsequent data-out is driven nominally $'CK/4$ later.

The output drivers are enabled and begin driving either HIGH or LOW nominally two WDQS cycles (odd bytes) and one WDQS cycle (even bytes) prior to the first valid data bit. See Internal DBIac States with Read for bus pre-conditioning with RDBI enabled and disabled modes on a first READ command.

The output drivers will drive High-Z after the completion of the burst provided no other READ command has been issued.

The internal WDQS clock tree is optimized for lowest signal skew among signals within a group as outlined in Table 47. The grouping is aligned with the physical location of signals in a DWORD (see HBM3 microbump matrix) with no lane being repaired.

Each group contains 6 to 8 signals. The internal WDQS clock tree however compensates the different loading by e.g. adding dummy loads. The per-group de-skew is also deterministic and not frequency dependent. A larger signal skew should be expected between different groups. A per-group de-skew is recommended to achieve the largest signaling margin for read data.

In this context RDQS_t and RDQS_c are treated as regular data out signals within group T4.


Table 47: Signal Groups for Read Data De-Skew

Group	Signal List (DWORD0)	Signal List (DWORD1)
T0	DQ0, DQ1, DQ2, DQ8, DQ9, DQ10, ECC0, ECC1	DQ32, DQ33, DQ34, DQ40, DQ41, DQ42, ECC2, ECC3
T1	DQ3, DQ4, DQ11, DQ12, RD0, DPAR0	DQ35, DQ36, DQ43, DQ44, RD2, DPAR1
T2	DQ5, DQ6, DQ7, DQ13, DQ14, DQ15, DBI0, DBI1	DQ37, DQ38, DQ39, DQ45, DQ46, DQ47, DBI4, DBI5
T3	DQ16, DQ17, DQ18, DQ24, DQ25, DQ26, SEV0, SEV1	DQ48, DQ49, DQ50, DQ56, DQ57, DQ58, SEV2, SEV3
T4	DQ19, DQ20, DQ27, DQ28, RD1, RDQS0_t, RDQS0_c	DQ51, DQ52, DQ59, DQ60, RD3, RDQS1_t, RDQS1_c
T5	DQ21, DQ22, DQ23, DQ29, DQ30, DQ31, DBI2, DBI3	DQ53, DQ54, DQ55, DQ61, DQ62, DQ63, DBI6, DBI7

Clock to Write Data Strobe Timing: the CK clock to write data strobe (WDQS) relationship is the same as for writes and shown in the figure below. Related parameter:

- $t_{DQSS}(\text{min/max})$ describes the allowed range for a rising or falling WDQS edge relative to the rising or falling CK clock edge;
- t_{DQSS} is the actual position of a WDQS edge relative to CK;
- t_{WQSH} describes the WDQS HIGH pulse width;
- t_{WQSL} describes the WDQS LOW pulse width;
- t_{WPRE2} describes the WDQS preamble for reads;
- t_{WPST2} describes the WDQS postamble for reads.

Write Data Strobe to Read Data Timings: the write data strobe (WDQS) to read data and read data strobe (RDQS) relationship is shown in the figure below. Related parameters:

- $t_{WDQS2DQ_O}(\text{min/max})$ describes the allowed range for a rising or falling RDQS edge relative to the rising or falling WDQS edge; $t_{WDQS2DQ_O}$ may span more than one CK clock cycle;
- $t_{WDQS2DQ_O}$ is the actual position of an RDQS edge relative to WDQS;
- t_{QSH} describes the RDQS HIGH pulse width;
- t_{QSL} describes the RDQS LOW pulse width;
- t_{RPRE} describes the RDQS preamble for reads;
- t_{RPST} describes the RDQS postamble for reads;
- $t_{LZ}(\text{min/max})$ describes the allowed range for the data output High-Z to low impedance transition relative to WDQS;
- $t_{HZ}(\text{min/max})$ describes the allowed range for the data output low impedance to Hi-Z transition relative to WDQS.

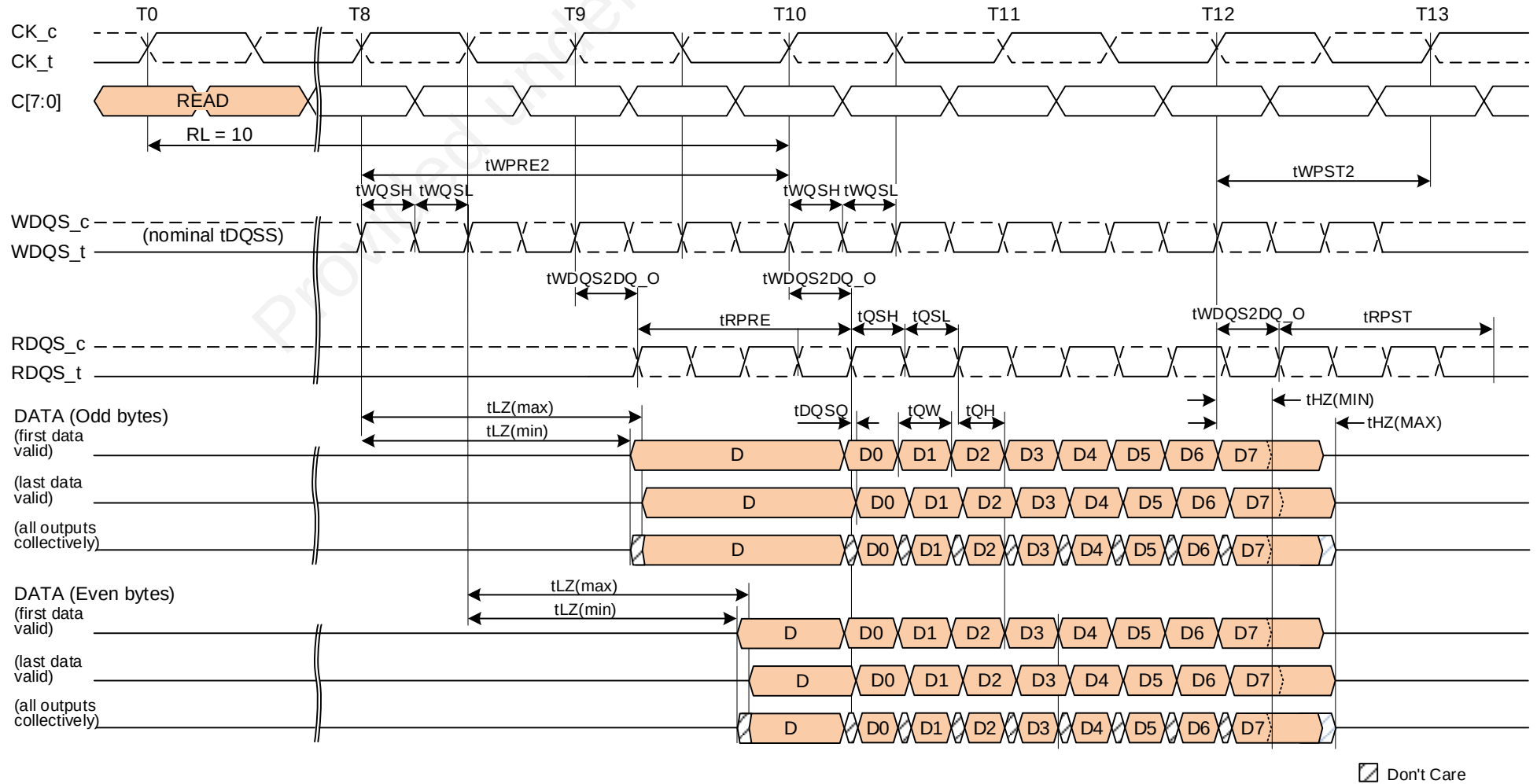
Read Data Strobe to Read Data Timings: the read data strobe (RDQS) to read data and relationship is shown in the figure below. Related parameters:



- t_{DQSQ} describes the latest valid transition of any associated DQ, DBI, ECC or SEV output for both rising and falling RDQS edges;
- t_{QH} describes the earliest invalid transition of any associated DQ, DBI, ECC or SEV output for both rising and falling RDQS edges;
- t_{QW} describes the valid data output window of any associated DQ, DBI, ECC or SEV output for both rising and falling RDQS edges;
- t_{DQ2DQ} describes DQ to DQ skew of any associated DQ, DBI, ECC or SEV output for both rising and falling RDQS edges.

Common notes: the following notes apply to all subsequent timing diagrams related to the READ command. See also each figure for additional notes.

- $RL = 10$, $WL = 6$ and $PL = 3$ (if applicable) are shown as examples;
- $DATA = DQ[31:0]$, $DBI[3:0]$, $ECC[1:0]$, $SEV[1:0]$ for PC0, and $DQ[63:32]$, $DBI[7:4]$, $ECC[3:2]$, $SEV[3:2]$ for PC1. $DPAR = DPAR0$ for PC0 and $DPAR 1$ for PC1 (if applicable);
- $D0 \dots D7 = BL8$ data burst. $P0 \dots P7 = BL8$ parity data burst;
- Read data timings are shown with $t_{DQSS} = 0ns$ and $t_{WDQS2DQ_O} = 0ns$;
- D or $D7 =$ last data-out from a previous READ command (not if a first READ after reset, MRS, self refresh or write-to-read);
- $WDQS_t_c$ is $WDQS0_t_c$ for PC0, and $WDQS1_t_c$ for PC1. $RDQS_t_c$ is $RDQS0_t_c$ for PC0, and $RDQS1_t_c$ for PC1;
- $RDBI$ could be on or off and is controlled with $MR0 OP0$.

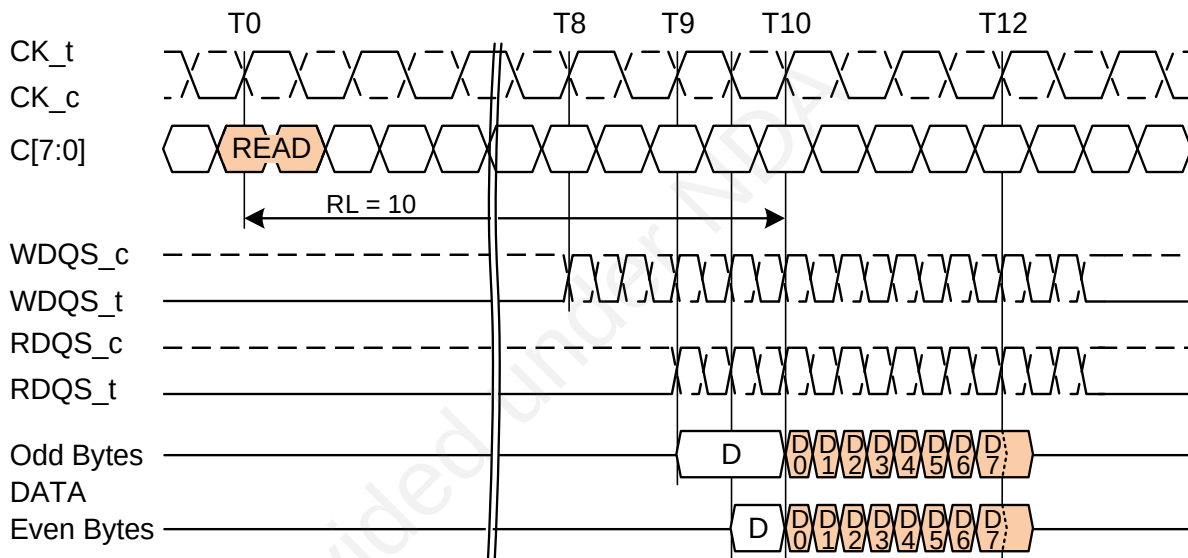

Figure 50: Clock to RDQS and Data Output Timings




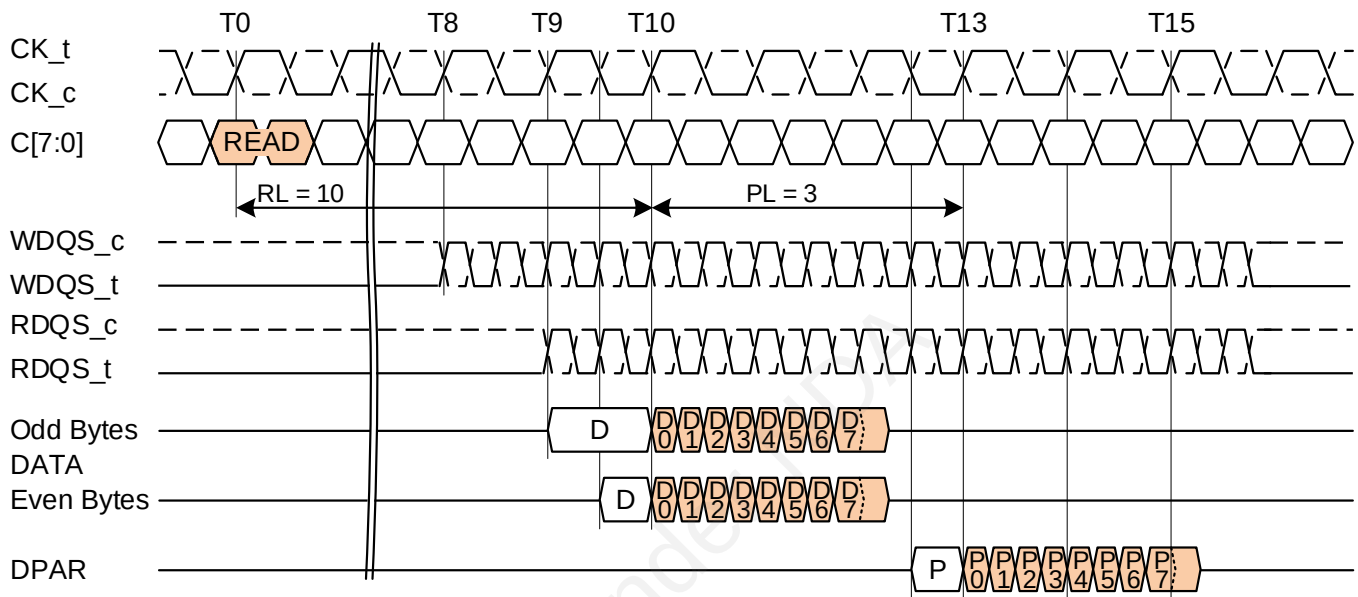
A basic single read burst is shown in Figure 44 . Four WDQS preamble pulses are received at T8 and T9, and two RDQS preamble pulses are returned at T9. Two WDQS postamble pulses are received and two RDQS postamble pulses are returned at T12.

The data bus is preconditioned with the last UI of the preceding read burst one clock cycle (even bytes) and half a clock cycle (odd bytes) prior to the first valid data element of the read burst regardless whether RDBI is enabled in MR0 or not.

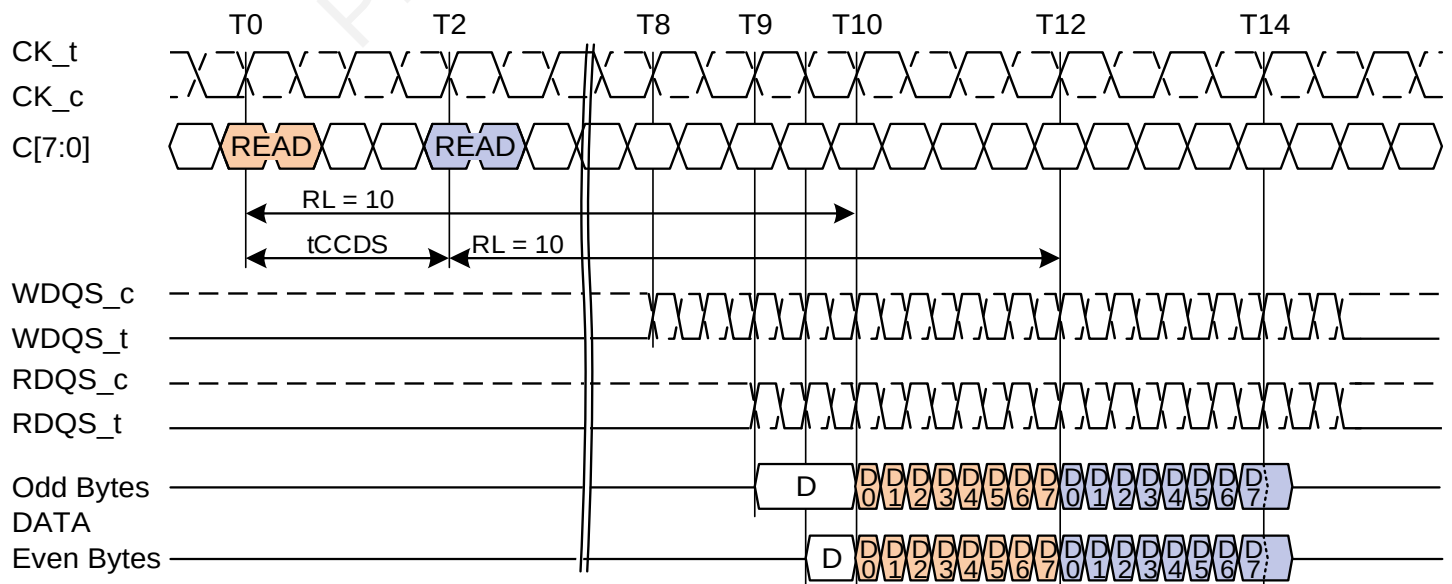
Figure 51: Single Read Burst



An example of a single read burst with data parity enabled is shown in the figure below for PL = 3. The DPAR output is preconditioned over half a clock cycle like for the even data bytes. With PL = 3 six additional WDQS and RDQS pulses for DPAR are received and returned at cycles T13, T14 and T15.


Figure 52: Single Read Burst with PL = 3


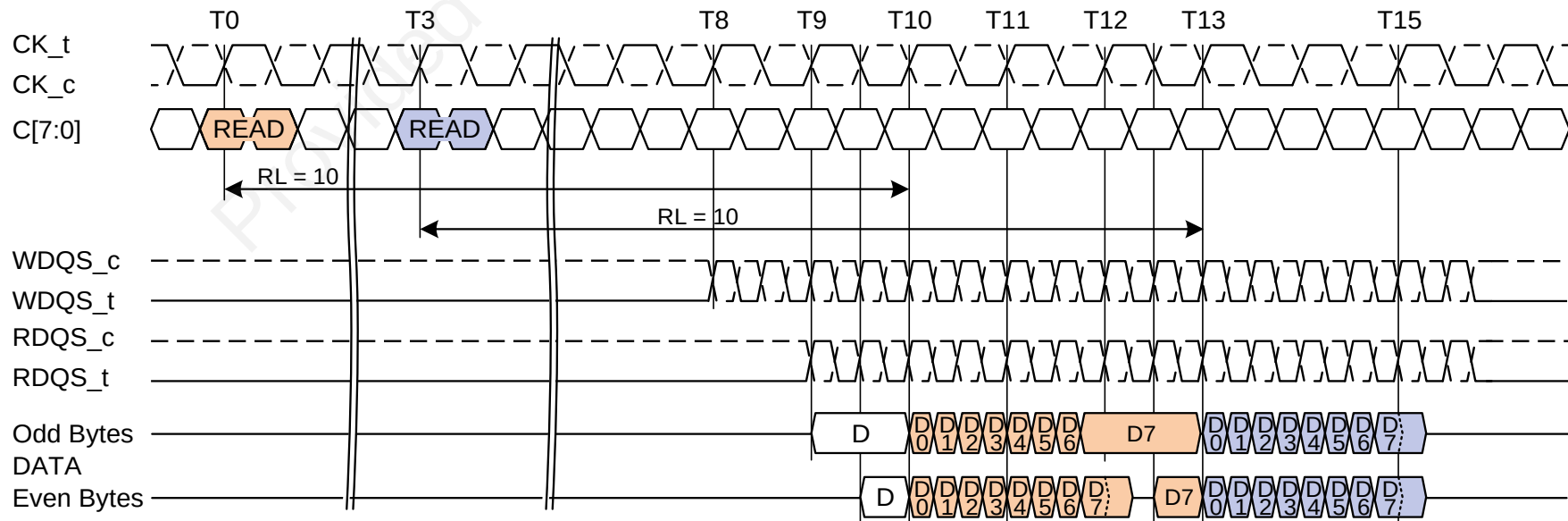
Data from any read burst may be concatenated with data from a subsequent READ command. A continuous flow of data can be maintained as shown in Figure 53. The first data element from the new burst follows the last element of a completed burst. The new READ command should be issued after the previous READ command according to the t_{CCDS} timing. If that READ command is to another idle bank then an ACTIVATE command must precede the READ command and t_{RCDRD} also must be met.

Figure 53: Seamless Read Bursts ($t_{CCDS} = 2$)


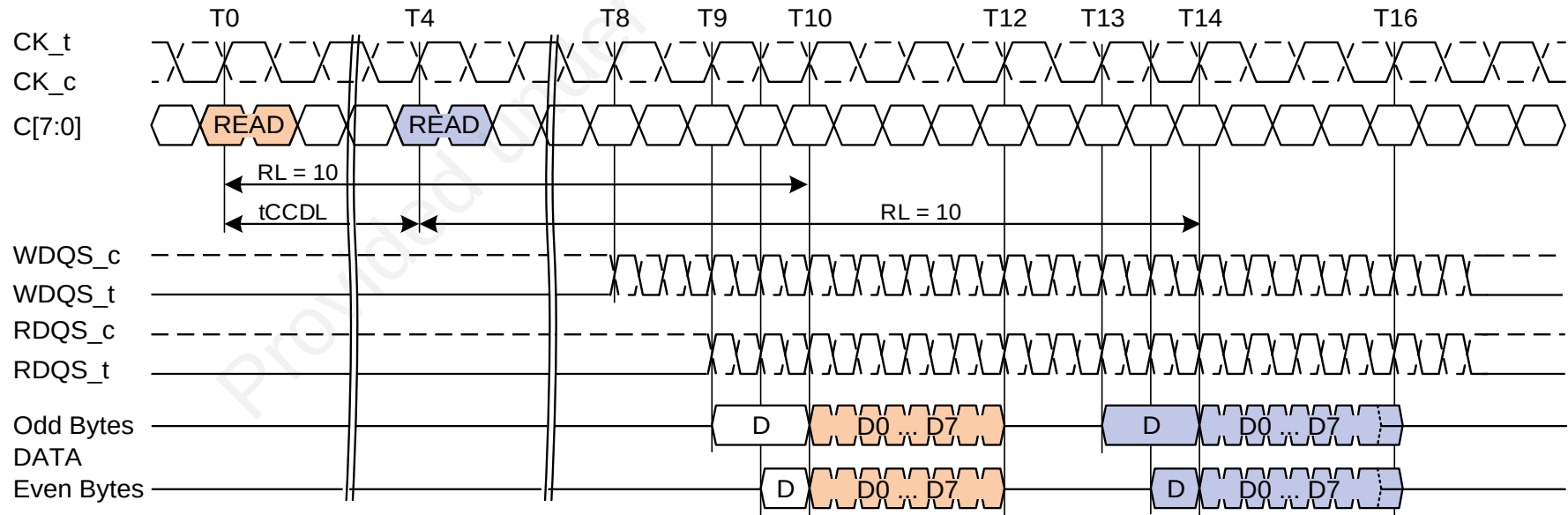


Examples of non-seamless read bursts are shown in Figure 54. for $t_{CCD} = 3$ and in Figure 55 for $t_{CCDL} = 4$. With $t_{CCD} = 3$ the data bus for odd bytes does not return to High-Z state between the two reads bursts; instead, the last UI of the first read burst is extended at T12 as the 1 t_{CK} long bus preconditioning for the second read burst. On the contrary, the data bus for even bytes returns to High-Z state for a short period of time between the two reads bursts before the last data out of the first read burst is re-driven preceding the second read burst.

Figure 54: Non-Seamless Read Bursts ($t_{CCDL} = 3$)

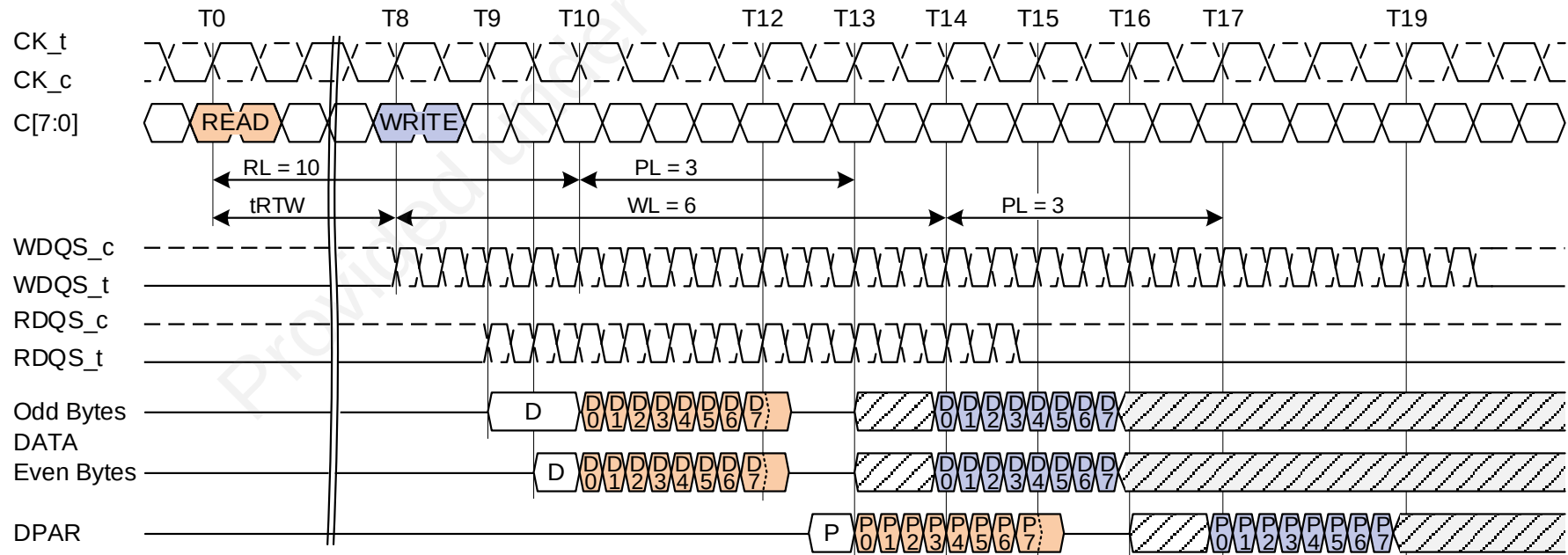


With $t_{CCD} = 4$ the data bus timing of each of the two read bursts is identical to a single read burst. The data bus returns to High-Z between the read bursts, and the last data out of the first read burst is re-driven at T13 preceding the second read burst. The pre- and postamble pulses of $WDQS$ and $RDQS$ overlap in this case, resulting in continuous $WDQS$ and $RDQS$ pulses over the two read bursts.


Figure 55: Non-Seamless Read Bursts ($t_{CCDL} = 4$)


A WRITE can be issued any time after a READ command as long as the bus turnaround time t_{RTW} is met as shown in Figure 56. If that WRITE command is to another idle bank, then an ACTIVATE command must precede the WRITE command and t_{RCDWR} also must be met.

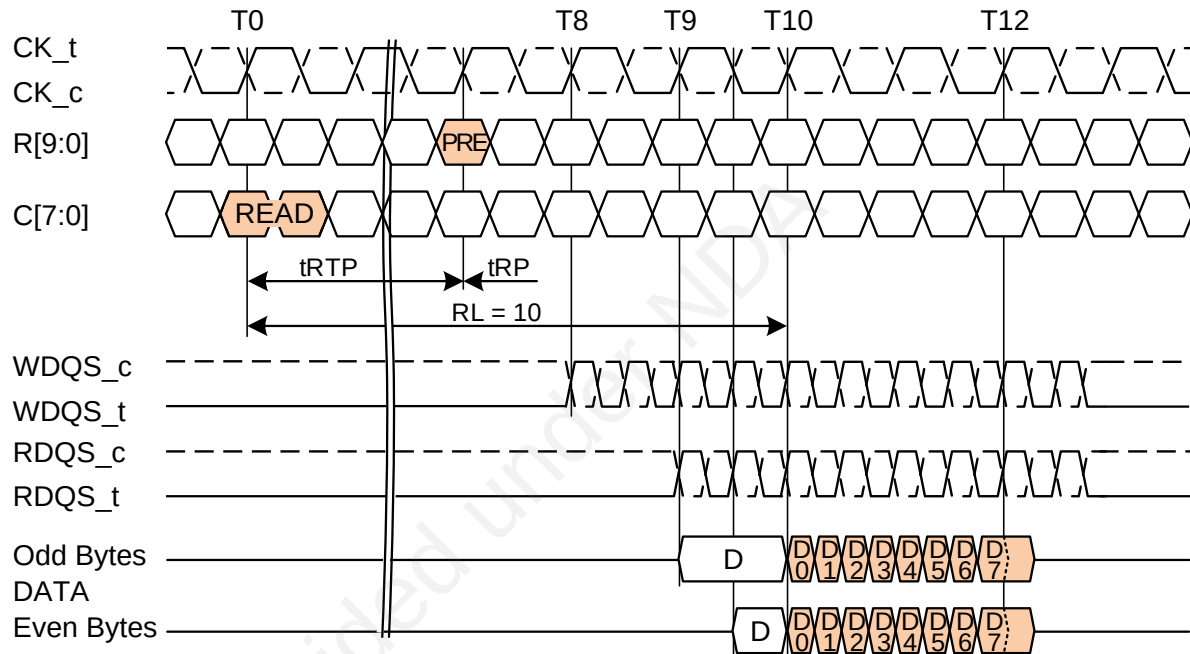
The parity latency PL has no impact on the minimum bus turnaround time because the DPAR signal is delayed by the same number of cycles for reads and writes and therefore the bus turnaround for DPAR is only delayed by PL as compared to the data bytes.


Figure 56: Read To Write with PL = 3


Note: 1. ¹tRTW is not a device limit but determined by the system bus turnaround time.

A PRECHARGE can be issued $tRTP$ after the READ command as shown in Figure 57. After the PRECHARGE command, a subsequent ACTIVATE command to the same bank cannot be issued until tRP is met.

Figure 57: Read To Precharge



Internal DBIac State with Read

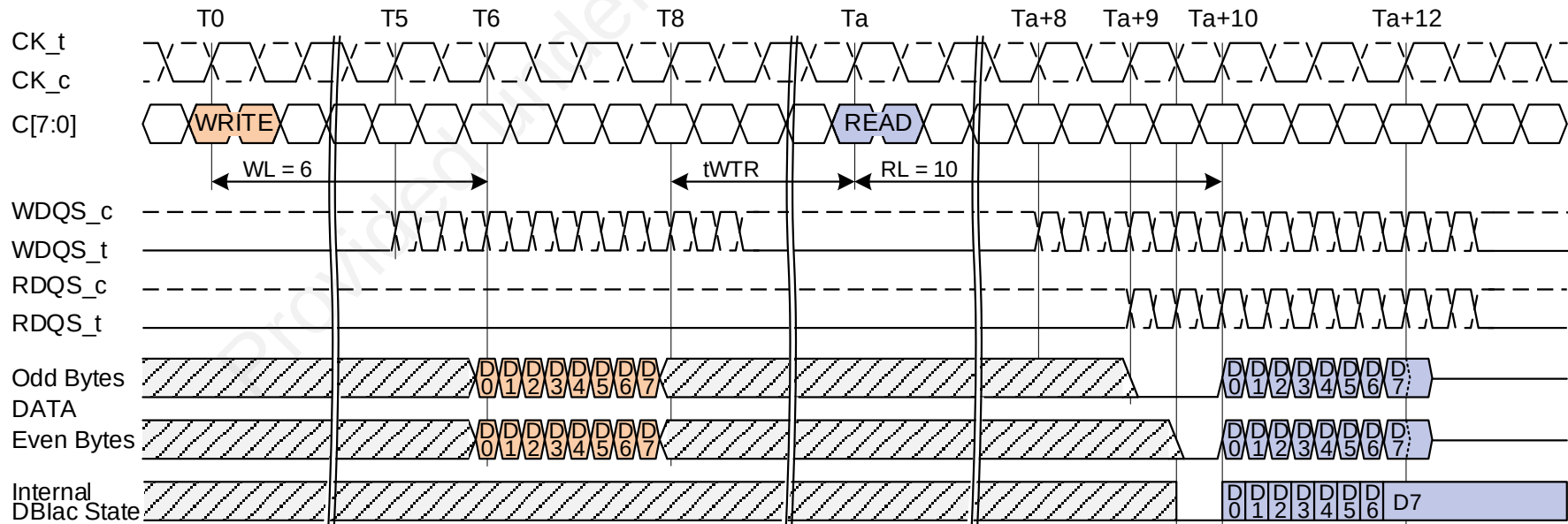
The HBM3 DRAM resets the internal DBIac state to LOW whenever any of the following events occur:

- RESET_n de-assertion;
- a MODE REGISTER SET (MRS) command is received, or the mode registers have been updated via the IEEE1500 MODE_REGISTER_DUMP_SET instruction;
- a write-to-read bus turnaround;
- self refresh exit.

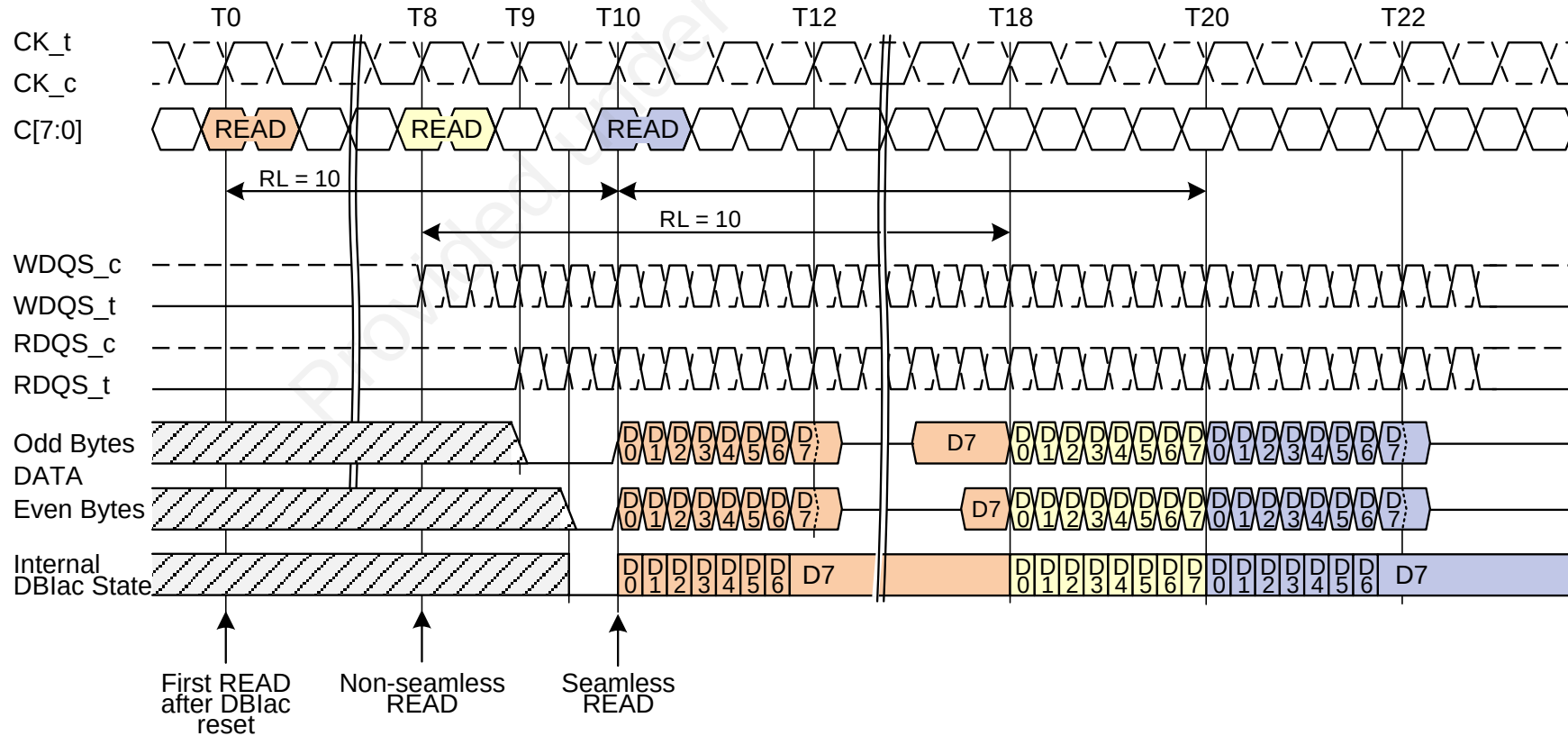
For all other events or commands, the internal DBIac state is not reset and the HBM3 DRAM will use its previous state for DBIac calculation.

When a first READ command is registered after a DBI reset, the HBM3 DRAM preconditions the bus to LOW prior to read data regardless whether RDBI is enabled or disabled in the mode register, as shown in Figure 58 in case of a write-to-read bus turnaround. The internal state D7 corresponding to the last UI of the read burst is internally stored as a seed value for a subsequent read burst.

The DPAR signal is not included in the DBI calculation and not preconditioned to LOW; its initial state is undefined (LOW or HIGH).


Figure 58: Internal DBIac State Reset for Write to Read


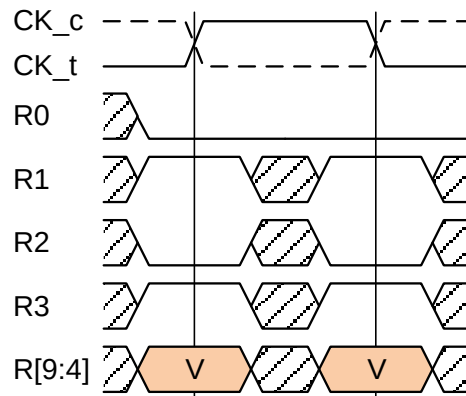
Once the read burst is complete, the HBM3 DRAM tri-states all DQ, DBI and ECC output drivers. However, the HBM3 DRAM internally stores the last data-out of the DQ, DBI, ECC and SEV outputs to pre-condition the bus prior to a subsequent read; it also uses the last data-out of the DQ and DBI outputs for DBIac calculation for any subsequent read operation barring a condition to DBI reset. For non-gapless read operations, the HBM3 DRAM pre-conditions all data outputs to the last data-out of the previous burst nominally two WDQS cycles (odd bytes) and one WDQS cycle (even bytes) prior to the first valid data bit as shown in the figure below.


Figure 59: Bus Preconditioning and DBI States for Read


Power-Down

Power-down is entered with the POWER-DOWN ENTRY command. POWER-DOWN ENTRY is a one-cycle command received on the row command inputs R[9:0] and latched with the rising and falling CK clock edges as shown in Figure 60. The command requires a CNOP command on the column command inputs C[7:0].

Figure 60: POWER_DOWN ENTRY Command



Note: 1. V = Valid (H or L)

Power-down entry must not be issued when read or write operations are in progress on either PC. A read operation is completed when the last data element including parity (when enabled) and RDQS postamble have been transmitted on the outputs. A write operation is completed when the last data element including parity (when enabled) has been written to the memory array with t_{WR} satisfied; for writes with auto-precharge, the number of clock cycles programmed in the mode register for WR must have elapsed instead.

Power-down entry is allowed while any other operations such as row activation, precharge, auto precharge, or refresh are in progress, but the power-down I_{DD} specification will not apply until such operations are complete.

If power-down occurs when all banks are idle, this mode is referred to as precharge power-down; if power-down occurs when there is a row active in any bank, this mode is referred to as active power-down.

To ensure that there is enough time to internally process the power-down entry, POWER-DOWN ENTRY and CNOP commands have to be maintained for t_{CPDED} period. Also, the CK clock must be held stable for t_{CKPDE} cycles.

Once t_{CPDED} and t_{CKPDE} have been met, the signals shall have the following states (see Table 48):

- The R0 input receivers remains active, and R0 = LOW must be maintained to keep the HBM3 DRAM in power-down;
- The CK clock receiver remains active. The clock may be stopped with CK_t and CK_c being driven to static LOW and HIGH levels, respectively; in that case the clock must be stable again with $t_{CH(min)}$ and $t_{CL(min)}$ satisfied at least t_{CKPDX} cycles prior to power-down exit;
- RDQS_t and RDQS_c continue driving static LOW and HIGH levels, respectively;



- AERR, DERR continue driving static LOW levels;
- TEMP and CATTRIP continue driving valid HIGH or LOW levels
- All other input and output buffers are deactivated.

Table 48: Signal States in Power-Down

Signal Group	Signal State	Signal Group	Signal State
RESET_n	HIGH	DQ, DBI, ECC, SEV, DPAR	X
CK_t / CK_c	LOW / HIGH or toggling	WDQS_t / WDQS_c	X
R0	LOW	RDQS_t / RDQS_c	LOW / HIGH
R[9:1], C[7:0], ARFU, APAR	X	DERR	LOW
AERR	LOW	TEMP, CATTRIP	Valid

Note: 1. X = Don't Care (HIGH, LOW or floating); Valid = HIGH or LOW (periodically updated)

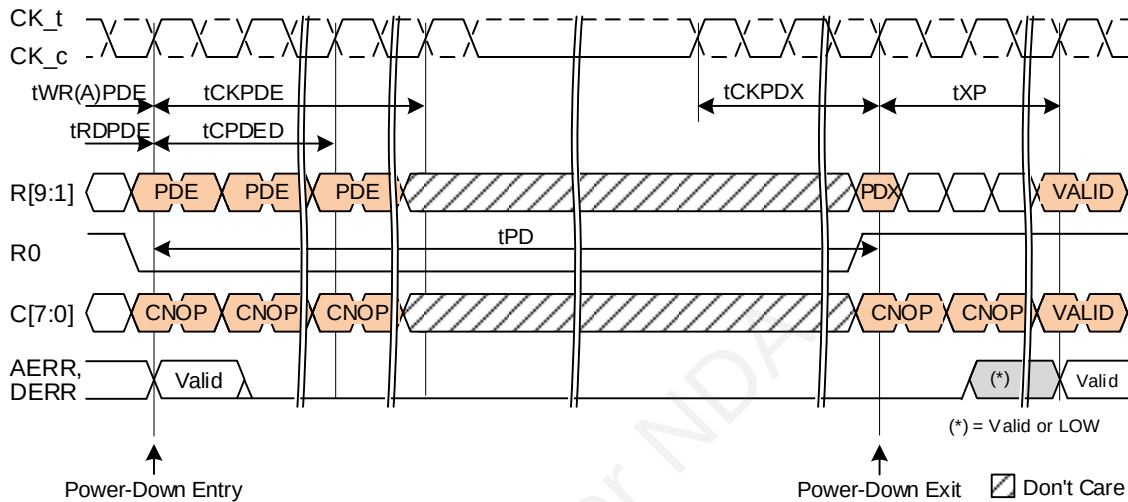
No refresh operations are performed in power-down mode. The maximum duration in power-down mode is limited by the refresh requirements of the device.

While in power-down the device will maintain the internal DBI state for the DBIac calculation when DBI is enabled in the mode register.

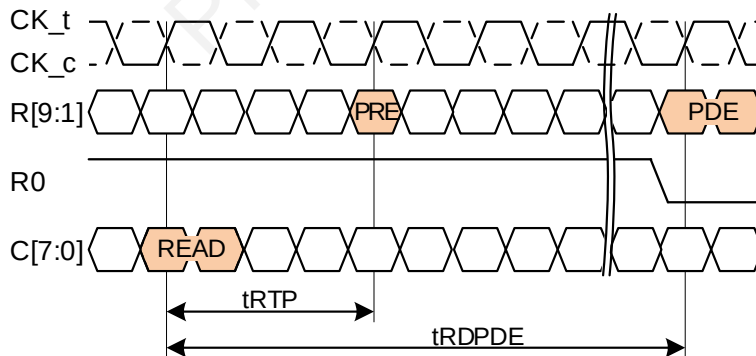
Power-down is synchronously exited when R0 is registered HIGH in conjunction with CNOP commands. A valid executable command may be applied ^tXP cycles later. The minimum power-down duration is specified by ^tPD.

If CA parity is enabled in MR0 OP6, parity is evaluated for the POWER-DOWN ENTRY command. The HBM3 DRAM requires POWER-DOWN ENTRY, RNOP and CNOP commands with valid parity for the entire ^tCPDED period, while it will suspend parity checking after power-down entry and drive AERR to a static LOW. DERR remains LOW as there are no data bursts in progress at this time.

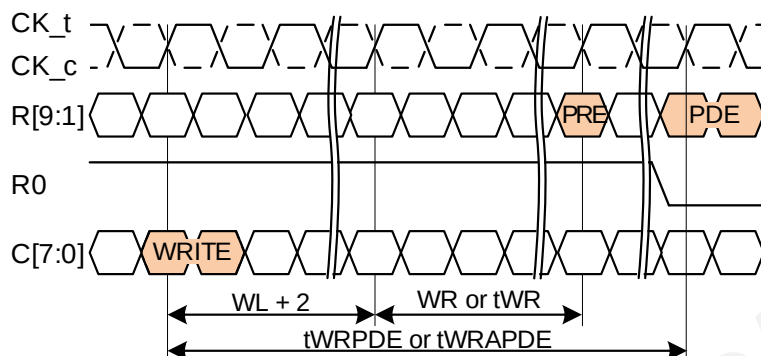
Parity is not evaluated for the POWER-DOWN EXIT command. The HBM3 DRAM requires RNOP and CNOP commands with valid parity for the entire ^tXP period, while within ^tXP period it will resume parity checking and indicating parity errors on AERR. DERR remains LOW as there are no data bursts in progress at this time.

Figure 61: Power-Down Entry and Exit


- Notes:
1. Only PDE and CNOP commands are allowed during 'CPDED period; only RNOP and CNOP commands allowed during 'XP period.
 2. Write bursts must have been completed with 'WRPDE (or 'WRAPDE in case of a WRITE with auto precharge) satisfied prior to power-down entry.
 3. Read bursts must have been completed with 'RDPDE satisfied prior to power-down entry.
 4. Address inputs are „Don't Care“ for power-down entry and exit.
 5. AERR, DERR are driven LOW when parity check is suspended during power-down. Signals are shown with 'PARAC = 0 and 'PARDQ = 0 for illustration purpose.
 6. The CK clock may be stopped during power-down as shown, or toggling.

Figure 62: READ or READ with Auto Precharge to Power-Down Entry Timing


- Notes:
1. PRE indicates the internal auto-precharge for RDA commands.
 2. R0 must be held high until the end of the read burst operation.

Figure 63: WRITE or WRITE with Auto Precharge to Power-Down Entry Timing


- Notes:
1. PRE indicates the internal auto-precharge for WRA commands.
 2. R0 must be held high until the end of the write burst operation.
 3. t_{WR} is the analog value used with WR commands.
 4. nWR is the number of clock cycles programmed for WR in the mode register and used with WRA commands.

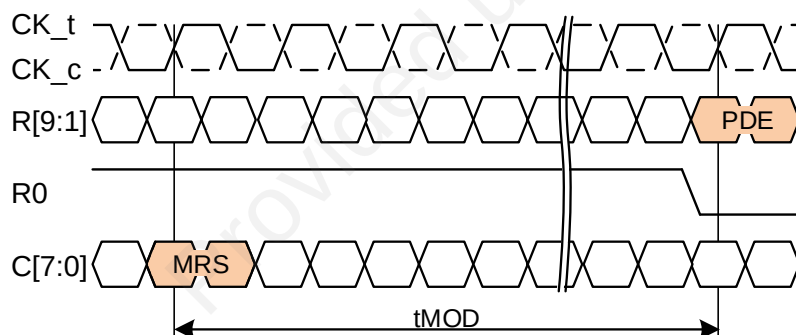
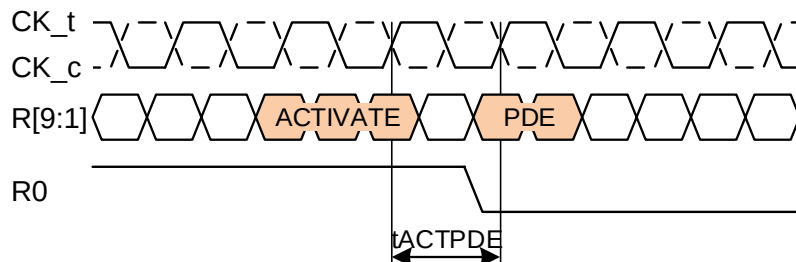
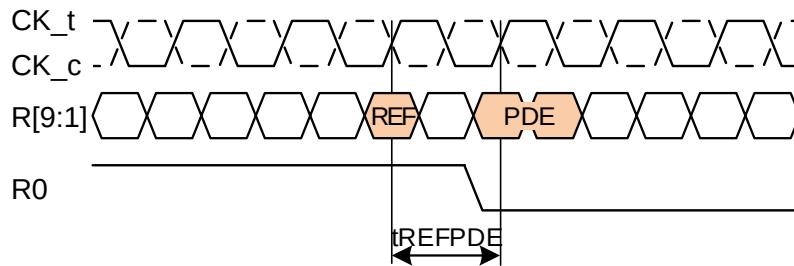
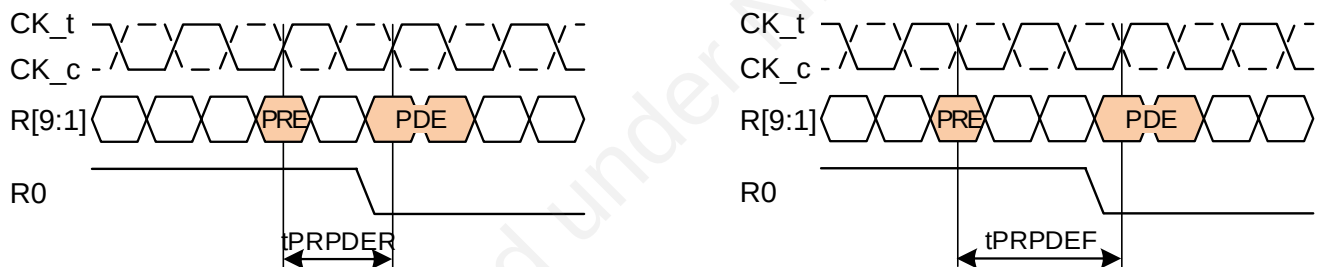
Figure 64: MODE REGISTER SET to Power-Down Entry Timing

Figure 65: ACTIVATE to Power-Down Entry Timing


Figure 66: Refresh to Power-Down Entry Timing


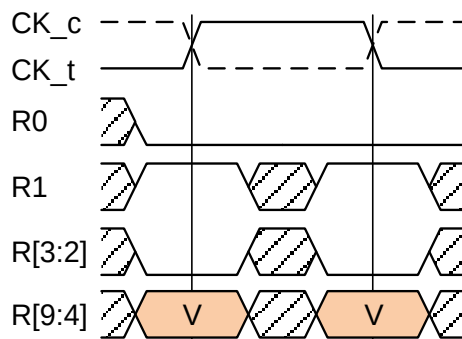
Note: 1. REF could be REFab, REFpb, RFMab or RFMpb.

Figure 67: PRECHARGE to Power-Down Entry Timing


Self Refresh

Self refresh can be used to retain data in the HBM3 DRAM, even if the rest of the system is powered down. When in the self refresh mode, the HBM3 DRAM retains data without external clocking.

SELF REFRESH ENTRY is a one-cycle command received on the row command inputs R[9:0] and latched with the rising and falling CK clock edges as shown in Figure 68. The command requires a CNOP command on the column command inputs C[7:0].

Figure 68: SELF REFRESH ENTRY Command


Note: 1. V = Valid (H or L)

Self refresh entry is only allowed when all banks in both PCs are precharged with t_{RP} satisfied, and when the last data elements including parity from a preceding READ command have been pushed out (t_{RDSRE}). Also, all timings from other preceding commands must be satisfied (e.g. t_{MOD} , t_{RFCab} , t_{RFCpb}).



Once the SELF REFRESH ENTRY command is registered, R0 must be held LOW to keep the device in self refresh mode. For proper self refresh operation, all power supplies (V_{DDC} , V_{DDQ} , V_{DDQL} , V_{PP}) must be at valid levels. The HBM3 DRAM initiates a minimum of one internal refresh within t_{CKSR} period once it enters self refresh mode.

To ensure that there is enough time to internally process the self refresh entry, POWER-DOWN ENTRY and CNOP commands have to be maintained for t_{CPDED} period following the SELF REFRESH ENTRY command. Also, the CK clock must be held stable for t_{CKSRE} cycles.

The clocks are internally disabled during self refresh to save power. The minimum time that the HBM3 device must remain in self refresh mode is t_{CKSR} . The user may halt the external clock or change the external clock frequency t_{CKSRE} after self refresh entry is registered. However, the clock must be restarted and stable t_{CKSRX} before the device can exit self refresh operation.

Once t_{CPDED} and t_{CKSRE} have been met, the signals shall have the following states (see Table 49):

- The R0 input receivers remains active, and R0 = LOW must be maintained to keep the HBM3 DRAM in self refresh mode;
- The CK clock receiver is disabled; the clock may be stopped, or the clock frequency may be changed; the clock must be stable again with $t_{CH(min)}$ and $t_{CL(min)}$ satisfied at least t_{CKSRX} cycles prior to self refresh exit;
- RDQS_t and RDQS_c continue driving static LOW and HIGH levels, respectively;
- AERR, DERR continue driving static LOW levels;
- TEMP and CATTRIP continue driving valid HIGH or LOW levels
- All other input and output buffers are deactivated.

Table 49: Signal States in Self Refresh

Signal Group	Signal State	Signal Group	Signal State
RESET_n	HIGH	DQ, DBI, ECC, SEV, DPAR	X
CK_t / CK_c	X	WDQS_t / WDQS_c	X
R0	LOW	RDQS_t / RDQS_c	LOW / HIGH
R[9:1], C[7:0], ARFU, APAR	X	DERR	LOW
AERR	LOW	TEMP, CATTRIP	Valid

Note: 1. X = Don't Care (HIGH, LOW or floating); Valid = HIGH or LOW (periodically updated)

If CA parity is enabled in MR0 OP6, parity is evaluated for the SELF REFRESH ENTRY command. The HBM3 DRAM requires PDE and CNOP commands with valid parity for the entire t_{CPDED} period, while it will suspend parity checking after self refresh entry and drive AERR to a static LOW. DERR remains LOW as there are no data bursts in progress at this time.



Temperature Compensated Refresh Reporting

Temperature Compensated Refresh Trip Points

The HBM3 DRAM provides temperature compensated refresh related information to the controller via an encoding on the TEMP[1:0] outputs. The gray-coded encoding as in Table 50 defines the required refresh rate as determined by the hottest device in the stack to maintain data integrity. The reported refresh rate is updated when the junction temperature exceeds the specific trip-point levels appropriate for each refresh rate.

Table 50: Temperature Compensated Refresh Trip Points

TEMP[1:0]	Refresh Rate	Temperature Range	Notes
LOW, LOW	$1.0 \times 'REFI$	$T \leq +95^{\circ}\text{C}$	
LOW, HIGH	$0.5 \times 'REFI$	$+95^{\circ}\text{C} < T \leq +105^{\circ}\text{C}$	
HIGH, HIGH	Not supported	$T > +105^{\circ}\text{C}$	1

Notes: 1. TEMP[1:0] = High, High indicates that the device is operated outside the specified operating temperature range. Device functionality is not guaranteed in this case.

Catastrophic Temperature Sensor

The CATTRIP sensor logic detects if the junction temperature of any die in the HBM3 stack exceeds a catastrophic trip-point level. The level is set to a value below the temperature that would result in permanent damage of the device. If the junction temperature anywhere in the stack exceeds that catastrophic trip-point level, the HBM3 device will drive CATTRIP to HIGH.

Table 51: CATTRIP Temperature Trip Point

CATTRIP Temperature Trip Point
+125°C

The CATTRIP output is sticky in that device power-off is required to clear the CATTRIP output to LOW. Sufficient time should be allowed for the device to cool after a CATTRIP event. See HBM3 Power-Up and Initialization Sequence for the initialization of the CATTRIP output.

The circuits associated with the CATTRIP output will operate correctly even if the catastrophic trip-point level has been exceeded, and regardless of whether the external or internal clocks have stopped. The functionality of CATTRIP can be verified by writing a "1" to MR7 OP7 to force CATTRIP to HIGH, and "0" to set CATTRIP back to LOW.



Interconnect Redundancy Remapping

The HBM3 DRAM supports interconnect lane remapping to help improve SIP assembly yield and recover functionality of the HBM3 DRAM. The `SOFT_LANE_REPAIR` and `HARD_LANE_REPAIR` instructions are used to perform lane remapping. Lane remapping is independent for each channel.

The HBM3 DRAM can be programmed to retain the remapped lane information even when power is completely removed from the HBM3 DRAM. `SOFT_LANE_REPAIR` and `HARD_LANE_REPAIR` instructions can only be issued as part of the device initialization and before normal memory operation has commenced, e.g. before the CK clock has started to toggle.

AWORD Remapping

There is one redundant AWORD lane per channel to either repair one lane in the row command bus or one lane in the column command bus. APAR and ARFU are associated with the column command bus repair as shown in Table 53. CK_c, CK_t, and AERR signals cannot be remapped. After a lane is remapped, the input buffer associated with the broken lane is turned off and the input buffer associated with the redundant bump (RA) is turned on. All functionalities are preserved with row or column bus lane remapping.

Table 52: AWORD Row Command Bus Remapping

Description	Register Encoding	R1	R2	R3	R0	R4	R5	R6	R7	R8	R9	RA
Repair R1 Lane	0001	XX	R1	R2	R3	R0	R4	R5	R6	R7	R8	R9
Repair R2 Lane	0010	R1	XX	R2	R3	R0	R4	R5	R6	R7	R8	R9
Repair R3 Lane	0011	R1	R2	XX	R3	R0	R4	R5	R6	R7	R8	R9
Repair R0 Lane	0000	R1	R2	R3	XX	R0	R4	R5	R6	R7	R8	R9
Repair R4 Lane	0100	R1	R2	R3	R0	XX	R4	R5	R6	R7	R8	R9
Repair R5 Lane	0101	R1	R2	R3	R0	R4	XX	R5	R6	R7	R8	R9
Repair R6 Lane	0110	R1	R2	R3	R0	R4	R5	XX	R6	R7	R8	R9
Repair R7 Lane	0111	R1	R2	R3	R0	R4	R5	R6	XX	R7	R8	R9
Repair R8 Lane	1000	R1	R2	R3	R0	R4	R5	R6	R7	XX	R8	R9
Repair R9 Lane	1001	R1	R2	R3	R0	R4	R5	R6	R7	R8	XX	R9
Reserved	1010 to 1110	R1	R2	R3	R0	R4	R5	R6	R7	R8	R9	RA
Default - No Repair	1111	R1	R2	R3	R0	R4	R5	R6	R7	R8	R9	RA

- Notes:
1. XX = Lane is remapped.
 2. The register encoding refers to the `AWORD_RA[3:0]` field in the `LANE_REPAIR` WDR and requires that the `AWORD_CA[3:0]` field is set to `0FH` (no column command bus remapping).


Table 53: AWORD Column Command Bus Remapping

Description	Register Encoding	C0	C1	C2	C3	C4	C5	C6	C7	APAR	ARFU	RA
Repair C0 Lane	0000	XX	C0	C1	C2	C3	C4	C5	C6	C7	APAR	ARFU
Repair C1 Lane	0001	C0	XX	C1	C2	C3	C4	C5	C6	C7	APAR	ARFU
Repair C2 Lane	0010	C0	C1	XX	C2	C3	C4	C5	C6	C7	APAR	ARFU
Repair C3 Lane	0011	C0	C1	C2	XX	C3	C4	C5	C6	C7	APAR	ARFU
Repair C4 Lane	0100	C0	C1	C2	C3	XX	C4	C5	C6	C7	APAR	ARFU
Repair C5 Lane	0101	C0	C1	C2	C3	C4	XX	C5	C6	C7	APAR	ARFU
Repair C6 Lane	0110	C0	C1	C2	C3	C4	C5	XX	C6	C7	APAR	ARFU
Repair C7 Lane	0111	C0	C1	C2	C3	C4	C5	C6	XX	C7	APAR	ARFU
Repair APAR Lane	1000	C0	C1	C2	C3	C4	C5	C6	C7	XX	APAR	ARFU
Repair ARFU Lane	1001	C0	C1	C2	C3	C4	C5	C6	C7	APAR	XX	ARFU
Reserved	1010 to 1110	C0	C1	C2	C3	C4	C5	C6	C7	APAR	ARFU	RA
Default - No Repair	1111	C0	C1	C2	C3	C4	C5	C6	C7	APAR	ARFU	RA

- Notes:
1. XX = Lane is remapped.
 2. The register encoding refers to the AWORD_CA[3:0] field in the LANE_REPAIR WDR and requires that the AWORD_RA[3:0] field is set to 0F_H (no row command bus remapping).

In a first example, C4 is the broken lane in the column command bus with no broken lanes on the row command bus. The lane is remapped by programming the channel's LANE_REPAIR WDR bits AWORD_CA[3:0] to 4h and AWORD_RA[3:0] to Fh.

Table 54: Original Lane Assignment – AWORD Column Repair

	ARFU		C7		C5		C4		C2		C0
RA		APAR		C6		CK _t		C3		C1	
	R9		R7		CK _c		R4		R3		R1
AERR		R8		R6		R5		R0		R2	

Table 55: Remapped Lane Assignment – AWORD Column Repair

	APAR		C6		C4		XX		C2		C0
ARFU		C7		C5		CK _t		C3		C1	
	R9		R7		CK _c		R4		R3		R1
AERR		R8		R6		R5		R0		R2	



In a second example, R0 is the broken lane in the row command bus with no broken lanes on the column command bus. The lane is remapped by programming the channel's LANE_REPAIR WDR bits AWORD_RA[3:0] to 0h and AWORD_CA[3:0] to Fh.

Table 56: Original Lane Assignment – AWORD Row Repair

	ARFU		C7		C5		C4		C2		C0
RA		APAR		C6		CK_t		C3		C1	
	R9		R7		CK_c		R4		R3		R1
AERR		R8		R6		R5		R0		R2	

Table 57: Remapped Lane Assignment – AWORD Row Repair

	ARFU		C7		C5		C4		C2		C0
R9		APAR		C6		CK_t		C3		C1	
	R8		R6		CK_c		R0		R3		R1
AERR		R7		R5		R4		XX		R2	

DWORD Remapping

HBM3 DRAMs support remapping of one broken data bus lane per double byte. Two adjacent bytes (e.g. DQ[15:0], DQ[31:16], DQ[47:32], DQ[63:48]) are treated as a pair (double byte), but each double byte is treated independently.

After a lane is remapped, the input and output buffers associated with the broken lane are turned off and the input and output buffers associated with the redundant lane (RD) are additionally turned on.

It is required to program “1111b” for the intact byte within the double byte while the remapping for the broken lane in the other byte is encoded according to the table.

DBI functionality is preserved as long as the mode register setting for DBI function is enabled. There is no impact on the data parity function. WDQS_c, WDQS_t, RDQS_c, RDQS_t, DPAR and DERR signals cannot be remapped.

During reads, the RD output drivers are enabled along with the DQ, DBI and ECC/SEV lanes of the physical byte the lane is located in: RD0 and RD2 are located within even bytes and thus enabled half a clock cycle prior to the first valid data bit, and RD1 and RD3 are located within odd bytes and thus enabled one clock cycle prior to the first valid data bit.


Table 58: DWORD Remapping for One Byte

Description	Register Encoding	ECC0 (ECC1, SEV0, SEV1)	DQ0 (DQ8, DQ16, DQ24)	DQ1 (DQ9, DQ17, DQ25)	DQ2 (DQ10, DQ18, DQ26)	DQ3 (DQ11, DQ19, DQ27)	DQ4 (DQ12, DQ20, DQ28)	DQ5 (DQ13, DQ21, DQ29)	DQ6 (DQ14, DQ22, DQ30)	DQ7 (DQ15, DQ23, DQ31)	DBI0 (DBI1, DBI2, DBI3)	RD0 (RD0, RD1, RD1)
Repair Lane 0	0000	XX	ECC0	DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7	DBI0
Repair Lane 1	0001	ECC0	XX	DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7	DBI0
Repair Lane 2	0010	ECC0	DQ0	XX	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7	DBI0
Repair Lane 3	0011	ECC0	DQ0	DQ1	XX	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7	DBI0
Repair Lane 4	0100	ECC0	DQ0	DQ1	DQ2	XX	DQ3	DQ4	DQ5	DQ6	DQ7	DBI0
Repair Lane 5	0101	ECC0	DQ0	DQ1	DQ2	DQ3	XX	DQ4	DQ5	DQ6	DQ7	DBI0
Repair Lane 6	0110	ECC0	DQ0	DQ1	DQ2	DQ3	DQ4	XX	DQ5	DQ6	DQ7	DBI0
Repair Lane 7	0111	ECC0	DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	XX	DQ6	DQ7	DBI0
Repair Lane 8	1000	ECC0	DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	XX	DQ7	DBI0
Repair Lane 9	1001	ECC0	DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7	XX	DBI0
Reserved	1010 to 1110	ECC0	DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7	DBI0	RD0
Default - No Repair	1111	ECC0	DQ0	DQ1	DQ2	DQ3	DQ4	DQ5	DQ6	DQ7	DBI0	RD0

- Notes:
1. XX = Lane is remapped.
 2. DWORD0_BYTE0 is shown as an example.
 3. ECC is associated with DWORD0_BYTE0, DWORD0_BYTE1, DWORD1_BYTE0 and DWORD1_BYTE1.
 4. SEV is associated with DWORD0_BYTE2, DWORD0_BYTE3, DWORD1_BYTE2 and DWORD1_BYTE3.

As an example, ECC0 is a broken lane for byte 0 while all lanes for byte 1 are intact. The lane is remapped as illustrated in Table 9 by programming the LANE REPAIR WDR bits DWORD0_BYTE0[3:0] to 0h and bits DWORD0_BYTE1[3:0] to Fh.

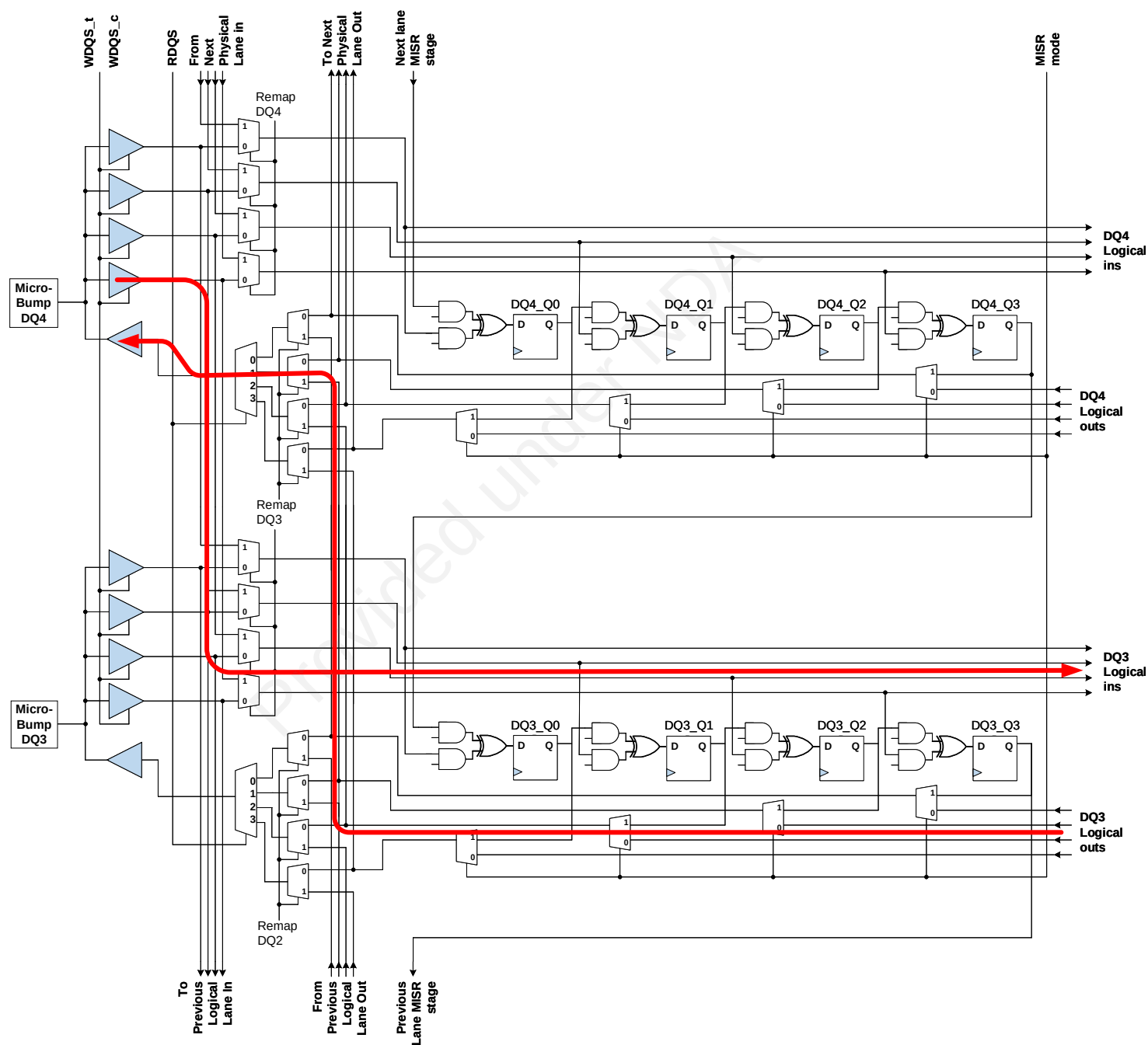
Table 59: Original DWORD Lane Assignment - Bytes [1:0]

	DQ7		DQ5		RD0		DQ3		DQ1		ECC0
DBI0		DQ6		DQ4		PAR0		DQ2		DQ0	
	VDDQL		VDDQL		VDDQL		VDDQL		VDDQL		VDDQL
DBI1		DQ14		DQ12		WDQS0_t		DQ10		DQ8	
	DQ15		DQ13		WDQS0_c		DQ11		DQ9		ECC1


Table 60: Remapped DWORD Lane Assignment - Bytes [1:0]

	DQ6		DQ4		DBI0		DQ2		DQ0		XX
DQ7		DQ5		DQ3		PAR0		DQ1		ECC0	
	VDDQL		VDDQL		VDDQL		VDDQL		VDDQL		VDDQL
DBI1		DQ14		DQ12		WDQS0_t		DQ10		DQ8	
	DQ15		DQ13		WDQS0_c		DQ11		DQ9		ECC1

The circuit diagram in the figure below illustrates the DQ lane remapping in more detail. Physical micro-bump DQ3 will be connected to internal logical DQ3 input and output paths when the DQ3 lane is not remapped; with remapping the internal DQ3 input and output paths would be routed to the physical DQ4 micro-bump.



Duty Cycle Adjuster (DCA)

HBM3 DRAMs support Duty Cycle Adjuster (DCA) that allows the memory controller to adjust the DRAM internally generated WDQS to compensate for a systemic duty cycle error. The DCA is located before the WCK divider or equivalent (see Figure 10). The DCA will affect the WDQS duty cycle for both Write and Read operations.

A separate DCA is provided for each WDQS (see Table 22):

- the DCA for WDQS0 (PC0) is controlled via MR11 OP[3:0];
- the DCA for WDQS1 (PC1) is controlled via MR11 OP[7:4];

A range of -7 steps to +7 steps is supported as shown in Figure 71 and changes the effective internal WDQS duty cycle as follows:

- a positive value increases the effective t_{WQSH} time and decreases the effective t_{WQSL} time;
- a negative value decreases the effective t_{WQSH} time and increases the effective t_{WQSL} time.

The use of the DCA is optional for the memory controller and is not supported at CK clock frequencies lower than f_{CKDCA} ; at those frequencies it is required to disable the DCA by setting the DCA code to the default value (0000_B).

A duty cycle adjustment with or without a duty cycle monitor sequence shall be performed prior to WDQS-to-CK Alignment Training. The duty cycle adjustment shall be performed at the maximum operating frequency used by the system.

An example of the effect of a DCA code change to the WDQS duty cycle is shown in Figure 72. The maximum offset and step size (in ps) are given in Table 61.

Figure 71: Duty Cycle Adjuster Range

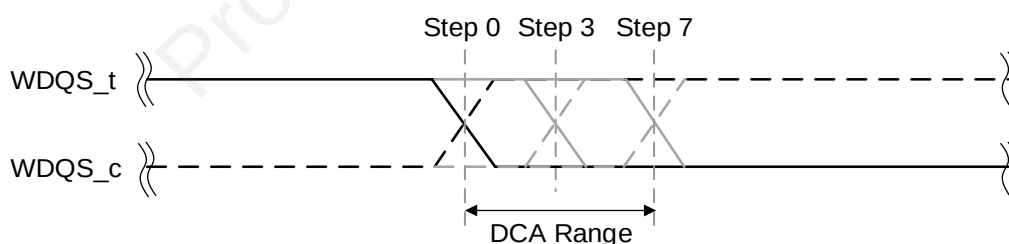
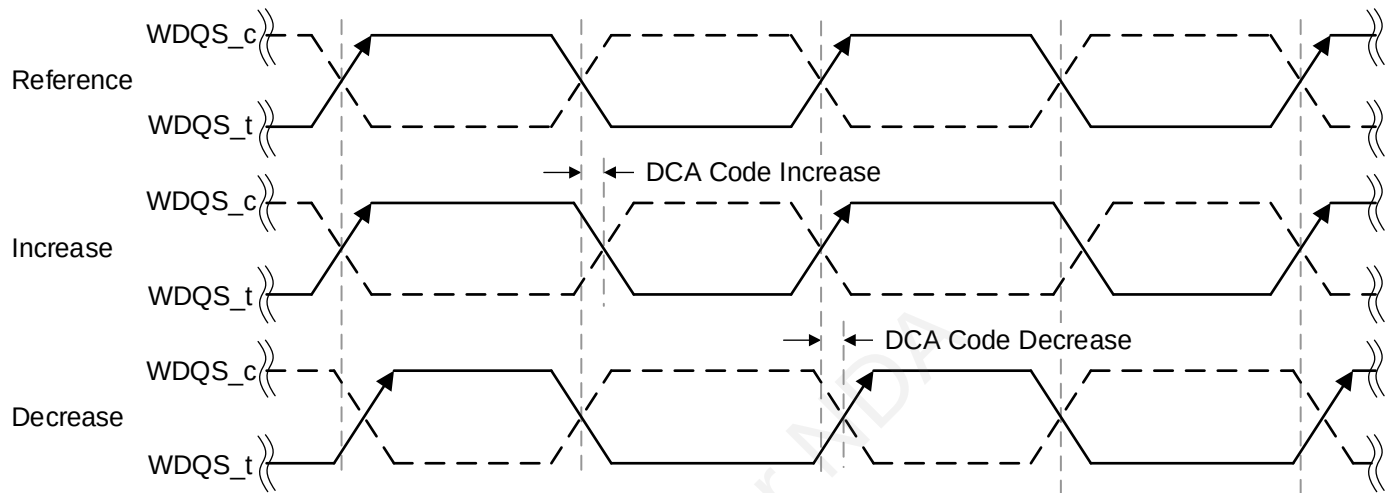


Table 61: DCA Maximum Offset and Step Size

Parameter	Min	Max	Unit	Notes
Duty cycle adjuster maximum offset	15	35	ps	1, 2
Duty cycle adjuster single step size	2	5	ps	1, 3

- Notes:
- These values are guaranteed by design.
 - The parameter describes the absolute maximum offset from step 0 to step +7 or from step 0 to step -7.
 - The single step size reflects the non-linearity of each step.

If the duty cycle distortion in the system changes with variations in voltage and temperature, it is suggested to repeat the duty cycle adjustment as needed. It is recommended to repeat the WDQS-to-CK Alignment Training after any DCA code change.


Figure 72: Relationship Between WDQS Waveform and DCA Code Change (Example)


Note: 1. Refer to the AC Timings section for the definition of t_{WQSH} , t_{WQSL} and t_{WQS} .

Duty Cycle Monitor (DCM)

The HBM3 DRAM includes a duty cycle monitor (DCM) that allows the memory controller to observe the DRAM internal WDQS clock tree duty cycle distortion on the DERR0 output for DWORD0 (PC0) and on the DERR1 output for DWORD 1 (PC1).

The DCM is controlled via MR8 OP1 (see Table 19). Once DCM is enabled by setting MR8 OP1 to 1_B , the DCM will start the WDQS duty cycle distortion measurement and provide the result on DERR0 and DERR1 after waiting at least t_{DCMM} time. An even number of continuous WDQS pulses will be required for the complete duration of the measurement cycle, from the MRS command that initiates the measurement until the t_{DCMM} timing has been met. The result will be available until the DCM is disabled by setting MR8 OP1 back to 0_B . The DERR outputs will then return to their default state latest after t_{MOD} has elapsed. The use of the DCM is not supported at CK clock frequencies lower than f_{CKDCA} .

Command allowed while in this mode are REFab, REFpb, RFMab, RFMpb, RNOP, CNOP and MRS to disable the duty cycle monitor respectively. Internal current spikes generated by the use of REFab, REFpb, RFMab and RFMpb commands in this mode may negatively impact the training result. Controllers that cannot account for this impact should avoid use of REFab, REFpb, RFMab and RFMpb commands in this mode.

Table 62: DCM Measurement Result

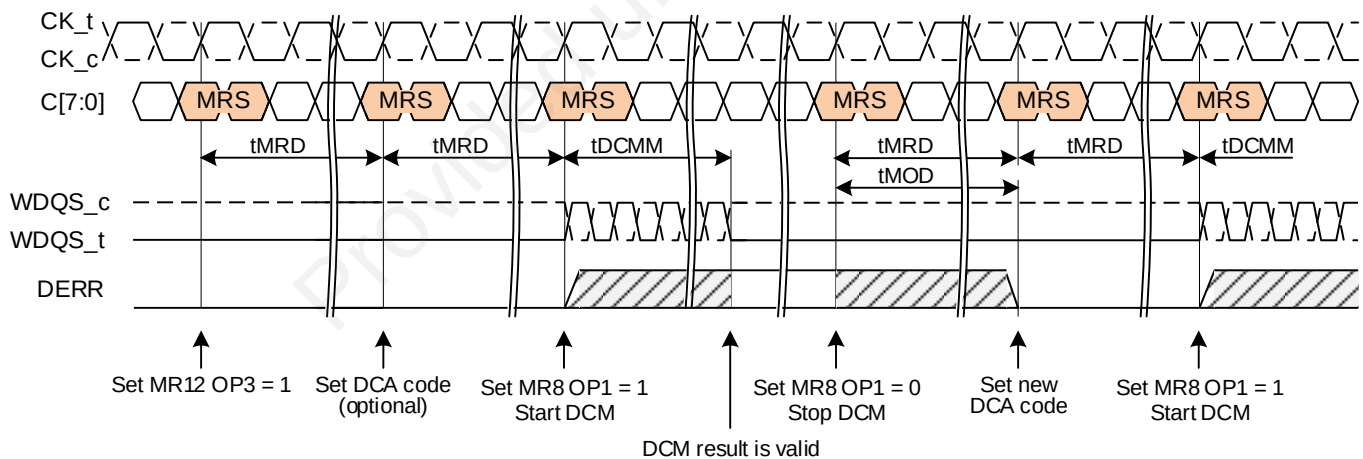
WDQS Duty Cycle	Result (DERR0, DERR1)	Note
<50%	LOW	1
≥50%	HIGH	

Note: 1. The result is valid a time t_{DCMM} after enabling DCM.

The following command sequence shall be used for WDQS duty cycle adjustment (see also Figure 73):

1. Set MR12 OP3 = 1 to enter Duty Cycle Training Mode
2. enable both WDQS strobes;
3. enable DCM and wait for t_{DCMM} ;
4. observe the measurement result via DERR0 and DERR1 outputs;
5. disable DCM and wait for t_{MOD} ; DERR0 and DERR1 outputs return to their default state;
6. Disable both WDQS strobes;
7. issue an MRS command to set an appropriate DCA codes for both WDQS strobes and wait for t_{MOD} ;
8. repeat steps 2 to 7 as needed;
9. Set MR12 OP3 = 0 to exit Duty Cycle Training Mode
10. perform WDQS-to-CK alignment training

Figure 73: Example Sequence for WDQS Duty Cycle Correction



Note: 1. The host may send continuous WDQS pulses throughout the whole duty cycle adjustment procedure, in addition to the required WDQS pulses as shown in the figure.

WDQS-to-CK Alignment Training

WDQS-to-CK alignment training allows the host to observe the phase offset between the WDQS strobes in both PCs and the CK clock to aid in keeping the phase relationship within the limits given by the t_{DQSS} specification. The WDQS2CK bit in MR8 OP3 is associated with this training mode.

WDQS-to-CK alignment training is required to be performed at least once after device initialization if adherence to the t_{DQSS} timing cannot be guaranteed without performing this alignment training. WDQS-to-CK alignment training is not required if the t_{DQSS} timing is met. An effort to further narrow the WDQS-to-CK phase offset by using this training mode will not improve the stable device operation.

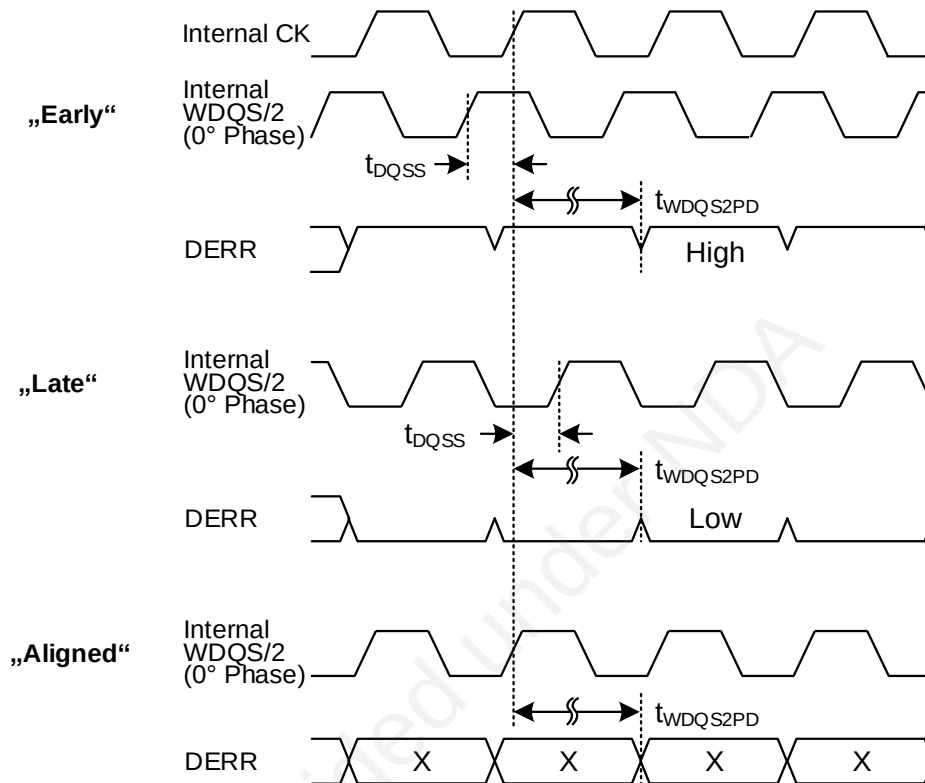
The following steps are required for WDQS-to-CK alignment training:



1. Enter WDQS-to-CK alignment training mode by setting the WDQS2CK bit to 1 and wait t_{MOD} . Commands allowed while in this mode are REFab, REFpb, RFMab and RFMpb, RNOP, CNOP and MRS to exit WDQS-to-CK alignment training, respectively. Internal current spikes generated by the use of REFab, REFpb, RFMab and RFMpb commands in this mode may negatively impact the training result. Controllers that cannot account for this impact should avoid use of REFab, REFpb, RFMab and RFMpb commands in this mode.
2. Enable both WDQS0 and WDQS1 strobes; keep both strobes constantly running in order to generate a valid read-out at both phase detectors with each CK clock cycle;
3. Slowly sweep the WDQS0 and WDQS1 phases with respect to the CK clock, and monitor both DERR0 and DERR1 signals for the phase detector's result as shown in Table 63 and Figure 74; each phase detector latches the 0° phase of the internally divided WDQS strobe with each rising CK clock edge and provides the result on the DERR0 signal for WDQS0 and the DERR1 signal for WDQS1 after $t_{WDQS2PD}$;
4. The ideal alignment is indicated by the phase detector output transitioning from "early" to "late" when the delay of the WDQS phase is continuously increased;
5. When the phase relationship between WDQS and CK meets the t_{DQSS} specification, stop both WDQS strobes; ensure that the number of WDQS pulses issued while in this training mode is an even number such that the internal WDQS state is back at its reset state once the training has finished. With that, no specific synchronization between CK and WDQS is required for correct write and read operation;
6. Exit WDQS-to-CK alignment training mode by setting the WDQS2CK bit to 0_B and wait t_{MOD} .

Table 63: Phase Detector and DERR Signal Behavior

Internal WDQS/2 (0° Phase) Sampled by CK	WDQS Phase	DERR0 DERR1	Recommended Action
HIGH	Early	HIGH	Increase delay on WDQS
LOW	Late	LOW	Decrease delay on WDQS


Figure 74: DERR Signal Behavior in WDQS-to-CK Alignment Training




WDQS Interval Oscillator

As voltage and temperature change on the HBM3 DRAM, the WDQS clock tree delay will shift and may require re-training. The HBM3 DRAM includes an internal WDQS clock-tree oscillator to measure the amount of delay over a given time interval (determined by the controller), allowing the controller to compare the trained delay value to the delay value seen at a later time. The WDQS interval oscillator (“oscillator”) will provide the controller with important information regarding the need to re-train, and the magnitude of potential error. The oscillator is not associated with any channel and operates fully independent of any channel’s operating frequency or state (e.g. bank active, bank idle, power-down or self refresh). Also, no CK, WDQS or WRCK clock is required while the oscillator is counting. The oscillator is disabled by default upon power-up.

The IEEE1500 instructions WOSC_RUN and WOSC_COUNT are associated with the oscillator. Setting the WOSC_START_STOP bit in the WOSC_RUN WDR to 1 will start an internal ring oscillator that counts the number of times a signal propagates through a copy of the WDQS clock tree. The oscillator is stopped by setting the WOSC_START_STOP bit back to 0. The maximum count is $2^{24} - 1$, and the longest run time for the oscillator to not overflow the counter can be calculated as follows:

$$\text{Longest Run Time Interval} = 2^{24} \times t_{RX_WDQS2DQ}(\text{min})$$

The validity of the clock count is indicated by the WOSC_COUNT_VALID bit in the WOSC_COUNT WDR. The default state of 0 indicates an invalid count. The state is also set to 0 when the oscillator is started. When the oscillator stops, the WOSC_COUNT_VALID bit is set to 1_B to indicate a valid count, and the result of the counter is stored in the WOSC_COUNT_VALUE field of the WOSC_COUNT WDR. The WOSC_COUNT_VALID bit will remain 0_B (invalid) if the counter overflows (2^{24} or more cycles) or if the oscillator is interrupted by pulling RESET_n to LOW. On the other hand, pulling WRST_n to LOW does not impact the oscillator’s operation. After the oscillator stops the host may issue the WOSC_COUNT instruction to read out the count.

The controller may adjust the accuracy of the result by running the oscillator for shorter (less accurate) or longer (more accurate) duration. The accuracy of the result for a given temperature and voltage is determined by the following equation:

$$\text{WDQS Oscillator Granularity Error} = \frac{2 \times (\text{WDQS delay})}{\text{Run Time}}$$

where:

- Run Time = total time between the oscillator starting and automatically stopping;
- WDQS delay = the value of the WDQS clock tree delay [$t_{RX_WDQS2DQ}(\text{min/max})$]

Additional matching error must be included, which is the difference between WDQS training circuit and the actual WDQS clock tree across voltage and temperature. Therefore, the total accuracy of the DQS Oscillator counter is given by:

$$\text{WDQS Oscillator Accuracy} = 1 - \text{Granularity Error} - \text{Matching Error}$$

Example: if the total time between start and stop is 100ns, and the maximum WDQS clock tree delay is 400ps [$t_{RX_WDQS2DQ}(\text{max})$], then the WDQS Oscillator Granularity Error is:

$$\text{WDQS Oscillator Granularity Error} = \frac{2 \times (0.4\text{ns})}{100\text{ns}} = 0.8\%$$

This equates to a granularity timing error of 3.2ps.

Assuming a circuit Matching Error of 5.5ps across voltage and temperature, then the accuracy is:

$$\text{WDQS Oscillator Accuracy} = 1 - \frac{(3.2 + 5.5)\text{ps}}{400\text{ps}} = 97.8\%$$

Example: Running the WDQS Oscillator for a longer period improves the accuracy. If the total time between start and stop is 250ns, and the maximum WDQS clock tree delay is 400ps [$t_{RX_WDQS2DQ}(\text{max})$], then the WDQS Oscillator Granularity Error is:

$$\text{WDQS Oscillator Granularity Error} = \frac{2 \times (0.4\text{ns})}{250\text{ns}} = 0.32\%$$

This equates to a granularity timing error of 1.28ps.

Assuming a circuit Matching Error of 5.5ps across voltage and temperature, then the accuracy is:

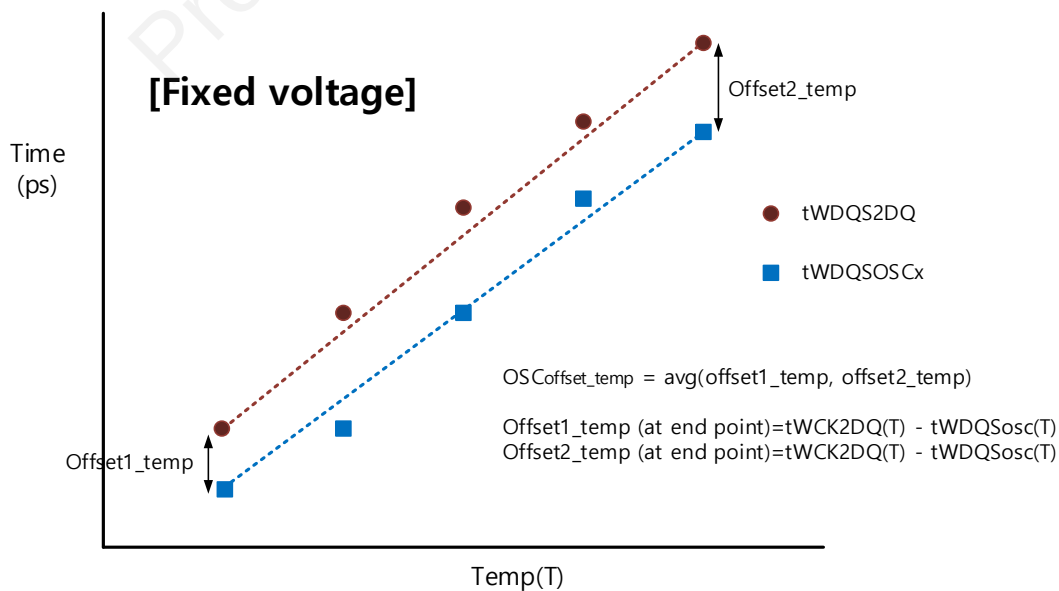
$$\text{WDQS Oscillator Accuracy} = 1 - \frac{(1.28 + 5.5)\text{ps}}{400\text{ps}} = 98.3\%$$

The WDQS Interval Oscillator matching error is defined as the difference between the WDQS training circuit (interval oscillator) and the actual WDQS clock tree across voltage and temperature.

Parameters:

- t_{RX_DQS2DQ} : Actual DQS clock tree delay
- t_{DQSosc} : Training circuit (interval oscillator) delay
- $WOSCOffset(V)$: Average delay difference over voltage
- $WOSCOffset(T)$: Average delay difference over temperature
- $WOSCMATCH(V)$: WDQS oscillator matching error over voltage
- $WOSCMATCH(T)$: WDQS oscillator matching error over temperature

Figure 75: Oscillator Offset (OSCOffset_volt)

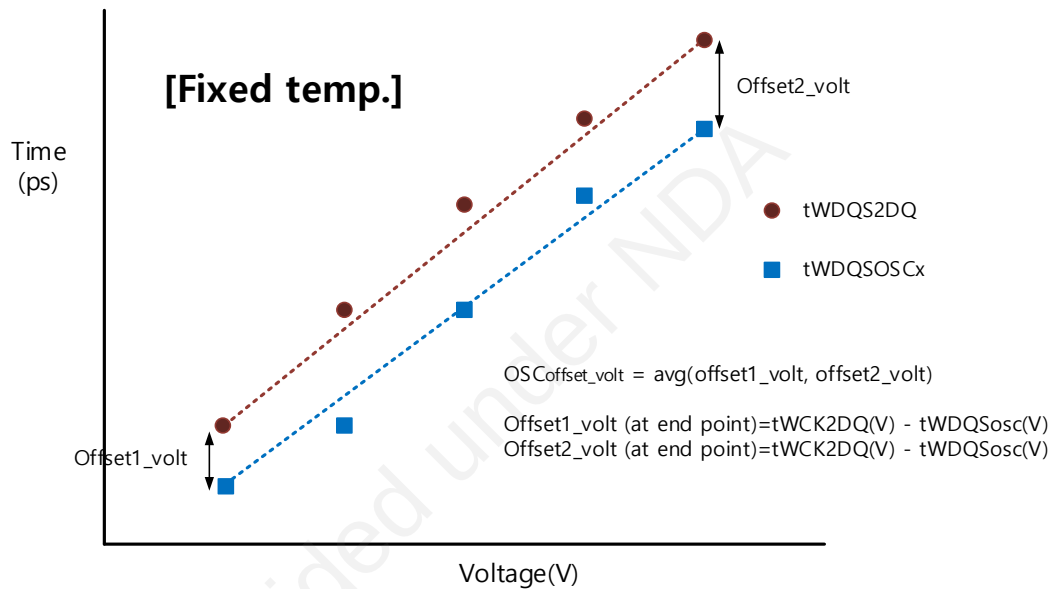


$WOSCMATCH(V)$:

$$WOSCMATCH(V) = [t_{RX_WDQS2DQ}(V) - t_{WDQSosc}(V) - WOSCOffset(V)]$$

$t_{WDQSosc}(V):$

$$t_{WDQSosc}(V) = \frac{\text{Run Time}}{2 \times \text{Count}}$$

Figure 76: Oscillator Offset (OSCOffset_temp)

 $WOSCMATCH(T):$

$$WOSCMATCH(T) = [t_{RX_WDQS2DQ}(T) - t_{WDQSosc}(T) - WOSCOFFSET(T)]$$

 $t_{WDQSosc}(T):$

$$t_{WDQSosc}(T) = \frac{\text{Run Time}}{2 \times \text{Count}}$$

Table 64: WDQS Oscillator Matching Error Specification

Symbol	Parameter	Min	Max	Unit	Notes
WOSCMATCH(V)	WDQS Oscillator Matching Error: voltage variation			ps	1, 2, 3, 5
WOSCMATCH(T)	WDQS Oscillator Matching Error: temperature variation			ps	1, 2, 3, 5
WOSCOFFSET(V)	WDQS Oscillator Offset for voltage variation			ps	2, 5
WOSCOFFSET(T)	WDQS Oscillator Offset for temperature variation			ps	2, 5

- Notes:
1. The WOSCMATCH is the matching error per between the actual WDQS and WDQS interval oscillator over voltage or temperature.
 2. This parameter will be characterized or guaranteed by design.
 3. The input stimulus for $t_{RX_WDQS2DQ}$ will be consistent over voltage and temperature conditions.
 4. $t_{RX_WDQS2DQ}(V, \text{ or } T)$ delay will be the average of WDQS to DQ delay over the runtime period.
 5. The matching error and offset of WOSC came from WDQS Interval oscillator.
 6. These parameters are defined per device.



tWDQS2DQ_I Offset due to Temperature and Voltage Variation

As temperature and voltage change on the HBM3 DRAM, the WDQS clock tree will shift and may require retraining. The oscillator is usually used to measure the amount of delay over a given time interval (determined by the controller), allowing the controller to compare the trained delay value to the delay value seen at a later time. The tWDQS2DQ_I offset due to temperature and voltage variation specification can be used for instances when the oscillator cannot be used to control the tWDQS2DQ_I (see AC Timings).

Soft and Hard Repair

The HBM3 DRAM allows to repair failing rows in the memory array using two methods:

- Soft repair allows a temporary row repair without using a permanent fusing mechanism. It is intended to enable validation that the intended repair works as expected. The IEEE1500 SOFT_REPAIR instruction is associated with this method. A channel must be in bank idle state as long as the SOFT_REPAIR instruction is loaded in the WIR.
- Once a soft repair is validated, the user may choose to perform a fused hard repair. A Hard repair is permanent and must be followed by a chip reset (RESET_n pulled LOW). The IEEE1500 HARD_REPAIR instruction is associated with this method. All channels must be in bank idle state as long as the HARD_REPAIR instruction is loaded in the WIR.

Repair resources (redundant rows) are provided per PC and per bank. The number and availability of repair resources are provided via the HS_RES_CAP instruction. The use of soft repair will not decrement the HS_RES_CAP register so the host controller must track the resources used. If there is no repair resource available in a certain bank then the host controller should not issue a soft repair to that bank and the DRAM will ignore the programming sequence.

In case several soft repairs shall be performed within one PC it is required to program these soft repairs in ascending bank order.

The repair granularity indicating the number of repaired rows per a single soft or hard repair is sixteen rows. The row address bits associated with each repair are indicated in the PPR_RA field of the DEVICE_ID WDR and listed in Table 65.

Table 65: Associated Row Addresses

Associated Row Addresses			
RA12	RA2	RA1	RA0

The soft repair supports an Undo and Lock function. The soft repair Undo will restore a previously used repair resource back to its unused state and at the same time reactivate the original (unrepaired) row instead. A locked repair resource cannot be set back to the unused state using the Undo function.

For this HBM3 DRAM the availability of the Undo and Lock functions and the number of available repair resources for soft repair are configurable via the SPPRUL bit in MR10 OP0.

- with the bit set to 0_B (default), four repair resource are available per PC and per bank; in this case the Undo and Lock feature is not supported and the corresponding bits in the SOFT_REPAIR WDR are ignored.
- with the bit set to 1_B, one repair resource is available per PC and per bank; in this case the Undo and Lock feature is supported.



The selection shall be made during device initialization: once Soft Repair Undo / Lock has been enabled, it must remain enabled; disabling the feature must be followed by a chip reset (RESET_n = LOW).

The host controller can lock down a used soft repair resource by issuing a soft repair with the SOFT_REPAIR_LOCK and SOFT_REPAIR_START bits as 1_B. A locked repair resource cannot be set back to the unused state using the Undo function. Only a chip reset (RESET_n pulled LOW) or power-cycling can unlock a locked repair resource

The Undo and Lock functions are mutually exclusive and therefore must not be set 1_B at the same time when issuing a soft repair.

When using soft repair specifically with the Undo function, the host controller must manage and schedule the refresh operation properly on the valid data of a row address. If a row has been repaired, all refresh commands will exclude the original row from being refreshed and refresh the repair row instead. Similarly, unused repair resources will not be refreshed which includes those resources that had been allocated but were then set back to the unused state using the Undo operation. Especially when switching back and forth between an original and a repair row, regular refresh commands may not hit both rows within the required refresh interval. A possible method to prevent a potential data loss is to explicitly issue ACTIVATE and PRECHARGE commands to the mapped-out rows before and after the soft repair operations.

Self Repair

The HBM3 DRAM supports self repair to help improve SIP assembly yield or to achieve a high level of system reliability by scanning for and repairing failures in the HBM3 during the initialization process.

IEEE1500 instructions SELF_REP and SELF_REP_RESULTS are associated with the self repair functionality. Self repair is initiated by loading the SELF_REP instruction. Since the instruction works on 8 channels at a time, WIR [12:8] must be set to either 1C_H or 1D_H to select one half of the 16 channels to run on. A parallel operation of Self Repair on both groups of 8 channels is not supported. See Table 75 for the mapping of channels for 1C_H and 1D_H.

The clock sources for SELF_REP is the WRCK clock which must be constantly running while the SELF_REP instruction is loaded in the WIR.

Figure 77 provides a flow chart showing the 3 flows for the REP_TYPE field (bits[3:2]) of the SELF_REP instruction. Setting REP_TYPE to 11_B will instruct the HBM3 DRAM to start the first phase of the self repair process which is self-test to identify any hard failures. The SELF_REP instruction works on one SID at a time and must be run on each SID separately by using the SID_SELECT field (bits [5:4]). The number of SELF_REP instructions required to check all channels and SIDs is listed in Table 66. The SELFR_REF_RATE field (bits [7:6]) must be set by the host to control temperature compensated refresh rate.

Table 66: SELF_REP Instructions vs. Stack Height

Stack Height	SID_SELECT	Minimum Number of SELF_REP Instructions
8-High	00 _B , 01 _B	4
12-High	00 _B , 01 _B , 10 _B	6

The self-test will use vendor specific pattern(s) that detect hard failures in the HBM3 DRAM. Once the self-test phase is complete the HBM3 DRAM will proceed to the auto-repair phase. The auto-



repair automatically repairs up to 8 failed addresses (up to 2 per memory slice, see Table 4) from the self-test phase.

SELF_REP may be issued any time after the device has been properly initialized, specifically after 'INIT3 has been met, and the HBM3 DRAM is in the all banks idle state.

This HBM3 DRAM shares repair resources between soft repair, hard repair and self repair as indicated by the SHARED_REP_RES field of the DEVICE_ID WDR being set to 1_B. If the DRAM shares the resources the host can use the HS_REP_CAP instruction to tell how many resources are available in each channel for self repair. Any repair done by self repair will update the resources per bank in that channel.

Because of the shared resources, the DRAM must not use all the available resources in a bank. One resource per bank must be left for the host to perform soft repair. If the host desires to allow self repair to use all of the shared resources then the SHARED_OVERRIDE field can be set to 1_B.

Before any self repair, the host is recommended to clear any soft repairs using the “undo” function on the channels the host plans to run self repair, or a chip reset. Failure to do so could result in the DRAM using the shared resources and operation is then not guaranteed. After the self repair is complete the host can perform soft repair(s) after checking if resources are available.

Table 67 illustrates the expected behavior of the HBM3 DRAM and the expected SELF_REP_RESULTS when SHARED_OVERRIDE is set to 0_B and 1_B.

Table 67: Self Repair Expected DRAM Behavior for Shared Repair Resources

Case	Resources vs Fails	Examples						
		Resource(s) before Self Repair	Fails	Override	DRAM Behavior	Results	Resource(s) after Self Repair	Subsequent Soft / Hard Repair
1	Resources < Fails	1	2	0 _B (No)	No repair ¹	Unrepairable fails remain	1	Yes
				1 _B (Yes)	Auto-repair	Unrepairable fails remain	0	No
2	Resources = Fails	2	2	0 _B (No)	No repair ¹	Unrepairable fails remain	2	Yes
				1 _B (Yes)	Auto-repair ²	1. repair: fails remain 2. repair: no fails remain ³	1 / 0 ⁴	Yes / No ⁴
3	Resources > Fails	3	2	0 _B (No)	Auto-repair ²		2 / 1 ³	Yes
				1 _B (Yes)	Auto-repair ²			

- Notes:
1. The number of available resources for SELF_REP is not sufficient to repair all fails, and the OVERRIDE bit is set to 0_B. In that case no repair is performed.
 2. This HBM3 DRAM performs up to 8 repairs per SELF_REP (up to 2 repairs per memory slice).
 3. After the initial SELF_REP the DRAM will report 'fails remain'; the host will need to issue a second SELF_REP to repair the other fail. After that the results will be 'no fails remain'.
 4. The host has the option to do no further repair and leave the remaining resource for soft repair, or use up the one remaining resource with an additional SELF_REP.

During the self repair process the host can poll the DRAM for the status using the SR_PROGRESS field of the SELF_REP instruction. The DRAM will report whether the self-test is in progress (01_B), the auto-repair is in progress (10_B) or the self repair process has completed (11_B) or is not running (00_B). This reporting is the same for all 8 channels in the channel group selected by WIR[12:8]. The SELF_REP instruction must be kept in the WIR during this time.



Once the self repair process is complete, the SELF_REP_RESULTS instruction can be issued to read out the results. The DRAM will report the results for each SID and will indicate whether fails remain (01_B), whether unrepairable fails remain (10_B), whether SELF_REP should be run again (11_B), or whether self repair has not run since the initialization or no fails remain after most recent run (00_B).

If after running both the self-test and auto-repair phases the results indicate that fails remain, the SELF_REP instruction can be issued to run only the auto-repair phase to repair additional fails by setting the REP_TYPE to 10_B. With REP_TYPE set to 10_B the DRAM will repair one additional row address from the previous self-test phase. If additional fails remain, the host can continue to issue SELF_REP instructions with REP_TYPE = 10_B, followed by SELF_REP_RESULTS, until the DRAM reports 00_B to indicate that there are no fails remaining. If the DRAM reports 11_B, this is an indication to the host to run SELF_REP again with either REP_TYPE set to 01_B (self-test only) or 11_B (self-test and auto-repair) to load internal fail addresses from the self-test phase.

With REP_TYPE set to 01_B, the SELF_REPAIR instruction will only run the self-test phase and the host can check the results after completion to decide on next steps.

If the host runs the auto-repair only without previously having run the self test phase, then the SELF_REP_RESULTS instruction will report 00_B as there are no failing address(es).

If the DRAM reports that unrepairable fails remain (10_B) on a channel, this indicates that there are not enough repair elements remaining to repair fail addresses latched during the self-test phase. The host can decide whether to run self repair again to repair other channels or complete the repair process.

Once the self repair process is complete, the host must issue a reset by driving RESET_n to LOW and then following the Initialization Sequence with Stable Power.

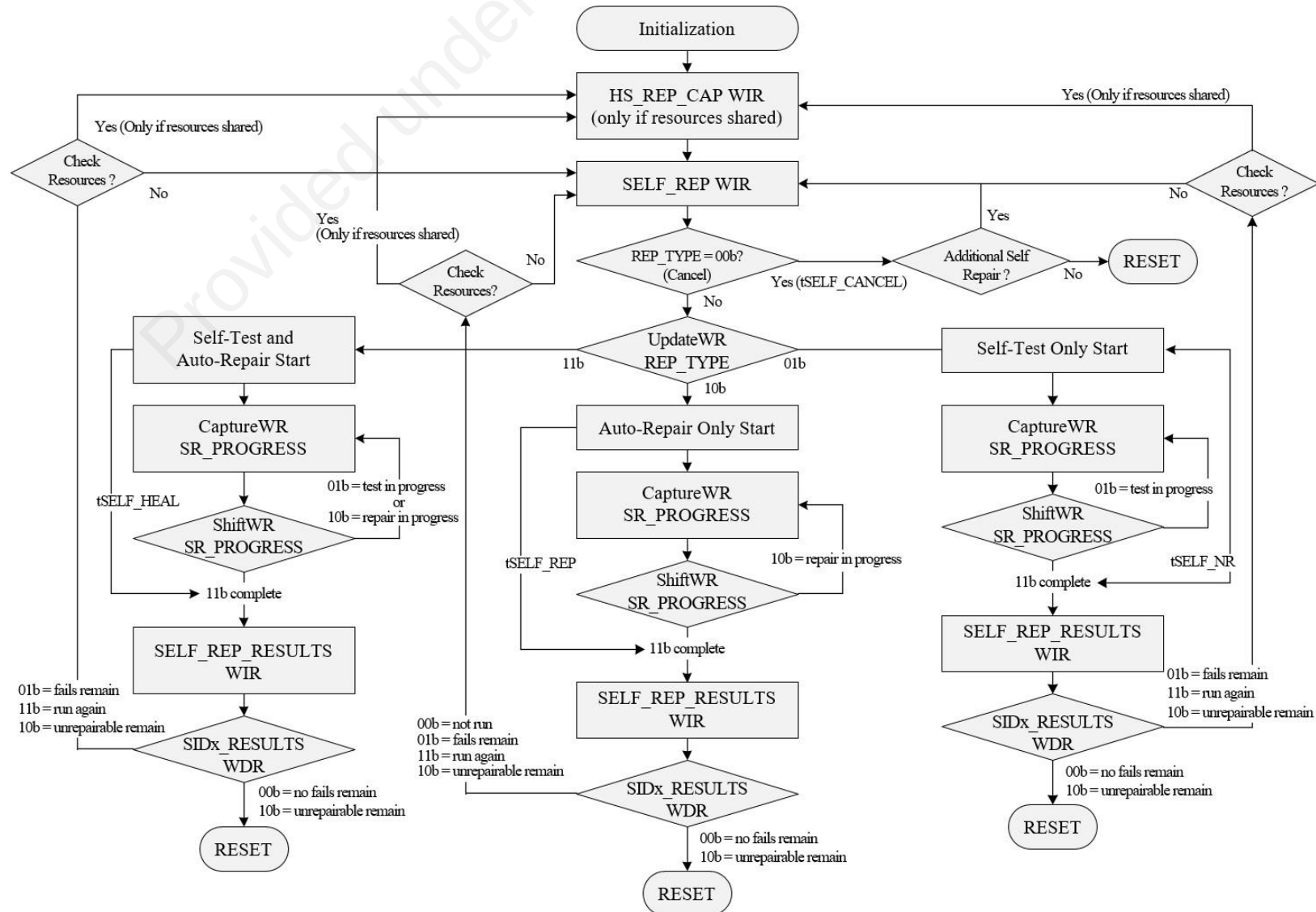
The host is able to cancel the self repair in progress by using the SELF_REP instruction with REP_TYPE set to 00_B, however only the self-test phase can be cancelled. The SR_PROGRESS field of the SELF_REP will be set to 00_B and the host must wait tSELF_CANCEL before any additional self repair. If no further repair is needed the host must reset the DRAM.

If the host does not use the polling to determine completion then the following timing parameters will indicate the completion of the REP_TYPE (see also Table 104).

Table 68: Self Repair Timings

Parameter	REP_TYPE	Phase	Max.	Unit
tSELF_HEAL	11 _B	Self-test and auto-repair	tSELF_REP + tSELF_NR	s
tSELF_REP	10 _B	Auto-repair	5	s
tSELF_NR	01 _B	Self-test	12	s
tSELF_CANCEL	00 _B	Self-test cancel time	10	ms

The HS_REP_CAP instruction can be used to determine how many resources are available for self repair. Any repairs done by self repair will update the resources per bank.


Figure 77: Self Repair Flow Chart


Loopback Test Modes

A Multiple-Input Shift Register (MISR) / Linear Feedback Shift Register (LFSR) circuit is defined within the HBM3 AWORD and DWORD I/O blocks. These circuits are intended for testing and training the link between the host and the HBM3 DRAM.

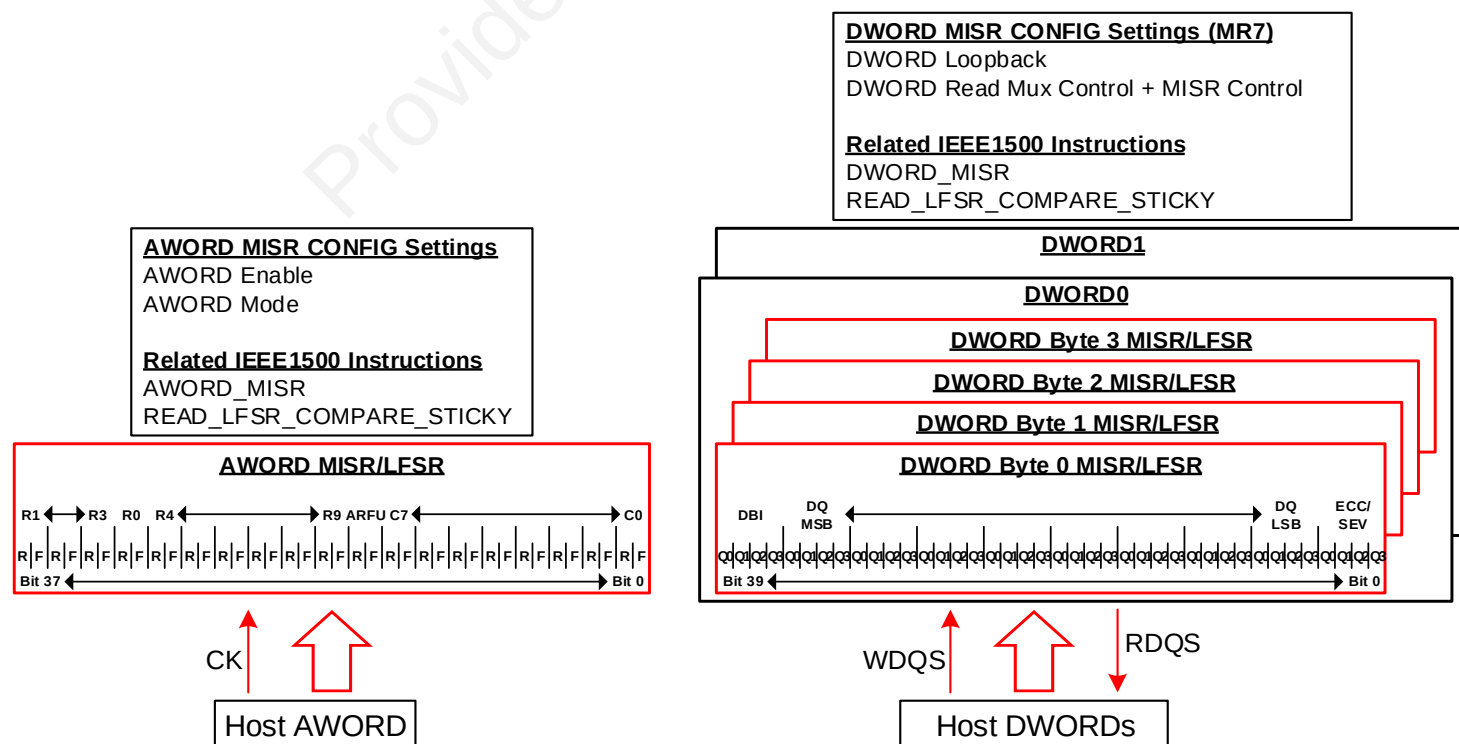
Referring to Figure 78, each byte within a DWORD implements a 40-bit MISR/LFSR circuit comprised of 4 QDR bits (Q0 to Q3) for each of the 10 data signals (8 DQ, DBI and ECC/SEV). These 4 QDR bits represent the 4 UI per CK clock cycle latched by WDQS on writes or driven along with RDQS on reads. Note that a BL8 write or read burst is 2 ^tCK long and thus corresponds to two steps of the DWORD MISR/LFSR. The MISR/ LFSR circuits operate independently across the bytes.

The AWORD implements a 38-bit MISR/LFSR circuit comprised of two DDR bits (Rise/ Fall) for each of the 18 row and column command/address inputs including ARFU.

When the MISR content is read via the IEEE 1500 port DWORD_MISR instruction, the four bytes per DWORD (160 bits) for the two DWORDS within a channel are serially shifted out, providing a total of 320 bits. The 38-bit AWORD MISR content is read via the AWORD_MISR instruction. See Table 86 and Table 87, respectively, for the bit-order for both registers.

The term MISR modes collectively refers to all of the modes - LFSR mode, register mode, MISR mode, and LFSR compare mode. AWORD MISR modes and DWORD MISR modes refer to all of the modes defined for the specific bus.

Figure 78: MISR Features Block Diagram



HBM Polynomial Structures

Figure 79 shows an example of a 4-bit Galois type MISR/LFSR structure that implements the following polynomial:

$$f(x) = X^4 + X^3 + 1$$

The example circuit and function table are for illustration only, and this circuit's modes are not fully representative of the actual DWORD and AWORD MISR definitions as outlined below. For example, the circuit implements a reset function, while the AWORD and DWORD MISRs instead implement a preset function, where specific bits are set to logic 1.

Figure 79: Example of a 4-bit MISR-LFSR Implementing $f(x) = X^4 + X^3 + 1$

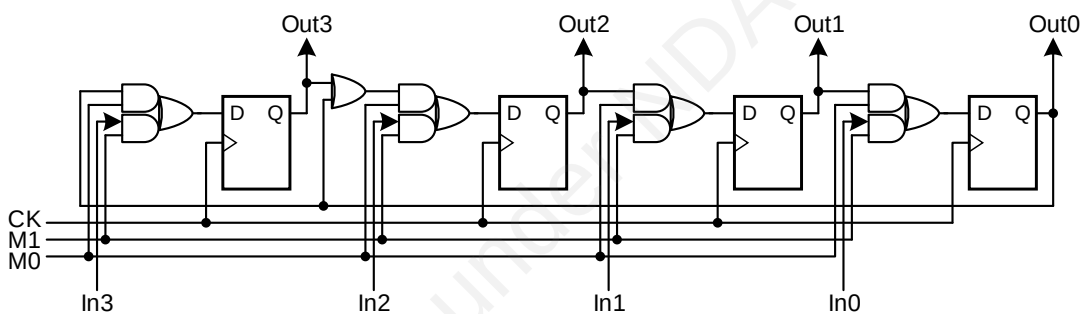


Table 69: MISR Function Table

M1	M0	Function
0	0	Reset
0	1	LFSR
1	0	Register
1	1	MISR

AWORD MISR Polynomial

The AWORD MISR structure is a 38-bit MISR/LFSR with the following polynomial:

$$f(x) = X^{38} + X^6 + X^5 + X + 1$$

The AWORD MISR may be serially accessed via the AWORD_MISR instruction.

DWORD MISR Polynomial

The DWORD MISR structure is a 40-bit MISR/LFSR per byte with the following polynomial:

$$f(x) = X^{40} + X^{38} + X^{21} + X^{19} + 1$$

The DWORD MISRs are accessed via the DWORD_MISR instruction. The 4 bytes per DWORD and the two DWORDs per channels are concatenated into a 320-bit WDR.



General Loopback Modes Features and Behavior

This section addresses features and behaviors that generally apply to all of the MISR modes.

- a) **Entering the MISR modes:** MISR modes may be enabled after t_{INIT3} within the initialization sequence; they may also be entered any time after completing the initialization (see Initialization). DWORD MISR modes are controlled via Mode Register 7 (MR7), while AWORD MISR modes are controlled via the IEEE1500 port AWORD_MISR_CONFIG instruction. AWORD and DWORD MISR modes cannot be used simultaneously since the DWORD MISR modes are driven via READ and WRITE commands on the AWORD bus.
- b) **Entering and exiting AWORD MISR modes:** HBM3 DRAMs allow the AWORD MISR modes to be utilized on one or more channels while the other channels continue to operate normally. After normal initialization, to enter the AWORD MISR modes the host must put the desired channel(s) into either precharge power-down or self refresh mode. Self refresh mode may be used in order to retain memory content while using the AWORD MISR modes, as needed. Enabling the AWORD MISR modes re-enables the AWORD input buffers that are normally disabled in power-down and self refresh modes, which may result in increased current draw over the IDD2P, IDD2P0 and IDD6x specifications. If returning to normal operation is not required, the host may assert an initialization sequence per section Initialization after operating the AWORD MISR modes. The sequence for entering AWORD MISR modes, and then exiting back to normal operation is as follows:
 - 1) At any time after initializing the HBM3 DRAM enters the all banks idle state.
 - 2) Enter either the precharge power-down or self refresh. Maintain R0 = LOW while in these modes.
 - 3) Stop the CK clock with CK_t = LOW and CK_c = HIGH.
 - 4) Enable/enter and operate the AWORD MISR modes (AWORD_MISR_CONFIG ENABLE = 1_B). Finish these operations with CK stopped (CK_t = LOW, CK_c = HIGH) and R0 = LOW.
 - 5) Disable the AWORD MISR modes and follow the power-down or self refresh exit procedures.
 - 6) When using the AWORD MISR modes after t_{INIT3} within the initialization sequence, power-down or self refresh entry and exit does not apply.

If the DRAM is not required to continue with mission mode operation after AWORD MISR test, there is no requirement on the row/column command bus and the precharge power-down or self refresh state after loopback test. The AWORD_MISR_CONFIG enable bit can be reset by WRST_n during a subsequent initialization sequence.

- c) **Entering and exiting DWORD MISR modes:** HBM3 DRAMs allow the DWORD MISR modes to be utilized on one or more channels while the other channels continue to operate normally. After normal initialization, to enter the DWORD MISR modes the host must put the desired channel(s) into the all banks idle state, enable the DWORD MISR modes (MR7 OP0 = 1_B; see Table 17), and then enter precharge power-down or self refresh mode. Self refresh mode may be used in order to retain memory content while using the DWORD MISR modes, as needed. Enabling the DWORD MISR modes before entering precharge power-down or self refresh keeps the AWORD and DWORD I/O buffers enabled and may result in increased current draw over the IDD2P, IDD2P0 and IDD6x specifications. DWORD MISR modes may also be enabled after t_{INIT3} within the initialization sequence. Also see items h), k), and n) for related DWORD MISR modes configuration setting. On the column command bus only READ (RD), WRITE (WR), and CNOP commands may be issued which operate the DWORD MISR modes, MR7 MRS commands to select the DWORD MISR modes, and MR15 MRS command to adjust the



V_{REFD} level. On the row command bus R0 must be driven to static LOW. The sequence for entering DWORD MISR modes, and then exiting back to normal operation is as follows:

- 1) At any time after initializing the HBM3 DRAM enters the all banks idle state.
- 2) Set all configuration mode registers as needed for use in the DWORD MISR modes (see items h), k), and n)).
- 3) Enable the DWORD loopback by setting MR7 OP0 = 1_B and wait t_{MOD}.
- 4) Enter either the precharge power-down or self refresh. Maintain R0 = LOW while in these modes.
- 5) Select the DWORD MISR mode via MR7 while keeping R0 pulled LOW to maintain the power-down or self refresh state. Operate the DWORD MISR by issuing READ, WRITE, and CNOP commands. After completing DWORD MISR operations, send CNOP commands.
- 6) Follow the power-down exit (PDX) or self refresh exit (SRX) procedures.
- 7) Disable the DWORD loopback testing by setting MR7 OP0 = 0_B and wait for t_{MOD} before continuing normal operation.

MRS commands are not supported until after t_{INIT5} in the initialization sequences; therefore, to configure and control the mode registers for DWORD MISR modes usage after t_{INIT3} the MODE_REGISTER_DUMP_SET instruction must be used. The sequence for entering and operating the DWORD MISR modes after t_{INIT3} in the initialization sequence is as follows:

- 1) Start the CK clock and enter power-down mode with PDE and CNOP on the command busses.
- 2) Use the MODE_REGISTER_DUMP_SET instruction to set all mode registers as needed for use in the DWORD MISR modes (see items h), k), and n)), enable DWORD loopback (MR7 OP0 = 1_B) and wait t_{MOD}.
- 3) Select and operate the DWORD MISR modes via MR7 settings (using MODE_REGISTER_DUMP_SET) and sending READ, WRITE and CNOP commands.
- 4) After completing DWORD MISR operations, send CNOP commands, disable DWORD loopback test mode by setting MR7 OP0 = 0_B using MODE_REGISTER_DUMP_SET, then wait t_{MOD}.
- 5) The CK clock may be stopped if desired.
- 6) Proceed to other IEEE1500 instructions or proceed with the initialization sequence from Figure 8, time Td.

If the HBM3 DRAM shall not continue with mission mode operation after DWORD MISR test, there is no requirement to follow the power-down exit (PDX) or self refresh exit (SRX) procedures and disabling DWORD loopback testing in MR7. The loopback enable bit can be reset by a subsequent initialization sequence with RESET_n = LOW.

- d) **Command decode is disabled in AWORD MISR modes:** When AWORD MISR modes are enabled (AWORD MISR CONFIG ENABLE = 1_B - On), the traffic sent on the AWORD bus is not limited to valid commands and the command decoder is disabled to prevent undefined states and operations.
- e) **With lane repairs the MISR bit positions remain with their logical signals:** the MISR bits are associated with their logic signals, not the physical microbumps (see Figure 70). For



example, if DQ3 has been repaired (which routes the DQ3 data to the DQ4 microbump) the data received on the DQ4 microbump is routed to the DQ3 MISR bits. Effectively, the behaviors for all MISR modes are unchanged - all 10 bits of the byte are captured in the MISR in the same bit locations, as if no lane repair were active.

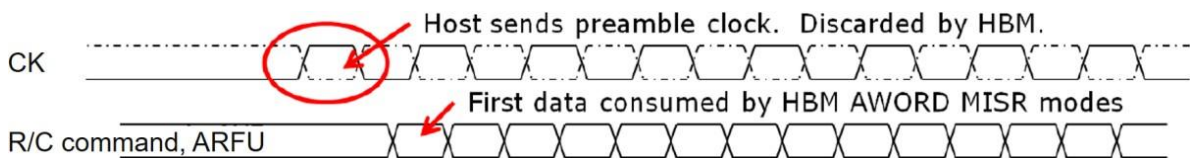
- f) **DBI, and ECC/SEV logic circuits are not functional in DWORD MISR modes** (see also Table 70 and Table 71): the DBI and ECC/SEV signals are treated as pure data signals. Their raw values are captured, compared, or sent without regard to their normal bus inversion or ECC/SEV functional meaning.
- It is required to enable WDBI and RDBI in MR0 OP[1:0] in order to enable the I/O buffers on the DBI signals. A value of 0_B is internally assumed for all DBI write data in case WDBI is disabled.
 - Regardless whether meta data and severity reporting are enabled in MR9 or not, setting MR7 DWORD Loopback Enable = 1_B will enable the ECC/SEV signal's I/O buffers. Note that the SEV signals are bidirectional I/Os in loopback test mode only.
 - The host may write DBI encoded or non-encoded data to the HBM3 DRAM. In MISR mode or register mode, the raw (not DBI decoded) write data will be directly captured into the MISR.
 - For LFSR compare mode to match, the host must send the LFSR generated raw data on all 10 signals of the byte without write DBI encoding. The HBM3 DRAM will not DBI decode the received data, and thus the host must send the raw LFSR data in order for LFSR compare to match.
 - In LFSR mode, the HBM3 DRAM will generate non-DBI encoded read data.
- g) **DWORD read path parity traffic generation** (see also Table 70 and Table 71): the parity logic is not active when loopback test mode is enabled and the RPAR setting in MR0 has no effect. To generate traffic on the DPAR signal a nearby DQ signal is copied on the DPAR signal: DQ2 is sent on DPAR0 (DWORD0), and DQ34 is sent on DPAR1 (DWORD1), irrespective of any lane repairs. The parity signals are driven simultaneously with the DQ data - effectively with PL = 0. A suggested host-side implementation is to use signature register circuits for checking the validity of the received parity signal. When reading back the LFSR_COMPARE_STICKY error bits, the parity signal output is unspecified.
- h) **AWORD and DWORD write parity checking** (see also Table 70 and Table 71): the HBM3 DRAM's parity evaluation logic is active when loopback test mode is enabled and outputs results on AERR after 'PARAC and on DERR after 'PARDQ, respectively, if the parity calculation is enabled in MR0 (see Table 10). The parity latency in MR1 (see Table 11) must be set to a supported PL value. The HBM3 DRAM will process write parity per the PL setting and protocol, including any required additional WDQS cycles. A suggested host-side implementation is to use signature register circuits for checking the correctness of the AERR and DERR signals. It is also suggested that the host generate data on the DWORD DPAR signals in order to exercise these signal paths and logic.
- i) **Preset state AAAAAAAAAAh and 2AAAAAAAAAh**: the preset state for the DWORD MISR is AAAAAAAAAAh. This is a useful state for producing an alternating 0-1 pattern on all 10 bits associated with a DWORD byte when put into Register mode (read direction). This basic pattern may be used by the host for RDQS eye centering. READ commands from the DWORD MISR are supported in Preset state. WRITE and READ commands to and from the DWORD MISR do not change the DWORD MISR content in this mode.

The AWORD MISR is also preset to the same 0-1 pattern (2AAAAAAAh for implementation consistency; although the AWORD cannot be enabled to drive this data pattern back to the host).

The preset state may be overridden using the register modes (see AWORD and DWORD Register Modes (write direction)).

- j) **DWORD MISR is writeable via IEEE1500:** the normal method for writing the DWORD MISR is through the functional interface (see Test Method for DWORD Write MISR mode). Seed values for the DWORD MISR may also be written using the DWORD_MISR IEEE 1500 port instruction.
- k) **DWORD read and write latencies must be set properly:** READ and WRITE commands generate DWORD MISR modes traffic. The normal mode DWORD read and write protocol is followed using the latency settings as supported by the operating frequency being used.
- l) **DWORD write preamble and postamble strobes adhere to the normal protocol:** for DWORD Register, MISR and LFSR compare modes (write direction), WDQS preamble and postamble strobes must be issued, and the HBM3 DRAM samples the DWORD data consistent with the write protocols defined in the Write Command section.
- m) **DWORD read preamble and postamble strobes adhere to the normal protocol:** for DWORD Register and LFSR modes (read direction), and when returning the LFSR_COMPARE_STICKY bits, the HBM3 DRAM will produce RDQS preamble and postamble strobes and send DWORD data consistent with the read protocols defined in the Read Command section.
- n) **AWORD MISR modes preamble clock filter:** in the AWORD MISR modes, the host is expected to stop CK toggling, enable the desired AWORD MISR mode, and then start sending CK toggles and AWORD data. To avoid timing impairment on the CK startup cycle, the HBM3 DRAM will treat the first received CK cycle as a preamble clock cycle and not process the data on the AWORD signals in MISR or register mode, nor compare them in LFSR compare mode. The MISR block will keep its state unchanged during this filter cycle. The first clock cycle filter circuit is enabled by setting AWORD_MISR_CONFIG bits [1:0] = 00_B. The first data sampled by the HBM3 DRAM is on the second CK clock cycle. Only the very first CK clock cycle will be filtered. If the host were to stop and restart CK clocking while remaining in an AWORD MISR mode without applying another preset, the AWORD data will be sampled on the starting clock cycle, with possible CK edge timing impairment.

Figure 80: AWORD MISR Modes Preamble Clock Filter Behavior



- o) **Cycles processed in the MISR modes:** AWORD MISR modes rely on stopping CK clocks before and after the test sequence. All AWORD cycles sent to the HBM3 DRAM after the filtered preamble clock cycle are processed into the MISR (MISR mode and register mode) or compared (LFSR compare mode), including the last cycle before CK is stopped. An even number of AWORD cycles – not counting the filtered preamble cycle – provides that the test result is the same regardless of the specific AWORD MISR implementation. For DWORD MISR modes, all valid data write cycles are processed into the MISR (MISR mode and register mode)



or compared (LFSR compare mode) while the DWORD MISR modes are enabled, consistent with the write protocols defined in the Write Command section. For example, if 10 non-seamless WRITE commands are sent to the HBM3 DRAM in DWORD MISR mode, a total of 80 data bit times (UI) will be processed into the MISR.

Table 70 and Table 71 summarize the behavior of the DBI, ECC, SEV and DPAR signals when loopback test mode is enabled in MR7 OP0.

Table 70: DWORD I/O Pins in Loopback Test Mode Write Operations

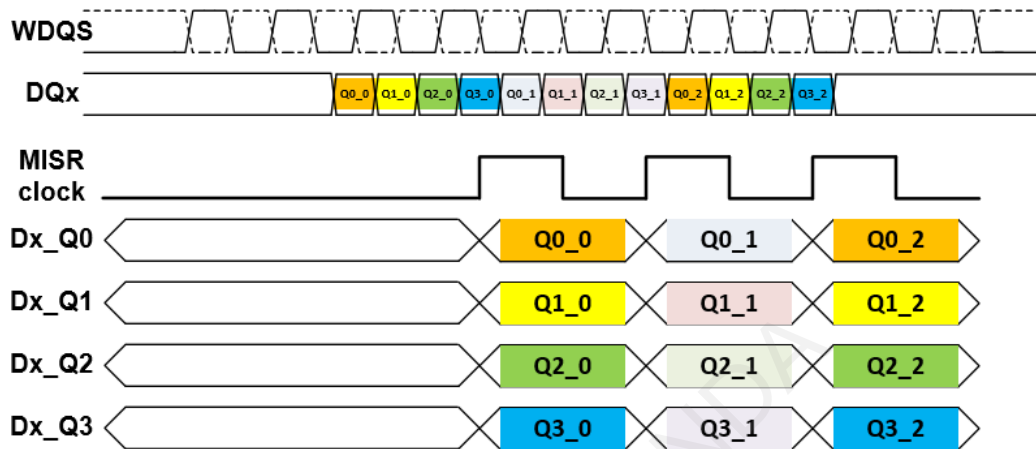
WDBI (MR0 OP1)	WPAR (MR0 OP5)	MD (MR9 OP0)	SEVR (MR9 OP1)	Requirement
Off	X	X	X	DBI input is disabled and a 0 _B is internally forced
On				DBI input is enabled and treated as a normal data input; no DBI decode
X	Off	X	X	DPAR input is disabled; no parity check
	On			DPAR input is enabled; parity is checked with PL = 3 as in normal mode
X	X	On or Off	On or Off	MD and SEVR bits are ignored; ECC and SEV are treated as normal data inputs (note that no write is defined for SEV in normal mode)

Table 71: DWORD I/O Pins in Loopback Test Mode Read Operations

RDBI (MR0 OP0)	RPAR (MR0 OP4)	MD (MR9 OP0)	SEVR (MR9 OP1)	Requirement
Off	X	X	X	DBI output is disabled
On				DBI output is enabled and treated as a normal data output; no DBI encode
X	On or Off	X	X	RPAR bit is ignored; DPAR output is enabled and a copy of DQ2 (for DPAR0) or DQ34 (for DPAR1) is transmitted with no additional latency (PL = 0); no parity generation
X	X	On or Off	On or Off	MD and SEVR bits are ignored; ECC and SEV are treated as a normal data outputs

AWORD and DWORD MISR Modes (Write direction)

When the AWORD or DWORD MISR modes are active, the data on the AWORD or DWORD data signals is received based on the CK clock or WDQS strobes respectively and compressed in the MISRs. The host is in complete control of the number of data cycles that are sent, and if successfully received by the HBM3 DRAM the values captured in the respective MISRs will be repeatable and deterministic.

Figure 81: DWORD MISR Write Behavior**Test Method for AWORD MISR Mode**

- After device initialization, set the HBM3 DRAM into precharge power-down or self refresh mode and stop the CK clock with CK_t = LOW, CK_c = HIGH.
- Enable and preset the AWORD MISR by setting AWORD_MISR_CONFIG ENABLE = 1_B and MODE[1:0] = 00_B. The preset also enables the preamble clock filter.
- Set the AWORD to MISR mode by setting MODE[1:0] = 11_B.
- Send three or more CK clock cycles and AWORD data. The first received CK clock cycle is discarded as a preamble clock and shall be followed by an even number of clock cycles. The HBM3 DRAM clocks the received data into the AWORD MISR and evaluates AWORD parity, if enabled. Stop the CK clock with CK_t = LOW, CK_c = HIGH.
- The host reads the MISR content via the AWORD_MISR instruction.

Test Method for DWORD MISR Mode (Write direction)

- Enable DWORD MISR mode and preset the DWORD MISR by setting MR7 OP0 = 1_B and OP[5:3] = 000_B. Alternate seed values can be loaded via the functional interface or the DWORD_MISR instruction.
- Enable DWORD MISR mode by setting MR7 OP[5:3] = 011_B.
- Send one or more DWORD write cycles consistent with the write protocol. The HBM3 DRAM clocks the received data into the DWORD MISRs and evaluates parity, if enabled.
- The host reads the MISR content via the DWORD_MISR instruction. The MISR content is also readable via the functional interface (see DWORD Register Mode (Read direction)).

AWORD and DWORD Register Modes (Write direction)

When the AWORD or DWORD register modes are active, the data on the AWORD or DWORD data signals is received based on the CK clock or WDQS strobes respectively and stored directly into the respective MISR without compression. Effectively the AWORD MISR operates as a 2-bit storage register and will store the AWORD data of the last CK cycle, and the DWORD MISR operate will store the last 4 UI of the last write burst.



Register modes are intended for basic, quick link testing and training, and for setting a seed for the DWORD MISR.

Test method for AWORD Register Mode

- After device initialization, set the HBM3 DRAM into precharge power-down or self refresh mode and stop the CK clock with CK_t = LOW, CK_c = HIGH.
- Enable and preset the AWORD MISR by setting AWORD_MISR_CONFIG ENABLE = 1_B and MODE[1:0] = 00_B. The preset also enables the preamble clock filter.
- Set the AWORD to register mode by setting MODE[1:0] = 10_B.
- Send three or more CK clock cycles and AWORD data. The first received CK clock cycle is discarded as a preamble clock and shall be followed by an even number of clock cycles. The HBM3 DRAM clocks the received data into the AWORD MISR and evaluates AWORD parity, if enabled. Stop the CK clock with CK_t = LOW, CK_c = HIGH. The data of the last clock cycle is retained in the AWORD MISR.
- The host reads the MISR content via the AWORD_MISR instruction.

Note that the AWORD write register mode cannot practically be used to set an alternate seed value into the AWORD MISR. In the above procedure in step d) the preamble clock filter circuit is exercised and cleared. At this point while it is allowed for the host to then stop sending AWORD cycles, set the AWORD_MISR_CONFIG MODE to MISR mode or LFSR compare mode, and then send additional AWORD cycles, there may be timing impairment for the beginning of the second set of AWORD cycles. The preamble clock filter circuit cannot be re-enabled for these additional AWORD cycles without applying the AWORD MISR preset function, which would overwrite the alternate seed value applied by a preceding AWORD write register operation.

Test Method for DWORD Register Mode (Write direction)

- Enable DWORD register mode by setting MR7 OP0 = 1_B and OP[5:3] = 010_B. A preset is not required prior to using register mode.
- Send one or more DWORD write cycles consistent with the write protocol. The HBM clocks the raw write data into the DWORD MISRs without compression and evaluates parity, if enabled. The DWORD MISRs will store the last 4 UI of the last write burst.
- The host reads the MISR content via the IEEE 1500 DWORD_MISR instruction. The MISR content is also readable via the functional interface (see DWORD Register Mode (Read direction)).

DWORD Register Mode (Read direction)

The content of DWORD MISRs may be read via the data bus. The DWORD read mux control bit in MR7 selects either the MISR or the LFSR Compare sticky error bits as data source. Issue READ commands and the HBM3 DRAM will produce RDQS preamble and postamble strobes and send DWORD data consistent with the read protocols.

Intended uses for the various read data sources include the following:

- Reading the sticky error bits after an LFSR Compare mode test sequence by setting MR7 OP1 = 1_B. Sticky error data is a single data bit per signal and is output as static values on the interface for a full read burst.
- Reading a basic clock pattern on all or select signals for DWORD read link training from MISR by setting MR7 OP1 = 0_B; with the data pattern being set before using the DWORD MISR

preset or a DWORD register write (see AWORD and DWORD Register Modes (Write direction)).

- c) Reading the MISR result at the end of a MISR mode test sequence by setting MR7 OP1 = 0_B. The results of a DWORD MISR mode test sequence may also be read via the DWORD_MISR instruction. The MISR content is sent on UI 0 to 3 and then repeated on UI 4 to 7.

Test Method for DWORD Register Mode (Read direction)

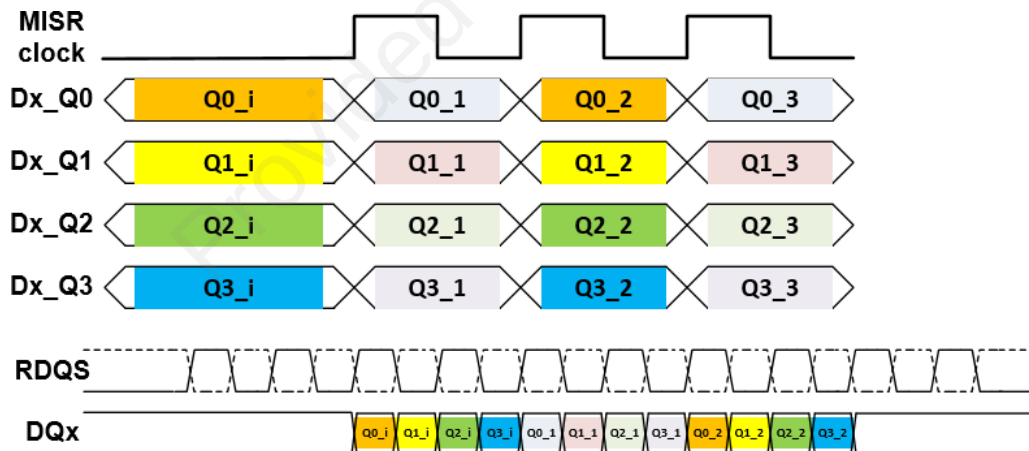
- a) Enable the test mode and select the desired read-back register by setting MR7 OP0 = 1_B and OP[5:3] = 010_B (for register mode) and select the DWORD MISR as source via MR7 OP1 = 0_B.
- b) Send one or more READ commands. The HBM3 DRAM responds following the programmed RL setting and the normal read protocol.

DWORD LFSR Mode (Read direction)

When in DWORD LFSR mode, the HBM3 DRAM generates DWORD data from the LFSR in response to READ commands. LFSR data is generated consistent with the read protocol, along with RDQS preamble and postamble strobes. The first data cycle (UI 0 to 3) generated will be the LFSR initial state, based on preset or an alternate seed value if loaded.

Note that there is no AWORD LFSR mode since the AWORD bus is input-only.

Figure 82: DWORD Read LFSR Mode Behavior



Note: 1. The figure does not illustrate the actual MISR implementation which clocks the MISR every 8 UI.

Test Method for DWORD LFSR Mode

- a) Enable DWORD loopback by setting MR7 OP0 = 1 and preset the MISR by setting MR7 OP[5:3] = 000_B. Alternate seed values can be loaded via the functional interface or the DWORD_MISR instruction (see AWORD and DWORD Register Modes (Write direction)).
- b) Enable DWORD LFSR mode by setting MR7 OP[5:3] = 001_B and OP1 = 0_B.
- c) Send one or more READ commands. The HBM3 DRAM responds following the programmed RL setting and the normal read protocol with read data produced by the LFSR.

AWORD and DWORD LFSR Compare Modes (Write direction)

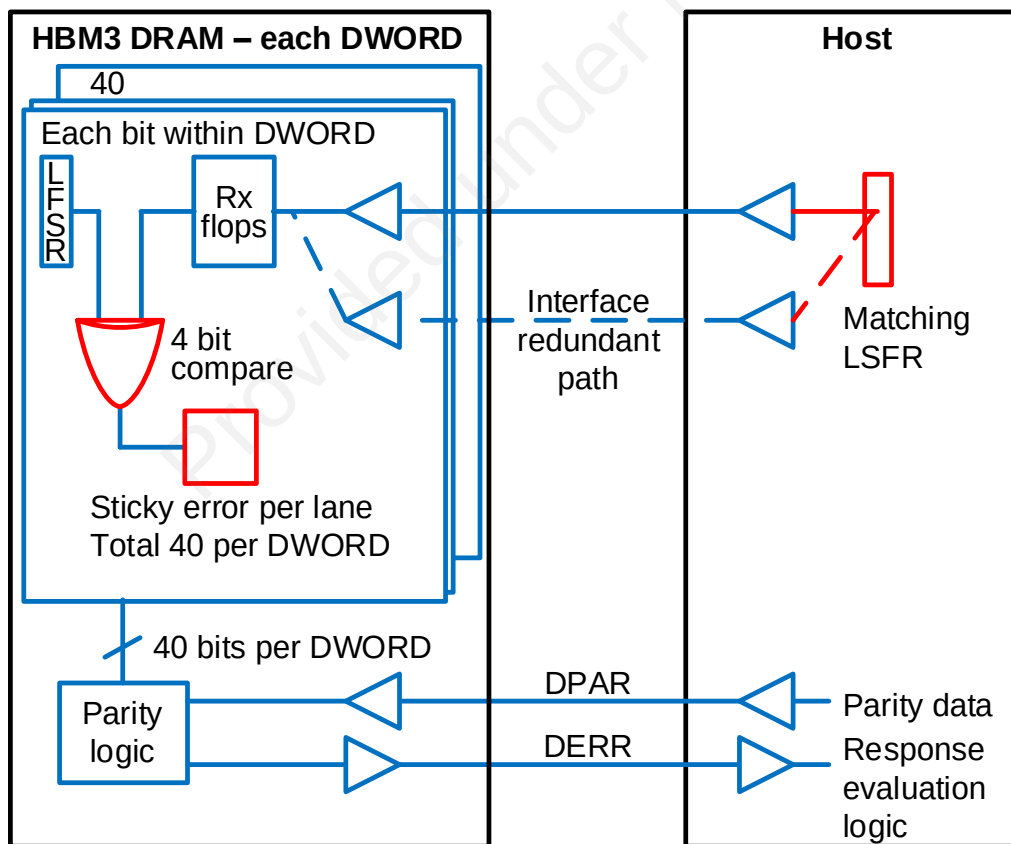
The LFSR compare modes enable direct identification of failing signal connections between the host and HBM. It is assumed that the host implements LFSR data generators that match the

lengths and polynomials of the HBM3 DRAM's LFSRs, and that the host and HBM3 DRAM LFSRs start and run in synch. The LFSRs generate data on each signal, and the compare circuitry check for matching data for each data unit interval (UI). Any mismatch between the data received and the data predicted by the LFSR will set the sticky error bit for the respective signal(s). The first data cycle expected from the host and compared by the HBM3 DRAM will be the LFSR's initial state, based on preset or an alternate seed value if loaded.

Once a miscompare is found on a signal, its sticky error bit is set to 1_B for the remainder of the test. The sticky error bits may be read via the READ_LFSR_COMPARE_STICKY instruction. DWORD related sticky error bits may also be read via the data bus while AWORD sticky error bits can only be read via the IEEE1500 test port. The sticky error bits are reset to 0_B via the MR7 OP[5:3] = 000_B for the DWORD related bits, and AWORD_MISR_CONFIG MODE[1:0] = 00_B.

The figure below illustrates the system-level configuration for LFSR compare mode.

Figure 83: LFSR Compare Mode Block Diagram



Note that data produced on the DPAR signals from the host to the HBM3 DRAM is an implementation suggestion for exercising the parity signal paths and HBM3 input timing and logic. The host-side implementation for parity signal generation is not specified. This figure also illustrates that a host-driven logical signal is compared with the matching logical signal data by the HBM3 DRAM compare circuit, regardless of any active lane repairs which may shift the physical signal routing. The AWORD LFSR compare circuit matches the DWORD circuit except for the APAR signal that is not included in the test.


Test method for AWORD LFSR Compare Mode

- a) After device initialization, set the HBM3 DRAM into precharge power-down or self refresh mode and stop the CK clock with CK_t = LOW, CK_c = HIGH.
- b) Enable and preset the AWORD MISR by setting AWORD_MISR_CONFIG ENABLE = 1_B and MODE[1:0] = 00_B. The preset operation also clears the AWORD sticky error bits and enables the preamble clock filter circuit. The host-side LFSR data generator should also be preset to the same value.
- c) Enable the AWORD LFSR compare mode by setting AWORD_MISR_CONFIG MODE[1:0] = 01_B.
- d) The host sends three or more CK clock cycles with LFSR-generated data on the AWORD signals. The first received CK clock cycle is discarded as a preamble clock and shall be followed by an even number of clock cycles. The HBM3 DRAM's LFSR predicts expected AWORD data from the host, based on matching LFSR polynomials and seeds at both ends. Any mismatch sets the sticky error for the respective signal. Parity is evaluated, if enabled. Stop the CK clock with CK_t = LOW, CK_c = HIGH.
- e) The host reads the sticky error bits via the READ_LFSR_COMPARE_STICKY instruction to determine which signals failed.

Test Method for DWORD LFSR Compare mode

- a) Enable and preset the DWORD MISR by setting MR7 OP0 = 1_B OP[5:3] = 000_B. The preset operation also clears the DWORD per-signal sticky error bits. Alternate seed values can be loaded via the functional interface or the DWORD_MISR instruction (see AWORD and DWORD Write Register Modes). The host-side LFSR data generator should also be preset to the same value.
- b) Enable DWORD LFSR compare mode by setting MR7 OP[5:3] = 100_B.
- c) Send one or more DWORD write cycles with LFSR-generated data on the DWORD signals following the programmed WL setting and the normal write protocol. The HBM3 DRAM's LFSRs predict expected DWORD data, based on matching LFSR polynomials and seeds at both ends. Any mismatch set the sticky error for the respective signal. Parity is evaluated, if enabled.
- d) The host reads the sticky error bits to determine which signals failed. The bits are readable via the READ_LFSR_COMPARE_STICKY instruction or the data bus (see DWORD Register Mode (Read direction)).

Direct Access (DA) Test Port

A direct access (DA) test port is available via DA[39:0] for vendor specific test implementations. Two microbumps are associated with each DA signal. A depopulated area for probing is located close to the DA port region in columns 21 to 36 of the HBM3 bump matrix (see Microbump Matrix).

Access to the DA test port is controlled via signal DA0. When DA0 = LOW, DA[39:1] drivers are in Hi-Z and input receivers are disabled allowing the bus to float. When DA0 = HIGH, DA[39:1] are enabled for vendor specific test features and the IEEE 1500 port is disabled. CATTRIP and TEMP[1:0] outputs remain active, but their state may not be valid and shall be ignored. The DA0 input is equipped with an internal pull-down resistor which ensures that DA0 is held LOW and the test port remains inactive even if DA0 is left floating.

If the DA test port is not used in the system, it is recommended to tie all DA port signals to V_{SS}.

The DA test port may be enabled at any time after the power ramp has been completed and all supply voltages are within their defined ranges ('INIT0), and after waiting for at least 'INIT1 time. The level of the RESET_n input shall be irrelevant for DA test port enabling.

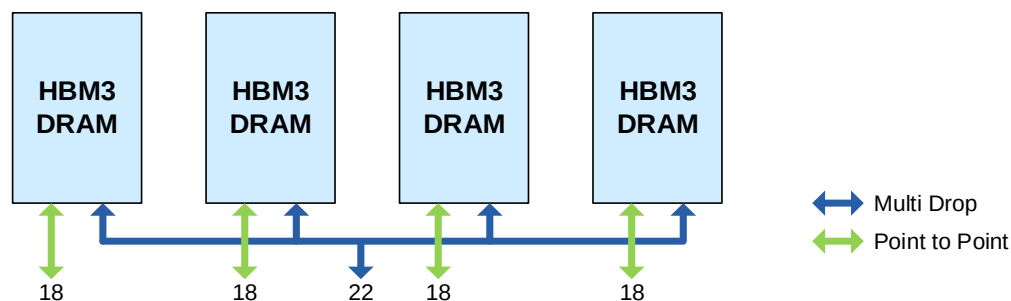
The DA test port may be disabled at any time by pulling DA0 to LOW. The HBM3 DRAM may then resume normal operation by after performing a chip reset followed by a device initialization as described in the Initialization Sequence with Stable Power section.

18 DA signals are designated to connect point-to-point to each HBM3 DRAM. 22 signals are designated to connect in parallel to up to four HBM3 DRAM devices on a multi drop bus as shown in Figure 84. The function of each of these signals is vendor specific. Table 72 defines which DA signals are allocated for point-to-point and for multi drop.

Table 72: DA Test Port Signal Allocation

Signal Group	DA Port Signal List	Count
Point-to-Point	DA[17:0]	18
Multi-Drop	DA[39:18]	22

Figure 84: DA Port Connection Diagram For Multiple HBM3 DRAMs



DA Test Port Lockout

The DA test port can be disabled (locked) by setting MR8 OP0 bit to 1_B. The bit is defined for channels **a** or **e** only. Once the bit is set to 1_B, the DA test port will remain disabled unless power is removed from the HBM3 DRAM. Any chip reset through pulling RESET_n LOW or via IEEE1500 HBM_RESET instruction or writing a 0_B via an MRS command or IEEE1500 instruction MODE_REGISTER_DUMP_SET will not clear the locked state.



IEEE1500 Test Port

The IEEE Standard 1500 compliant test access port provides a direct test connection between a host and the HBM3 DRAM. The HBM3 DRAM's test port extends the standard specification and replicates the WSO output per channel. This allows some instructions to be executed in parallel across channels and eliminates the need for cross-channel arbitration for WSO.

IEEE1500 operations may be asserted at any time after device initialization and during normal memory operation including when the HBM3 DRAM is in power-down or self refresh mode. See section Interaction with Mission Mode Operation for how the various instructions interact with normal operation, and requirements for returning to normal operation. See also section Initialization Sequence For Use Of IEEE1500 Instructions Including Lane Repair for a subset of operations that are allowed before the device initialization has been completed.

Please refer to ieee.org for more details about the IEEE1500 standard.

Interaction Between DA Test Port and IEEE1500 Test Access Port

DA0 = LOW selects the IEEE1500 test access port and DA0 = HIGH selects the DA test port. It is possible to operate the HBM3 DRAM without using the test ports. In this case both DA0 and WRST_n must be pulled LOW to prevent the device from entering test modes.

Table 73 summarizes the status of the test access port signals.

Table 73: Test Access Port Signal Status

WRST_n	DA0, MR8 OP0	Signal Name	Type	Status
LOW	DA0 = LOW or MR8 OP0 = 1 _B	Other IEEE1500 inputs ¹	Input	X
		WSO	Output	V
		DA[39:1]	I/O	X
HIGH	DA0 = LOW or MR8 OP0 = 1 _B	Other IEEE1500 inputs ¹	Input	Active
		WSO	Output	V
		DA[39:1]	I/O	X
Don't Care	DA0 = HIGH and MR8 OP0 = 0 _B	Other IEEE1500 inputs ¹	Input	X
		WSO	Output	V
		DA[39:1]	I/O	Direct test access

Notes: 1. WRCK, SELECTWR, CAPTUREWR, SHIFTW, UPDATWR, WSI.
 2. V = Valid signal (either HIGH or LOW, but not floating); X = Don't Care.



IEEE1500 Test Access Port I/O Signals

Table 74: Test Access Port I/O Signals

Symbol	Type	Description
WRCK	Input	Dedicated clock used to operate the IEEE1500 test port
WRST_n	Input	When pulled LOW, WRST_n asynchronously puts the IEEE1500 test port into its normal system mode. No WRCK clocks are required when WRST_n is LOW. See WDR Reset State
WSI	Input	IEEE1500 test port serial input
SELECTWIR	Input	SELECTWIR determines whether the instruction register (WIR) or a wrapper data register is being accessed
CAPTUREWR	Input	Controls a Capture operation in the selected wrapper register (WR)
SHIFTWR	Input	Controls a Shift operation in the selected wrapper register (WR)
UPDATEWR	Input	Controls an Update operation in the selected wrapper register (WR)
WSO[a:p]	Output	IEEE1500 test port per-channel serial output

IEEE1500 Test Access Port Functional Description

Figure 85 shows the HBM3 DRAM's IEEE1500 compliant architecture that uses an asymmetrical WSP (Wrapper Serial Port) with a single WSI and sixteen per channel WSOs. The standard compliant register stack is shown in the figure, including the Wrapper Bypass Register (WBY), Wrapper Boundary Register (WBR), and Wrapper Data Registers (WDR). The C, S and U notation for the registers refer to Capture, Shift and Update respectively, and indicate for each of the registers which functions are supported by that register. For example, the WBY only provides a Shift stage, whereas the WDRs provide Shift/ Capture and Update stages.

The WSO[a:p] output drivers are permanently enabled with their drive state being LOW, HIGH, or undefined based on the current instruction loaded into the WIR. For example, if BYPASS is the current instruction, then WSO output data is defined only after one or more WRCK clock cycles have been applied. A WSO output will drive a LOW when a channel is disabled via the CHANNEL_DISABLE instruction or marked as “not present / not working” in the DEVICE_ID WDR.

The Wrapper Instruction Register (WIR) logic is included in Figure 85, and Figure 86 shows further details of the WIR implementation. The WIR is described in the IEEE1500 Test Access Port Instruction Register section, and the instructions supported by this HBM3 DRAM in the Test Instructions section.

The five channel select bits of the WIR shift stage in Figure 86 are decoded to generate the CHSelect[a:p] outputs which control the per channel operation of the instructions. When a channel is not selected for an active instruction, then the CAPTUREWDR[a:p], SHIFTWDR[a:p] and UPDATEWDR[a:p] enables of the WSP are gated off. This will disable the WDRs of unselected channels for the decoded instruction. This gating is shown by the logic AND gates at the output of the de-multiplexer in Figure 85.

Figure 85: IEEE1500 Logic Diagram

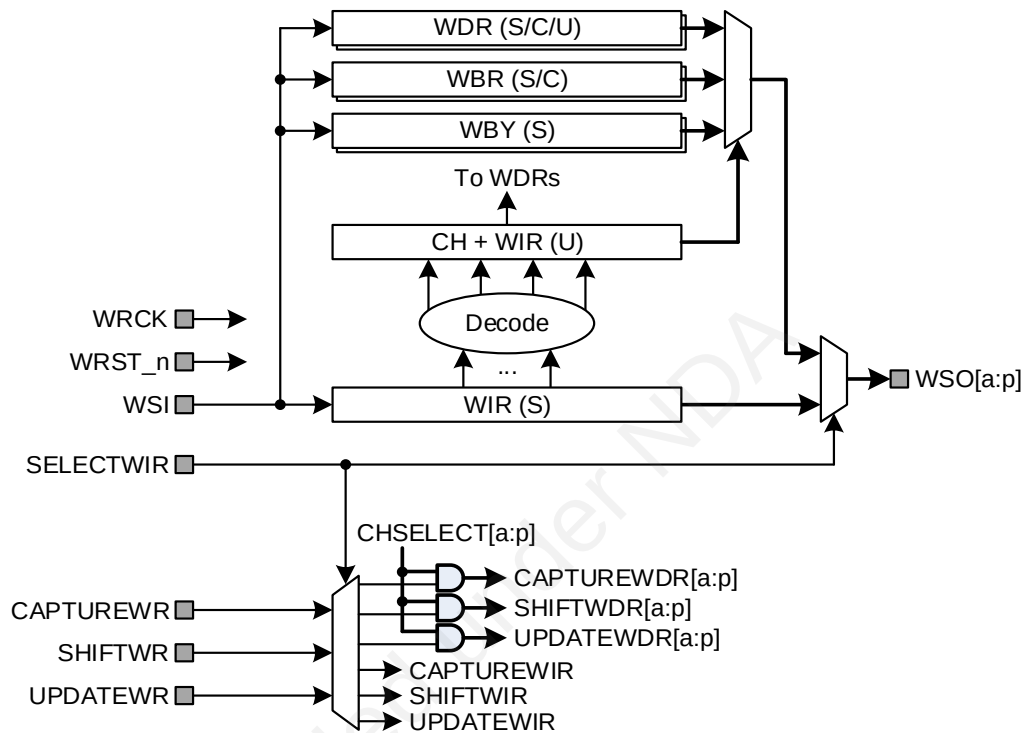
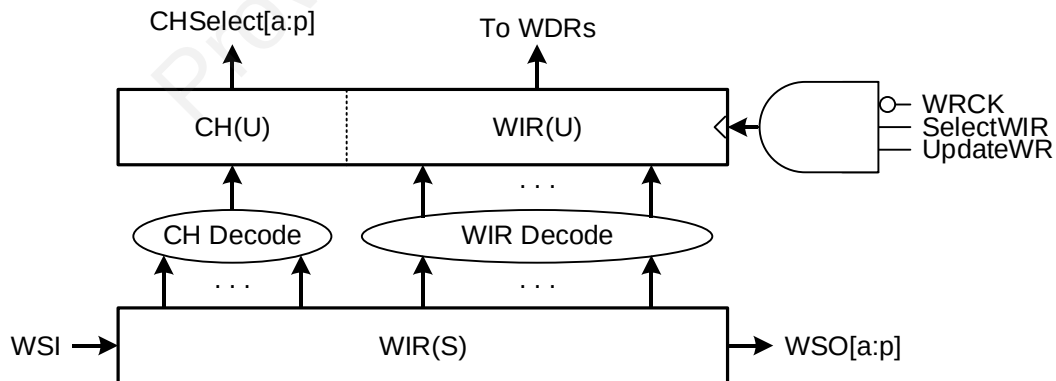


Figure 86: WIR Channel Select Logic Diagram



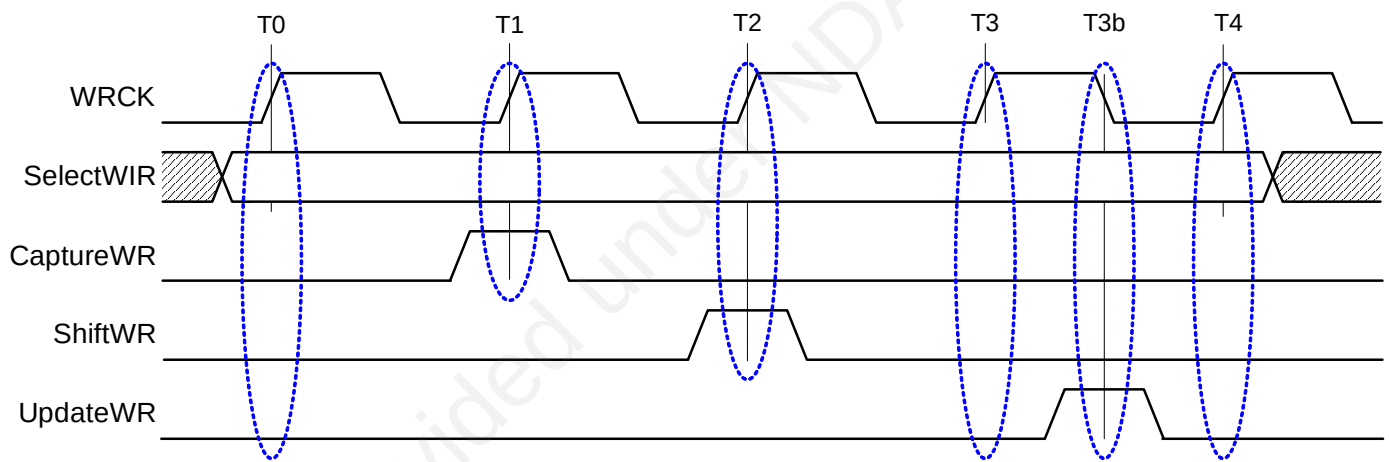
HBM3 DRAMs are allowed to support less than 16 channels. The availability of each channel is coded in DEVICE_ID WDR bits [23:8]. Unavailable channels do not respond to IEEE1500 instructions.

Figure 87 illustrates an IEEE1500 port operation sequence with a minimum number of WRCK cycles:

- Signal SELECTWR is set at clock edge T0. Control signals CAPTUREWR, ShiftWR and UpdateWR are all inactive as they are not allowed to change coincident with SELECTWR. SELECTWR must be kept stable until after completion of the complete sequence which spans until clock edge T4.



- A WDR capture operation is performed at clock edge T1 with CAPTUREWR sampled High at T1.
- A single WDR shift operation is performed at clock edge T2 with SHIFTWDR sampled High at T2.
- A WDR update operation is performed at clock edge T3b with UPDATEWR sampled High at T3b. Please note that the update operation occurs on the falling WRCK clock edge.
- For some IEEE1500 port instructions a capture, shift or update event may not be specified; please refer to the description of each instruction for details.

Figure 87: IEEE1500 Port Operation


Test Access Port Wrapper Data Register (WDR) Types

Read-Only WDRs

WDR bit fields that are specified as read-only capture data into the shift stage register when a CaptureWR event is performed. The read-only WDRs keep their state during an UpdateWR event and do not have an update stage register. Read-only WDRs shift out their content during a ShiftWR event. Data shifted into WSI during the ShiftWR event is ignored.

Write-Only WDRs

WDR bit fields that are specified as write-only copy all data bits into the update stage register when an UpdateWR event is performed. When a write-only WDR is connected between WSI and WSO, any CaptureWR event has no effect on the WDR. Write-only WDRs shift out their content during the ShiftWR event.

Read and Write WDRs

Read/Write WDRs operate as merged function of write-only and read-only WDRs. They capture data bits into the shift stage register during a CaptureWR event and copy bits from the shift stage into the update stage simultaneously when the UpdateWR event is performed.



WDR Bit Order

The LSB (bit 0) of each WDR is located nearest to the serial output (WSO) and shifted out first. WSO output timing and valid data window are defined in Table 104.

WDR Reset State

Asserting WRST_n to LOW asynchronously asserts these states on the HBM3 DRAM's IEEE1500 test port logic:

- all WDRs place their update and / or shift stages (where applicable) into a state that ensures that the HBM3 DRAM returns to mission mode operation and all test modes are disabled;
- the WIR is set to BYPASS, effectively clearing any prior EXTEST_RX, EXTEST_TX, or CHANNEL_ID instruction, thus returning all functional signals to their normal functional mode. Boundary scan chain content is undefined;
- No change to any previously loaded SOFT_REPAIR, HARD_REPAIR, SOFT_LANE_REPAIR, or HARD_LANE_REPAIR register content;
- the content of the DWORD_MISR and AWORD_MISR is undefined;
- the AWORD MISR is disabled by setting bit 2 in the AWORD_MISR_CONFIG WDR to 0;
- the CHANNEL_DISABLE WDR is reset (refer to the CHANNEL_DISABLE instruction for conditions to re-enable a disabled channel);
- any ongoing MBIST operation will be terminated.

IEEE1500 Test Access Port Instruction Register

The HBM3 DRAM supports a 13-bit Wrapper Instruction Register (WIR). Bits WIR[12:8] select the channel and bits WIR[7:0] encode the test instruction. When SELECTWIR is asserted, the WIR will not respond to the CaptureWR event, and nothing will be captured into the WIR.

The WIR channel selection definition applies only to instructions defined in Table 75 (WIR[7:0] = 00h to 1Fh). The definition does not apply to vendor specific instructions, and vendors may use bits WIR[12:8] for different purposes.

Table 75: WIR Channel Selection Definition

WIR[12:8]	Channel Select	WIR[12:8]	Channel Select	WIR[12:8]	Channel Select
00 _H	Channel a	08 _H	Channel i	1C _H	Channels a, b, e, f, i, j, m and n (8 channels)
01 _H	Channel b	09 _H	Channel j		
02 _H	Channel c	0A _H	Channel k	1D _H	Channels c, d, g, h, k, l, o and p (8 channels)
03 _H	Channel d	0B _H	Channel l		
04 _H	Channel e	0C _H	Channel m	1F _H	All channels
05 _H	Channel f	0D _H	Channel n	X _H	Ignored (all channels selected)
06 _H	Channel g	0E _H	Channel o		
07 _H	Channel h	0F _H	Channel p	All others	Reserved



Test Instructions

Test instructions supported by this HBM3 DRAM are listed in Table 76 and subsequently described in detail.

Table 76: IEEE1500 Port Test Instructions

WIR [12:8] ³	WIR [7:0]	Instruction ^{1,2}	Description	Reg. Type	WDR Length
X _H	00 _H	BYPASS	Bypass	R/W	1
1F _H , 0F _H - 00 _H	01 _H	EXTEST_RX	Microbump boundary scan RX test (open/short)	R	122
1F _H , 0F _H - 00 _H	02 _H	EXTEST_TX	Microbump boundary scan TX test (open/short)	W	122
	03 _H	(Reserved)			
	04 _H	(Reserved)			
X _H	05 _H	HBM_RESET	Functional reset excluding Wrapper Data Registers (WDRs) and any IEEE1500 test port logic or I/Os	W	1
1F _H , 0F _H - 00 _H	06 _H	MBIST	HBM3 DRAM resident memory BIST engine test	R/W	4
0F _H - 00 _H	07 _H	SOFT_REPAIR	Soft repair of a failing memory array physical row	W	26
0F _H - 00 _H	08 _H	HARD_REPAIR	Hard repair of a failing memory array physical row	W	26
1F _H , 0F _H - 00 _H	09 _H	DWORD_MISR	Read back for DWORD MISR and write of a seed value	R/W	320
1F _H , 0F _H - 00 _H	0A _H	AWORD_MISR	Read back for AWORD MISR	R	38
1F _H , 0F _H - 00 _H	0B _H	CHANNEL_ID	All TX I/Os go HIGH (except I/Os in MIDSTACK region)	W	1
	0C _H	(Reserved)			
1F _H , 0F _H - 00 _H	0D _H	AWORD_MISR_CONFIG	Allows IEEE1500 test port access to configure the AWORD MISR test feature	W	8
1F _H , 0F _H - 00 _H	0E _H	DEVICE_ID	Returns the HBM3 DRAM's unique identification code	R	160
X _H	0F _H	TEMPERATURE	Returns an 9-bit binary temperature code	R	9
1F _H , 0F _H - 00 _H	10 _H	MODE_REGISTER_DUMP_SET	Returns and set the HBM3 DRAM's Mode Register values	R/W	128
1F _H , 0F _H - 00 _H	11 _H	READ_LFSR_COMPARE_STICKY	Reads the sticky bit error for LFSR Compare feature	R	99
0F _H - 00 _H	12 _H	SOFT_LANE_REPAIR	Soft Lane Remapping	R/W	40
0F _H - 00 _H	13 _H	HARD_LANE_REPAIR	Hard Lane Remapping	R/W	40
1F _H , 0F _H - 00 _H	14 _H	CHANNEL_DISABLE	Disables a single channel or all channels	W	1
1F _H , 0F _H - 00 _H	15 _H	CHANNEL_TEMPERATURE	Returns an 9-bit binary maximum channel temperature code per SID	R	36
X _H	16 _H	WOSC_RUN	WDQS interval oscillator start/stop	W	1
X _H	17 _H	WOSC_COUNT	WDQS interval oscillator count	R	25
0F _H - 00 _H	18 _H	ECS_ERROR_LOG	Error Check and Scrub (ECS) Error Log Info	R	216
0F _H - 00 _H	19 _H	HS_REP_CAP	Returns whether banks have repair resources or not	R	256



WIR [12:8] ³	WIR [7:0]	Instruction ^{1,2}	Description	Reg. Type	WDR Length
1C _H , 1D _H	1A _H	SELF_REP	Self repair	R/W	9
1C _H , 1D _H	1B _H	SELF_REP_RESULTS	Self repair results	R	8
	1C _H to 3F _H	(Reserved)			
	40 _H to 8F _H	(Reserved for vendor test purposes)			
1F _H	90 _H	MBIST_SUMMARY	Read MBIST result summary	R	6
1F _H	91 _H	MBIST_REPAIR_VECTOR	Read MBIST repair vector	R	208
	92 _H to FF _H	(Reserved for vendor test purposes)			

- Notes:
- Reserved instruction codes 03_H, 04_H, 0C_H and 1C_H to 3F_H and will default to the BYPASS instruction when the WIR is loaded with the reserved encoding. Instruction codes 40_H to 8F_H and 92_H to FF_H are reserved for DRAM vendor purposes and should not be used.
 - Channels that are not selected by WIR[12:8] do not respond to the instruction and ignore any Update, Capture and Shift events.
 - WDRs shift out the least significant bit on the WSO port at the first WRCK of the shift sequence. WSO output timing and valid data window are defined in Table 104.
 - Global test instructions (WIR[12:8] = Xh) drive the same data on the WSO outputs of all active channels during ShiftWR events. Inactive channels (channels that are marked as “not present / not working” in the DEVICE_ID WDR and channels that have been disabled using the CHANNEL_DISABLE instruction) drive a static LOW on their WSO outputs instead.

BYPASS

The BYPASS instruction places a single bit WDR between WSI and each channel's WSO. Data is shifted from WSI to WSO through the one-bit WDR by WRCK.

BYPASS is the default instruction after asserting WRST_n to LOW.

Wrapper Data Register: when BYPASS is the current instruction, the 1-bit shift register as shown in Table 77 is connected between WSI and WSO[a:p], and the WSO outputs of all active channels drive the same data during ShiftWR events.

CaptureWR: when BYPASS is the current instruction, the CaptureWR event will have no effect.

UpdateWR: when BYPASS is the current instruction, the UpdateWR event will have no effect.

Table 77: BYPASS Wrapper Data Register

Bits	Field	Type	Description
0	BYPASS	–	Single bit bypass shift register per IEEE1500 standard



EXTEST_RX and EXTEST_TX

EXTEST_RX and EXTEST_TX are both intended for DC I/O connectivity testing similar to board level boundary scan. The receive notation in EXTEST_RX designates that the HBM3 I/O will sample the logic value and capture into the data register the value that is present at the micro bump interface. The transmit notation in EXTEST_TX designates that the HBM3 I/O will drive the logic value shifted into the data register at the micro bump interface. All HBM3 bidirectional I/O, inputs and outputs support both instructions. Differential inputs and outputs (CK_t/CK_c, WDQS_t/WDQS_c and RDQS_t/RDQS_c) also support both instructions on both the true and complement signals. Note that CK_t/CK_c, WDQS_t/WDQS_c, and RDQS_t/RDQS_c require a differential input signal (L/H or H/L) in EXTEST_RX.

While EXTEST_RX is the current instruction, all functional signals of the selected channel(s) enter a High-Z state, including the output-only signals AERR, DERR, RDQS_t/RDQS_c, TEMP[1:0] and CATTRIP. See also the Boundary Scan section.

I/O signals power up in input mode by default. The host will put all AWORD and DWORD drivers into High-Z state prior to loading the EXTEST_TX instruction into the WIR. As soon as EXTEST_TX becomes the current instruction, all AWORD and DWORD signals will change to output mode and remain in output mode until reset of the test logic or until a different instruction is updated on the channel.

A channel disabled either via the corresponding CHANNEL_AVAILABLE bit in the DEVICE ID WDR or via the CHANNEL_DISABLE instruction will not respond to the EXTEST_TX or EXTEST_RX instructions.

Wrapper Data Register: when EXTEST_RX or EXTEST_TX is the current instruction, the Wrapper Boundary Register (WBR) as shown in Table 78 is connected between WSI and WSO.

CaptureWR: when EXTEST_RX is the current instruction, the CaptureWR event will capture the input values into the shift stage of the WBR. The captured data is shifted out on WSO during a subsequent ShiftWR event. Inputs must be stable for the setup and hold times ^tSEXT and ^tHEXT.

When EXTEST_TX is the current instruction, the CaptureWR event will have no effect.

ShiftWR: the Wrapper Boundary Register (WBR) does not provide an update stage. When EXTEST_TX is the current instruction, the value driven on the outputs is directly derived from the WBR's shift stage and will update with each ShiftWR event. The new data will be stable after ^tOEXT time.

UpdateWR: when EXTEST_RX or EXTEST_TX is the current instruction, the UpdateWR event will have no effect.

Table 78: Wrapper Boundary Register (WBR)

Bits	Field	Type	Description
121	RESET_n TEMP0 0	I I/O I	Signals in MIDSTACK region channel b only channel n only (see note 1) channels a, c to m, o and p (reserved bit)
120	TEMP1 CATTRIP 0	I/O I/O I	Signals in MIDSTACK region channel n only (see note 1) channel o only (see note 1) channels a to m and p (reserved bit)



Bits	Field	Type	Description
119	DBI7	I/O	DWORD1 (PC1)
118	DQ63	I/O	
117	DQ62	I/O	
116	DQ61	I/O	
115	DQ60	I/O	
114	RD3	I/O	
113	DERR1	I/O	
112	DQ59	I/O	
111	DQ58	I/O	
110	DQ57	I/O	
109	DQ56	I/O	
108	SEV3	I/O	
107	DBI6	I/O	
106	DQ55	I/O	
105	DQ54	I/O	
104	DQ53	I/O	
103	DQ52	I/O	
102	RDQS1_c	I/O	
101	RDQS1_t	I/O	
100	DQ51	I/O	
99	DQ50	I/O	
98	DQ49	I/O	
97	DQ48	I/O	
96	SEV2	I/O	
95	DBI5	I/O	
94	DQ47	I/O	
93	DQ46	I/O	
92	DQ45	I/O	
91	DQ44	I/O	
90	WDQS1_c	I/O	
89	WDQS1_t	I/O	
88	DQ43	I/O	
87	DQ42	I/O	
86	DQ41	I/O	
85	DQ40	I/O	



Bits	Field	Type	Description
84	ECC3	I/O	DWORD1 (PC1) (cont'd)
83	DBI4	I/O	
82	DQ39	I/O	
81	DQ38	I/O	
80	DQ37	I/O	
79	DQ36	I/O	
78	RD2	I/O	
77	DPAR1	I/O	
76	DQ35	I/O	
75	DQ34	I/O	
74	DQ33	I/O	
73	DQ32	I/O	
72	ECC2	I/O	
71	AERR	I/O	AWORD
70	R9	I/O	
69	R8	I/O	
68	R7	I/O	
67	R6	I/O	
66	CK_c	I/O	
65	R5	I/O	
64	R4	I/O	
63	R0	I/O	
62	R3	I/O	
61	R2	I/O	
60	R1	I/O	
59	RA	I/O	
58	ARFU	I/O	
57	APAR	I/O	
56	C7	I/O	
55	C6	I/O	
54	C5	I/O	
53	CK_t	I/O	
52	C4	I/O	
51	C3	I/O	
50	C2	I/O	



Bits	Field	Type	Description
49	C1	I/O	AWORD (cont'd)
48	C0	I/O	
47	DBI3	I/O	DWORD0 (PC0)
46	DQ31	I/O	
45	DQ30	I/O	
44	DQ29	I/O	
43	DQ28	I/O	
42	RD1	I/O	
41	DERR0	I/O	
40	DQ27	I/O	
39	DQ26	I/O	
38	DQ25	I/O	
37	DQ24	I/O	
36	SEV1	I/O	
35	DBI2	I/O	
34	DQ23	I/O	
33	DQ22	I/O	
32	DQ21	I/O	
31	DQ20	I/O	
30	RDQS0_c	I/O	
29	RDQS0_t	I/O	
28	DQ19	I/O	
27	DQ18	I/O	
26	DQ17	I/O	
25	DQ16	I/O	
24	SEV0	I/O	
23	DBI1	I/O	
22	DQ15	I/O	
21	DQ14	I/O	
20	DQ13	I/O	
19	DQ12	I/O	
18	WDQS0_c	I/O	
17	WDQS0_t	I/O	
16	DQ11	I/O	
15	DQ10	I/O	



Bits	Field	Type	Description
14	DQ9	I/O	DWORD0 (PC0) (cont'd)
13	DQ8	I/O	
12	ECC1	I/O	
11	DBI0	I/O	
10	DQ7	I/O	
9	DQ6	I/O	
8	DQ5	I/O	
7	DQ4	I/O	
6	RD0	I/O	
5	DPAR0	I/O	
4	DQ3	I/O	
3	DQ2	I/O	
2	DQ1	I/O	
1	DQ0	I/O	
0	ECC0	I/O	

Note: 1. The EXTEST_RX instruction captures the internally generated output state (internal loopback).

HBM_RESET

The HBM_RESET instruction initiates an asynchronous functional reset of the HBM3 DRAM, equivalent to asserting RESET_n to LOW.

The HBM_RESET condition is not self-clearing. Instead, the reset state must explicitly be set and cleared. To accomplish an HBM3 reset, the HBM_RESET bit must be held as 1_B for a minimum duration of 'RES which equals 'PW_RESET (see Initialization Sequence with Stable Power Sequence with Stable Power).

It is pointed out that the Wrapper Serial Port (WSP) itself including the associated control logic and WDRs is not reset by the HBM_RESET instruction. The DA port signals are also not affected by the HBM_RESET instruction.

Wrapper Data Register: when HBM_RESET is the current instruction, the data register as shown in Table 79 is connected between WSI and WSO[a:p], and the WSO outputs of all active channels drive the same data during ShiftWR events.

CaptureWR: when HBM_RESET is the current instruction, the CaptureWR event will have no effect.

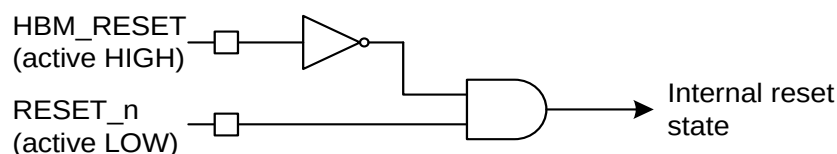
UpdateWR: when HBM_RESET is the current instruction, the UpdateWR event will load the value from the shift stage into the update stage and initiate or clear the functional reset.

Table 79: HBM_RESET Wrapper Data Register

Bits	Field	Type	Description
0	HBM_RESET	W	0 _B - Clear the functional reset 1 _B - Initiate the functional reset

Internally, the RESET_n signal and the HBM_RESET instruction are logically combined such that when either is true then the internal reset state is true. During power-up it is required that WRST_n be driven LOW, thus ensuring that the uninitialized IEEE1500 test port logic does not interfere with the power-up initialization sequence.

Figure 88: RESET_n and HBM_RESET Logic



After the power-up initialization, the RESET_n input is HIGH, and subsequent stable power resets may be asserted by either driving the RESET_n input LOW, or by using the HBM_RESET instruction. Note that the HBM_RESET instruction does not bring the HBM3 DRAM out of reset while the external RESET_n input is driven LOW. Similarly, the HBM3 DRAM cannot be brought out of reset using the RESET_n input while reset is asserted using the HBM_RESET instruction.

Table 80: RESET_n and HBM_RESET Truth Table

RESET_n	HBM_RESET WDR	Internal Reset State
LOW	0 _B	Reset asserted by RESET_n
LOW	1 _B	Reset asserted by both RESET_n and HBM_RESET
HIGH	0 _B	Out of reset state
HIGH	1 _B	Reset asserted by HBM_RESET

MBIST

The MBIST instruction controls the device hosted memory built in self-test along with the MBIST_SUMMARY and MBIST_REPAIR_VECTOR instructions.

Either a stable CK clock on channels **g** and **h** (CKg_t/c, CKh_t/c), or a stable WRCK clock are required as reference clock source while MBIST is active.

Wrapper Data Register: when MBIST is the current instruction, the data register as shown in Table 81 will be connected between WSI and WSO.

CaptureWR: when MBIST is the current instruction, the CaptureWR event will capture the read bit fields into the shift stage of the WDR.

UpdateWR: when MBIST is the current instruction, the UpdateWR event will simultaneously load all write bit fields from the shift stage to the update stage of the WDR.


Table 81: MBIST Wrapper Data Register

Bits	Field	Type	Description
3	Reserved	W	0 _B
2	MBIST_STOP	W	0 _B – No action 1 _B – Terminate any ongoing MBIST or repair check operation
1	REPAIR_CHK	W	0 _B – No action 1 _B – Run the repair capability check. This bit is self-clearing.
0	MBIST_RUN	W	0 _B – No action 1 _B – Run MBIST. This bit is self-clearing.
[3:2]	MBIST_ERROR_REPORT[1:0]	R	00 _B – No fail (pass) 01 _B – 1 to 4 fail(s) found 10 _B – More than 4 fails found (ECM overflow) 11 _B – Massive fails found, unrepairable
[1:0]	MBIST_STATUS[1:0]	R	00 _B – MBIST off or has not run since initialization 01 _B – MBIST in operation 10 _B – MBIST run completed 11 _B – Calculation of remaining repair resources ongoing

MBIST_SUMMARY

The MBIST_SUMMARY instruction summarizes the result of a preceding MBIST instruction. The result is provided per channel, therefore all WSO outputs must be observed to determine the test result across all channels of the device.

Wrapper Data Register: when MBIST_SUMMARY is the current instruction, the data register as shown in Table 82 will be connected between WSI and WSO.

CaptureWR: when MBIST_SUMMARY is the current instruction, the CaptureWR event will load the MBIST summary into the shift stage of the WDR.

UpdateWR: when MBIST_SUMMARY is the current instruction, the UpdateWR event will have no effect.


Table 82: MBIST_SUMMARY Wrapper Data Register

Bits	Field	Type	Description
[5:3]	DWORD1_SUMMARY [2:0]	R	Error summary for DWORD1 000 _B – 0 fails in ECM (pass) 001 _B – 1 fail in ECM 010 _B – 2 fails in ECM 011 _B – 3 fails in ECM 100 _B – 4 fails in ECM 101 _B – 5 or more fails in ECM (ECM overflow) 110 _B – Massive fails, unrepairable 111 _B – Reserved
[2:0]	DWORD0_SUMMARY [2:0]	R	Error summary for DWORD0 000 _B – 0 fails in ECM (pass) 001 _B – 1 fail in ECM 010 _B – 2 fails in ECM 011 _B – 3 fails in ECM 100 _B – 4 fails in ECM 101 _B – 5 or more fails in ECM (ECM overflow) 110 _B – Massive fails, unrepairable 111 _B – Reserved

MBIST_REPAIR_VECTOR

The MBIST_REPAIR_VECTOR instruction provides additional repair information as the result of a preceding MBIST instruction. The address order in the 26-bit wide fields is the same as defined for the REPAIR WDR (see Table 85).

Wrapper Data Register: when MBIST_REPAIR_VECTOR is the current instruction, the data register as shown in Table 83 will be connected between WSI and WSO.

CaptureWR: when MBIST_REPAIR_VECTOR is the current instruction, the CaptureWR event will load the MBIST repair vector information into the shift stage of the WDR.

UpdateWR: when MBIST_REPAIR_VECTOR is the current instruction, the UpdateWR event will have no effect.

Table 83: MBIST_REPAIR_VECTOR Wrapper Data Register

Bits	Field	Type	Description
[207:182]	DWORD1_FAIL_4[25:0]	R	Same bit order as DWORD1_FAIL_1
[181:156]	DWORD1_FAIL_3[25:0]	R	Same bit order as DWORD1_FAIL_1
[155:130]	DWORD1_FAIL_2[25:0]	R	Same bit order as DWORD1_FAIL_1
129	DWORD1_FAIL_1[25:0]	R	0 _B (fixed)
128		R	0 _B – Repairable 1 _B – Not repairable
127		R	1 _B – DWORD1 (PC1)
[126:125]		R	SID[1:0]
[124:121]		R	BA[3:0]
[120:105]		R	RA[15:0]



Bits	Field	Type	Description
104	DWORD1_FAIL_1[25:0] (cont'd)	R	0 _B – Fail status is invalid 1 _B – Fail status is valid
[103:78]	DWORD0_FAIL_4[25:0]	R	Same bit order as DWORD0_FAIL_1
[77:52]	DWORD0_FAIL_3[25:0]	R	Same bit order as DWORD0_FAIL_1
[51:26]	DWORD0_FAIL_2[25:0]	R	Same bit order as DWORD0_FAIL_1
25	DWORD0_FAIL_1[25:0]	R	0 _B (fixed)
24		R	0 _B – Repairable 1 _B – Not repairable
23		R	0 _B – DWORD0 (PC0)
[22:21]		R	SID[1:0]
[20:17]		R	BA[3:0]
[16:1]		R	RA[15:0]
0		R	0 _B – Fail status is invalid 1 _B – Fail status is valid

SOFT_REPAIR

The SOFT_REPAIR instruction allows a temporary row repair without using a permanent fusing mechanism. It is intended to enable validation that the intended repair works as expected. If power is removed from the device or a chip reset occurs with RESET_n pulled LOW, the soft repair will revert to the un-repaired state.

A channel must be in all bank idle state as long as the SOFT_REPAIR instruction is loaded in the WIR.

Wrapper Data Register: when SOFT_REPAIR is the current instruction, the data register as shown in Table 84 will be connected between WSI and WSO.

CaptureWR: when SOFT_REPAIR is the current instruction, the CaptureWR event will have no effect.

UpdateWR: when SOFT_REPAIR is the current instruction, the UpdateWR event will simultaneously load all bit fields from the shift stage into the update stage. Completion of the update event will initiate the soft repair sequence which will be completed after 'SREP time. A stable WRCK clock is required during 'SREP time.

Table 84: SOFT_REPAIR Wrapper Data Register

Bits	Field	Type	Description
25	SOFT_REPAIR_LOCK ¹	W	0 _B – Soft repair is open 1 _B – Soft repair is locked
24	SOFT_REPAIR_UNDO ¹	W	0 _B – Do a soft repair 1 _B – Undo a previous soft repair
23	SOFT_PC	W	0 _B – DWORD 0 (PC0) 1 _B – DWORD 1 (PC1)
[22:21]	SOFT_SID[1:0]	W	SID[1:0]



Bits	Field	Type	Description
[20:17]	SOFT_BK[3:0]	W	BA[3:0]
[16:1]	SOFT_ROW[15] SOFT_ROW[14:0]	W	0 _B RA[14:0]
0	SOFT_REPAIR_START	W	0 _B – Repair vector is invalid (ignore command) 1 _B – Repair vector is valid

Note: 1. The SOFT_REPAIR_LOCK and SOFT_REPAIR_UNDO bits are ignored when bit MR10 OP0 is set to 0_B.

HARD_REPAIR

The HARD_REPAIR instruction permanently repairs failing bit cells in the HBM3 DRAM by blowing respective fuses. The repair will be initiated on update of the data register. Completion of HARD_REPAIR requires a subsequent chip reset with RESET_n pulled LOW as described in the Interaction with Mission Mode Operation section.

All channels of the HBM3 DRAM must be in all bank idle state as long as the HARD_REPAIR instruction is loaded in the WIR.

Wrapper Data Register: when HARD_REPAIR is the current instruction, the data register as shown in Table 85 will be connected between WSI and WSO.

CaptureWR: when HARD_REPAIR is the current instruction, the CaptureWR event will have no effect.

UpdateWR: when HARD_REPAIR is the current instruction, the UpdateWR event will load the content from the shift stage into the update stage and initiate the hard repair sequence. The hard repair is completed after a waiting time of ^tHREP. A stable WRCK clock is required during ^tHREP time.

Table 85: HARD_REPAIR Wrapper Data Register

Bits	Field	Type	Description
[25:24]	RFU	W	00 _B (ignored)
23	HARD_PC	W	0 _B – DWORD 0 (PC0) 1 _B – DWORD 1 (PC1)
[22:21]	HARD_SID[1:0]	W	SID[1:0]
[20:17]	HARD_BK[3:0]	W	BA[3:0]
[16:1]	HARD_ROW[15] HARD_ROW[14:0]	W	0 _B RA[14:0]
0	HARD_REPAIR_START	W	0 _B – Repair vector is invalid (ignore command) 1 _B – Repair vector is valid



DWORD_MISR

This instruction captures and shifts out the DWORD MISR value on the WSO output. The instruction may also be used to preload data for use in LFSR mode. The DWORD MISR is associated with the DWORD I/O test feature.

Wrapper Data Register: when DWORD_MISR is the current instruction, the data register as shown in Table 86 will be connected between WSI and WSO. The notation is "..._Q0" to "..._Q3" for the 4 UI per CK clock cycle latched by WDQS in MISR mode or driven along with RDQS in LSFR mode.

Note that the MISR content is not specified after shifting out the MISR content. The host should reinitialize the MISR before continuing with additional testing, e.g. by using the MISR Preset function in Mode Register 7 (MR7). See HBM3 Loopback Test Modes for MISR mode features and usage.

CaptureWR: when DWORD_MISR is the current instruction, the CaptureWR event will load the respective MISR values into the shift stage of the WDR. A minimum waiting time of 'SMISR between the last data capture into the DWORD MISR and this CaptureWR event must be observed.

UpdateWR: when DWORD_MISR is the current instruction, the UpdateWR event will load the bits from the shift stage of the WDR into the DWORD MISR.

Table 86: DWORD_MISR Wrapper Data Register

Bits	Field	Type	Description
[319:160]	DWORD1	R/W	DWORD1: DQ[63:32], DBI[7:4], ECC[3:2] and SEV[3:2] (same bit ordering as DWORD0)
[159:120]	DWORD0_BYTE3	R/W	Byte 3 of DWORD0: DQ[31:24], DBI3 and SEV1 (same ordering as Byte 0)
[119:80]	DWORD0_BYTE2	R/W	Byte 2 of DWORD0: DQ[23:16], DBI2 and SEV0 (same ordering as Byte 0)
[79:40]	DWORD0_BYTE1	R/W	Byte 1 of DWORD0: DQ[15:8], DBI1 and ECC1 (same ordering as Byte 0)
39	DWORD0_DBI0_Q0	R/W	Byte 0 of DWORD0
38	DWORD0_DBI0_Q1	R/W	
37	DWORD0_DBI0_Q2	R/W	
36	DWORD0_DBI0_Q3	R/W	
35	DWORD0_DQ7_Q0	R/W	
34	DWORD0_DQ7_Q1	R/W	
33	DWORD0_DQ7_Q2	R/W	
32	DWORD0_DQ7_Q3	R/W	
31	DWORD0_DQ6_Q0	R/W	
30	DWORD0_DQ6_Q1	R/W	
29	DWORD0_DQ6_Q2	R/W	
28	DWORD0_DQ6_Q3	R/W	



Bits	Field	Type	Description
27	DWORD0_DQ5_Q0	R/W	Byte 0 of DWORD0 (cont'd)
26	DWORD0_DQ5_Q1	R/W	
25	DWORD0_DQ5_Q2	R/W	
24	DWORD0_DQ5_Q3	R/W	
23	DWORD0_DQ4_Q0	R/W	
22	DWORD0_DQ4_Q1	R/W	
21	DWORD0_DQ4_Q2	R/W	
20	DWORD0_DQ4_Q3	R/W	
19	DWORD0_DQ3_Q0	R/W	
18	DWORD0_DQ3_Q1	R/W	
17	DWORD0_DQ3_Q2	R/W	
16	DWORD0_DQ3_Q3	R/W	
15	DWORD0_DQ2_Q0	R/W	
14	DWORD0_DQ2_Q1	R/W	
13	DWORD0_DQ2_Q2	R/W	
12	DWORD0_DQ2_Q3	R/W	
11	DWORD0_DQ1_Q0	R/W	
10	DWORD0_DQ1_Q1	R/W	
9	DWORD0_DQ1_Q2	R/W	
8	DWORD0_DQ1_Q3	R/W	
7	DWORD0_DQ0_Q0	R/W	
6	DWORD0_DQ0_Q1	R/W	
5	DWORD0_DQ0_Q2	R/W	
4	DWORD0_DQ0_Q3	R/W	
3	DWORD0_ECC0_Q0	R/W	
2	DWORD0_ECC0_Q1	R/W	
1	DWORD0_ECC0_Q2	R/W	
0	DWORD0_ECC0_Q3	R/W	



AWORD_MISR

This instruction captures and shifts out the AWORD MISR value on the WSO output. The MISR in this instruction is associated with the AWORD loopback test feature.

Wrapper Data Register: when AWORD_MISR is the current instruction, the data register as shown in Table 87 will be connected between WSI and WSO. The notation is "..._R" for bits latched on the rising CK clock edge and "..._F" for bits latched on the falling CK clock edge.

Note that the content of the MISR is not specified after shifting out the MISR content. The host should reinitialize the MISR using the AWORD_MISR_CONFIG instruction before continuing with additional testing. See Loopback Test Modes for MISR mode features and usage.

CaptureWR: when AWORD_MISR is the current instruction, the CaptureWR event will load the respective MISR values into the shift stage of the WDR. A minimum waiting time of 'SMISR between the last data capture into the AWORD MISR and this CaptureWR event must be observed.

UpdateWR: when AWORD_MISR is the current instruction, the UpdateWR event will have no effect.

Table 87: AWORD_MISR Wrapper Data Register

Bits	Field	Type	Description
37	R1_R	R	AWORD
36	R1_F	R	
35	R2_R	R	
34	R2_F	R	
33	R3_R	R	
32	R3_F	R	
31	R0_R	R	
30	R0_F	R	
29	R4_R	R	
28	R4_F	R	
27	R5_R	R	
26	R5_F	R	
25	R6_R	R	
24	R6_F	R	
23	R7_R	R	
22	R7_F	R	
21	R8_R	R	
20	R8_F	R	
19	R9_R	R	
18	R9_F	R	
17	ARFU_R	R	



Bits	Field	Type	Description
16	ARFU_F	R	AWORD (cont'd)
15	C7_R	R	
14	C7_F	R	
13	C6_R	R	
12	C6_F	R	
11	C5_R	R	
10	C5_F	R	
9	C4_R	R	
8	C4_F	R	
7	C3_R	R	
6	C3_F	R	
5	C2_R	R	
4	C2_F	R	
3	C1_R	R	
2	C1_F	R	
1	C0_R	R	
0	C0_F	R	

CHANNEL_ID

This instruction enables the HBM channel identification by driving all bidirectional DWORD I/Os to HIGH unless a channel is disabled either via the corresponding CHANNEL_AVAILABLE bit in the DEVICE ID WDR or via the CHANNEL_DISABLE instruction. In these cases a channel will not respond to the CHANNEL_ID instruction.

Wrapper Data Register: when CHANNEL_ID is the current instruction, the data register as shown in Table 88 is connected between WSI and WSO.

CaptureWR: when CHANNEL_ID is the current instruction, the CaptureWR event will have no effect.

UpdateWR: when CHANNEL_ID is the current instruction, the UpdateWR event will load the enable bit from the shift stage into the update stage of the WDR. All bidirectional DWORD I/Os (DQ, DBI, RD, ECC/SEV and DPAR) will drive a HIGH latest after 'OVCHN when the enable bit is 1_B, and return to their default state latest after 'OZCHN when the enable bit is 0_B or a different instruction has been loaded in the WIR. Output pins RDQS_t/c and DERR maintain their default states regardless of the CHANNEL_ID instruction. DBI, ECC/SEV and DPAR will drive a HIGH even if the respective function is disabled in the mode register.

Table 88: CHANNEL_ID Wrapper Data Register

Bits	Field	Type	Description
0	ENABLE	W	0 _B –TX return to their default state 1 _B –TX drive a HIGH



AWORD_MISR_CONFIG

This instruction configures the AWORD MISR for subsequent tests. See Loopback Test Modes for MISR mode features and usage.

Wrapper Data Register: when AWORD_MISR_CONFIG is the current instruction, the data register as shown in Table 89 will be connected between WSI and WSO.

CaptureWR: when AWORD_MISR_CONFIG is the current instruction, the CaptureWR event will have no effect.

UpdateWR: when AWORD_MISR_CONFIG is the current instruction, the UpdateWR event will simultaneously load the configuration bits from the shift stage into the update stage of the WDR and configure the AWORD MISR into the desired mode.

Table 89: AWORD MISR Configuration Wrapper Data Register

Bits	Field	Type	Description
[7:3]	RFU	W	Reserved
2	ENABLE	W	0 _B – Off 1 _B – On
[1:0]	MODE[1:0]	W	00 _B – Preset: a) the AWORD MISR is preset to 0x2AAAAAAAAAh; b) the AWORD LFSR_COMPARE_STICKY bits are all cleared to 0; c) the AWORD preamble clock filter circuit is enabled. 01 _B – LFSR Compare mode 10 _B – Register mode: AWORD transfers are captured directly to the MISR. Note that register mode cannot be used to set an alternate seed value (see Test Method for AWORD (Write) Register Mode) 11 _B – MISR mode

DEVICE_ID

This instruction allows to shift out of the DEVICE_ID that provides various information about the HBM3 DRAM including a device specific unique serial number.

Wrapper Data Register: when DEVICE_ID is the current instruction, the data register as shown in Table 90 will be connected between WSI and WSO.

CaptureWR: when DEVICE_ID is the current instruction, the CaptureWR event will load the respective identification field values into the shift stage of the WDR.

UpdateWR: when DEVICE_ID is the current instruction, the UpdateWR event will have no effect.


Table 90: DEVICE_ID Wrapper Data Register

Bits	Field	Type	Description
[159:155]	OPT_FEATURES[4:0]	R	00000 _B (fixed)
154	SHARED_REP_RES	R	Sharing of HS_REP_CAP with self repair: 1 _B – Self repair resources are shared with hard/soft resources (fixed)
153	PPR_RSVD	R	0 _B (fixed)
[152:138]	PPR_RA[14:0]	R	Row addresses associated with a single soft or hard repair 1007 _H – Row addresses RA[12,2:0] are ignored (fixed)
137	RAADEC_C	R	RAA Counter Decrement per REF command for RFM level C (MR8 OP[5:4] = 11 _B). 1 _B – $0.5 \times \text{RAAIMT}$
[136:135]	RAAMMT_C[1:0]	R	RAA Maximum Management Threshold (RAAMMT) for RFM level C (MR8 OP[5:4] = 11 _B). 00 _B – $3 \times \text{RAAIMT}$
[134:132]	RAAIMT_C[2:0]	R	RAA Initial Management Threshold (RAAIMT) for RFM level C (MR8 OP[5:4] = 11 _B) 000 _B – 32
131	RAADEC_B	R	RAA Counter Decrement per REF command for RFM level B (MR8 OP[5:4] = 10 _B). 1 _B – $0.5 \times \text{RAAIMT}$
[130:129]	RAAMMT_B[1:0]	R	RAA Maximum Management Threshold (RAAMMT) for RFM level B (MR8 OP[5:4] = 10 _B). 00 _B – $3 \times \text{RAAIMT}$
[128:126]	RAAIMT_B[2:0]	R	RAA Initial Management Threshold (RAAIMT) for RFM level B (MR8 OP[5:4] = 10 _B). 001 _B – 40
125	RAADEC_A	R	RAA Counter Decrement per REF command for RFM level A (MR8 OP[5:4] = 01 _B). 1 _B – $0.5 \times \text{RAAIMT}$
[124:123]	RAAMMT_A[1:0]	R	RAA Maximum Management Threshold (RAAMMT) for RFM level A (MR8 OP[5:4] = 01 _B). 00 _B – $3 \times \text{RAAIMT}$
[122:120]	RAAIMT_A[2:0]	R	RAA Initial Management Threshold (RAAIMT) for RFM level A (MR8 OP[5:4] = 01 _B). 010 _B – 48
119	RAADEC	R	Default RAA Counter Decrement per REF command The field shall be ignored when the RFM bit is 0 _B . 1 _B – $0.5 \times \text{RAAIMT}$
[118:117]	RAAMMT[1:0]	R	Default RAA Maximum Management Threshold (RAAMMT) The field shall be ignored when the RFM bit is 0 _B . 00 _B – $3 \times \text{RAAIMT}$
[116:114]	RAAIMT[2:0]	R	Default RAA Initial Management Threshold (RAAIMT) The field shall be ignored when the RFM bit is 0 _B . 011 _B – 56



Bits	Field	Type	Description
113	ARFM	R	Adaptive Refresh Management (ARFM) 1 _B – ARFM is supported
112	RFM	R	Refresh Management (RFM) 0 _B – RFM is not required
[111:104]	MANUFACTURING_YEAR[7:0]	R	Binary encoded year 00000000 _B = 2020 ... 00000100 _B = 2024
[103:96]	MANUFACTURING_WEEK[7:0]	R	Binary encoded week 00000001 _B = ww1 ... 00110100 _B = ww52
[95:32]	SERIAL_NO[63:0]	R	64-bit unique device ID
[31:28]	MANUFACTURER_ID [3:0]	R	1111 _B – Micron (fixed)
[27:24]	DENSITY[3:0]	R	Memory density per channel (see Addressing table) 1001 _B –8-High 1010 _B –12-High
[23:8]	CHANNEL_AVAILABLE[15:0] ¹	R	Channel Available 0 _B – Channel not present / not working 1 _B – Channel present / working Channel encoding (1 bit per channel): bit 0: channel a bit 1: channel b ... bit 14: channel o bit 15: channel p Note: For channels marked as “not present / not working”, please refer to the CHANNEL_DISABLE instruction for details of that channel's signal status
[7:0]	MODEL_PART_NUMBER[7:0]	R	TBD

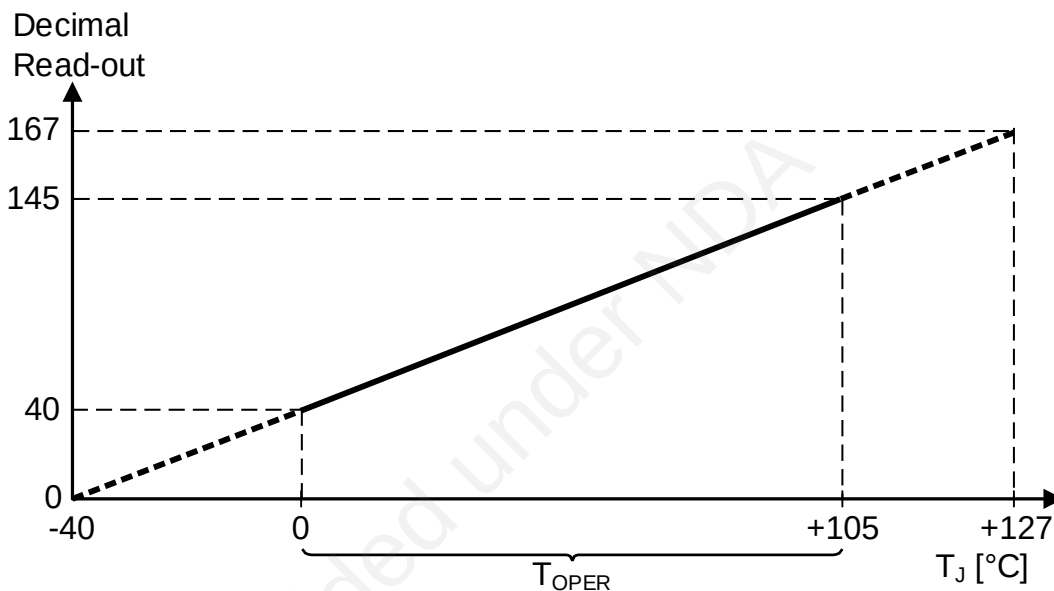
Note: 1. A channel marked as “not present / not working” keeps all AWORD and DWORD input and output buffers permanently disabled and drives that channel's WSO to LOW.



TEMPERATURE

This instruction captures the HBM3 DRAM's junction temperature. Temperature reporting is specified as an 9-bit field: bit 0 indicates the validity of the temperature sensor read-out, and bits [8:1] indicate the temperature in degrees Celsius.

Figure 89: Temperature Read-out



Wrapper Data Register: when TEMPERATURE is the current instruction, the data register as shown in Table 91 is connected between WSI and WSO[a:p], and the WSO outputs of all active channels drive the same data during ShiftWR events.

CaptureWR: when TEMPERATURE is the current instruction, CaptureWR event will load the temperature field values into the shift stage of the WDR.

UpdateWR: when TEMPERATURE is the current instruction, the UpdateWR event will have no effect.

Table 91: TEMPERATURE Wrapper Data Register

Bits	Field	Type	Description
[8:1]	TEMP[7:0]	R	Temperature in °C. Examples: 0000 0000 _B = -40°C 0010 1000 _B = 0°C 0100 0001 _B = 25°C 1000 0111 _B = 95°C 1010 0111 _B = 127°C
0	VALID	R	1 _B – Valid (fixed)

Note: 1. Refer to Table 107 for the supported operating temperature range.



MODE_REGISTER_DUMP_SET

This instruction provides read (dump) and write (set) access to the HBM3 mode registers.

Wrapper Data Register: when MODE_REGISTER_DUMP_SET is the current instruction, the data register as shown in Table 92 will be connected between WSI and WSO.

CaptureWR: when MODE_REGISTER_DUMP_SET is the current instruction, the CaptureWR event will capture the mode register content into the shift stage of the WDR. An 'X' value will be captured for reserved mode registers or mode register fields (see Mode Registers section). A waiting time $t_{MRSS} = t_{MOD}$ from a preceding MRS command must have elapsed before executing the CaptureWR event.

UpdateWR: when MODE_REGISTER_DUMP_SET is the current instruction, the UpdateWR event will simultaneously load the bit fields from the shift stage to the update stage of the WDR. The updated mode register content will be valid for subsequent mission mode operation after t_{UPDMRS} .

Table 92: MODE_REGISTER_DUMP_SET Wrapper Data Register

Bits	Field	Type	Description
[127:120]	MR15	R/W	MR15[7:0]
[119:112]	MR14	R/W	MR14[7:0]
[111:104]	MR13	R/W	MR13[7:0]
[103:96]	MR12	R/W	MR12[7:0]
[95:88]	MR11	R/W	MR11[7:0]
[87:80]	MR10	R/W	MR10[7:0]
[79:72]	MR9	R/W	MR9[7:0]
[71:64]	MR8	R/W	MR8[7:0]
[63:56]	MR7	R/W	MR7[7:0]
[55:48]	MR6	R/W	MR6[7:0]
[47:40]	MR5	R/W	MR5[7:0]
[39:32]	MR4	R/W	MR4[7:0]
[31:24]	MR3	R/W	MR3[7:0]
[23:16]	MR2	R/W	MR2[7:0]
[15:8]	MR1	R/W	MR1[7:0]
[7:0]	MR0	R/W	MR0[7:0]

READ_LFSR_COMPARE_STICKY

This instruction is used to capture the LFSR Compare Sticky error data to be shifted out on the WSO output. The instruction is associated with the AWORD and DWORD I/O loopback test features. Data register bit positions are specified in Table 93. Note that the content of the MISR and LFSR Compare Sticky error data registers is not specified after shifting out the sticky error content. The host should reinitialize the MISR registers (such as with MR7 Preset and AWORD_MISR_CONFIG preset) before continuing with additional testing. See Loopback Test Modes for MISR mode features and usage.



While both the AWORD and DWORD sticky error bits share a common WDR, the bits are set and cleared only by their respective AWORD or DWORD Preset and LFSR Compare operations. For example, an AWORD_MISR_CONFIG preset operation clears the AWORD sticky error bits, and the state of the DWORD sticky error bits is undefined; therefore, the host should ignore the DWORD sticky error bits when operating the AWORD LFSR Compare mode. Conversely, the DWORD_MISR_CONFIG preset operation clears the DWORD sticky error bits, and the state of the AWORD sticky error bits is undefined; therefore, the host should ignore the AWORD sticky error bits when operating the DWORD LFSR Compare mode.

Wrapper Data Register: when READ_LFSR_COMPARE_STICKY is the current instruction, the data register as shown in Table 93 is connected between WSI and WSO.

CaptureWR: when READ_LFSR_COMPARE_STICKY is the current instruction, the CaptureWR event will load the sticky error values into the shift stage of the WDR.

UpdateWR: when READ_LFSR_COMPARE_STICKY is the current instruction, the UpdateWR event will have no effect.

Table 93: READ_LFSR_COMPARE_STICKY Wrapper Data Register

Bits	Field	Type	Description
[98:59]	DWORD1	R	DWORD1: DQ[63:32], DBI[7:4], ECC[3:2] and SEV[3:2] (same ordering as DWORD0)
58	AWORD_R1	R	AWORD
57	AWORD_R2	R	
56	AWORD_R3	R	
55	AWORD_R0	R	
54	AWORD_R4	R	
53	AWORD_R5	R	
52	AWORD_R6	R	
51	AWORD_R7	R	
50	AWORD_R8	R	
49	AWORD_R9	R	
48	AWORD_ARFU	R	
47	AWORD_C7	R	AWORD (cont'd)
46	AWORD_C6	R	
45	AWORD_C5	R	
44	AWORD_C4	R	
43	AWORD_C3	R	
42	AWORD_C2	R	
41	AWORD_C1	R	
40	AWORD_C0	R	
[39:30]	DWORD0_BYTE_3	R	Byte 3 of DWORD0: DQ[31:24], DBI3 and SEV1 (same ordering as Byte 0)



Bits	Field	Type	Description
[29:20]	DWORD0_BYTE_2	R	Byte 2 of DWORD0: DQ[23:16], DBI2 and SEV0 (same ordering as Byte 0)
[19:10]	DWORD0_BYTE_1	R	Byte 1 of DWORD0: DQ[15:8], DBI1 and ECC1 (same ordering as Byte 0)
9	DWORD0_DBI0	R	Byte 0 of DWORD0 (PC0)
8	DWORD0_DQ7	R	
7	DWORD0_DQ6	R	
6	DWORD0_DQ5	R	
5	DWORD0_DQ4	R	
4	DWORD0_DQ3	R	
3	DWORD0_DQ2	R	
2	DWORD0_DQ1	R	
1	DWORD0_DQ0	R	
0	DWORD0_ECC0	R	

SOFT_LANE_REPAIR and HARD_LANE_REPAIR

The SOFT_LANE_REPAIR and HARD_LANE_REPAIR instructions convey lane remapping and repair information. Both instructions share the same LANE_REPAIR WDR. SOFT_LANE_REPAIR and HARD_LANE_REPAIR instructions can only be issued as part of the device initialization and before normal memory operation has commenced, e.g. before the CK clock has started to toggle.

Wrapper Data Register: the LANE_REPAIR wrapper data register as shown in Table 94 will be connected between WSI and WSO.

Figure 90 illustrates the interaction between SOFT_LANE_REPAIR and HARD_LANE_REPAIR instructions and the associated registers. It is pointed out that the actual I/O lane remapping is derived from the content of the lane repair shadow register.

CaptureWR: when either SOFT_LANE_REPAIR or HARD_LANE_REPAIR is the current instruction, the CaptureWR event will load the lane remapping data from the lane repair shadow register into the shift stage of the WDR. This internal lane repair shadow register is pre-loaded with the repair data from a preceding HARD_LANE_REPAIR operation upon HBM3 DRAM initialization (RESET_n pulled LOW). The memory controller may use these data to configure the lane repair accordingly at the host; it may also use these data as a seed value for subsequent lane repair operations.

UpdateWR (SOFT_LANE_REPAIR): when SOFT_LANE_REPAIR is the current instruction, the UpdateWR event will load the lane remapping data from the shift stage of the WDR into the lane repair shadow register and force the I/O lanes to be remapped accordingly. This remapping is non-persistent; it will be lost when RESET_n is pulled LOW or the device loses power. Pulling WRST_n LOW does not reset the lane repair shadow register.

UpdateWR (HARD_LANE_REPAIR): when HARD_LANE_REPAIR is the current instruction, the UpdateWR event will load the lane remapping data from the shift stage of the WDR into the hard lane repair register. The controller must wait 'HLREP to allow the device to complete this operation and permanently store the repair vector.



Only a single broken lane can be repaired at a time, in order to limit the current constraint of the associated circuits. If multiple lanes are to be repaired, it is required to shift in the repair vectors for each broken lane sequentially, with all other lane repair setting = F_H, and initiate each actual lane repair with a separate UpdateWR event.

The UpdateWR event itself does not lead to an actual re-mapping of the I/O lanes. For such re-mapping to get effective it is required to initiate a chip reset by pulling RESET_n LOW for at least t_{PW_RESET}, which copies the repair vector from the hard lane repair register into the lane repair shadow register as shown in Figure 90.

A stable WRCK clock is required during t_{SLREP} and t_{HLREP} times.

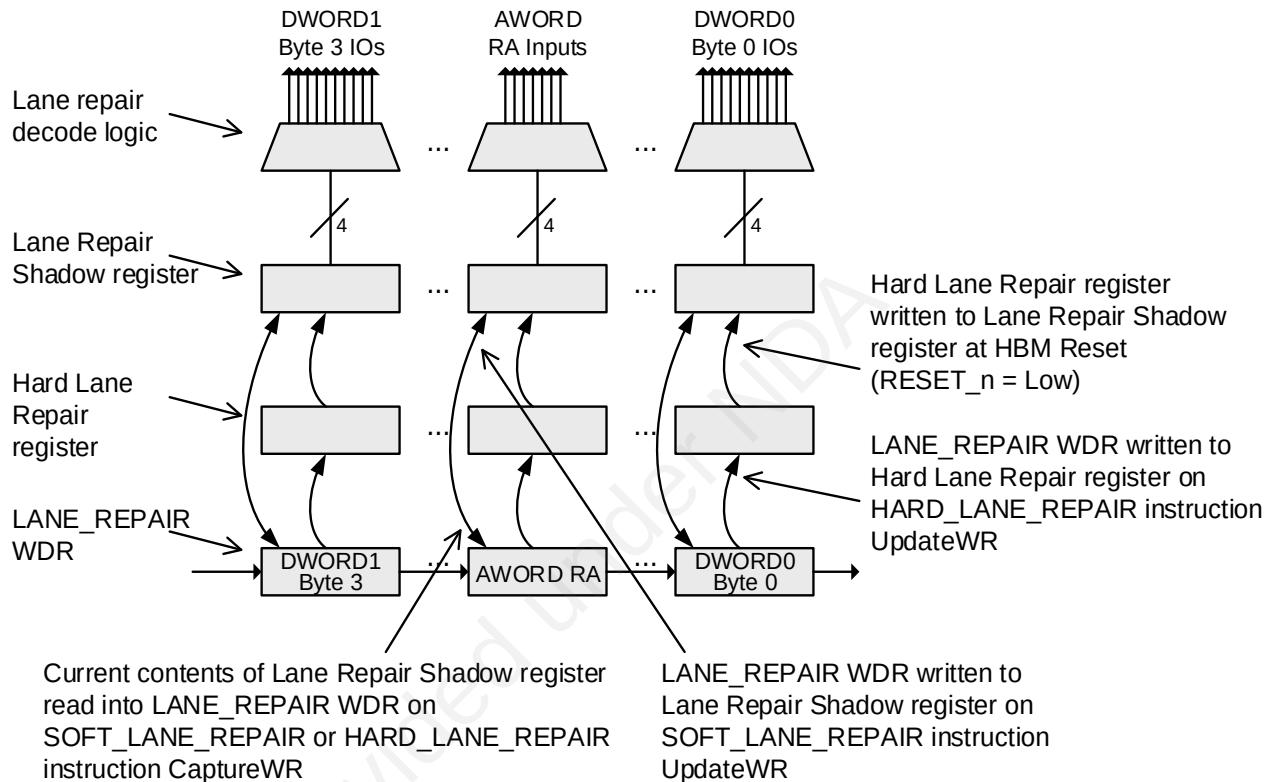
Table 94: LANE_REPAIR Wrapper Data Register

Bits	Field	Type	Description
[39:36]	DWORD1_BYTE3[3:0]	R/W	Lane remapping applied to DWORD1 byte 3 0 _H : SEV3 1 _H - 8 _H : DQ56 - DQ63 9 _H : DBI7 A _H - E _H : Reserved F _H : No lane remapping (default)
[35:32]	DWORD1_BYTE2[3:0]	R/W	Lane remapping applied to DWORD1 byte 2 0 _H : SEV2 1 _H - 8 _H : DQ48 - DQ55 9 _H : DBI6 A _H - E _H : Reserved F _H : No lane remapping (default)
[31:28]	DWORD1_BYTE1[3:0]	R/W	Lane remapping applied to DWORD1 byte 1 0 _H : ECC3 1 _H - 8 _H : DQ40 - DQ47 9 _H : DBI5 A _H - E _H : Reserved F _H : No lane remapping (default)
[27:24]	DWORD1_BYTE0[3:0]	R/W	Lane remapping applied to DWORD1 byte 0 0 _H : ECC2 1 _H - 8 _H : DQ32 - DQ39 9 _H : DBI4 A _H - E _H : Reserved F _H : No lane remapping (default)
[23:20]	AWORD_RA[3:0]	R/W	Lane remapping to AWORD 0 _H - 9 _H : R0 - R9 A _H - E _H : Reserved F _H : No lane remapping (default)
[19:16]	AWORD_CA[3:0]	R/W	Lane remapping to AWORD 0 _H - 7 _H : C0 - C7 8 _H : APAR 9 _H : ARFU A _H - E _H : Reserved F _H : No lane remapping (default)



Bits	Field	Type	Description
[15:12]	DWORD0_BYTE3[3:0]	R/W	Lane remapping applied to DWORD0 byte 3 0 _H : SEV1 1 _H - 8 _H : DQ24 - DQ31 9 _H : DBI3 A _H - E _H : Reserved F _H : No lane remapping (default)
[11:8]	DWORD0_BYTE2[3:0]	R/W	Lane remapping applied to DWORD0 byte 2 0 _H : SEV0 1 _H - 8 _H : DQ16 - DQ23 9 _H : DBI2 A _H - E _H : Reserved F _H : No lane remapping (default)
[7:4]	DWORD0_BYTE1[3:0]	R/W	Lane remapping applied to DWORD0 byte 1 0 _H : ECC1 1 _H - 8 _H : DQ8 - DQ15 9 _H : DBI1 A _H - E _H : Reserved F _H : No lane remapping (default)
[3:0]	DWORD0_BYTE0[3:0]	R/W	Lane remapping applied to DWORD0 byte 0 0 _H : ECC0 1 _H - 8 _H : DQ0 - DQ7 9 _H : DBI0 A _H - E _H : Reserved F _H : No lane remapping (default)

Figure 90: Registers associated with Lane Repair Instructions



CHANNEL DISABLE

This instruction disables one channel or all channels as specified in WIR[12:8]. The instruction may only be issued after INIT3 timing has been met following a chip reset with RESET_n being pulled LOW and only before the CK clock is started for the first time. The disabled channel(s) will transition into a safe low-power state where they do not respond to commands. All AWORD and DWORD input and output buffers including CK_t/CK_c and WDQS_t/WDQS_c inputs of the disabled channel(s) will be turned off, thus allowing all external signals to float. The channel(s) will also not respond to EXTEXT_RX, EXTEST_TX and CHANNEL_ID instructions. Both RESET_n and WRST_n must be maintained HIGH during the all-channel disable state. The channel's WSO output(s) will remain active and drive a LOW.

The all-channel disable option shall be used during host-side HTOL and/or other reliability tests to prevent unintentional degradation of the HBM3 DRAM. This option may only be issued after !INIT3 timing has been met, before the CK clock is started for the first time after exit from reset state, and only if no other IEEE1500 instruction except BYPASS or DEVICE ID has been loaded before.

A disabled channel can be enabled again by pulling both RESET_n and WRST_n to LOW and following the procedure described in the Initialization Sequence with Stable Power section.

Wrapper Data Register: when CHANNEL_DISABLE is the current instruction, the data register as shown in Table 95 will be connected between WSI and WSO.

CaptureWR: when CHANNEL_DISABLE is the current instruction, the CaptureWR event will have no effect.

UpdateWR: when CHANNEL_DISABLE is the current instruction, the UpdateWR event will load the disable bit from the shift stage into the update stage of the WDR and asynchronously transition into a safe low power state when the bit is 1. Input and output buffers will be disabled latest after tCHDIS.

Table 95: CHANNEL_DISABLE Wrapper Data Register

Bits	Field	Type	Description
0	CH_DIS	W	0 _B – Channel is enabled (default) 1 _B – Channel is disabled Note: clearing the bit to 0 _B does <u>not</u> re-enable a channel

CHANNEL_TEMPERATURE

This instruction captures the channel's junction temperature. Temperature reporting is specified as an 9-bit field per SID: the LSB indicates the validity of the temperature sensor read-out, and the 8-bit fields indicate the temperature in degrees Celsius.

The device reports the same temperature value for all the channels within the same core-die. Refer to Figure 91 for the per-channel read-out to memory die mapping.

Wrapper Data Register: when CHANNEL_TEMPERATURE is the current instruction, the data register as shown in Table 96 will be connected between WSI and WSO.

CaptureWR: when CHANNEL_TEMPERATURE is the current instruction, the CaptureWR event load the temperature field values into the shift stage of the WDR.

UpdateWR: when CHANNEL_TEMPERATURE is the current instruction, the UpdateWR event will have no effect.

Figure 91: Per-Channel Temperature Read-out to Memory Die Mapping

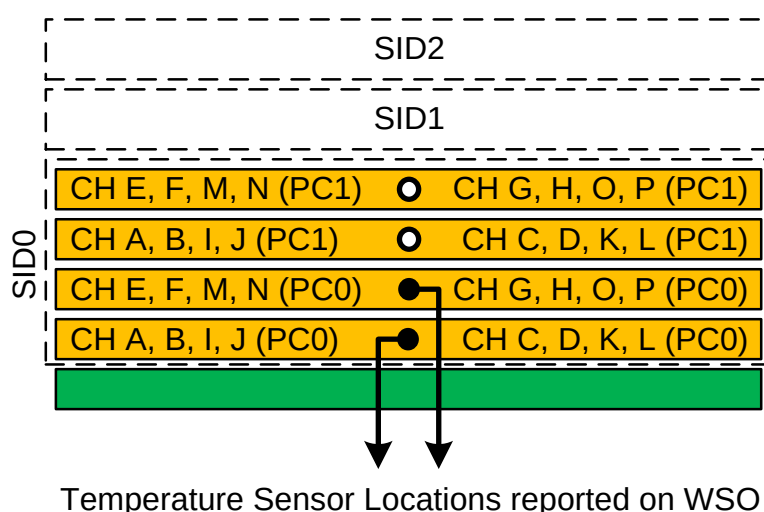



Table 96: CHANNEL_TEMPERATURE Wrapper Data Register

Bits	Field	Type	Description
[35:28]	CHANNEL_SID3_TEMP[7:0]	R	0000000 _B
27	CHANNEL_SID3_VALID	R	0 _B – Invalid
[26:19]	CHANNEL_SID2_TEMP[7:0]	R	0000000 _B (24GB device) Maximum channel temperature in °C for SID2 Same encoding as in CHANNEL_SID0_TEMP[7:0] (36GB device)
18	CHANNEL_SID2_VALID	R	Temperature sensor output valid for SID[1:0] = 10 0 _B – Invalid (24GB device) 1 _B – Valid (36GB device)
[17:10]	CHANNEL_SID1_TEMP[7:0]	R	Maximum channel temperature in °C for SID1. Same encoding as in CHANNEL_SID0_TEMP[7:0]
9	CHANNEL_SID1_VALID	R	Temperature sensor output valid for SID[1:0] = 01 1 _B – Valid
[8:1]	CHANNEL_SID0_TEMP[7:0]	R	Maximum channel temperature in °C for SID0. The binary value represents the device's junction temperature in °C. Examples: 0000 0000 _B = -40°C 0010 1000 _B = 0°C 0100 0001 _B = 25°C 1000 0111 _B = 95°C 1010 0111 _B = 127°C
0	CHANNEL_SID0_VALID	R	Temperature sensor output valid for SID[1:0] = 00 1 _B – Valid

- Notes:
1. For unsupported SID fields the HBM3 DRAM will report the sensor output as invalid and the temperature as all zeros.
 2. As each channel is distributed across two dies within the same SID, the device reports the maximum channel temperature value between the core-dies across which the channel is distributed.
 3. Refer to Table 107 for the supported operating temperature range.

WOSC_RUN and WOSC_COUNT

The WOSC_RUN and WOSC_COUNT instructions are associated with the WDQS Interval Oscillator in the HBM3 DRAM.

Wrapper Data Register: when WOSC_RUN is the current instruction, the WOSC_RUN data register as shown in Table 97 is connected between WSI and WSO[a:p], and the WSO outputs of all active channels drive the same data during ShiftWR events.

When WOSC_COUNT is the current instruction, the WOSC_COUNT data register as shown in Table 98 is connected between WSI and WSO[a:p], and the WSO outputs of all active channels drive the same data during ShiftWR events.

CaptureWR: when WOSC_RUN is the current instruction, the CaptureWR event will have no effect. When WOSC_COUNT is the current instruction, the CaptureWR event will load the WDQS oscillator count value into the shift stage of the WDR and update the WOSC_COUNT_VALID field of the WDR.



UpdateWR: when WOSC_RUN is the current instruction, the UpdateWR event will start or stop the WDQS Interval Oscillator. When WOSC_COUNT is the current instruction, the UpdateWR event will have no effect.

Table 97: WOSC_RUN Wrapper Data Register

Bits	Field	Type	Description
0	WOSC_START_STOP	W	0 _B – Stop WDQS Interval Oscillator 1 _B – Start WDQS Interval Oscillator

Table 98: WOSC_COUNT Wrapper Data Register

Bits	Field	Type	Description
[24:1]	WOSC_COUNT_VALUE	R	Oscillator count value. Range: 0 to 2 ²⁴ - 1
0	WOSC_COUNT_VALID	R	0 _B – Count is invalid (default) 1 _B – Count is valid

- Notes:
1. The WDQS oscillator count value is used to train WDQS to the data valid window. The value reported in this WDR can be used by the memory controller to periodically adjust the phase of WDQS relative to data.
 2. The oscillator count value (bits [24:1]) is reset by starting the oscillator.

ECS_ERROR_LOG

This instruction is used to provide ECS error log information to the host. The encoding of error types (NE, CEs, CEm, UE) is the same as on the SEV outputs during read bursts (UI 4 to 7, see Table 30). The error log updates follow the priority rules as in Table 34. Refer to the Error Check and Scrub (ECS) section for details.

Wrapper Data Register: when ECS_ERROR_LOG is the current instruction, the data register as shown in Table 99 is connected between WSI and WSO.

CaptureWR: when ECS_ERROR_LOG is the current instruction, the CaptureWR event will load the respective error log field values into the shift stage of the WDR.

UpdateWR: when ECS_ERROR_LOG is the current instruction, the UpdateWR event will have no effect.

Table 99: ECS_ERROR_LOG Wrapper Data Register

Bits	Field	Type	Description
[215:190]	SID3_PC1_ECS[25:0]	R	000 0000 _H
189	SID3_PC1_ECS_VALID	R	0 _B
[188:163]	SID3_PC0_ECS[25:0]	R	000 0000 _H
162	SID3_PC0_ECS_VALID	R	0 _B
[161:136]	SID2_PC1_ECS[25:0]	R	000 0000 _H (24GB device) ECS error log for SID2, PC1 (36GB device) Same encoding as in SID0_PC0_ECS
135	SID2_PC1_ECS_VALID	R	0 _B (24GB device) ECS error log valid for SID2, PC1 (36GB device) Same encoding as in SID0_PC0_ECS_VALID



Bits	Field	Type	Description
[134:109]	SID2_PC0_ECS[25:0]	R	000 0000 _H (24GB device) ECS error log for SID2, PC0 (36GB device) Same encoding as in SID0_PC0_ECS
108	SID2_PC0_ECS_VALID	R	0 _B (24GB device) ECS error log valid for SID2, PC0 (36GB device) Same encoding as in SID0_PC0_ECS_VALID
[107:82]	SID1_PC1_ECS[25:0]	R	ECS error log for SID1, PC1 Same encoding as in SID0_PC0_ECS
81	SID1_PC1_ECS_VALID	R	ECS error log valid for SID1, PC1 Same encoding as in SID0_PC0_ECS_VALID
[80:55]	SID1_PC0_ECS[25:0]	R	ECS error log for SID1, PC0 Same encoding as in SID0_PC0_ECS
54	SID1_PC0_ECS_VALID	R	ECS error log valid for SID1, PC0 Same encoding as in SID0_PC0_ECS_VALID
[53:28]	SID0_PC1_ECS[25:0]	R	ECS error log for SID0, PC1 Same encoding as in SID0_PC0_ECS
27	SID0_PC1_ECS_VALID	R	ECS error log valid for SID0, PC1 Same encoding as in SID0_PC0_ECS_VALID
[26:1]	SID0_PC0_ECS[25:0]	R	ECS error log for SID0, PC0 [25:22] – Bank address BA[3:0] [21:7] – Row address RA[14:0] [6:2] – Column address CA[4:0] [1:0] – Error type [1:0]. Encoding: 00 _B – NE (No error) 01 _B – CEs (Corrected single-bit error) 10 _B – UE (Uncorrectable error) 11 _B – CEm (Corrected multi-bit error)
0	SID0_PC0_ECS_VALID	R	ECS error log valid for SID0, PC0 0 _B – Invalid (default) 1 _B – Valid

SELF_REP and SELF_REP_RESULTS

The SELF_REP and SELF_REP_RESULTS instructions are associated with HBM3 Self Repair.

Wrapper Data Register: when SELF_REP is the current instruction, the SELF_REP data register as shown in Table 100 is connected between WSI and WSO.

When SELF_REP_RESULT is the current instruction, the SELF_REP_RESULT data register as shown in Table 101 is connected between WSI and WSO.

CaptureWR: when SELF_REP is the current instruction, the CaptureWR event will capture the SR_PROGRESS to the shift stage register.

When SELF_REP_RESULTS is the current instruction, the CaptureWR event will capture the SID[3:0]_RESULTS to the shift stage register.

UpdateWR: when SELF_REP is the current instruction, the UpdateWR event will copy the SELF_REP_RATE, SID_SELECT and REP_TYPE into the update stage register.



When SELF_REP_RESULTS is the current instruction, the UpdateWR event will have no effect.

Table 100: SELF_REP Wrapper Data Register

Bits	Field	Type	Description
8	SHARED_OVERRIDE	W	0 _B – DRAM must leave 1 of the shared resources 1 _B – DRAM must use all resources for Self Repair
[7:6]	SELFR_REF_RATE[1:0]	W	00 _B –Reserved
[5:4]	SID_SELECT	W	SID[1:0]
[3:2]	REP_TYPE[1:0]	W	00 _B – Cancel ¹ 01 _B – Run test only, no repair 10 _B – Auto repair only 11 _B – Run test and auto repair
[1:0]	SR_PROGRESS[1:0]	R	00 _B – Not running (default) 01 _B – Test in progress 10 _B – Repair in progress 11 _B – Complete

- Notes:
- Setting REP_TYPE = 00_B will cancel an ongoing self-test (SR_PROGRESS = 01_B), but not any ongoing repair; it will be treated as a NOP when no self-test is in progress. The SR_PROGRESS is set to 00_B (not running) after the cancellation.
 - The SHARED_OVERRIDE field allows the DRAM to use all the resources for self repair.
 - The same status are reported for all 8 channels in the channel group selected by WIR[12:8].

Table 101: SELF_REP_RESULT Wrapper Data Register

Bits	Field	Type	Description
[7:6]	SID3_RESULTS[1:0]	R	00 _B
[5:4]	SID2_RESULTS[1:0]	R	00 _B (24GB device) Valid (36GB device). Same encoding as in SID0_RESULTS
[3:2]	SID1_RESULTS[1:0]	R	Valid. Same encoding as in SID0_RESULTS
[1:0]	SID0_RESULTS[1:0]	R	00 _B – SELF_REPAIR test has not run since device initialization or no fails remain after the most recent run (default) 01 _B – Fail(s) remain 10 _B – Unrepairable fail(s) remain 11 _B – SELF_REPAIR should be run again

- Notes:
- For unsupported SID fields, the DRAM will report the results as 00_B (SELF-REPAIR test has not run).
 - The same results are reported for all 8 channels in the channel group selected by WIR[12:8].

HS_REP_CAP

The HS_REP_CAP instruction tells the host whether a bank(s) has resources for either hard repair, soft repair or self repair. For each bank the gray-coded encoding indicates whether there are no resources, 1 resource, or 2 or more resources.

This instruction shall be used by the host before performing any soft, hard or self repair. This register will be updated with the completion of a hard repair or a self repair.

Wrapper Data Register: when HS_REP_CAP is the current instruction, the data register as shown in Table 102 is connected between WSI and WSO.



CaptureWR: when HS_REP_CAP is the current instruction, the CaptureWR event will load the related information into the shift stage of the WDR.

UpdateWR: when HS_REP_CAP is the current instruction, the UpdateWR event will have no effect.

Table 102: HS_REP_CAP Wrapper Data Register

Bits	Field	Type	Description
[255:192]	HS_REPAIR_RES_SID3[63:0]	R	0000 0000 0000 0000 _H (fixed value)
[191:128]	HS_REPAIR_RES_SID2[63:0]	R	0000 0000 0000 0000 _H (24GB device) Resource(s) available for SID2, PC0 banks [15:0] and PC1 banks [15:0] (36GB device)
[127:64]	HS_REPAIR_RES_SID1[63:0]	R	Resource(s) available for SID1, PC0 banks [15:0] and PC1 banks [15:0]
[63:32]	HS_REPAIR_RES_SID0[63:0]	R	Resource(s) available for SID0, PC1 banks [15:0]
[31:2]			Resource(s) available for SID0, PC0 banks [15:1]
[1:0]			Resource(s) available for SID0, PC0, bank 0 00 _B – No resource(s) available 01 _B – 1 resource available 10 _B – Reserved 11 _B – 2 or more resources available



Interaction with Mission Mode Operation

Table 103 defines the interaction of the various IEEE 1500 port instructions with mission mode operation, and any instruction exit requirements.

Table 103: IEEE1500 Port Instruction Interactions

Instruction	Interaction with Mission Mode	Post Instruction Requirements
BYPASS DWORD_MISR ¹ AWORD_MISR ¹ READ_LFSR_COMPARE_STICKY ¹ DEVICE_ID TEMPERATURE MODE_REGISTER_DUMP_SET (dump) CHANNEL_TEMPERATURE WOSC_RUN WOSC_COUNT ECS_ERROR_LOG SELF_REP_RESULTS HS_REP_CAP	Instructions may be used at any time. Core memory content is retained if refresh specifications are met.	None
SOFT_REPAIR ^{2, 5} SOFT_LANE_REPAIR ² MODE_REGISTER_DUMP_SET (set) AWORD_MISR_CONFIG ³	Core memory content is retained if refresh specifications are met.	Meet IEEE1500 Port AC Timings
EXTEST_RX, EXTEST_TX INTEST_RX, INTEST_TX MBIST CHANNEL_ID HARD_REPAIR ⁴ HARD_LANE_REPAIR ⁴ CHANNEL_DISABLE ⁶ SELF_REP ⁴	HBM3 DRAM interface state and core memory content is not defined.	Reset

- Notes:
1. While accessing these MISR-related registers has no interaction with mission mode, operating the AWORD and DWORD MISR modes may result in memory content loss unless the channel is put into self refresh mode. See Loopback Test Modes.
 2. Soft memory array and lane repairs imply that memory content is at least partially incorrect. While the HBM3 DRAM imposes no restrictions on the interface state and memory content, the host should consider the health of the memory content based on the repair(s) being applied.
 3. See Loopback Test Modes for proper sequencing of the AWORD MISR test modes
 4. Self / hard memory array and hard lane repairs involve blowing fuses. Normal operation on any channel is not supported when the self / hard repair operations are used. A chip reset with RESET_n pulled LOW is required after self / hard repair operations before returning to normal operation.
 5. For a soft repair operation it is required that the channel is held in bank idle state from the time when the SOFT_REPAIR instruction is loaded in the WIR until the SOFT_REPAIR instruction is unloaded.
 6. In case of the all-channel disable option, refer to the CHANNEL_DISABLE instruction for conditions on how to re-enable a disabled channel.



IEEE1500 Test Port AC Timing Parameters

Table 104: IEEE1500 Test Port AC Timing Parameters

Symbol	Parameter	Min	Max	Unit	Notes
IEEE1500 Port I/O Timings					
t_{CKTP}	WRCK clock period	20	—	ns	
t_{CKTPH}	WRCK clock high pulse width	0.45	—	t_{CKTP}	
t_{CKTPL}	WRCK clock low pulse width	0.45	—	t_{CKTP}	
t_{WRSTL}	WRST_n pulse width low	100	—	ns	
t_{WINIT1}	IEEE1500 port operation after WRST_n deassertion	3	—	t_{CKTP}	
t_{WINIT2}	IEEE1500 port input signals stable before WRST_n deassertion	10	—	ns	
t_{SWRST}	Rising WRST_n edge to WRCK setup time	100	—	ns	
t_{SR}	WSP input setup time to WRCK rising edge	5	—	ns	3
t_{HR}	WSP input hold time from WRCK rising edge	5	—	ns	3
t_{SF}	WSP input setup time to WRCK falling edge	5	—	ns	4
t_{HF}	WSP input hold time from WRCK falling edge	5	—	ns	4
t_{OVWSO}	WSO output valid time from WRCK falling edge	—	10	ns	5
EXTEST_RX Instruction Related Timings					
t_{SEXT}	Input setup time to WRCK rising edge	5	—	ns	6
t_{HEXT}	Input hold time from WRCK rising edge	5	—	ns	6
EXTEST_TX Instruction Related Timing					
t_{OVEXT}	Output valid time from WRCK rising edge	—	5	ns	7
HBM_RESET Instruction Related Timing					
t_{RES}	HBM_RESET instruction minimum active time	t_{PW_RESET} (MIN)	—	μ s	8
SOFT_REPAIR and HARD_REPAIR Instruction Related Timings					
t_{SREP}	SOFT_REPAIR minimum waiting time	1	—	ms	9
t_{HREP}	HARD_REPAIR minimum waiting time	1	—	s	10
DWORD_MISR and AWORD_MISR Instruction Related Timing					
t_{SMISR}	DWORD and AWORD MISR data capture to WDR data capture delay	20	—	ns	11
CHANNEL_ID Instruction Related Timings					
t_{OVCHN}	Output HIGH time from WRCK falling edge	—	20	ns	12
t_{OZCHN}	Output return to default state delay	—	20	ns	13



Symbol	Parameter	Min	Max	Unit	Notes
MODE_REGISTER_DUMP_SET Instruction Related Timings					
^t UPDMRS	WDR update to Mode Register valid delay	100	—	ns	14
^t MRSS	MRS command to WDR data capture delay	^t MOD (MIN)	—	ns	15
AWORD_MISR_CONFIG Instruction Related Timing					
^t CMISR	AWORD MISR configuration to MISR operation delay	20	—	ns	16
SOFT_LANE_REPAIR and HARD_LANE_REPAIR Instruction Related Timings					
^t SLREP	SOFT_LANE_REPAIR minimum waiting time	1	—	μs	17
^t HLREP	HARD_LANE_REPAIR minimum waiting time	1	—	s	18
CHANNEL_DISABLE Instruction Related Timings					
^t CHDIS	Channel disable to input and output buffer disable delay	—	50	ns	19
SELF_REPAIR Instruction Related Timings					
^t SELF_HEAL	Self-test and auto-repair	—	^t SELF_REP + ^t SELF_NR	s	20
^t SELF_REP	Auto-repair	—	5	s	20
^t SELF_NR	Self-test	—	12	s	20
^t SELF_CANCEL	Self-test cancel time	—	10	ms	20

Notes:

- AC timing parameters apply to each channel of the HBM3 DRAM independently except for timings related to IEEE1500 inputs that are common to all channels. No timing parameters are specified across channels, and all channels operate independently of each other.
- All parameters assume proper device initialization.
- Parameter applies to WSI, SELECTWIR, SHIFTWIR and CAPTUREWR inputs.
- Parameter applies to UPDATEWR input.
- Parameter applies to WSO output changes resulting from Wrapper Instruction Register (WIR), Wrapper Bypass Register (WBY) or any Wrapper Data Register (WDR) shift operation.
- Parameter applies to all HBM inputs and bidirectional IOs in the CaptureWR cycle when the active instruction is EXTEST_RX.
- Parameter applies to all HBM outputs and bidirectional IOs in the ShiftWR cycle when the active instruction is EXTEST_TX.
- Parameter applies when the active instruction is HBM_RESET; it is measured from either the falling WRCK edge that loads the HBM_RESET instruction in the UpdateWIR cycle (in case no WDR is associated with the instruction) or the falling WRCK edge that sets the WDR bit to 1_B in the UpdateWR cycle (in case a WDR is associated with the instruction) until either the HBM_RESET instruction is invalidated or the WDR bit is set back to 0_B. The minimum value equals the RESET_n minimum low time with stable power (^tPW_RESET).
- Parameter applies when the active instruction is SOFT_REPAIR; it describes the minimum time for the HBM3 DRAM to perform the internal soft repair; it is measured from the falling WRCK edge that loads the repair vector and repair start bit in the UpdateWR cycle until the instruction is invalidated.
- Parameter applies when the active instruction is HARD_REPAIR; it describes the minimum time for the HBM3 DRAM to perform the internal hard repair; it is measured from the falling WRCK edge that loads the repair vector and repair start bit in the UpdateWR cycle until the instruction is invalidated.
- Parameter applies when the active instruction is DWORD_MISR or AWORD_MISR; it is measured from the last CK clock that updates the data in the respective MISR until the rising WRCK edge associated with the CaptureWR cycle that copies the MISR data into the WDR shift register.
- Parameter applies when the active instruction is CHANNEL_ID; it describes the maximum duration from either the falling WRCK edge that sets the CHANNEL_ID instruction in the UpdateWIR cycle (when no WDR is associated with



- the instruction) or the falling WRCK edge in the UpdateWR cycle that sets the enable bit in the WDR to 1_B (when a WDR is associated with the instruction) until the outputs and bidirectional IOs drive a High.
13. Parameter applies when the active instruction is CHANNEL_ID; it describes the maximum duration from either the falling WRCK edge that sets any instruction other than CHANNEL_ID instruction in the UpdateWIR cycle (when no WDR is associated with the instruction) or the falling WRCK edge in the UpdateWR cycle that sets the enable bit in the WDR to 0_B (when a WDR is associated with the instruction) until the outputs and bidirectional IOs return to their default state.
 14. Parameter applies when the active instruction is MODE_REGISTER_DUMP_SET; it describes the minimum required delay between the falling WRCK edge in the UpdateWR cycle that loads the Mode Registers from the WDR shift register until any valid command other than RNOP and CNOP can be issued at the command interface.
 15. Parameter applies when the active instruction is MODE_REGISTER_DUMP_SET; it describes the minimum required delay between the last MRS command that loads any Mode Register and the rising WRCK edge in the CaptureWR cycle that copies the Mode Register content into the WDR shift register.
 16. Parameter applies when the active instruction is AWORD_MISR_CONFIG; it describes the minimum required delay between the falling WRCK edge in the UpdateWR cycle that loads the AWORD MISR configuration until the MISR configuration is valid for any subsequent AWORD or DWORD MISR operation in the CK clock domain.
 17. Parameter applies when the active instruction is SOFT_LANE_REPAIR; it describes the minimum time for the HBM3 DRAM to perform the internal soft lane repair; it is measured from the falling WRCK edge that loads the repair vector in the UpdateWR cycle until the instruction is invalidated.
 18. Parameter applies when the active instruction is HARD_LANE_REPAIR; it describes the minimum time for the HBM3 DRAM to perform the internal hard lane repair; it is measured from the falling WRCK edge that loads the repair vector in the UpdateWR cycle until the instruction is invalidated.
 19. Parameter applies when the active instruction is CHANNEL_DISABLE; it describes the minimum waiting time for disabling the channel's input and output buffers after the UpdateWR event that sets the CHANNEL_DISABLE WDR to 1_B.
 20. Parameter represents the maximum time the operation will take to complete and therefore the minimum time the host must wait.

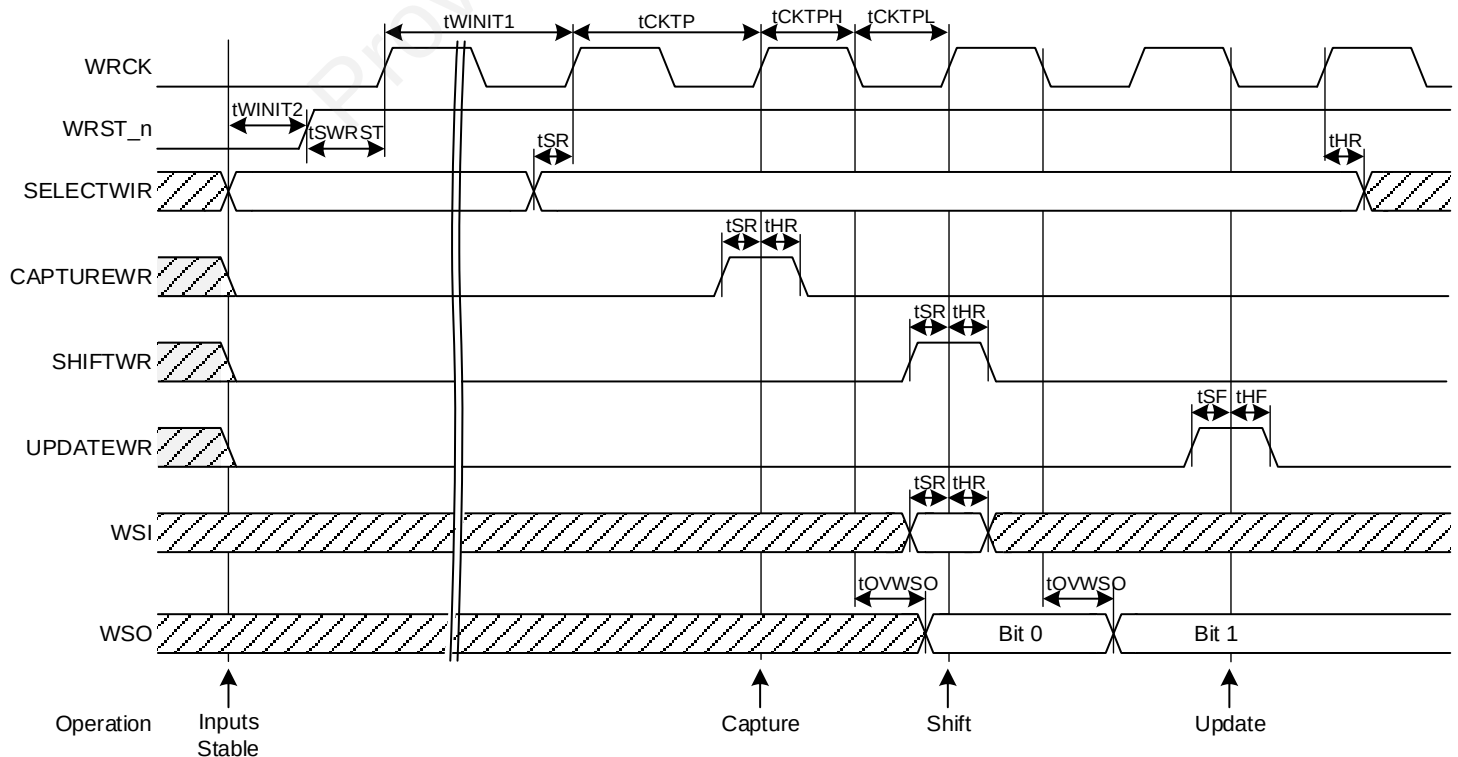
Figure 92: IEEE1500 Port Input and Output Timings


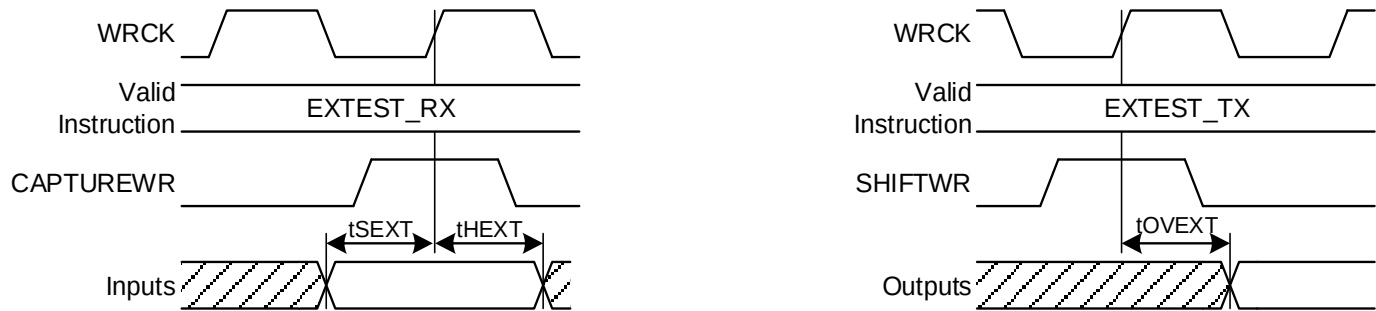
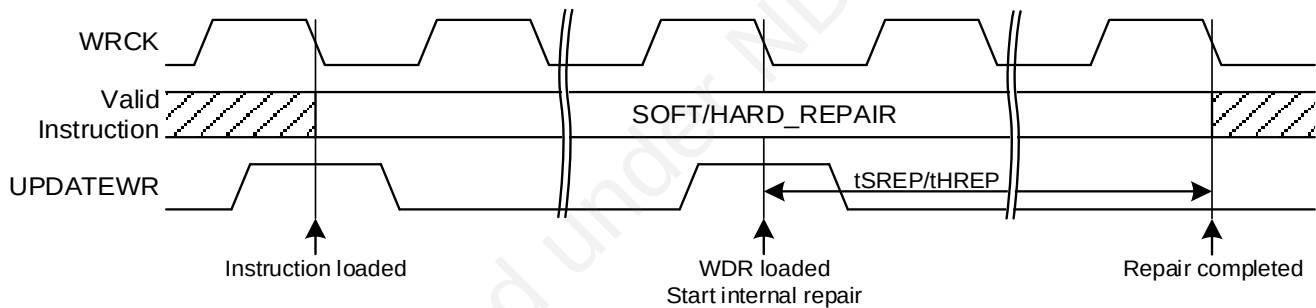
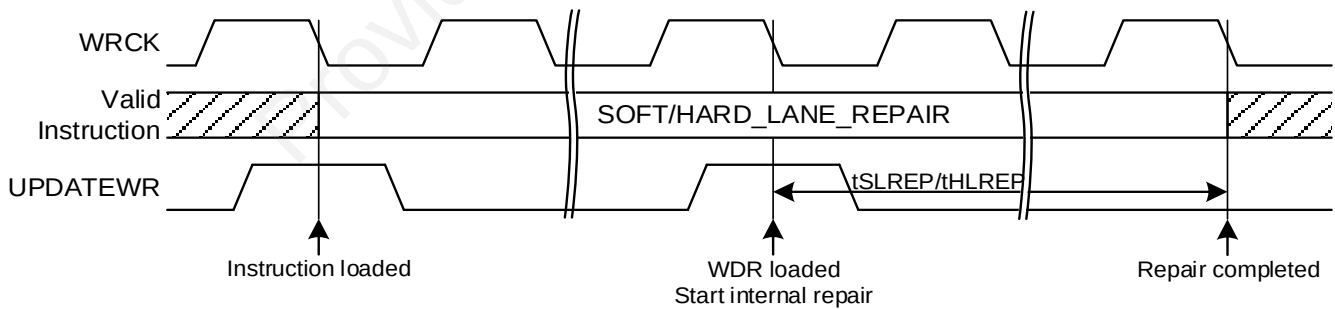
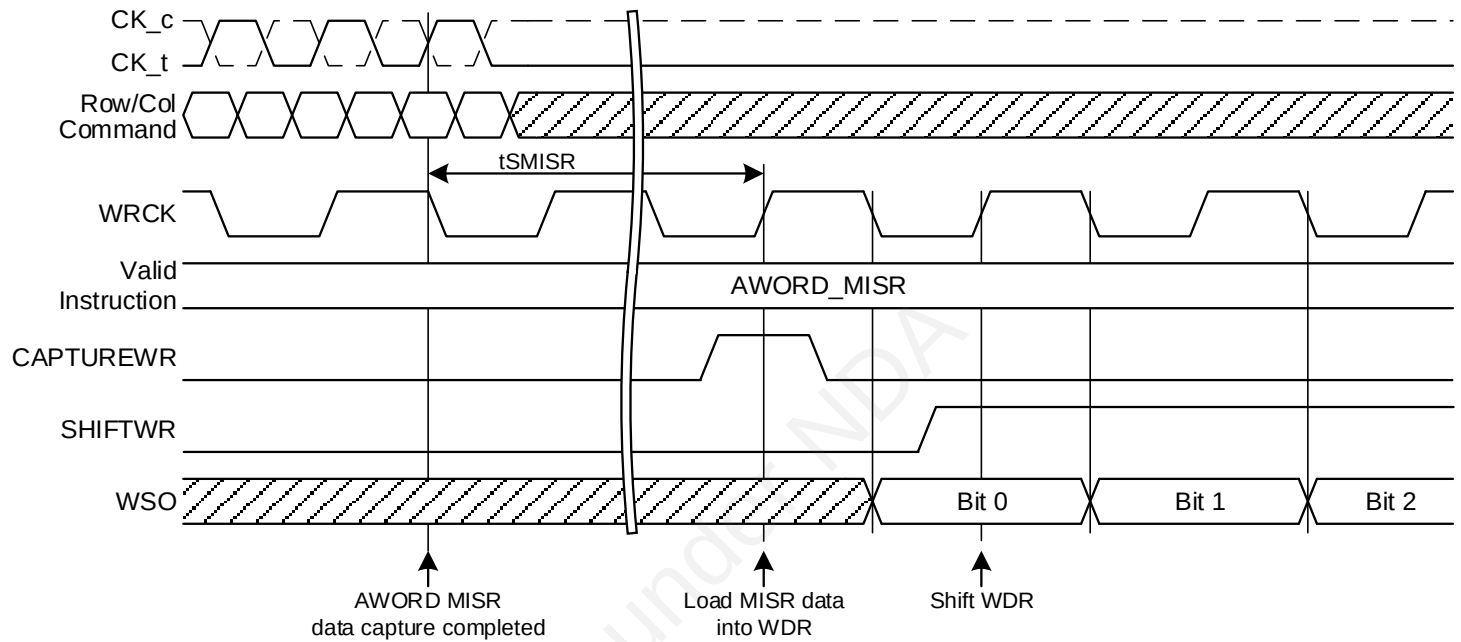
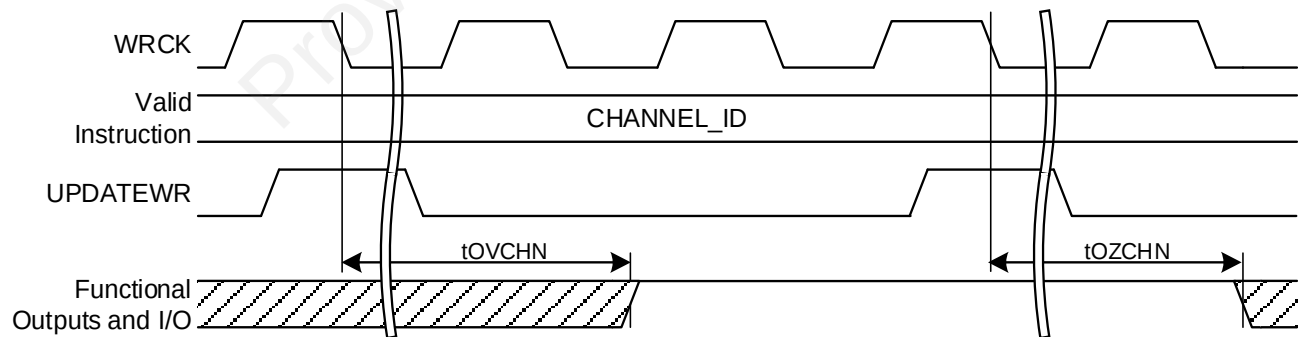
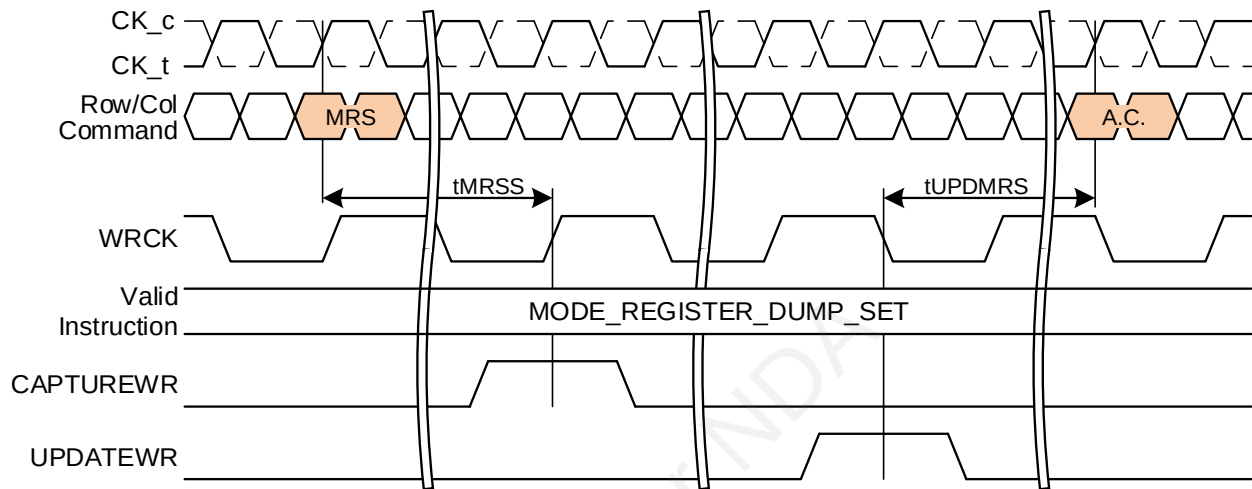
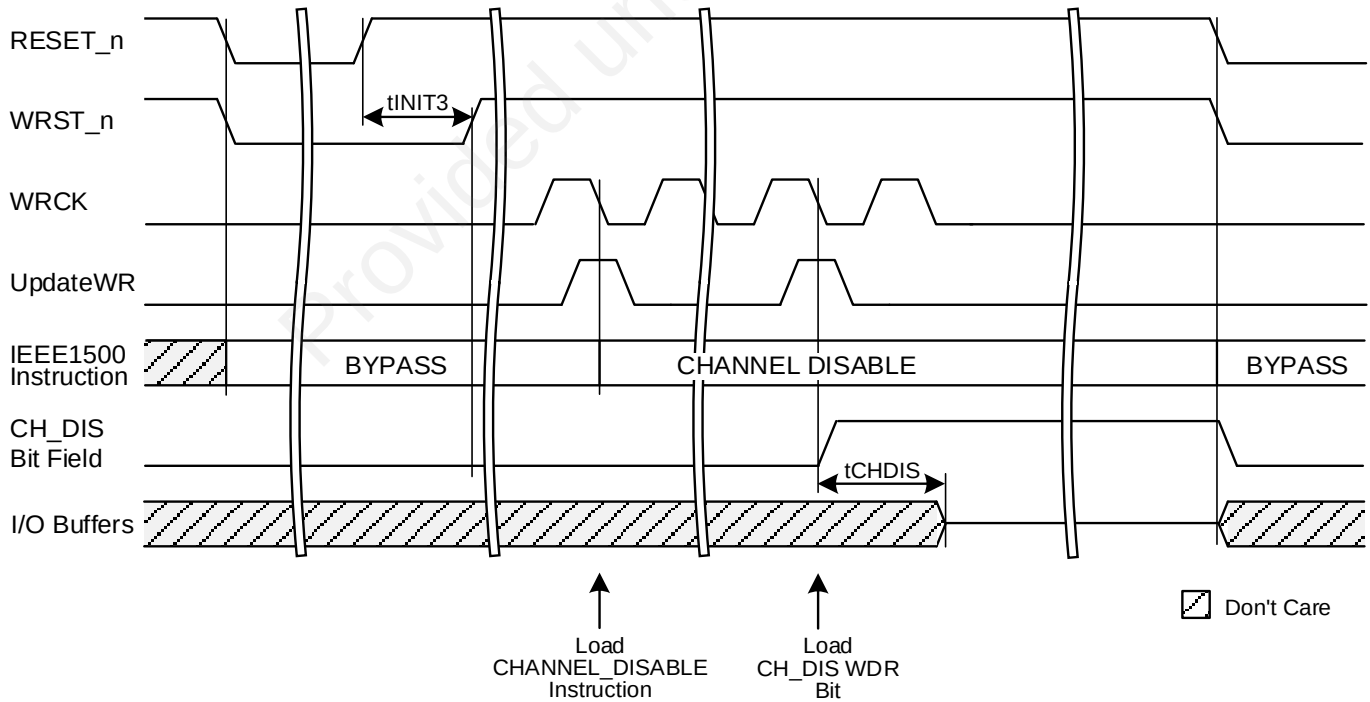

Figure 93: IEEE1500 EXTEST_RX and EXTEST_TX Instruction Related Timings

Figure 94: IEEE1500 SOFT_REPAIR and HARD_REPAIR Instruction Related Timings

Figure 95: IEEE1500 SOFT_LANE_REPAIR and HARD_LANE_REPAIR Instruction Related Timings



Figure 96: IEEE1500 DWORD_MISR / AWORD_MISR Instruction Related Timings


- Notes:
1. Same timings for data inputs and DWORD MISR with DWORD_MISR instruction.
 2. 'OVWSO = 0 for illustration purpose.

Figure 97: IEEE1500 CHANNEL_ID Instruction Related Timings


- Note:
1. 'OVCHN and 'OZCHN timings refer to set/reset of the ENABLE bit in the WDR.


Figure 98: IEEE1500 MODE_REGISTER_DUMP_SET Instruction Related Timings

Figure 99: IEEE1500 CHANNEL_DISABLE Instruction Related Timings




Boundary Scan

The HBM3 DRAM supports a boundary scan chain per channel via the IEEE1500 test port. The boundary scan operation is associated with IEEE1500 test port instructions EXTEST_RX and EXTEST_TX.

Scan data is shifted in through WSI and out through the respective WSO, based on the active channel selections in the WIR (see Table 75). All functional signals are included in the boundary scan chains. Table 6 lists the micro-bump boundary scan chain order. Bit position 0 is the first bit shifted in on WSI and out on the WSOs.

Three global signals in the MIDSTACK area are routed to the channel n and o scan chains: TEMP[1:0] are associated with channel n, and CATTRIP is associated with channel o. The boundary scan chain length for all channels is 122 bits (see Table 78), with dummy WDR bit padding as needed on the MSB end of the chains. Matched length boundary scan chains allow all channel chains to be loaded with matching data with one shift operation when WIR[12:8] = 1Fh.

All input-only and output-only signals are implemented as bi-directional I/Os to aid in SIP package level testing and fault isolation. Effectively, all signals support both EXTEST_RX and EXTEST_TX instructions.

I/O signals power up in input mode by default. The host will put all AWORD and DWORD drivers into High-Z state prior to loading the EXTEST_TX instruction into the WIR. As soon as EXTEST_TX becomes the current instruction, all AWORD and DWORD signals will change to output mode and the outputs will drive the values shifted into the WDR shift stage.

When EXTEST_RX is the current instruction, all functional signals of the selected channel(s) enter a High-Z state, including the output-only signals AERR, DERR, RDQS_t/RDQS_c, TEMP and CATTRIP. On a subsequent CaptureWR event the signals will capture the input values into the WDR shift stage.

Boundary scan mode entry may be asserted at any time after device initialization and before normal memory operation has commenced, e.g. before the CK clock has started to toggle. Upon exiting the boundary scan mode, the state of the HBM3 DRAM is unknown and the integrity of the data content of the memory array is not guaranteed and therefore the reset initialization sequence is required before returning to normal operation.



Operating Conditions

Absolute Maximum Ratings

Stresses greater than those listed may cause permanent damage to the device. This is a stress rating only, and functional operation of the device at these or any other conditions above those indicated in the operational sections of this specification is not implied. Exposure to absolute maximum rating conditions for extended periods may adversely affect reliability.

Table 105: Absolute Maximum Ratings

Symbol	Parameter	Min	Max	Unit	Notes
V _{DDC}	Voltage on V _{DDC} relative to V _{SS}	-0.3	1.4	V	1
V _{DDQ}	Voltage on V _{DDQ} relative to V _{SS}	-0.3	1.4	V	1
V _{DDQL}	Voltage on V _{DDQL} relative to V _{SS}	-0.3	0.8	V	1
V _{PP}	Voltage on V _{PP} relative to V _{SS}	-0.3	2.1	V	
V _{IN} /V _{OUT}	Voltage on any signal relative to V _{SS}	-0.3	1.4	V	
T _{STG}	Storage temperature	-55	+125	°C	2

- Notes:
1. See HBM3 Power-up and Initialization Sequence for the relationship between power supplies.
 2. Storage temperature is the case surface temperature on the center/top side of the HBM3 DRAM. For the measurement conditions, please refer to JESD51-2 standard.

DC Operating Conditions

Table 106: DC Operating Conditions

Symbol	Parameter	Min	Typ	Max	Unit	Notes
V _{DDC}	Core supply voltage	1.067	1.1	1.177	V	1,2
V _{DDQ}	Output supply voltage	1.067	1.1	1.177	V	1,2
V _{DDQL}	Supply Voltage for Tx Driver Output Stage	0.38	0.4	0.44	V	1,2
V _{PP}	Pump voltage	1.746	1.8	1.95	V	2

- Notes:
1. V_{DDC} and V_{DDQ} supplies are independent and must not be tied together internally on the HBM3 DRAM.
 2. The voltage ranges are defined at the HBM3 DRAM micropillars. DC bandwidth is limited to 20 MHz.



Operating Temperature

Table 107: Operating Temperature

Symbol	Parameter	Value	Unit	Notes
T _{OPER}	Operating temperature	0 to +105	°C	1, 2

- Notes:
1. The operating temperature refers to the junction temperature of all memory dies and the logic die of the HBM3 DRAM. The host is required to monitor the operating temperature via the IEEE1500 test port instructions TEMPERATURE and CHANNEL_TEMPERATURE. The host is also required to monitor the CATTRIP output that signals if the junction temperature of any die in the HBM3 DRAM exceeds a catastrophic trip-point level that could result in permanent damage of the device.
 2. During operation of the device, the user will not force the junction temperature to remain at 0°C by cooling the device from the outside to temperatures below 0°C. At any time the device is powered-up, T_{OPER} (MIN) = T_{CASE} (MIN) = T_{JUNCTION} (MIN) is assumed. During normal operation the junction temperature is expected to increase above T_{OPER} (MIN) due to self-heating of the device.

Electrostatic Discharge Sensitivity Characteristics

Table 108: Electrostatic Discharge Sensitivity Characteristics

Symbol	Parameter ¹	Min	Max	Unit	Notes
EDC _{HBM}	Human body model (HBM) for all signals except DA port Human body model (HBM) for DA port signals	N/A 1000	–	V	2
EDC _{CDM}	Charged-device model (CDM) for all signals except DA port Charged-device model (CDM) for DA port signals	100 250	–	V	3

- Notes:
1. State-of-the-art basic ESD control measures have to be in place when handling the device.
 2. Refer to ESDA/JEDEC Joint Standard JS-001-2014 for measurement procedures.
 3. Refer to ESDA/JEDEC Joint Standard JS-002-2014 for measurement procedures.

Pad Capacitances

Table 109: Silicon Pad Capacitances

Symbol	Parameter	Min	Max	Unit	Notes
C _{IO}	Pad Capacitance – DQs, DBI, ECC, SEV, DPAR	–	0.4	pF	1
C _{CA}	Pad Capacitance – R, C, ARFU, APAR	–	0.4	pF	1
C _{RDQS}	Pad Capacitance – RDQS_t/RDQS_c	–	0.4	pF	1
C _{WDQS}	Pad Capacitance – WDQS_T/WDQS_c	–	0.4	pF	1
C _{CK}	Pad Capacitance – CK_t/CK_c	–	0.4	pF	1
C _{ERROR}	Pad Capacitance – AERR, DERR	–	0.4	pF	1

- Note:
1. This parameter is not subject to production test.

Input Leakage Current Specification

Table 110: Input Leakage Current Specification

Symbol	Parameter	Min	Max	Unit	Notes
I_{L1}	Input leakage current for AWORD and DWORD inputs	-5	5	μA	1
I_{L2}	Input leakage current for AWORD and DWORD outputs and I/O signals	-15	15	μA	1

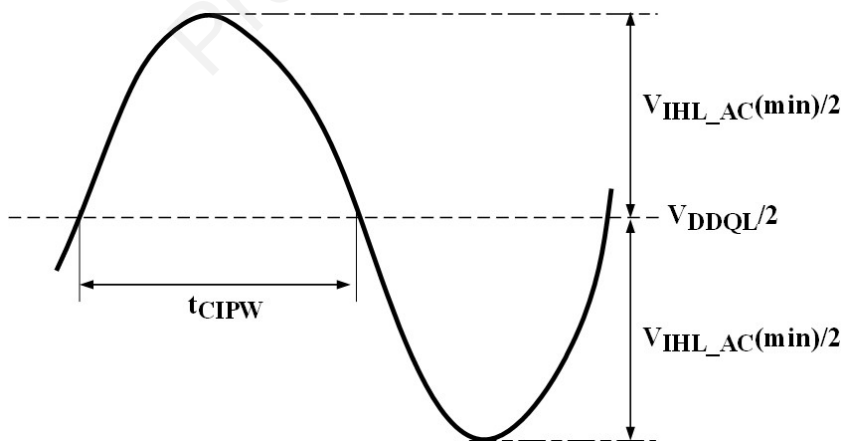
Note: 1. Any input $0\text{V} \leq V_{IN} \leq V_{DDQL}$; all other inputs including IEEE1500 port inputs not under test = 0V.

AWORD Input Voltage Level Specification

Table 111: Command/Address Input Voltage Level Specification

Symbol	Parameter	Min	Max	Unit	Notes
V_{IHCA}	Input HIGH voltage	$V_{REFCA} + 0.1$	–	V	1, 3
V_{ILCA}	Input LOW voltage	–	$V_{REFCA} - 0.1$	V	1, 3
V_{ILHCA_AC}	Command/address input single pulse amplitude	240	–	mV	2, 3

- Notes:
1. Applies to all AWORD inputs (R, C, ARFU and APAR). V_{REFCA} is determined by the V_{REFCA} bits in MR14.
 2. CA single input pulse amplitude has to meet or exceed V_{IHL_AC} at any point over the total UI. No timing requirement above level. V_{IHL_AC} is the peak-to-peak voltage centered around $V_{DDQL}/2$ such that $V_{IHLCA_AC}/2$ min has to be met both above and below $V_{DDQL}/2$.
 3. Parameter applies to all speed bins.

Figure 100: AWORD Input Single Pulse Amplitude and Width




CK, WDQS and RDQS Input and Output Voltage Level Specification

Table 112: Clock and Data Strobe Input and Output Voltage Level Specification

Symbol	Parameter	Min	Max	Unit	Notes
V_{IDCK}	CK clock input differential input voltage	0.16	–	V	1, 7
V_{IXCK}	CK clock differential input crosspoint voltage	$V_{DDQL}/2 - 0.04$	$V_{DDQL}/2 + 0.04$	V	2, 7
V_{IDWDQS}	WDQS differential input voltage	0.15	–	V	3, 7
V_{IXWDQS}	WDQS differential input crosspoint voltage	$V_{DDQL}/2 - 0.03$	$V_{DDQL}/2 + 0.03$	V	4, 7
V_{OXRDS}	RDQS differential output crosspoint voltage	$V_{DDQL}/2 - TBD$	$V_{DDQL}/2 - TBD$	V	5, 6, 7
SR_WDQS	WDQS differential input slew rate	TBD	TBD	V/ns	7

- Notes:
1. V_{IDCK} is the magnitude of the difference between the input level on CK_t and the input level on CK_c.
 2. The input reference level for timings referenced to CK is the point at which CK_t and CK_c cross.
 3. V_{IDWDQS} is the magnitude of the difference between the input level on WDQS_t and the input level on WDQS_c.
 4. The input reference level for timings referenced to WDQS is the point at which WDQS_t and WDQS_c cross.
 5. The output reference level for timings referenced to RDQS is the point at which RDQS_t and RDQS_c cross. This parameter is guaranteed by design with the output timing reference load and RDBI enabled. It includes V_{DDQL} , V_{DDQ} AC noise impact of TBD mV (pk pk) at the supply microbumps; AC noise includes system PDN impact.
 6. This parameter is guaranteed by design at the DRAM micropillars with output timing reference load and Read DBI enabled.
 7. Parameter applies to all speed bins.

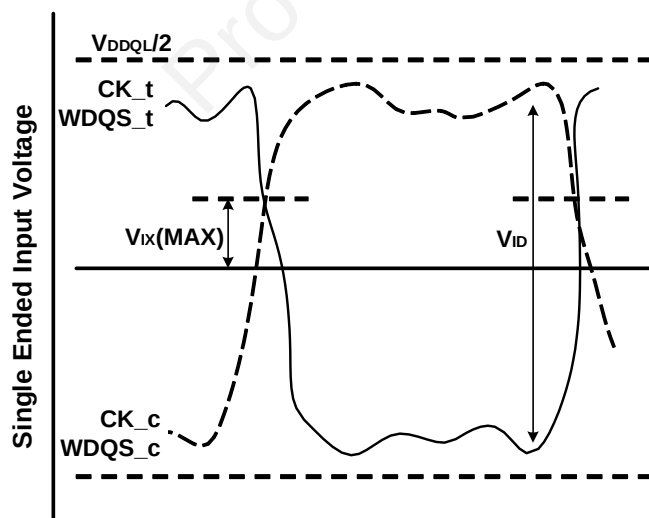
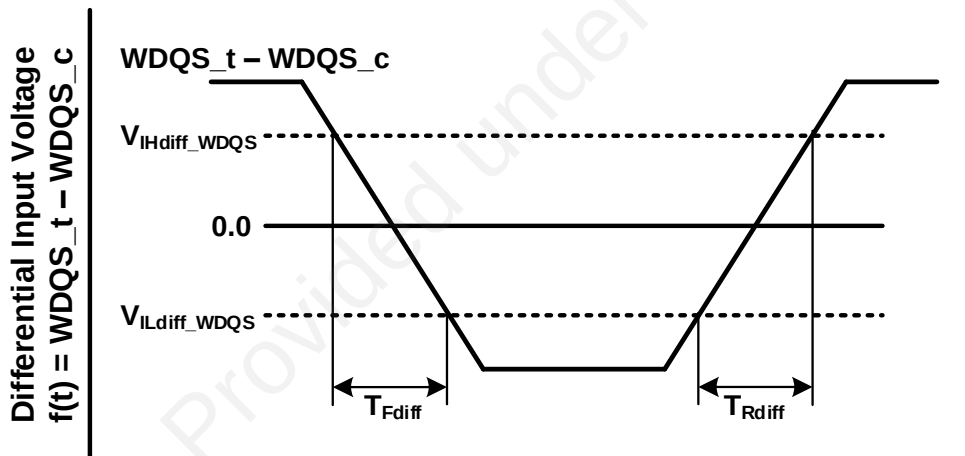
Figure 101: CK and WDQS Input Waveform



Table 113: Differential Input Levels for WDQS_t, WDQS_c

Symbol	Parameter	Min	Unit	Notes
V_{IHdiff_WDQS}	WDQS differential input HIGH voltage	TBD	V	
V_{ILdiff_WDQS}	WDQS differential input LOW voltage	TBD	V	

Table 114: Differential Input Slew Rate Definition for WDQS_t, WDQS_c

Description	From	To	Defined By
WDQS Differential Input Slew Rate for Rising Edge (WDQS_t – WDQS_c)	V_{ILdiff_WDQS}	V_{IHdiff_WDQS}	$ V_{ILdiff_WDQS} - V_{IHdiff_WDQS} / T_{Rdiff}$
WDQS Differential Input Slew Rate for Falling Edge (WDQS_t – WDQS_c)	V_{IHdiff_WDQS}	V_{ILdiff_WDQS}	$ V_{ILdiff_WDQS} - V_{IHdiff_WDQS} / T_{Fdiff}$

Figure 102: WDQS Differential Input Slew Rate Definition


DWORD Input Voltage Level Specification

Table 115: Data Input Voltage Level Specification

Symbol	Parameter	Min	Max	Unit	Notes
V_{DIVW}	Data input eye mask voltage (peak-to-peak)	120	–	mV	1, 2
SR_{IN_DIVW}	Input slew rate over V_{DIVW}	1	7	V/ns	3
V_{IHLAQ_AC}	Data input single pulse amplitude	190	–	mV	4

- Notes:
1. Data Rx mask voltage and timing parameters are applied per pin and includes DRAM DQ to WDQS voltage AC noise impact for frequencies > 20MHz at a fixed temperature on a die.
 2. Rx mask voltage V_{DIVW} has to be centered around $V_{cent_DQ(pin_mid)}$.
 3. Input slew rate over V_{DIVW} mask centered at $V_{cent_DQ(pin_mid)}$.
 4. DQ single input pulse amplitude into the receiver has to meet or exceed V_{IHLAQ_AC} at any point over the total UI. No timing requirement above level. V_{IHLAQ_AC} is the peak to peak voltage centered around $V_{cent_DQ(pin_mid)}$ such that $V_{IHL_AC}/2$ min has to be met both above and below $V_{cent_DQ(pin_mid)}$.

The data input receiver mask for voltage and timing as shown in the figure below applies to each input receiver. The eye mask (V_{DIVW} , t_{DIVW}) defines the area the input signal must not encroach in order for the data input receiver to successfully capture an input signal with a BER of lower than TBD. The mask is a receiver property.

$V_{cent_DQ(pin_mid)}$ is defined as the midpoint between the largest V_{cent_DQ} voltage level and the smallest V_{cent_DQ} voltage level across all DQ pins for a given DRAM TBD (determined by DRAM V_{REFD} training granularity, i.e. component/channel/DWORD) level. Each DQ V_{cent} is defined by the center, i.e. widest opening, of the cumulative data input eye as depicted in Figure 104. This clarifies that any DRAM TBD level variation must be accounted for within the DRAM data input receiver mask.

Figure 103: Data Input Receiver Voltage and Timing

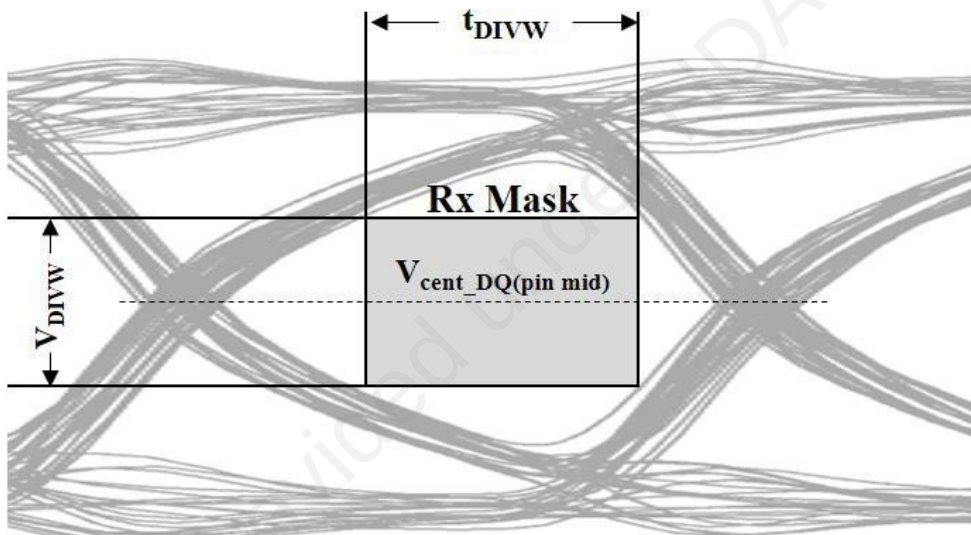
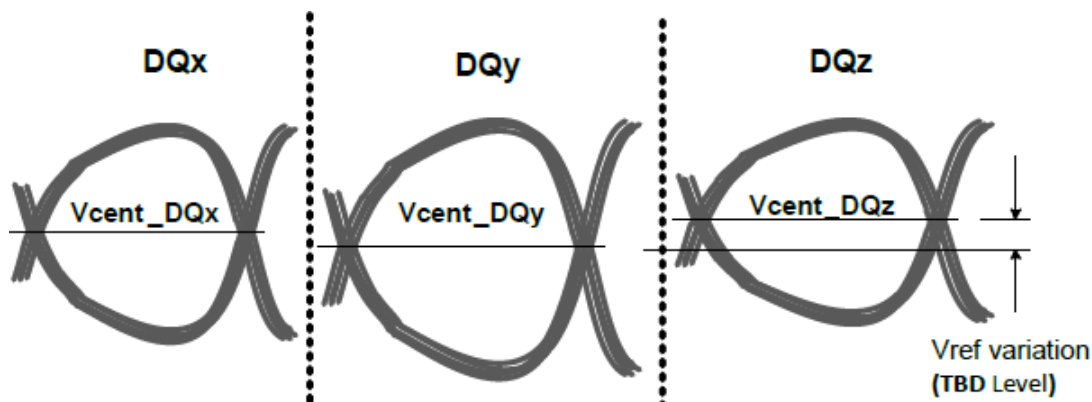


Figure 104: Across Pin V_{REFD} Voltage Variation





MIDSTACK Signal Input and Output Voltage Level Specification

Table 116: MIDSTACK Signal Input and Output Voltage Level Specification

Symbol	Parameter	Min	Max	Unit	Notes
V_{IHR}	Input HIGH voltage for MIDSTACK inputs	$0.7 \times V_{DDQ}$	–	V	1
V_{ILR}	Input LOW voltage for MIDSTACK inputs	–	$0.2 \times V_{DDQ}$	V	1
V_{OHR}	Output HIGH voltage for MIDSTACK outputs	$0.7 \times V_{DDQ}$	–	V	2
V_{OLR}	Output LOW voltage for MIDSTACK outputs	–	$0.3 \times V_{DDQ}$	V	2

- Notes:
1. Applies to RESET_n, WRST_n, WRCK, SELECTWIR, SHIFTWR, CAPTUREWR, UPDATEWR and WSI inputs.
 2. Applies to CATTRIP, TEMP and WSO outputs.

Transmit Driver Currents

HBM3 drivers have current settings programmable separately for pullup and pulldown drivers with 20% accuracy. Driver targets are shown in the table below.

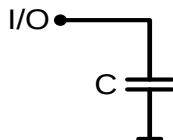
Table 117: Transmit Driver Current Specification

Nominal	Min	Max	Unit
8	6.4	9.6	mA
10	8.0	12.0	mA
12	9.6	14.4	mA
14	11.2	16.8	mA

Note: 1. Transmit driver currents are measured at $0.5 \times V_{DDQL}$.

Output Timing Reference Load

$C = C_{TOTAL} - C_{IO}$; where $C_{TOTAL} = 2.5\text{pF}$

Figure 105: Timing Reference Load

Table 118: Data Output Voltage Level Specification

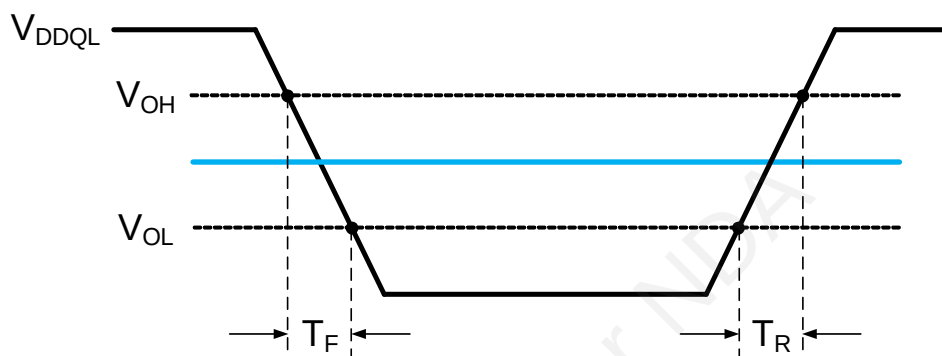
Symbol	Parameter	Min	Max	Unit	Notes
V_{OH}	Output HIGH voltage	$V_{DDQL}/2 + 75$	–	mV	
V_{OL}	Output LOW voltage	–	$V_{DDQL}/2 - 75$	mV	

Output Rise and Fall Times

$$T_R = \{ C_{TOTAL} \times (V_{OH} - V_{OL}) \} / I; T_F = \{ C_{TOTAL} \times (V_{OH} - V_{OL}) \} / I;$$

I = transmit drive current in mA

Figure 106: Output Rise and Fall Time Definition

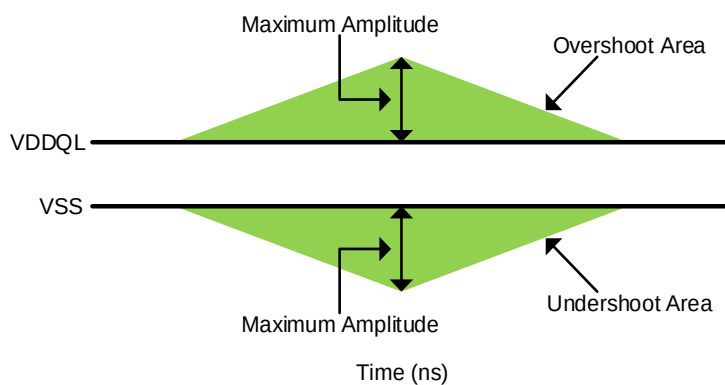


Overshoot / Undershoot

Table 119: Overshoot/Undershoot Specification

Parameter	-64	-72	-80	Unit
Maximum peak amplitude allowed for overshoot area	0.12	0.12	0.12	V
Maximum peak amplitude allowed for undershoot area	0.12	0.12	0.12	V
Maximum overshoot area above V_{DDQ}	9	8	7.5	mV-ns
Maximum undershoot area below V_{SS}	9	8	7.5	mV-ns

Figure 107: Overshoot / Undershoot Definition





IDD Specification

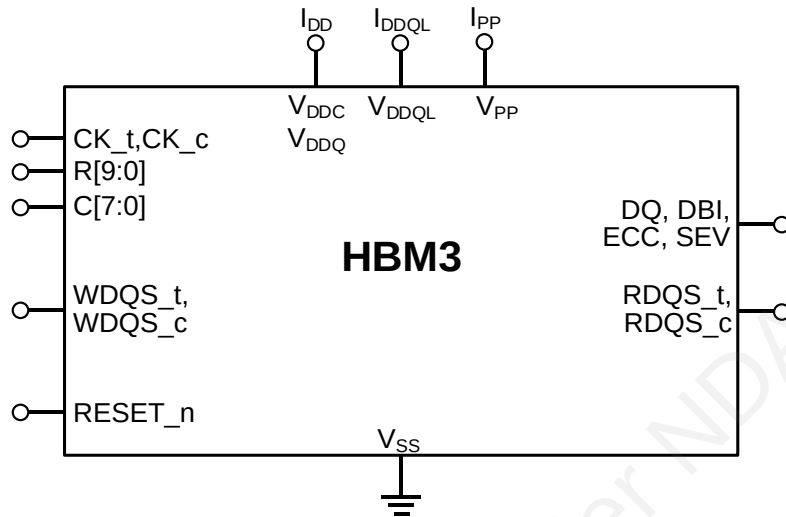
IDD, IDDQL and IPP Specification Parameters and Test Conditions

This chapter defines operating current measurement conditions.

- I_{DD} currents are measured as time-averaged currents with all V_{DDC} and V_{DDQ} microbumps of the HBM3 DRAM under test tied together.
- I_{DDQL} currents use the same definitions as I_{DD} except that the current on the V_{DDQL} supply is measured. All V_{DDQL} microbumps of the HBM3 DRAM under test are tied together for I_{PP} current measurements. The output reference load C_{TOTAL} is 2.5pF.
- I_{PP} currents use the same definitions as I_{DD} except that the current on the V_{PP} supply is measured. All V_{PP} microbumps of the HBM3 DRAM under test are tied together for I_{PP} current measurements.
- I_{DD} , I_{DDQL} and I_{PP} measurements are taken with all channels of the HBM3 DRAM simultaneously executing the same pattern.

For I_{DD} measurements, the following definitions apply:

- “0_B” and “LOW” are defined as $V_{IN} \leq V_{IL}(\text{max})$;
- “1_B” and “HIGH” are defined as $V_{IN} \geq V_{IH}(\text{min})$;
- WL, RL and RTP fields in MR1, MR2 and MR5, respectively, are programmed to appropriate values;
- Write DBI and read DBI are enabled in MR0 OP[1:0];
- Command/address parity and data parity are disabled in MR0 OP[6:4];
- Meta data is enabled in MR9 OP0;
- Error severity reporting are disabled in MR9 OP1;
- CNOP/RNOP commands and all address inputs are stable during idle command cycles;
- Basic I_{DD} measurement conditions are described in Table 120. I_{DD} measurements are done after properly initializing the HBM3 DRAM. This includes the pre-load of the memory array with data pattern used with I_{DD4R} and I_{DD7} measurements.
- Timing parameters t_{RAS} , t_{RP} , t_{RREFD} , t_{RFCab} and t_{RFCpb} are used as given in Table 121. If not already an integer, round up to the next integer.
- Data pattern used with I_{DD4R} , I_{DD4W} and I_{DD7} measurements (BL8, per data byte):
 Pattern A: 00h, 3Ch, F0h, 69h, AAh, FFh, C3h, 55h
 Pattern B: F0h, A5h, 96h, 0Fh, C3h, 55h, CCh, 5Ah

Figure 108: Measurement Setup for IDD and IPP Measurements

Table 120: Basic IDD Measurement Conditions

Symbol	Parameter / Condition
I _{DD0} , I _{DDQL0} , I _{PP0}	One Bank Activate Precharge Current: t _{CK} = t _{CK(min)} ; t _{RC} , t _{RAS} and t _{RP} as defined in Table 121; R and C inputs are HIGH between valid commands; DQ, ECC and DBI inputs are LOW; bank and row addresses with ACT and PRE commands as defined in Table xx
I _{DD2P} , I _{DDQL2P} , I _{PP2P}	Precharge Power-down Current: Device is in precharge power-down; t _{CK} = t _{CK(min)} ; all banks are idle; R0 input is LOW; R[9:1] and C inputs are HIGH; DQ, ECC/SEV and DBI inputs are LOW
I _{DD2P0} , I _{DDQL2P0} , I _{PP2P0}	Precharge Power-down Current with clock stop: Device is in precharge power-down; CK_t is LOW; CK_c is HIGH; all banks are idle; R0 input is LOW; R[9:1] and C inputs are HIGH; DQ, ECC/SEV and DBI inputs are LOW
I _{DD2N} , I _{DDQL2N} , I _{PP2N}	Precharge Standby Current: t _{CK} = t _{CK(min)} ; all banks are idle; R and C inputs are HIGH; DQ, ECC/SEV and DBI inputs are LOW
I _{DD3P} , I _{DDQL3P} , I _{PP3P}	Active Power-down Current: Device is in active power-down; t _{CK} = t _{CK(min)} ; one bank is active; R0 input is LOW; R[9:1] and C inputs are HIGH; DQ, ECC/SEV and DBI inputs are LOW
I _{DD3P0} , I _{DDQL3P0} , I _{PP3P0}	Active Power-down Current with clock stop: Device is in active power-down; CK_t is LOW; CK_c is HIGH; one bank is active; R0 input is LOW; R[9:1] and C inputs are HIGH; DQ, ECC/SEV and DBI inputs are LOW
I _{DD3N} , I _{DDQL3N} , I _{PP3N}	Active Standby Current: t _{CK} = t _{CK(min)} ; one bank is active; R and C inputs are HIGH; DQ, ECC and DBI inputs are LOW
I _{DD4R} , I _{DDQL4R} , I _{PP4R}	Read Burst Current: t _{CK} = t _{CK(min)} ; all banks activated; continuous read burst across bank groups as defined in Table 123; C _{TOTAL} = 2.5pF



Symbol	Parameter / Condition
I _{DD4W} , I _{DDQL4W} , I _{PP4W}	Write Burst Current: t _{CK} = t _{CK} (min); all banks activated; continuous write burst across bank groups as defined in Table 124
I _{DD5B} , I _{DDQL5B} , I _{PP5B}	All-bank Refresh Burst Current: t _{CK} = t _{CK} (min); t _{RFCab} as defined in Table 121; R and C inputs are HIGH between valid commands; DQ, ECC and DBI inputs are LOW
I _{DD5P} , I _{DDQL5P} , I _{PP5P}	Per-Bank Refresh Burst Current: t _{CK} = t _{CK} (min); all banks refreshed as defined in Table 125; t _{RFCpb} and t _{RREFD} as defined in Table 121; R and C inputs are HIGH between valid commands; DQ, ECC and DBI inputs are LOW
I _{DD6} , I _{DDQL6} , I _{PP6}	Self Refresh Current: R and C inputs are LOW; DQ, ECC and DBI inputs are LOW
I _{DD7} , I _{DDQL7} , I _{PP7}	All-bank Interleave Read Current: t _{CK} = t _{CK} (min); all banks activated and precharged as defined in Table 126; continuous read burst across bank groups as defined in Table 126; C _{TOTAL} = 2.5pF
I _{DD8} , I _{DDQL8} , I _{PP8}	Reset Low Current: RESET _n is LOW; All AWORD and DWORD inputs including CK _t , CK _c , WDQS _t , WDQS _c are LOW; Note: Reset low current reading is valid once power is stable and RESET _n has been LOW for at least 1ms

Table 121: Timings used for IDD Measurement-Loop Pattern

Symbol	Parameter		Value	Unit
t _{RC}	ACTIVATE to ACTIVATE command period		48	ns
t _{RAS}	ACTIVATE to PRECHARGE command period		32	ns
t _{RP}	PRECHARGE command period		16	ns
t _{RFCab}	REFRESH command period	8-High	410	ns
		12-High	450	ns
t _{RFCpb}	PER-BANK REFRESH command period (all densities)		240	ns
t _{RREFD}	PER-BANK REFRESH command period (different bank) and PER-BANK REFRESH to ACTIVATE (different bank) command delay		8	ns


Table 122: IDD0 Measurement-Loop Pattern

Sub-Loop	Cycle Number	Row Command	Column Command	Row Bank Address (SID[1:0], BA[3:0])	Row Address (RA[13:0])
0	0	ACT – PC0	CNOP	00h, 00h	01555h
	1		CNOP		
	2	ACT – PC1	CNOP	00h, 00h	01555h
	3		CNOP		
	4	RNOP	CNOP	N/A	N/A
	...	Repeat RNOP/CNOP until cycle (nRAS-1)			
	nRAS	PRE – PC0	CNOP	00h, 00h	N/A
	nRAS + 1	RNOP	CNOP	N/A	N/A
	nRAS + 2	PRE – PC1	CNOP	00h, 00h	N/A
	nRAS + 3	RNOP	CNOP	N/A	N/A
	...	Repeat RNOP/CNOP until cycle (nRC-1)			
1	nRC	Repeat sub-loop 0 pattern until cycle (2 × nRC-1); use BA = 05h and RA = 02AAAh instead			
2	2 × nRC	Repeat sub-loop 0 pattern until cycle (3 × nRC-1); use BA = 02h and RA = 01555h instead			
3	3 × nRC	Repeat sub-loop 0 pattern until cycle (4 × nRC-1); use BA = 07h and RA = 02AAAh instead			
4	4 × nRC	Repeat sub-loop 0 pattern until cycle (5 × nRC-1); use BA = 01h and RA = 01555h instead			
5	5 × nRC	Repeat sub-loop 0 pattern until cycle (6 × nRC-1); use BA = 06h and RA = 02AAAh instead			
6	6 × nRC	Repeat sub-loop 0 pattern until cycle (7 × nRC-1); use BA = 03h and RA = 01555h instead			
7	7 × nRC	Repeat sub-loop 0 pattern until cycle (8 × nRC-1); use BA = 04h and RA = 02AAAh instead			
8 to 15	Repeat sub-loops 0 to 7 pattern; use BA3 = 1 instead				
16 to 31	For 8-High and 12-High devices: repeat sub-loops 0 to 15 pattern; use SID[1:0] = 01h instead				
32 to 47	For 12-High devices: repeat sub-loops 0 to 15 pattern; use SID[1:0] = 02h instead				


Table 123: IDD4R Measurement-Loop Pattern

Sub-Loop	Cycle Number	Row Command	Column Command	Col. Bank Address (SID[1:0], CBA[3:0])	Column Address (CA[4:0])	Data Pattern (1 Byte)
0	0	RNOP	READ – PC0	00h, 00h	0Ah	Pattern A
	1	RNOP	READ – PC1	00h, 00h	0Ah	Pattern A
	2	RNOP	READ – PC0	00h, 04h	15h	Pattern B
	3	RNOP	READ – PC1	00h, 04h	15h	Pattern B
	4	RNOP	READ – PC0	00h, 08h	1Ah	Pattern A
	5	RNOP	READ – PC1	00h, 08h	1Ah	Pattern A
	6	RNOP	READ – PC0	00h, 0Ch	05h	Pattern B
	7	RNOP	READ – PC1	00h, 0Ch	05h	Pattern B
	8	RNOP	READ – PC0	00h, 04h	0Ah	Pattern A
	9	RNOP	READ – PC1	00h, 04h	0Ah	Pattern A
	10	RNOP	READ – PC0	00h, 08h	15h	Pattern B
	11	RNOP	READ – PC1	00h, 08h	15h	Pattern B
	12	RNOP	READ – PC0	00h, 0Ch	1Ah	Pattern A
	13	RNOP	READ – PC1	00h, 0Ch	1Ah	Pattern A
	14	RNOP	READ – PC0	00h, 00h	05h	Pattern B
	15	RNOP	READ – PC1	00h, 00h	05h	Pattern B
	16	RNOP	READ – PC0	00h, 08h	0Ah	Pattern A
	17	RNOP	READ – PC1	00h, 08h	0Ah	Pattern A
	18	RNOP	READ – PC0	00h, 0Ch	15h	Pattern B
	19	RNOP	READ – PC1	00h, 0Ch	15h	Pattern B
	20	RNOP	READ – PC0	00h, 00h	1Ah	Pattern A
	21	RNOP	READ – PC1	00h, 00h	1Ah	Pattern A
	22	RNOP	READ – PC0	00h, 04h	05h	Pattern B
	23	RNOP	READ – PC1	00h, 04h	05h	Pattern B
	24	RNOP	READ – PC0	00h, 0Ch	0Ah	Pattern A
	25	RNOP	READ – PC1	00h, 0Ch	0Ah	Pattern A
	26	RNOP	READ – PC0	00h, 00h	15h	Pattern B
	27	RNOP	READ – PC1	00h, 00h	15h	Pattern B
	28	RNOP	READ – PC0	00h, 04h	1Ah	Pattern A
	29	RNOP	READ – PC1	00h, 04h	1Ah	Pattern A
	30	RNOP	READ – PC0	00h, 08h	05h	Pattern B
	31	RNOP	READ – PC1	00h, 08h	05h	Pattern B
1	32 to 63	Repeat sub-loop 0 pattern; use CBA0 = 1 instead; maintain SID[1:0] = 00h				
2	64 to 127	Repeat sub-loops 0 and 1 pattern; use CBA1 = 1 instead; maintain SID[1:0] = 00h				
	128 + 129	RNOP	CNOP	N/A	N/A	N/A
3	130 to 259	For 8-High and 12-High devices: repeat sub-loops 0 to 2 pattern; use SID[1:0] = 01h instead				
4	260 to 389	For 12-High devices: repeat sub-loops 0 to 2 pattern; use SID[1:0] = 10h instead				


Table 124: IDD4W Measurement-Loop Pattern

Sub-Loop	Cycle Number	Row Command	Column Command	Col. Bank Address (SID[1:0], CBA[3:0])	Column Address (CA[4:0])	Data Pattern (1 Byte)
0	0	RNOP	WRITE – PC0	00h, 00h	0Ah	Pattern A
	1	RNOP	WRITE – PC1	00h, 00h	0Ah	Pattern A
	2	RNOP	WRITE – PC0	00h, 04h	15h	Pattern B
	3	RNOP	WRITE – PC1	00h, 04h	15h	Pattern B
	4	RNOP	WRITE – PC0	00h, 08h	1Ah	Pattern A
	5	RNOP	WRITE – PC1	00h, 08h	1Ah	Pattern A
	6	RNOP	WRITE – PC0	00h, 0Ch	05h	Pattern B
	7	RNOP	WRITE – PC1	00h, 0Ch	05h	Pattern B
	8	RNOP	WRITE – PC0	00h, 04h	0Ah	Pattern A
	9	RNOP	WRITE – PC1	00h, 04h	0Ah	Pattern A
	10	RNOP	WRITE – PC0	00h, 08h	15h	Pattern B
	11	RNOP	WRITE – PC1	00h, 08h	15h	Pattern B
	12	RNOP	WRITE – PC0	00h, 0Ch	1Ah	Pattern A
	13	RNOP	WRITE – PC1	00h, 0Ch	1Ah	Pattern A
	14	RNOP	WRITE – PC0	00h, 00h	05h	Pattern B
	15	RNOP	WRITE – PC1	00h, 00h	05h	Pattern B
	16	RNOP	WRITE – PC0	00h, 08h	0Ah	Pattern A
	17	RNOP	WRITE – PC1	00h, 08h	0Ah	Pattern A
	18	RNOP	WRITE – PC0	00h, 0Ch	15h	Pattern B
	19	RNOP	WRITE – PC1	00h, 0Ch	15h	Pattern B
	20	RNOP	WRITE – PC0	00h, 00h	1Ah	Pattern A
	21	RNOP	WRITE – PC1	00h, 00h	1Ah	Pattern A
	22	RNOP	WRITE – PC0	00h, 04h	05h	Pattern B
	23	RNOP	WRITE – PC1	00h, 04h	05h	Pattern B
	24	RNOP	WRITE – PC0	00h, 0Ch	0Ah	Pattern A
	25	RNOP	WRITE – PC1	00h, 0Ch	0Ah	Pattern A
	26	RNOP	WRITE – PC0	00h, 00h	15h	Pattern B
	27	RNOP	WRITE – PC1	00h, 00h	15h	Pattern B
	28	RNOP	WRITE – PC0	00h, 04h	1Ah	Pattern A
	29	RNOP	WRITE – PC1	00h, 04h	1Ah	Pattern A
	30	RNOP	WRITE – PC0	00h, 08h	05h	Pattern B
	31	RNOP	WRITE – PC1	00h, 08h	05h	Pattern B
1	32 to 63	Repeat sub-loop 0 pattern; use CBA0 = 1 instead; maintain SID[1:0] = 00h				
2	64 to 127	Repeat sub-loops 0 and 1 pattern; use CBA1 = 1 instead; maintain SID[1:0] = 00h				
	128 + 129	RNOP	CNOP	N/A	N/A	N/A
3	130 to 259	For 8-High and 12-High devices: repeat sub-loops 0 to 2 pattern; use SID[1:0] = 01h instead				
4	260 to 389	For 12-High devices: repeat sub-loops 0 to 2 pattern; use SID[1:0] = 10h instead				


Table 125: IDD5P Measurement-Loop Pattern

Sub-Loop	Cycle Number	Row Command	Column Command	Row Bank Address (SID[1:0], BA[3:0])
0	0	PER_BANK REFRESH – PC0	CNOP	00h, 00h
	1	RNOP	CNOP	N/A
	2	PER_BANK REFRESH – PC1	CNOP	00h, 00h
	3	RNOP	CNOP	N/A
	...	Repeat RNOP/CNOP until cycle (nRREFD-1)		
1	nRREFD	Repeat sub-loop 0 pattern until cycle (2 × nRREFD -1); use BA = 05h instead		
2	2 × nRREFD	Repeat sub-loop 0 pattern until cycle (3 × nRREFD -1); use BA = 02h instead		
3	3 × nRREFD	Repeat sub-loop 0 pattern until cycle (4 × nRREFD -1); use BA = 07h instead		
4	4 × nRREFD	Repeat sub-loop 0 pattern until cycle (5 × nRREFD -1); use BA = 01h instead		
5	5 × nRREFD	Repeat sub-loop 0 pattern until cycle (6 × nRREFD -1); use BA = 06h instead		
6	6 × nRREFD	Repeat sub-loop 0 pattern until cycle (7 × nRREFD -1); use BA = 03h instead		
7	7 × nRREFD	Repeat sub-loop 0 pattern until cycle (8 × nRREFD -1); use BA = 04h instead		
8 to 15	Repeat sub-loops 0 to 7 pattern; use BA3 = 1 instead			
16 to 31	For 4-High devices: wait for (nRFCpb - 15 × nRREFD) For all other devices:: repeat sub-loops 0 to 15 pattern; use SID[1:0] = 01h instead			
32 to 47	For 12-High devices: repeat sub-loops 0 to 15 pattern; use SID[1:0] = 02h instead			


Table 126: IDD7 Measurement-Loop Pattern

Sub-Loop	Cycle Number	Row Command	Column Command	Row Bank Addr. (SID[1:0], RBA[3:0])	Col. Bank Addr. (SID[1:0], CBA[3:0])	Row Address (RA[13:0])	Column Address (CA[4:0])	Data Pattern (1 Byte)
0	0	ACT – PC0	READ – PC0	00h, 02h	00h, 00h	01555h	0Ah	Pattern A
	1		READ – PC1		00h, 00h		0Ah	Pattern A
	2	ACT – PC1	READ – PC0	00h, 02h	00h, 04h	01555h	15h	Pattern B
	3		READ – PC1		00h, 04h		15h	Pattern B
	4	RNOP	READ – PC0	N/A	00h, 08h	N/A	1Ah	Pattern A
	5	RNOP	READ – PC1	N/A	00h, 08h	N/A	1Ah	Pattern A
	6	RNOP	READ – PC0	N/A	00h, 0Ch	N/A	05h	Pattern B
	7	RNOP	READ – PC1	N/A	00h, 0Ch	N/A	05h	Pattern B
	8	ACT – PC0	READ – PC0	00h, 06h	00h, 04h	02AAAh	0Ah	Pattern A
	9		READ – PC1		00h, 04h		0Ah	Pattern A
	10	ACT – PC1	READ – PC0	00h, 06h	00h, 08h	02AAAh	15h	Pattern B
	11		READ – PC1		00h, 08h		15h	Pattern B
	12	RNOP	READ – PC0	N/A	00h, 0Ch	N/A	1Ah	Pattern A
	13	RNOP	READ – PC1	N/A	00h, 0Ch	N/A	1Ah	Pattern A
	14	RNOP	READ – PC0	N/A	00h, 00h	N/A	05h	Pattern B
	15	RNOP	READ – PC1	N/A	00h, 00h	N/A	05h	Pattern B
	16	ACT – PC0	READ – PC0	00h, 0Ah	00h, 08h	01555h	0Ah	Pattern A
	17		READ – PC1		00h, 08h		0Ah	Pattern A
	18	ACT – PC1	READ – PC0	00h, 0Ah	00h, 0Ch	01555h	15h	Pattern B
	19		READ – PC1		00h, 0Ch		15h	Pattern B
	20	RNOP	READ – PC0	N/A	00h, 00h	N/A	1Ah	Pattern A
	21	RNOP	READ – PC1	N/A	00h, 00h	N/A	1Ah	Pattern A
	22	RNOP	READ – PC0	N/A	00h, 04h	N/A	05h	Pattern B
	23	RNOP	READ – PC1	N/A	00h, 04h	N/A	05h	Pattern B
	24	ACT – PC0	READA – PC0	00h, 0Eh	00h, 0Ch	02AAAh	0Ah	Pattern A
	25		READA – PC1		00h, 0Ch		0Ah	Pattern A
	26	ACT – PC1	READA – PC0	00h, 0Eh	00h, 00h	02AAAh	15h	Pattern B
	27		READA – PC1		00h, 00h		15h	Pattern B
	28	RNOP	READA – PC0	N/A	00h, 04h	N/A	1Ah	Pattern A
	29	RNOP	READA – PC1	N/A	00h, 04h	N/A	1Ah	Pattern A
	30	RNOP	READA – PC0	N/A	00h, 08h	N/A	05h	Pattern B
	31	RNOP	READA – PC1	N/A	00h, 08h	N/A	05h	Pattern B
1	32 to 63	Repeat sub-loop 0 pattern; use RBA0 = 1 and CBA0 = 1 instead; maintain SID[1:0] = 00h						
2	64 to 127	Repeat sub-loops 0 and 1 pattern; use RBA1 = 0 and CBA1 = 1 instead; maintain SID[1:0] = 00h						
	128 + 129	RNOP	CNOP	N/A	N/A	N/A	N/A	N/A
3	130 to 259	For 8-High and 12-High devices: repeat sub-loops 0 to 2 pattern; use SID[1:0] = 01h instead						
4	260 to 389	For 12-High devices: repeat sub-loops 0 to 2 pattern; use SID[1:0] = 10h instead						


Table 127: Maximum Operating Current Ratings

Symbol	Parameter	8-High				12-High				Unit
		-64	-72	-80	-92	-64	-72	-80	-92	
I _{DD0}	One bank activate precharge current	2000	2060	2120	2200	2400	2440	2500	2580	mA
I _{DDQL0}		40	40	40	40	45	45	45	45	mA
I _{PP0}		250	258	264	274	302	311	318	328	mA
I _{DD2P}	Precharge power-down current	1400	1450	1490	1560	1900	1940	1980	2050	mA
I _{DDQL2P}		40	40	40	40	45	45	45	45	mA
I _{PP2P}		200	200	200	200	240	240	240	240	mA
I _{DD2P0}	Precharge power-down current with clock stop	1280	1320	1360	1420	1660	1690	1740	1800	mA
I _{DDQL2P0}		40	40	40	40	45	45	45	45	mA
I _{PP2P0}		200	200	200	200	240	240	240	240	mA
I _{DD2N}	Precharge standby current	1510	1570	1630	1730	1910	1970	2040	2140	mA
I _{DDQL2N}		40	40	40	40	45	45	45	45	mA
I _{PP2N}		200	200	200	200	240	240	240	240	mA
I _{DD3P}	Active power-down current	1530	1580	1630	1700	2010	2050	2100	2180	mA
I _{DDQL3P}		40	40	40	40	45	45	45	45	mA
I _{PP3P}		200	200	200	200	240	240	240	240	mA
I _{DD3P0}	Active power-down current with clock stop	1400	1440	1480	1540	1800	1840	1890	1950	mA
I _{DDQL3P0}		40	40	40	40	45	45	45	45	mA
I _{PP3P0}		200	200	200	200	240	240	240	240	mA
I _{DD3N}	Active standby current	1690	1750	1820	1910	2110	2170	2240	2340	mA
I _{DDQL3N}		40	40	40	40	45	45	45	45	mA
I _{PP3N}		200	200	200	200	240	240	240	240	mA



Symbol	Parameter	8-High				12-High				Unit
		-64	-72	-80	-92	-64	-72	-80	-92	
I _{DD4R}	READ burst current	17700	19400	21100	23700	18600	20400	22500	25600	mA
I _{DDQL4R}		1070	1150	1240	1370	1360	1600	2050	2730	mA
I _{PP4R}		494	522	545	580	610	670	707	763	mA
I _{DD4W}	WRITE burst current	17300	18900	20500	22900	18200	20000	21900	24700	mA
I _{DDQL4W}		597	648	697	770	999	1190	1530	2030	mA
I _{PP4W}		311	325	336	352	486	505	518	539	mA
I _{DD5B}	All-bank refresh burst current	7480	7510	7570	7660	10100	10200	10400	10800	mA
I _{DDQL5B}		40	40	40	40	45	45	45	45	mA
I _{PP5B}		901	906	913	923	1260	1290	1310	1340	mA
I _{DD5P}	Per-bank refresh burst current	8230	8420	8560	8780	10900	11000	11300	11600	mA
I _{DDQL5P}		40	40	40	40	45	45	45	45	mA
I _{PP5P}		934	977	1007	1052	1370	1390	1420	1450	mA
I _{DD6}	Self-refresh current	1890	1930	1980	2050	2720	2790	2820	2870	mA
I _{DDQL6}		40	40	40	40	45	45	45	45	mA
I _{PP6}		230	238	242	250	318	328	335	346	mA
I _{DD7}	All Bank Interleave Read Current	20900	23000	25200	28400	21500	24000	26500	30300	mA
I _{DDQL7}		989	1070	1150	1260	1340	1600	2050	2730	mA
I _{PP7}		930	1010	1090	1210	1090	1200	1330	1520	mA
I _{DD8}	Reset low current	2940	2940	2940	2940	4350	4350	4350	4350	mA
I _{DDQL8}		40	40	40	40	45	45	45	45	mA
I _{PP8}		83	83	83	83	112	112	112	112	mA

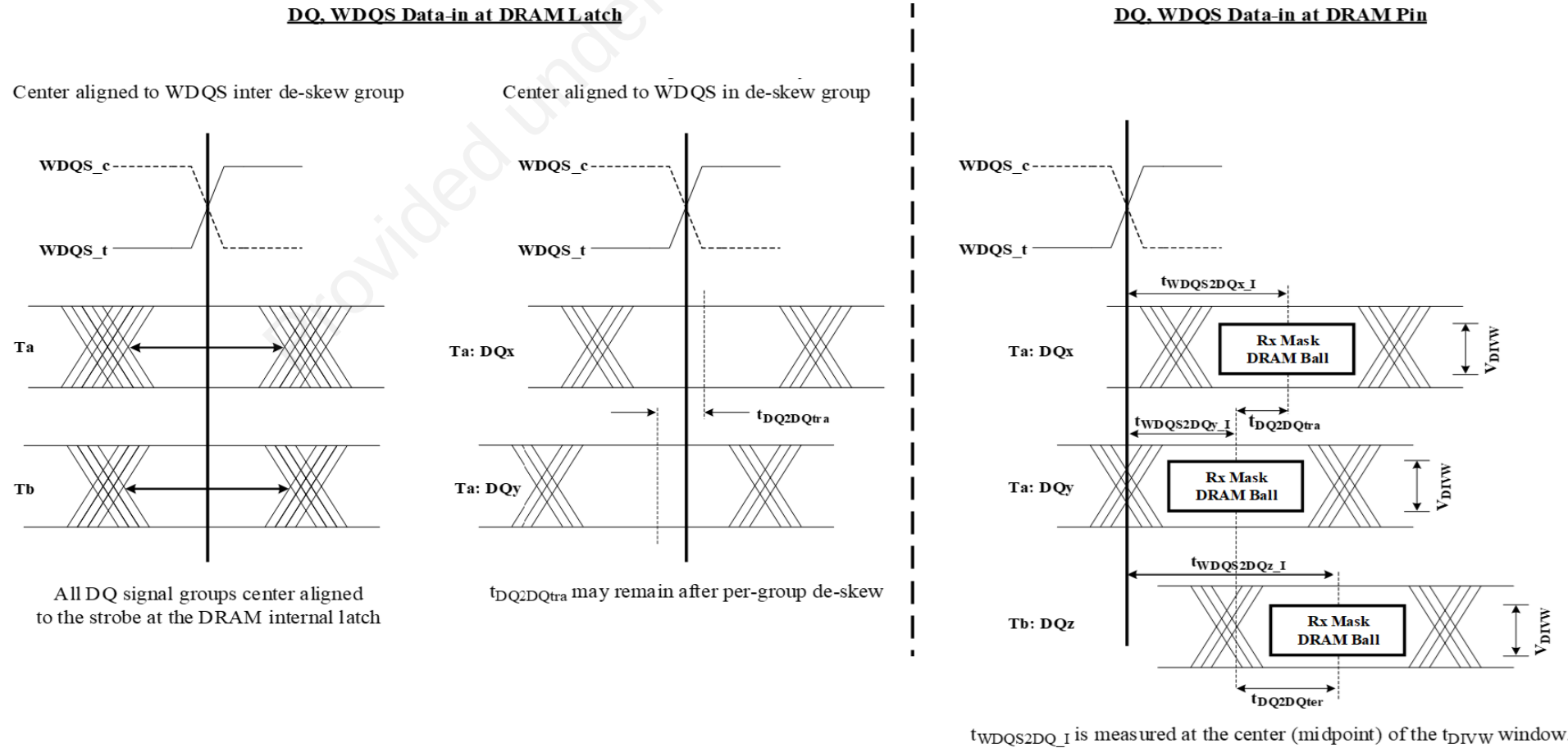


AC Timings

Data Input Timings Considerations

All WDQS-to-DQ timings are measured from WDQS to the center (midpoint) of the 'DIVW window taken at the V_{DIVW} voltage levels centered around $V_{cent_DQ(pin_mid)}$. The timings at the inputs are referenced with respect to all data signal groups center aligned to the DRAM internal latch. The data-to-data offset within a write de-skew group, 'DQ2DQtra_I, is defined as the difference between the minimum and maximum 'WDQS2DQ_I for a given de-skew group. The data-to-data offset between different de-skew groups, 'DQ2DQter_I, is defined as the difference between the minimum and maximum 'WDQS2DQ_I for a whole DWORD.

Provided under NDA

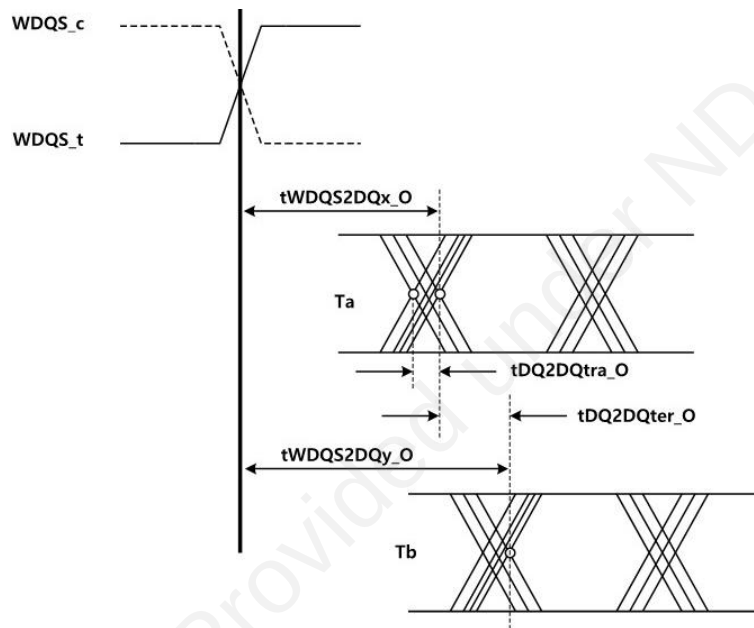
Figure 109: DQ to WDQS Timings at DRAM Inputs


- Notes:
1. DQx and DQy are in the same de-skewing group (Ta), and DQz is in another de-skewing group (Tb) within a DWORD.
 2. DQx represents the max $t_{WDQS2DQ_I}$ in group Ta, and DQy represents the min $t_{WDQS2DQ_I}$ in group Ta, also in DWORD.
 3. DQz represents the max $t_{WDQS2DQ_I}$ in a DWORD.
 4. Timing different between $t_{WDQS2DQx_I}$ and $t_{WDQS2DQy_I}$ represents the max $t_{DQ2DQtra}$.
 5. Timing different between $t_{WDQS2DQy_I}$ and $t_{WDQS2DQz_I}$ represents the max $t_{DQ2DQter}$.

Data Output Timings Considerations

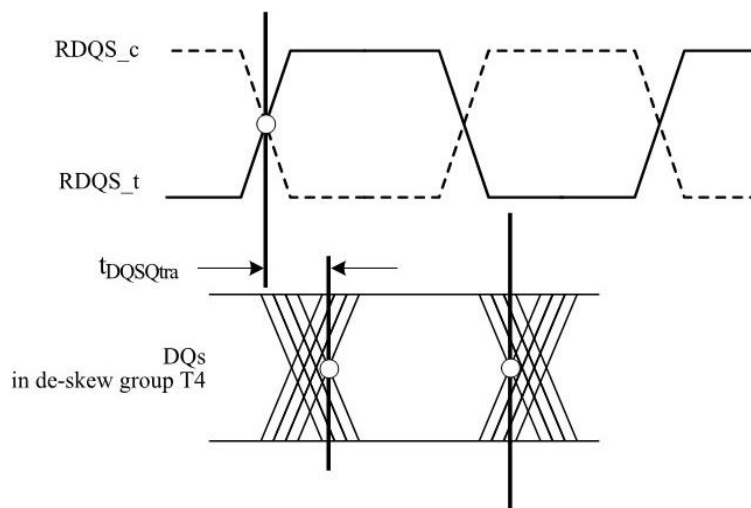
The data-to-data offset within a read de-skew group, $t_{DQ2DQtra_O}$, is defined as the difference between the earliest and the latest valid transition for a given de-skew group. The data-to-data offset between different de-skew groups, $t_{DQ2DQter_O}$, is defined as the difference between the earliest and the latest valid transition for a whole DWORD. $t_{WDQS2DQ_O}$ is defined as the WDQS to read data and RDQS offset.

Figure 110: Read Data Skew Timing Definitions

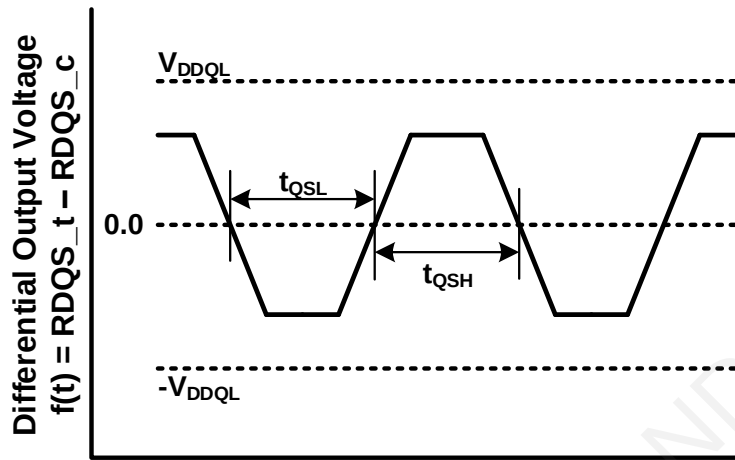


Note: 1. DQx and DQy are in different de-skewing groups (Ta and Tb in this example).

Figure 111: RDQS to Read Data Skew Timing Definitions



Note: 1. $t_{DQSQtra}$ defines the skew between RDQS and the last valid transition of any output in de-skew group T4.


Figure 112: RDQS Differential Output Voltage $f(t) = \text{RDQS}_t - \text{RDQS}_c$




AC Timings Part 1

Table 128: AC Timings Part 1 – Speed Bin Dependent

Symbol	Parameter	-64		-72		-80		-92		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max		
CK Clock Timings											
tCK	CK clock frequency	50	1600	50	1800	50	2000	50	2300	MHz	
tCK	CK period	0.625	20	0.556	20	0.5	20	0.435	20	ns	4
tCH	Absolute CK clock HIGH-level width	0.45	0.55	0.45	0.55	0.45	0.55	0.45	0.55	tCK	40
tCL	Absolute CK clock LOW-level width	0.45	0.55	0.45	0.55	0.45	0.55	0.45	0.55	tCK	40
fCKDCA	CK clock frequency with DCA enabled	TBD	—	1600	—	1600	—	1600	—	MHz	36
Command / Address Input Timings											
tIS	Command and address input setup time	69	—	61	—	55	—	48	—	ps	5
tIH	Command and address input hold time	69	—	61	—	55	—	48	—	ps	5
tCIPW	Command and address input pulse width	219	—	194	—	175	—	152	—	ps	27
WDQS Timings											
tDQSS	WDQS/2 (0° phase) rising edge to CK rising edge phase offset	MAX(-200, -0.2 × tCK)	MIN(200, 0.2 × tCK)	MAX(-200, -0.2 × tCK)	MIN(200, 0.2 × tCK)	MAX(-200, -0.2 × tCK)	MIN(200, 0.2 × tCK)	MAX(-200, -0.2 × tCK)	MIN(200, 0.2 × tCK)	ps	
tWDQS2CK	WDQS-to-CK phase search range during WDQS-to-CK alignment training	-0.4	0.4	-0.4	0.4	-0.4	0.4	-0.4	0.4	tCK	



Symbol	Parameter	-64		-72		-80		-92		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max		
^t WDQS2PD	CK clock to phase detector output delay in WDQS-to-CK alignment training mode	—	5	—	5	—	5	—	5	ns	
^t WDQS	WDQS frequency	100	3200	100	3600	100	4000	100	4600	MHz	
^t WDQS	WDQS period	0.3125	10	0.278	10	0.25	10	0.2175	10	ns	41
^t WQSH(avg)	Average WDQS differential input HIGH pulse width	0.47	0.53	0.47	0.53	0.47	0.53	0.47	0.53	^t WDQS	42
^t WQSL(avg)	Average WDQS differential input LOW pulse width	0.47	0.53	0.47	0.53	0.47	0.53	0.47	0.53	^t WDQS	42
^t WQSH(abs)	Absolute WDQS differential input HIGH pulse width	0.45	0.55	0.45	0.55	0.45	0.55	0.45	0.55	^t WDQS	43
^t WQSL(abs)	Absolute WDQS differential input LOW pulse width	0.45	0.55	0.45	0.55	0.45	0.55	0.45	0.55	^t WDQS	43
^t WPRE1	WDQS preamble for writes	2		2		2		2		nWDQS	
^t WPRE2	WDQS preamble for reads	4		4		4		4		nWDQS	
^t WPST1	WDQS postamble for writes	2		2		2		2		nWDQS	
^t WPST2	WDQS postamble for reads	2		2		2		2		nWDQS	
^t RPRE	RDQS preamble for reads	2		2		2		2		nWDQS	
^t RPST	RDQS postamble for reads	2		2		2		2		nWDQS	
Data Input Timings											
^t DIPW	Data input single pulse width	0.55	—	TBD	—	TBD	—	TBD	—	UI	27
^t DIVW	Data input valid window with PSIJ	0.30	—	TBD	—	TBD	—	TBD	—	UI	
^t WDQS2DQ_I	WDQS to write data offset	275	750	275	750	275	750	275	750	ps	33, 34, 45
^t WDQS2DQ_I_VOLT	WDQS to write data offset voltage variation	—	0.5	—	0.5	—	0.5	—	0.5	ps/mV	34, 46



Symbol	Parameter	-64		-72		-80		-92		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max		
^t WDQS2DQ_I_TEMP	WDQS to write data offset temperature variation	—	0.35	—	0.35	—	0.35	—	0.35	ps/°C	34, 46, 47
^t DQ2DQtra_I	DQ-to-DQ intra-group skew for writes	—	10	—	10	—	10	—	10	ps	
^t DQ2DQter_I	DQ-to-DQ inter-group skew for writes	—	30	—	30	—	30	—	30	ps	
Data Output Timings											
^t QSH	RDQS differential output HIGH time	^t WQSH(abs) - 0.08	—	^t WQSH(abs) - 0.08	—	^t WQSH(abs) - 0.08	—	^t WQSH(abs) - 0.08	—	^t WDQS	6, 38
^t QSL	RDQS differential output LOW time	^t WQSL(abs) - 0.08	—	^t WQSL(abs) - 0.08	—	^t WQSL(abs) - 0.08	—	^t WQSL(abs) - 0.08	—	^t WDQS	6, 39
^t QH	DQ, DBI, ECC, SEV output hold time	MIN(^t QSH, ^t QSL)	—	MIN(^t QSH, ^t QSL)	—	MIN(^t QSH, ^t QSL)	—	MIN(^t QSH, ^t QSL)	—	^t WDQS	6
^t QW	DQ, DBI, ECC, SEV output window per pin	MIN(^t QSH, ^t QSL) - 0.07	—	MIN(^t QSH, ^t QSL) - 0.07	—	MIN(^t QSH, ^t QSL) - 0.07	—	MIN(^t QSH, ^t QSL) - 0.07	—	UI	6, 31
^t DQSQtra	RDQS-to-DQ skew in byte T4	—	20	—	20	—	20	—	20	ps	6
^t WDQS2DQ_O	WDQS to read data and RDQS offset	500	1000	500	1000	500	1000	500	1000	ps	6, 33, 35
^t WDQS2DQ_O_VOLT	WDQS to read data and RDQS offset voltage variation	—	2.5	—	2.5	—	2.5	—	2.5	ps/mV	6, 35, 46
^t WDQS2DQ_O_TEMP	WDQS to read data and RDQS offset temperature variation	—	0.6	—	0.6	—	0.6	—	0.6	ps/°C	6, 35, 46, 47
^t DQ2DQtra_O	DQ-to-DQ intra-group skew for reads	—	10	—	10	—	10	—	10	ps	6
^t DQ2DQter_O	DQ-to-DQ inter-group skew for reads	—	30	—	30	—	30	—	30	ps	6



Symbol	Parameter	-64		-72		-80		-92		Unit	Notes
		Min	Max	Min	Max	Min	Max	Min	Max		
t _{LZ}	DQ, DBI, ECC high impedance to low impedance time from WDQS	Min: t _{WDQS2DQ_O} (MIN) - t _{QH} (MIN) Max: t _{WDQS2DQ_O} (MAX) + t _{DQSQtra} (MAX)								ns	6
t _{HZ}	DQ, DBI, ECC low impedance to high impedance time from WDQS	Min: t _{WDQS2DQ_O} (MIN) Max: t _{WDQS2DQ_O} (MAX) + t _{DQSQtra} (MAX)								ns	6



AC Timings Part 2

Table 129: AC Timings Part 2 – All Speed Bins

Symbol	Parameter	Min	Max	Unit	Notes
Latency Timings					
^f CKRL86	CK clock frequency with RL = 86	2150	2300	MHz	
^f CKRL85	CK clock frequency with RL = 85	2150	2300	MHz	
^f CKRL84	CK clock frequency with RL = 84	2130	2300	MHz	
^f CKRL83	CK clock frequency with RL = 83	2100	2300	MHz	
^f CKRL82	CK clock frequency with RL = 82	2075	2300	MHz	
^f CKRL81	CK clock frequency with RL = 81	2050	2300	MHz	
^f CKRL80	CK clock frequency with RL = 80	2020	2300	MHz	
^f CKRL79	CK clock frequency with RL = 79	1990	2300	MHz	
^f CKRL78	CK clock frequency with RL = 78	1960	2300	MHz	
^f CKRL77	CK clock frequency with RL = 77	1930	2300	MHz	
^f CKRL76	CK clock frequency with RL = 76	1900	2300	MHz	
^f CKRL75	CK clock frequency with RL = 75	1875	2300	MHz	
^f CKRL74	CK clock frequency with RL = 74	1850	2300	MHz	
^f CKRL73	CK clock frequency with RL = 73	1820	2300	MHz	
^f CKRL72	CK clock frequency with RL = 72	1790	2300	MHz	
^f CKRL71	CK clock frequency with RL = 71	1760	2300	MHz	
^f CKRL70	CK clock frequency with RL = 70	1730	2300	MHz	
^f CKRL69	CK clock frequency with RL = 69	1700	2300	MHz	
^f CKRL68	CK clock frequency with RL = 68	1675	2250	MHz	
^f CKRL67	CK clock frequency with RL = 67	1645	2200	MHz	
^f CKRL66	CK clock frequency with RL = 66	1615	2175	MHz	
^f CKRL65	CK clock frequency with RL = 65	1585	2150	MHz	
^f CKRL64	CK clock frequency with RL = 64	1560	2125	MHz	
^f CKRL63	CK clock frequency with RL = 63	1530	2100	MHz	
^f CKRL62	CK clock frequency with RL = 62	1500	2050	MHz	
^f CKRL61	CK clock frequency with RL = 61	1470	2000	MHz	
^f CKRL60	CK clock frequency with RL = 60	1445	1950	MHz	
^f CKRL59	CK clock frequency with RL = 59	1415	1925	MHz	
^f CKRL58	CK clock frequency with RL = 58	1385	1900	MHz	
^f CKRL57	CK clock frequency with RL = 57	1360	1850	MHz	



Symbol	Parameter	Min	Max	Unit	Notes
^f CKRL56	CK clock frequency with RL = 56	1330	1800	MHz	
^f CKRL55	CK clock frequency with RL = 55	1300	1750	MHz	
^f CKRL54	CK clock frequency with RL = 54	1270	1725	MHz	
^f CKRL53	CK clock frequency with RL = 53	1245	1700	MHz	
^f CKRL52	CK clock frequency with RL = 52	1215	1650	MHz	
^f CKRL51	CK clock frequency with RL = 51	1185	1600	MHz	
^f CKRL50	CK clock frequency with RL = 50	1155	1575	MHz	
^f CKRL49	CK clock frequency with RL = 49	1130	1550	MHz	
^f CKRL48	CK clock frequency with RL = 48	1100	1500	MHz	
^f CKRL47	CK clock frequency with RL = 47	1070	1475	MHz	
^f CKRL46	CK clock frequency with RL = 46	1045	1450	MHz	
^f CKRL45	CK clock frequency with RL = 45	1015	1400	MHz	
^f CKRL44	CK clock frequency with RL = 44	985	1350	MHz	
^f CKRL43	CK clock frequency with RL = 43	955	1325	MHz	
^f CKRL42	CK clock frequency with RL = 42	930	1300	MHz	
^f CKRL41	CK clock frequency with RL = 41	900	1250	MHz	
^f CKRL40	CK clock frequency with RL = 40	870	1200	MHz	
^f CKRL39	CK clock frequency with RL = 39	845	1175	MHz	
^f CKRL38	CK clock frequency with RL = 38	815	1150	MHz	
^f CKRL37	CK clock frequency with RL = 37	785	1100	MHz	
^f CKRL36	CK clock frequency with RL = 36	755	1075	MHz	
^f CKRL35	CK clock frequency with RL = 35	730	1050	MHz	
^f CKRL34	CK clock frequency with RL = 34	700	1000	MHz	
^f CKRL33	CK clock frequency with RL = 33	670	975	MHz	
^f CKRL32	CK clock frequency with RL = 32	645	950	MHz	
^f CKRL31	CK clock frequency with RL = 31	615	900	MHz	
^f CKRL30	CK clock frequency with RL = 30	585	875	MHz	
^f CKRL29	CK clock frequency with RL = 29	560	850	MHz	
^f CKRL28	CK clock frequency with RL = 28	530	800	MHz	
^f CKRL27	CK clock frequency with RL = 27	500	750	MHz	
^f CKRL26	CK clock frequency with RL = 26	475	725	MHz	
^f CKRL25	CK clock frequency with RL = 25	445	700	MHz	
^f CKRL24	CK clock frequency with RL = 24	415	675	MHz	
^f CKRL23	CK clock frequency with RL = 23	385	650	MHz	
^f CKRL22	CK clock frequency with RL = 22	360	600	MHz	



Symbol	Parameter	Min	Max	Unit	Notes
^f CKRL21	CK clock frequency with RL = 21	330	550	MHz	
^f CKRL20	CK clock frequency with RL = 20	50	525	MHz	
^f CKRL19	CK clock frequency with RL = 19	50	500	MHz	
^f CKRL18	CK clock frequency with RL = 18	50	450	MHz	
^f CKRL17	CK clock frequency with RL = 17	50	400	MHz	
^f CKRL16	CK clock frequency with RL = 16	50	340	MHz	
^f CKRL15	CK clock frequency with RL = 15	50	310	MHz	
^f CKRL14	CK clock frequency with RL = 14	50	280	MHz	
^f CKWL14	CK clock frequency with WL = 14	1850	2300	MHz	
^f CKWL13	CK clock frequency with WL = 13	1420	2000	MHz	
^f CKWL12	CK clock frequency with WL = 12	1220	1420	MHz	
^f CKWL11	CK clock frequency with WL = 11	965	1220	MHz	
^f CKWL10	CK clock frequency with WL = 10	820	965	MHz	
^f CKWL9	CK clock frequency with WL = 9	650	820	MHz	
^f CKWL8	CK clock frequency with WL = 8	510	650	MHz	
^f CKWL7	CK clock frequency with WL = 7	425	510	MHz	
^f CKWL6	CK clock frequency with WL = 6	50	425	MHz	
Row Access Timings					
^t RC	ACTIVATE to ACTIVATE command period	45	—	ns	
^t RAS	ACTIVATE to PRECHARGE command period	30	9 × ^t REFI (MAX)	ns	7
^t RCDRD	ACTIVATE to READ command delay	17	—	ns	
^t RCDWR	ACTIVATE to WRITE command delay	MAX(1 × ^t CK, ^t RCDRD (MIN) - WL × ^t CK)	—	ns	
^t RRDL	ACTIVATE bank A to ACTIVATE or PER-BANK REFRESH bank B command delay same bank group	MAX(3 × ^t CK, 2)	—	ns	8
^t RRDS	ACTIVATE bank A to ACTIVATE or PER-BANK REFRESH bank B command delay different bank groups	MAX(3 × ^t CK, 2)	—	ns	9
^t FAW	Four bank activate window	4 × ^t RRDS (MIN)	—	ns	10
^t RTP	READ to PRECHARGE command delay	MAX(4 × ^t CK, 4)	—	ns	11, 32
^t RP	PRECHARGE command period	MAX (1 × ^t CK, 15)	—	ns	
^t WR	WRITE recovery time	20	—	ns	
^t DAL	Auto precharge write recovery + precharge time	—	—	nCK	12
^t PPD	PRECHARGE to PRECHARGE command delay	0.5	—	nCK	



Symbol	Parameter	Min	Max	Unit	Notes
Column Access Timings					
t_{CCDL}	READ/WRITE bank A to READ/WRITE bank B command delay same bank group	4	—	nCK	13, 14
t_{CCDS}	READ/WRITE bank A to READ/WRITE bank B command delay different bank groups	2	—	nCK	15, 16
t_{CCDR}	READ SID A to READ SID B command delay	2	—	nCK	17
t_{WTRL}	Internal WRITE to READ command delay same bank group	$4 \times t_{CK} + 4$	—	ns	13
t_{WTRS}	Internal WRITE to READ command delay different bank groups	$2 \times t_{CK} + 4$	—	ns	15
t_{RTW}	READ to WRITE command delay	(see note)	—	nCK	18
Power-Down Timings					
t_{PD}	Power-down entry to exit time	$t_{CPDED} + 6 \times t_{CK}$	$9 \times t_{REFI} \text{ (MAX)}$	ns	
t_{XP}	Power-down exit time	$\text{MAX}(10, \text{RU}(7.5\text{ns} / t_{CK}))$	—	nCK	
t_{CKPDE}	Valid CK clocks required after power-down entry	$\text{RU}(t_{CPDED} / t_{CK}) + 1$	—	nCK	
t_{CKPDX}	Valid CK clocks required before power-down exit	5	—	nCK	
t_{CPDED}	Command path disable delay	$\text{MAX}(5 \times t_{CK}, 7.5)$	—	ns	
t_{ACTPDE}	ACTIVATE to POWER-DOWN ENTRY command delay	1	—	nCK	
t_{PRPDER}	PRECHARGE (rising CK edge) to POWER-DOWN-ENTRY command delay	1	—	nCK	
t_{PRPDEF}	PRECHARGE (falling CK edge) to POWER-DOWN-ENTRY command delay	1.5	—	nCK	
$t_{REFabPDE}$	REFRESH to POWER-DOWN ENTRY command delay	1	—	nCK	
$t_{REFpbPDE}$	PER-BANK REFRESH to POWER-DOWN ENTRY command delay	1	—	nCK	
t_{MRSPDE}	MODE REGISTER SET to POWER-DOWN ENTRY command delay	$t_{MOD} \text{ (MIN)}$	—	nCK	
t_{RDPDE}	READ or READ w/ AP to POWER-DOWN ENTRY command delay	$\text{RL} + \text{PL} + 2 + \text{RU}((\text{DQSS} \text{ (MAX)} + t_{WDQS2DQ_O} \text{ (MAX)}) / t_{CK})$	—	nCK	
t_{WRPDE}	WRITE to POWER-DOWN-ENTRY command delay	$\text{WL} + \text{PL} + 3 + \text{MAX}(\text{RU}(t_{WR} / t_{CK}), 6)$	—	nCK	20
t_{WRAPDE}	WRITE w/ AP to POWER-DOWN-ENTRY command delay	$\text{WL} + \text{PL} + 3 + \text{MAX}(\text{WR}, 6)$	—	nCK	21
Self Refresh Timings					



Symbol	Parameter		Min	Max	Unit	Notes
tCKSR	Self refresh entry to exit time		tCPDED + 6 × tCK	—	ns	
tCKSRE	Valid CK clocks required after self refresh entry		RU(tCPDED/CK) + 1	—	nCK	
tCKSRX	Valid CK clocks required before self refresh exit		5	—	nCK	
tRDSRE	READ or READ w/ AP to SELF REFRESH ENTRY command delay		RL + PL + 3	—	nCK	
tXS	Exit self refresh command delay		MAX(10 × tCK, tRFCab (MIN) +10)	—	ns	
tXSMRSF	Exit self refresh to MODE REGISTER SET command delay after a frequency change		tXSMRS (MIN)	—	ns	
tXSMRS	Exit self refresh to MODE REGISTER SET command delay		MAX(10 × tCK, 15)	—	ns	
Refresh Timings						
tRAASRF	Minimum time in self refresh for per-bank RAA count to be reset to 0			1	ms	29
tRFCab	REFRESH command period	8-High	410	—	ns	22
		12-High	450	—	ns	
tRFCpb	PER-BANK REFRESH command period (all densities)		240	—	ns	
tRREFD	PER-BANK REFRESH command period (different bank) and PER-BANK REFRESH to ACTIVATE (different bank) command delay		MAX(3 × tCK, 8)	—	ns	
tREFI	Average periodic refresh interval for REFRESH command	T ≤ +95°C	—	3.9	μs	23
		+95°C < T ≤ +105°C	—	1.95	μs	
tREFIpb	Average periodic refresh interval for PER-BANK REFRESH command	8-High	—	tREFI (MAX) / 32	μs	22, 24
		12-High	—	tREFI (MAX) / 48	μs	
Miscellaneous Timings						
tMOD	MODE REGISTER SET command update delay		10	—	nCK	
tMRD	MODE REGISTER SET command period		10	—	nCK	
tRDMRS	READ command to MODE REGISTER SET command delay		RL + PL + 2 + RU((tDQSS (MAX) + tWDQS2DQ_O (MAX)) / tCK)	—	nCK	
tWRMRS	Internal WRITE to MODE REGISTER SET command delay		MAX(RU((tWR + tRP) / tCK), PL + RU(tPARDQ / tCK), 7)	—	nCK	
tDCMM	Duty Cycle Monitor Measurement time		1	—	μs	37
tVREFCA	Internal VREFCA offset single step settling time		—	300	ns	48



Symbol	Parameter	Min	Max	Unit	Notes
$t_{FVREFCA}$	Internal VREFCA offset full range settling time	—	300	ns	48
t_{VREFD}	Internal VREFD offset single step settling time	—	300	ns	49
t_{FVREFD}	Internal VREFD offset full range settling time	—	300	ns	49
t_{PARAC}	Rising CK clock to ADD/CMD parity error output delay	$4 \times t_{CK} + 0.8$	$4 \times t_{CK} + 2.0$	ns	25
t_{PARDQ}	Rising CK clock to write data parity error output delay	$7 \times t_{CK} + 1.6$	$7 \times t_{CK} + 3.5$	ns	26

Notes:

- AC timing parameters apply to each channel of the HBM3 DRAM independently. No timing parameters are specified across channels, and all channels operate independently of each other.
- (blank)
- All parameters assume proper device initialization.
- Parameter t_{CK} is calculated as the average clock period across any consecutive 1,000 cycle window, where each clock period is calculated both from rising CK_t edge to rising CK_t edge and falling CK_t edge to falling CK_t edge.
- Parameter is based on V_{IHCA} and V_{ILCA} levels.
- Parameter is measured with the output timing reference load and Read DBI enabled.
- For Reads and Writes with auto precharge enabled the device will hold off the internal precharge until $t_{RAS(min)}$ has been satisfied.
- Parameter applies when consecutive commands access the same bank group.
- Parameter applies when consecutive commands access different bank groups.
- Not more than four ACTIVATE or PER-BANK REFRESH commands are allowed within t_{FAW} period.
- Parameter applies when READ and PRECHARGE commands access the same bank.
- $t_{DAL} = (t_{WR}/t_{CK}) + (t_{RP}/t_{CK})$. For each of the terms, if not already an integer, round up to the next integer.
- Parameter applies when consecutive commands access the same bank group.
- t_{CCDL} is either for seamless consecutive READ or seamless consecutive WRITE commands.
- Parameter applies when consecutive commands access different bank groups.
- t_{CCDS} is either for seamless consecutive READ or seamless consecutive WRITE commands.
- t_{CCDR} is for 8-High or 12-High HBM3 DRAMs and should be used for seamless consecutive READ commands between different stack IDs (SID) instead of t_{CCDS} . For seamless WRITE commands the normal t_{CCDS} parameter applies. t_{CCDR} does not apply to DWORD MISR operations when DWORD Loopback is enabled in MR7.
- t_{RTW} is not a DRAM device limit but determined by the system bus turnaround time. Avoid bus contention by setting $t_{RTW(MIN)} = (RL + 2 - WL + 0.5) \times t_{CK} + t_{WDQS2DQ_O(MAX)} - t_{WDQS2DQ_I(MIN)}$, and round up to the next integer.
- (blank)
- t_{WR} is defined in ns. For calculation of t_{WRPDE} calculate t_{WR}/t_{CK} and round up to the next integer.
- t_{WR} in clock cycles as programmed in MR3.
- Density is given per device.
- A maximum of 8 consecutive REFRESH commands can be posted to an HBM3 DRAM, meaning that the maximum absolute interval between any REFRESH command and the next REFRESH command is $9 \times t_{REFI}$. The same postponement applies to PRE-BANK REFRESH commands.
- $t_{REFIpb} = t_{REFI} / N$; N = no. of banks.
- The nominal AERR HIGH time in case of a parity error is 1 nCK.
- The nominal DERR HIGH time in case of a parity error is 1 nCK.
- t_{CIPW} is based on V_{REFCA} level, and t_{DIPW} is based on V_{REFDQ} level.
- (blank)
- Parameter applies only to HBM3 DRAMs that require the use of Refresh Management (RFM).
- (blank)
- This parameter is a function of the WDQS input clock jitter.
- PRECHARGE and PRECHARGE ALL commands can be issued on a rising or a falling CK edge.
- PVT variation is included.
- The minimum-to-maximum range does not exceed 400ps.



35. The minimum-to-maximum range does not exceed 1.5ns.
36. Parameter 'CKDCA applies when a duty correction code other than the default 0000 is programmed in the mode register.
37. 'DCMM is measured from the MRS command that enables the duty cycle measurement until the measurement result is valid.
38. 'QSH describes the instantaneous differential output HIGH pulse width on RDQS_t – RDQS_c as it measures the next falling edge from an arbitrary rising edge. 'QSH edge measurement is based on zero voltage. See Figure 112.
39. 'QSL describes the instantaneous differential output HIGH pulse width on RDQS_t – RDQS_c as it measures the next falling edge from an arbitrary rising edge. 'QSL edge measurement is based on zero voltage. See Figure 112.
40. 'CH is the absolute instantaneous clock high pulse width, as measured from one rising edge to the following falling edge. 'CL is the absolute instantaneous clock low pulse width, as measured from one falling edge to the following rising edge.
41. Parameter 'WDQS is calculated as the average write clock period across any consecutive 200 cycle window.
42. 'WQSH(avg) is defined as the average high pulse width as calculated across any consecutive 200 high pulses. 'WQSL(avg) is defined as the average low pulse width as calculated across any consecutive 200 low pulses.
43. 'WQSH(abs) is the absolute instantaneous write clock high pulse width, as measured from one rising edge to the following falling edge. 'WQSL(abs) is the absolute instantaneous write clock low pulse width, as measured from one falling edge to the following rising edge.
44. (blank)
45. 'WDQS2DQ_I max delay variation as a function of the DC voltage variation for V_{DDQ}. It includes V_{DDQ} AC noise impact for frequencies > 20MHz and max voltage of TBD mVpk-pk from DC to 20MHz at a fixed temperature on the package.
46. Actual values could be positive or negative numbers depending on design implementation and process.
47. The parameter is referenced to the IEEE1500 TEMPERATURE instruction.
48. The parameter applies when the VREFCA level has been changed in MR14. 'VREFCA and 'FVREFCA are referenced from the MRS command to when the 90% level of the delta between old and new V_{REFCA} voltage has been reached.
49. The parameter applies when the VREFD level has been changed in MR15. 'VREFD and 'FVREFD are referenced from the MRS command to when the 90% level of the delta between old and new V_{REFD} voltage has been reached.

Microbump Matrix

The microbump matrix of the HBM3 DRAM employs a staggered pattern as depicted in Figure 113 where a 'staggered' bump is located halfway between major row and column, hence its location is determined by $X/2$ and $Y/2$. Table 130 shows geometric parameters of the staggered micro bump pattern. Parameter P_{MIN} is the minimum bump pitch anywhere in the microbump field.

Figure 113: Staggered Microbump Pattern

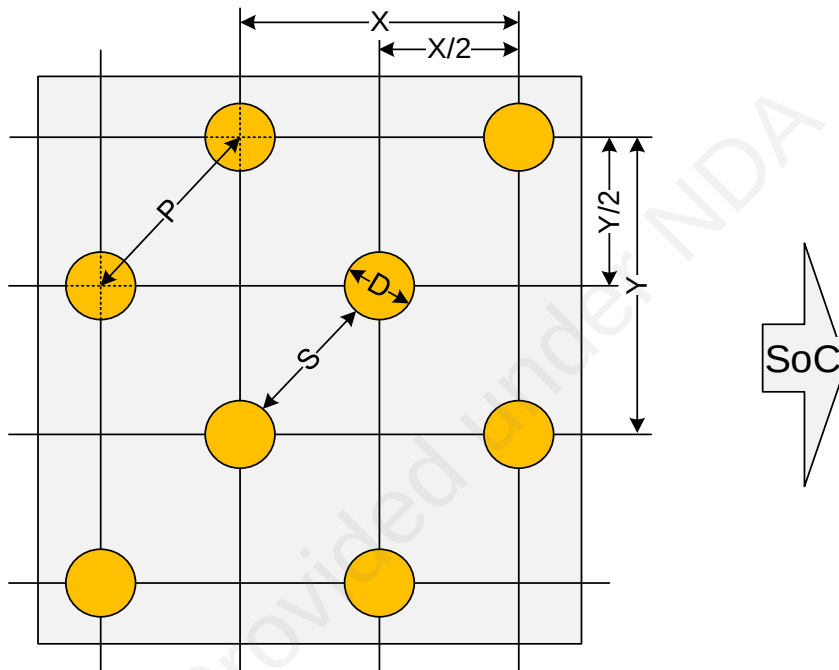


Table 130: Geometric Parameters of the Staggered Microbump Pattern

Label	Nominal Value	Description	
X	96μm	Horizontal pitch of two adjacent micro bumps	
Y	110μm	Vertical pitch of two adjacent micro bumps	
P_{MIN}	73μm	Minimum pitch of the bump field	
D	28μm	Microbump diameter	
S	$P_{MIN} - D$	Bump-to-bump air gap	

The bump matrix consists of a total of 7775 bumps arranged in 148 columns with a pitch of $X/2$ and 161 rows with a pitch of $Y/2$. The overall array size is $(147 \times X/2 + D)\mu\text{m} \times (160 \times Y/2 + D)\mu\text{m} = 7084\mu\text{m} \times 8828\mu\text{m}$.

The ball matrix is center aligned with the die. The ball array center is the origin of the ball location coordinates. Ball A1 is located at the top left at $X = -3528\mu\text{m}$, $Y = +4400\mu\text{m}$.

A geographical overview of the microbump matrix is provided in Figure 114, and the detailed bump matrices are provided in Table 132 (Part 1) thru Table 136 (Part 5). Due to space constraints these tables use abbreviations for specific functions as given in Table 131.



The orientation of the ballout shown is the bottom view looking at the microbumps.

Table 131: Legend

VPP	A	VSS	D
NC	B	VDDC	E
No Bump	C	VDDQ	M
		VDDQL	N

Figure 114: HBM3 Ballout – Geographical Overview (not to scale)

		Part 1		Part 2		Part 3	Part 4		Part 5			
Columns		1, 2	3 ... 20	21 ... 36	37 ... 51	52 ... 74	75 ... 90	91 ... 104	105 ... 118	119 ... 132	133 ... 146	147, 148
R o w s	A ... K	Mechanical Bumps	Power Supply Region	Depopulated Area for Probe Pads	Direct Access Test Port	Upper Left Power Supply Region	Upper Right Power Supply Region	Upper Right Power Supply Region				Mechanical Bumps
	L ... AD							DWORD0 Channel m	DWORD0 Channel i	DWORD0 Channel e	DWORD0 Channel a	
	AE ... AK							AWORD Channel m	AWORD Channel i	AWORD Channel e	AWORD Channel a	
	AL ... BD							DWORD1 Channel m	DWORD1 Channel i	DWORD1 Channel e	DWORD1 Channel a	
	BE ... BV							DWORD0 Channel n	DWORD0 Channel j	DWORD0 Channel f	DWORD0 Channel b	
	BW ... CD							AWORD Channel n	AWORD Channel j	AWORD Channel f	AWORD Channel b	
	CE ... CT							DWORD1 Channel n	DWORD1 Channel j	DWORD1 Channel f	DWORD1 Channel b	
	CU, CV					Depopulated Area for Probe Pads		Reset, IEEE1500 Port, Temperature, etc...				
	CW ... DC							DWORD1 Channel o	DWORD1 Channel k	DWORD1 Channel g	DWORD1 Channel c	
	DD, DE							AWORD Channel o	AWORD Channel k	AWORD Channel g	AWORD Channel c	
	DF ... DU							DWORD0 Channel o	DWORD0 Channel k	DWORD0 Channel g	DWORD0 Channel c	
	DW ... EC					Lower Left Power Supply Region	Lower Right Power Supply Region	DWORD1 Channel p	DWORD1 Channel l	DWORD1 Channel h	DWORD1 Channel d	
	ED ... EU							AWORD Channel p	AWORD Channel l	AWORD Channel h	AWORD Channel d	
	EV ... FL							DWORD0 Channel p	DWORD0 Channel l	DWORD0 Channel h	DWORD0 Channel d	
	FM ... FU							Lower Right Power Supply Region				
	FV ... GL											
	GM ... HA											


Table 132: HBM3 Ballout – Part 1 (Columns 1 to 36)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
A	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
B		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
C	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
D		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
E	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
F		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
G	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
H		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
J	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
K		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
L	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
M		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
N	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
P		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
R	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
T		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
U	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
V		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
W	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
Y		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
AA	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
AB		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
AC	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
AD		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
AE	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
AF		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
AG	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
AH		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
AJ	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
AK		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
AL	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
AM		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
AN	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
AP		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
AR	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
AT		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
AU	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
AV		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
AW	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
AY		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C


Table 132: HBM3 Ballout – Part 1 (Columns 1 to 36) (cont'd)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	
BA	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
BB		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
BC	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
BD		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
BE	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
BF		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
BG	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
BH		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
BJ	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
BK		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
BL	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
BM		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
BN	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
BP		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
BR	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
BT		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
BU	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
BV		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
BW	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
BY		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
CA	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
CB		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
CC	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
CD		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
CE	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
CF		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
CG	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
CH		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
CJ	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
CK		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
CL	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
CM		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
CN	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
CP		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
CR	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		
CT		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	


Table 132: HBM3 Ballout – Part 1 (Columns 1 to 36) (cont'd)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
CU	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
CV		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
CW	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
CY		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
DA	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
DB		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
DC	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	
DD		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C
DE	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C	


Table 132: HBM3 Ballout – Part 1 (Columns 1 to 36) (cont'd)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	
DF		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
DG	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
DH		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
DJ	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
DK		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
DL	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
DM		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
DN	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
DP		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
DR	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
DT		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
DU	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
DV		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
DW	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
DY		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
EA	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
EB		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
EC	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
ED		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
EE	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
EF		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
EG	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
EH		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
EJ	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
EK		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
EL	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
EM		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
EN	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
EP		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
ER	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
ET		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
EU	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
EV		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
EW	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
EY		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
FA	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
FB		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
FC	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C


Table 132: HBM3 Ballout – Part 1 (Columns 1 to 36) (cont'd)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	
FD		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
FE	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
FF		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
FG	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
FH		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
FJ	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
FK		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
FL	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
FM		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
FN	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
FP		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
FR	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
FT		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
FU	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
FV		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
FW	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
FY		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
GA	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
GB		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
GC	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
GD		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
GE	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
GF		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
GG	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
GH		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
GJ	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
GK		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
GL	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
GM		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
GN	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
GP		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
GR	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
GT		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
GU	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
GV		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
GW	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C
GY		B		D		C		D		D		C		E		D		C		E		C		C		C		C		C		C		C		C	
HA	B		C		D		D		C		D		E		C		D		E		C		C		C		C		C		C		C		C		C


Table 133: HBM3 Ballout – Part 2 (Columns 37 to 74)

	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74
A	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
B		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
C	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
D		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
E	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
F		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
G	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
H		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
J	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
K		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
L	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
M		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
N	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
P		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
R	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
T		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
U	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
V		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
W	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
Y		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
AA	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
AB		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
AC	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
AD		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
AE	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
AF		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
AG	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
AH		D		B		C		B		B		C		B		E		C		D		A		C		D		E		C		D		N		C		D
AJ	D		C		B		B		C		B		B		C		E		D		C		A		D		C		E		D		C		N		D	
AK		D		C		C		C		C		C		C		E		C		D		A		C		D		E		C		D		N		C		D
AL	D		C		D		E		C		D		E		C		E		D		C		A		D		C		E		D		C		N		D	
AM		D		D		C		E		D		C		E		E		C		D		A		C		D		E		C		D		N		C		D
AN	D		C		D		E		C		D		E		C		E		D		C		A		D		C		E		D		C		N		D	
AP		D		D		C		E		D		C		E		E		C		D		A		C		D		E		C		D		N		C		D
AR	D		C		C		C		C		C		C		C		E		D		C		A		D		C		E		D		C		N		D	
AT		D		DA0		C		DA1		B		C		B		E		C		D		A		C		D		E		C		D		N		C		D
AU	D		C		DA0		DA1		C		B		B		C		E		D		C		A		D		C		E		D		C		N		D	
AV		D		C		C		C		C		C		C		E		C		D		A		C		D		E		C		D		N		C		D
AW	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
AY		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D


Table 133: HBM3 Ballout – Part 2 (Columns 37 to 74) (cont'd)

	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74
BA	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
BB		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
BC	D		C		C		C		C		C		C		C		E		D		C		A		D		C		E		D		C		N		D	
BD		D		DA2		C		DA3		DA4		C		DA5		E		C		D		A		C		D		E		C		D		N		C		D
BE	D		C		DA2		DA3		C		DA4		DA5		C		E		D		C		A		D		C		E		D		C		N		D	
BF		D		C		C		C		C		C		C		E		C		D		A		C		D		E		C		D		N		C		D
BG	D		C		D		E		C		D		E		C		E		D		C		A		D		C		E		D		C		N		D	
BH		D		D		C		E		D		C		E		E		C		D		A		C		D		E		C		D		N		C		D
BJ	D		C		D		E		C		D		E		C		E		D		C		A		D		C		E		D		C		N		D	
BK		D		D		C		E		D		C		E		E		C		D		A		C		D		E		C		D		N		C		D
BL	D		C		C		C		C		C		C		C		E		D		C		A		D		C		E		D		C		N		D	
BM		D		DA6		C		DA7		DA8		C		DA9		E		C		D		A		C		D		E		C		D		N		C		D
BN	D		C		DA6		DA7		C		DA8		DA9		C		E		D		C		A		D		C		E		D		C		N		D	
BP		D		C		C		C		C		C		C		E		C		D		A		C		D		E		C		D		N		C		D
BR	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
BT		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
BU	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
BV		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
BW	D		C		C		C		C		C		C		C		E		D		C		A		D		C		E		D		C		N		D	
BY		D		DA10		C		DA11		DA12		C		DA13		E		C		D		A		C		D		E		C		D		N		C		D
CA	D		C		DA10		DA11		C		DA12		DA13		C		E		D		C		A		D		C		E		D		C		N		D	
CB		D		C		C		C		C		C		C		E		C		D		A		C		D		E		C		D		N		C		D
CC	D		C		D		E		C		D		E		C		E		D		C		A		D		C		E		D		C		N		D	
CD		D		D		C		E		D		C		E		E		C		D		A		C		D		E		C		D		N		C		D
CE	D		C		D		E		C		D		E		C		E		D		C		A		D		C		E		D		C		N		D	
CF		D		D		C		E		D		C		E		E		C		D		A		C		D		E		C		D		N		C		D
CG	D		C		C		C		C		C		C		C		E		D		C		A		D		C		E		D		C		N		D	
CH		D		DA14		C		DA15		DA16		C		DA17		E		C		D		A		C		D		E		C		D		N		C		D
CJ	D		C		DA14		DA15		C		DA16		DA17		C		E		D		C		A		D		C		E		D		C		N		D	
CK		D		C		C		C		C		C		C		E		C		D		A		C		D		E		C		D		N		C		D
CL	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
CM		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
CN	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
CP		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
CR	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
CT		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D



Table 133: HBM3 Ballout – Part 2 (Columns 37 to 74) (cont’d)

	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	
CU	D		C		E		D		C		E		D		C		C		C		C		C		C		C		C		C		C		C		C		
CV		D		E		C		D		E		C		D		C		C		C		C		C		C		C		C		C		C		C		C	
CW	D		C		E		D		C		E		D		C		C		C		C		C		C		C		C		C		C		C		C		
CY		D		E		C		D		E		C		D		C		C		C		C		C		C		C		C		C		C		C		C	
DA	D		C		E		D		C		E		D		C		C		C		C		C		C		C		C		C		C		C		C		
DB		D		E		C		D		E		C		D		C		C		C		C		C		C		C		C		C		C		C		C	
DC	D		C		E		D		C		E		D		C		C		C		C		C		C		C		C		C		C		C		C		
DD		D		E		C		D		E		C		D		C		C		C		C		C		C		C		C		C		C		C		C	
DE	D		C		E		D		C		E		D		C		C		C		C		C		C		C		C		C		C		C		C		


Table 133: HBM3 Ballout – Part 2 (Columns 37 to 74) (cont'd)

	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74
DF		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
DG	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
DH		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
DJ	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
DK		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
DL	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
DM		D		C		C		C		C		C		C		E		C		D		A		C		D		E		C		D		N		C		D
DN	D		C		DA18		DA19		C		DA20		DA21		C		E		D		C		A		D		C		E		D		C		N		D	
DP		D		DA18		C		DA19		DA20		C		DA21		E		C		D		A		C		D		E		C		D		N		C		D
DR	D		C		C		C		C		C		C		C		E		D		C		A		D		C		E		D		C		N		D	
DT		D		D		C		E		D		C		E		E		C		D		A		C		D		E		C		D		N		C		D
DU	D		C		D		E		C		D		E		C		E		D		C		A		D		C		E		D		C		N		D	
DV		D		D		C		E		D		C		E		E		C		D		A		C		D		E		C		D		N		C		D
DW	D		C		D		E		C		D		E		C		E		D		C		A		D		C		E		D		C		N		D	
DY		D		C		C		C		C		C		C		E		C		D		A		C		D		E		C		D		N		C		D
EA	D		C		DA22		DA23		C		DA24		DA25		C		E		D		C		A		D		C		E		D		C		N		D	
EB		D		DA22		C		DA23		DA24		C		DA25		E		C		D		A		C		D		E		C		D		N		C		D
EC	D		C		C		C		C		C		C		C		E		D		C		A		D		C		E		D		C		N		D	
ED		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
EE	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
EF		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
EG	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
EH		D		C		C		C		C		C		C		E		C		D		A		C		D		E		C		D		N		C		D
EJ	D		C		DA26		DA27		C		DA28		DA29		C		E		D		C		A		D		C		E		D		C		N		D	
EK		D		DA26		C		DA27		DA28		C		DA29		E		C		D		A		C		D		E		C		D		N		C		D
EL	D		C		C		C		C		C		C		C		E		D		C		A		D		C		E		D		C		N		D	
EM		D		D		C		E		D		C		E		E		C		D		A		C		D		E		C		D		N		C		D
EN	D		C		D		E		C		D		E		C		E		D		C		A		D		C		E		D		C		N		D	
EP		D		D		C		E		D		C		E		E		C		D		A		C		D		E		C		D		N		C		D
ER	D		C		D		E		C		D		E		C		E		D		C		A		D		C		E		D		C		N		D	
ET		D		C		C		C		C		C		C		E		C		D		A		C		D		E		C		D		N		C		D
EU	D		C		DA30		DA31		C		DA32		DA33		C		E		D		C		A		D		C		E		D		C		N		D	
EV		D		DA30		C		DA31		DA32		C		DA33		E		C		D		A		C		D		E		C		D		N		C		D
EW	D		C		C		C		C		C		C		C		E		D		C		A		D		C		E		D		C		N		D	
EY		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
FA	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	
FB		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D
FC	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		D	


Table 133: HBM3 Ballout – Part 2 (Columns 37 to 74) (cont'd)

	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	
FD		D		C		C		C		C		C		C		E		C		D		A		C		D		E		C		D		N		C		D	
FE	D		C		DA34		DA35		C		DA36		DA37		C		E		D		C		A		D		C		E		D		C		N		C		D
FF		D		DA34		C		DA35		DA36		C		DA37		E		C		D		A		C		D		E		C		D		N		C		D	
FG	D		C		C		C		C		C		C		C		E		D		C		A		D		C		E		D		C		N		C		D
FH		D		D		C		E		D		C		E		E		C		D		A		C		D		E		C		D		N		C		D	
FJ	D		C		D		E		C		D		E		C		E		D		C		A		D		C		E		D		C		N		C		D
FK		D		D		C		E		D		C		E		E		C		D		A		C		D		E		C		D		N		C		D	
FL	D		C		D		E		C		D		E		C		E		D		C		A		D		C		E		D		C		N		C		D
FM		D		C		C		C		C		C		C		E		C		D		A		C		D		E		C		D		N		C		D	
FN	D		C		DA38		DA39		C		B		B		C		E		D		C		A		D		C		E		D		C		N		C		D
FP		D		DA38		C		DA39		B		C		B		E		C		D		A		C		D		E		C		D		N		C		D	
FR	D		C		C		C		C		C		C		C		E		D		C		A		D		C		E		D		C		N		C		D
FT		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D	
FU	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		C		D
FV		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D	
FW	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		C		D
FY		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D	
GA	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		C		D
GB		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D	
GC	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		C		D
GD		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D	
GE	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		C		D
GF		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D	
GG	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		C		D
GH		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D	
GJ	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		C		D
GK		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D	
GL	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		C		D
GM		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D	
GN	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		C		D
GP		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D	
GR	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		C		D
GT		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D	
GU	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		C		D
GV		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D	
GW	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		C		D
GY		D		E		C		D		E		C		D		E		C		D		A		C		D		E		C		D		N		C		D	
HA	D		C		E		D		C		E		D		C		E		D		C		A		D		C		E		D		C		N		C		D


Table 134: HBM3 Ballout – Part 3 (Columns 75 to 90)

	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A	C		E		D		C		E		D		C		E	
B		E		C		D		E		C		D		E		C
C	C		E		D		C		E		D		C		E	
D		E		C		D		E		C		D		E		C
E	C		E		D		C		E		D		C		E	
F		E		C		D		E		C		D		E		C
G	C		E		D		C		E		D		C		E	
H		E		C		D		E		C		D		E		C
J	C		E		D		C		E		D		C		M	
K		E		C		D		E		C		D		E		C
L	C		E		D		C		E		D		C		M	
M		E		C		D		E		C		D		E		C
N	C		E		D		C		E		D		C		E	
P		E		C		D		E		C		D		E		C
R	C		E		D		C		E		D		C		E	
T		E		C		D		E		C		D		E		C
U	C		E		D		C		E		D		C		M	
V		E		C		D		E		C		D		E		C
W	C		E		D		C		E		D		C		E	
Y		E		C		D		E		C		D		E		C
AA	C		E		D		C		E		D		C		E	
AB		E		C		D		E		C		D		E		C
AC	C		E		D		C		E		D		C		M	
AD		E		C		D		E		C		D		E		C
AE	C		E		D		C		E		D		C		M	
AF		E		C		D		E		C		D		E		C
AG	C		E		D		C		E		D		C		E	
AH		E		C		D		E		C		D		E		C
AJ	C		E		D		C		E		D		C		M	
AK		E		C		D		E		C		D		E		C
AL	C		E		D		C		E		D		C		M	
AM		E		C		D		E		C		D		E		C
AN	C		E		D		C		E		D		C		E	
AP		E		C		D		E		C		D		E		C
AR	C		E		D		C		E		D		C		E	
AT		E		C		D		E		C		D		E		C
AU	C		E		D		C		E		D		C		M	
AV		E		C		D		E		C		D		E		C
AW	C		E		D		C		E		D		C		E	
AY		E		C		D		E		C		D		E		C


Table 134: HBM3 Ballout – Part 3 (Columns 75 to 90) (cont'd)

	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
BA	C		E		D		C		E		C		C		C	
BB		E		C		D		E		C		C		C		C
BC	C		E		D		C		E		C		C		M	
BD		E		C		D		E		C		C		M		C
BE	C		E		D		C		E		D		C		M	
BF		E		C		D		E		C		D		C		C
BG	C		E		D		C		E		D		C		C	
BH		E		C		D		E		C		C		C		C
BJ	C		E		D		C		E		C		C		C	
BK		E		C		D		E		C		C		M		C
BL	C		E		D		C		E		C		C		M	
BM		E		C		D		E		C		D		M		C
BN	C		E		D		C		E		D		C		C	
BP		E		C		D		E		C		D		C		C
BR	C		E		D		C		E		C		C		C	
BT		E		C		D		E		C		C		C		C
BU	C		E		D		C		E		C		C		M	
BV		E		C		D		E		C		C		M		C
BW	C		E		D		C		E		D		C		M	
BY		E		C		D		E		C		D		C		C
CA	C		E		D		C		E		D		C		C	
CB		E		C		D		E		C		C		C		C
CC	C		E		D		C		E		C		C		M	
CD		E		C		D		E		C		C		M		C
CE	C		E		D		C		E		D		C		M	
CF		E		C		D		E		C		D		C		C
CG	C		E		D		C		E		D		C		C	
CH		E		C		D		E		C		C		C		C
CJ	C		E		D		C		E		C		C		C	
CK		E		C		D		E		C		C		M		C
CL	C		E		D		C		E		C		C		M	
CM		E		C		D		E		C		D		M		C
CN	C		E		D		C		E		D		C		C	
CP		E		C		D		E		C		D		C		C
CR	C		E		D		C		E		C		C		C	
CT		E		C		D		E		C		C		C		C



Table 134: HBM3 Ballout – Part 3 (Columns 75 to 90) (cont’d)

	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
CU	C		C		C		C		C		C		C		C	
CV		C		C		C		C		C		C		C		C
CW	C		C		C		C		C		C		C		C	
CY		C		C		C		C		C		C		C		C
DA	C		C		C		C		C		C		C		C	
DB		C		C		C		C		C		C		C		C
DC	C		C		C		C		C		C		C		C	
DD		C		C		C		C		C		C		C		C
DE	C		C		C		C		C		C		C		C	


Table 134: HBM3 Ballout – Part 3 (Columns 75 to 90) (cont'd)

	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
DF		E		C		D		E		C		C		C		C
DG	C		E		D		C		E		C		C		C	
DH		E		C		D		E		C		D		C		C
DJ	C		E		D		C		E		D		C		C	
DK		E		C		D		E		C		D		M		C
DL	C		E		D		C		E		C		C		M	
DM		E		C		D		E		C		C		M		C
DN	C		E		D		C		E		C		C		C	
DP		E		C		D		E		C		C		C		C
DR	C		E		D		C		E		D		C		C	
DT		E		C		D		E		C		D		C		C
DU	C		E		D		C		E		D		C		M	
DV		E		C		D		E		C		C		M		C
DW	C		E		D		C		E		C		C		M	
DY		E		C		D		E		C		C		C		C
EA	C		E		D		C		E		D		C		C	
EB		E		C		D		E		C		D		C		C
EC	C		E		D		C		E		D		C		M	
ED		E		C		D		E		C		C		M		C
EE	C		E		D		C		E		C		C		M	
EF		E		C		D		E		C		C		C		C
EG	C		E		D		C		E		C		C		C	
EH		E		C		D		E		C		D		C		C
EJ	C		E		D		C		E		D		C		C	
EK		E		C		D		E		C		D		M		C
EL	C		E		D		C		E		C		C		M	
EM		E		C		D		E		C		C		M		C
EN	C		E		D		C		E		C		C		C	
EP		E		C		D		E		C		C		C		C
ER	C		E		D		C		E		D		C		C	
ET		E		C		D		E		C		D		C		C
EU	C		E		D		C		E		D		C		M	
EV		E		C		D		E		C		C		M		C
EW	C		E		D		C		E		C		C		M	
EY		E		C		D		E		C		C		C		C
FA	C		E		D		C		E		C		C		C	
FB		E		C		D		E		C		D		C		C
FC	C		E		D		C		E		D		C		C	


Table 134: HBM3 Ballout – Part 3 (Columns 75 to 90) (cont'd)

	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
FD		E		C		D		E		C		D		M		C
FE	C		E		D		C		E		C		C		M	
FF		E		C		D		E		C		C		M		C
FG	C		E		D		C		E		C		C		C	
FH		E		C		D		E		C		C		C		C
FJ	C		E		D		C		E		D		C		C	
FK		E		C		D		E		C		D		C		C
FL	C		E		D		C		E		D		C		M	
FM		E		C		D		E		C		C		M		C
FN	C		E		D		C		E		C		C		M	
FP		E		C		D		E		C		C		C		C
FR	C		E		D		C		E		D		C		C	
FT		E		C		D		E		C		D		C		C
FU	C		E		D		C		E		D		C		M	
FV		E		C		D		E		C		C		M		C
FW	C		E		D		C		E		C		C		M	
FY		E		C		D		E		C		C		C		C
GA	C		E		D		C		E		C		C		C	
GB		E		C		D		E		C		D		C		C
GC	C		E		D		C		E		D		C		C	
GD		E		C		D		E		C		D		M		C
GE	C		E		D		C		E		C		C		M	
GF		E		C		D		E		C		C		M		C
GG	C		E		D		C		E		C		C		C	
GH		E		C		D		E		C		C		C		C
GJ	C		E		D		C		E		D		C		C	
GK		E		C		D		E		C		D		C		C
GL	C		E		D		C		E		D		C		M	
GM		E		C		D		E		C		C		M		C
GN	C		E		D		C		E		C		C		M	
GP		E		C		D		E		C		C		C		C
GR	C		E		D		C		E		C		C		C	
GT		E		C		D		E		C		D		C		C
GU	C		E		D		C		E		D		C		C	
GV		E		C		D		E		C		D		E		C
GW	C		E		D		C		E		D		C		E	
GY		E		C		D		E		C		D		E		C
HA	C		E		D		C		E		D		C		E	


Table 135: HBM3 Ballout – Part 4 (Columns 91 to 118)

	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118
A	D		C		E		D		C		E		D		C		E		D		C		E		D		C	
B		D		E		C		D		E		C		D		E		C		D		E		C		D		E
C	D		C		E		D		C		E		D		C		E		D		C		E		D		C	
D		D		E		C		D		E		C		D		E		C		D		E		C		D		E
E	D		C		E		D		C		E		D		C		E		D		C		E		D		C	
F		D		E		C		D		E		C		D		E		C		D		E		C		D		E
G	C		C		C		C		C		C		C		C		C		C		C		C		C		C	
H		N		N		N		N		N		N		N		N		N		N		N		N		N		N
J	C		C		C		C		C		C		C		C		C		C		C		C		C		C	
K		M		M		M		M		M		M		M		M		M		M		M		M		M		M


Table 135 : HBM3 Ballout – Part 4 (Columns 91 to 118) (cont'd)

	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118
L	D		D		D		D		D		D		D		D		D		D		D		D		D		D	
M		C		DQm7		DQm5		RDm0		DQm3		DQm1		ECCm0		C		DQ17		DQ15		RD10		DQ13		DQ11		ECC10
N	C		DBIm0		DQm6		DQm4		DPARm0		DQm2		DQm0		C		DBI10		DQ16		DQ14		DPAR10		DQ12		DQ10	
P		N		N		N		N		N		N		N		N		N		N		N		N		N		N
R	C		DBIm1		DQm14		DQm12		WDQSm0_t		DQm10		DQm8		C		DBI11		DQ114		DQ112		WDQSi0_t		DQ110		DQ18	
T		C		DQm15		DQm13		WDQSm0_c		DQm11		DQm9		ECCm1		C		DQ115		DQ113		WDQSi0_c		DQ111		DQ19		ECC11
U	M		M		M		M		M		M		M		M		M		M		M		M		M		M	
V		D		D		D		D		D		D		D		D		D		D		D		D		D		D
W	C		DBIm2		DQm22		DQm20		RDQSm0_t		DQm18		DQm16		C		DBI12		DQ122		DQ120		RDQSi0_t		DQ118		DQ116	
Y		C		DQm23		DQm21		RDQSm0_c		DQm19		DQm17		SEVm0		C		DQ123		DQ121		RDQSi0_c		DQ119		DQ117		SEV10
AA	N		N		N		N		N		N		N		N		N		N		N		N		N		N	
AB		C		DQm31		DQm29		RDm1		DQm27		DQm25		SEVm1		C		DQ131		DQ129		RD11		DQ127		DQ125		SEV11
AC	C		DBIm3		DQm30		DQm28		DERRm0		DQm26		DQm24		C		DBI13		DQ130		DQ128		DERR10		DQ126		DQ124	
AD		M		M		M		M		M		M		M		M		M		M		M		M		M		M
AE	D		D		D		D		D		D		D		D		D		D		D		D		D		D	
AF		C		ARFUm		Cm7		Cm5		Cm4		Cm2		Cm0		C		ARFU1		C17		C15		C14		C12		C10
AG	C		RAm		APARm		Cm6		CKm_t		Cm3		Cm1		C		RA1		APAR1		C16		CK1_t		C13		C11	
AH		C		Rm9		Rm7		CKm_c		Rm4		Rm3		Rm1		C		R19		R17		CK1_c		R14		R13		R11
AJ	C		AERRm		Rm8		Rm6		Rm5		Rm0		Rm2		C		AERR1		R18		R16		R15		R10		R12	
AK		M		M		M		M		M		M		M		M		M		M		M		M		M		M
AL	D		D		D		D		D		D		D		D		D		D		D		D		D		D	
AM		C		DQm39		DQm37		RDm2		DQm35		DQm33		ECCm2		C		DQ139		DQ137		RD12		DQ135		DQ133		ECC12
AN	C		DBIm4		DQm38		DQm36		DPARm1		DQm34		DQm32		C		DBI14		DQ138		DQ136		DPAR11		DQ134		DQ132	
AP		N		N		N		N		N		N		N		N		N		N		N		N		N		N
AR	C		DBIm5		DQm46		DQm44		WDQSm1_t		DQm42		DQm40		C		DBI15		DQ146		DQ144		WDQSi1_t		DQ142		DQ140	
AT		C		DQm47		DQm45		WDQSm1_c		DQm43		DQm41		ECCm3		C		DQ147		DQ145		WDQSi1_c		DQ143		DQ141		ECC13
AU	M		M		M		M		M		M		M		M		M		M		M		M		M		M	
AV		D		D		D		D		D		D		D		D		D		D		D		D		D		D
AW	C		DBIm6		DQm54		DQm52		RDQSm1_t		DQm50		DQm48		C		DBI16		DQ154		DQ152		RDQSi1_t		DQ150		DQ148	
AY		C		DQm55		DQm53		RDQSm1_c		DQm51		DQm49		SEVm2		C		DQ155		DQ153		RDQSi1_c		DQ151		DQ149		SEV12
BA	N		N		N		N		N		N		N		N		N		N		N		N		N		N	
BB		C		DQm63		DQm61		RDm3		DQm59		DQm57		SEVm3		C		DQ163		DQ161		RD13		DQ159		DQ157		SEV13
BC	C		DBIm7		DQm62		DQm60		DERRm1		DQm58		DQm56		C		DBI17		DQ162		DQ160		DERR11		DQ158		DQ156	
BD		M		M		M		M		M		M		M		M		M		M		M		M		M		M


Table 135 : HBM3 Ballout – Part 4 (Columns 91 to 118) (cont'd)

	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118
BE	D		D		D		D		D		D		D		D		D		D		D		D		D		D	
BF		C		DQn7		DQn5		RDn0		DQn3		DQn1		ECCn0		C		DQj7		DQj5		RDj0		DQj3		DQj1		ECCj0
BG	C		DBIn0		DQn6		DQn4		DPARn0		DQn2		DQn0		C		DBIj0		DQj6		DQj4		DPARj0		DQj2		DQj0	
BH		N		N		N		N		N		N		N		N		N		N		N		N		N		N
BJ	C		DBIn1		DQn14		DQn12		WDQSn0_t		DQn10		DQn8		C		DBIj1		DQj14		DQj12		WDQSn0_c		DQj10		DQj8	
BK		C		DQn15		DQn13		WDQSn0_c		DQn11		DQn9		ECCn1		C		DQj15		DQj13		WDQSn0_c		DQj11		DQj9		ECCj1
BL	M		M		M		M		M		M		M		M		M		M		M		M		M		M	
BM		D		D		D		D		D		D		D		D		D		D		D		D		D		D
BN	C		DBIn2		DQn22		DQn20		RDQSn0_t		DQn18		DQn16		C		DBIj2		DQj22		DQj20		RDQSn0_c		DQj18		DQj16	
BP		C		DQn23		DQn21		RDQSn0_c		DQn19		DQn17		SEVn0		C		DQj23		DQj21		RDQSn0_c		DQj19		DQj17		SEVj0
BR	N		N		N		N		N		N		N		N		N		N		N		N		N		N	
BT		C		DQn31		DQn29		RDn1		DQn27		DQn25		SEVn1		C		DQj31		DQj29		RDj1		DQj27		DQj25		SEVj1
BU	C		DBIn3		DQn30		DQn28		DERRn0		DQn26		DQn24		C		DBIj3		DQj30		DQj28		DERRj0		DQj26		DQj24	
BV		M		M		M		M		M		M		M		M		M		M		M		M		M		M
BW	D		D		D		D		D		D		D		D		D		D		D		D		D		D	
BY		C		ARFUn		Cn7		Cn5		Cn4		Cn2		Cn0		C		ARFUj		Cj7		Cj5		Cj4		Cj2		Cj0
CA	C		RAn		APARn		Cn6		CKn_t		Cn3		Cn1		C		RAj		APARj		Cj6		CKj_t		Cj3		Cj1	
CB		C		Rn9		Rn7		CKn_c		Rn4		Rn3		Rn1		C		Rj9		Rj7		CKj_c		Rj4		Rj3		Rj1
CC	C		AERRn		Rn8		Rn6		Rn5		Rn0		Rn2		C		AERRj		Rj8		Rj6		Rj5		Rj0		Rj2	
CD		M		M		M		M		M		M		M		M		M		M		M		M		M		M
CE	D		D		D		D		D		D		D		D		D		D		D		D		D		D	
CF		C		DQn39		DQn37		RDn2		DQn35		DQn33		ECCn2		C		DQj39		DQj37		RDj2		DQj35		DQj33		ECCj2
CG	C		DBIn4		DQn38		DQn36		DPARn1		DQn34		DQn32		C		DBIj4		DQj38		DQj36		DPARj1		DQj34		DQj32	
CH		N		N		N		N		N		N		N		N		N		N		N		N		N		N
CJ	C		DBIn5		DQn46		DQn44		WDQSn1_t		DQn42		DQn40		C		DBIj5		DQj46		DQj44		WDQSn1_c		DQj42		DQj40	
CK		C		DQn47		DQn45		WDQSn1_c		DQn43		DQn41		ECCn3		C		DQj47		DQj45		WDQSn1_c		DQj43		DQj41		ECCj3
CL	M		M		M		M		M		M		M		M		M		M		M		M		M		M	
CM		D		D		D		D		D		D		D		D		D		D		D		D		D		D
CN	C		DBIn6		DQn54		DQn52		RDQSn1_t		DQn50		DQn48		C		DBIj6		DQj54		DQj52		RDQSn1_c		DQj50		DQj48	
CP		C		DQn55		DQn53		RDQSn1_c		DQn51		DQn49		SEVn2		C		DQj55		DQj53		RDQSn1_c		DQj51		DQj49		SEVj2
CR	N		N		N		N		N		N		N		N		N		N		N		N		N		N	
CT		C		DQn63		DQn61		RDn3		DQn59		DQn57		SEVn3		C		DQj63		DQj61		RDj3		DQj59		DQj57		SEVj3
CU	C		DBIn7		DQn62		DQn60		DERRn1		DQn58		DQn56		C		DBIj7		DQj62		DQj60		DERRj1		DQj58		DQj56	
CV		M		M		M		M		M		M		M		M		M		M		M		M		M		M



Table 135 : HBM3 Ballout – Part 4 (Columns 91 to 118) (cont’d)

	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118
CW	D		D		D		D		D		D		D		D		D		D		D		D		D		D	
CY		C		B		CATTRIP		CATTRIP		WSOe		WSOe		WSOm		WSOm		WSOk		WSOk		WSOi		WSOi		WSOg		WSOg
DA	M		M		M		M		M		M		M		M		M		M		M		M		M		M	
DB		C		B		TEMP1		TEMP0		WSOp		WSOp		WSOn		WSOn		WSOI		WSOI		WSOj		WSOj		WSOh		WSOh
DC	D		D		D		D		D		D		D		D		D		D		D		D		D		D	


Table 135 : HBM3 Ballout – Part 4 (Columns 91 to 118) (cont'd)

	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118
DD		M		M		M		M		M		M		M		M		M		M		M		M		M		M
DE	C		DBIo7		DQo62		DQo60		DERRo1		DQo58		DQo56		C		DBIk7		DQk62		DQk60		DERRk1		DQk58		DQk56	
DF		C		DQo63		DQo61		RDs3		DQo59		DQo57		SEVo3		C		DQk63		DQk61		RDk3		DQk59		DQk57		SEVk3
DG	N		N		N		N		N		N		N		N		N		N		N		N		N		N	
DH		C		DQo55		DQo53		RDQSo1_c		DQo51		DQo49		SEVo2		C		DQk55		DQk53		RDQSk1_c		DQk51		DQk49		SEVk2
DJ	C		DBIo6		DQo54		DQo52		RDQSo1_t		DQo50		DQo48		C		DBIk6		DQk54		DQk52		RDQSk1_t		DQk50		DQk48	
DK		D		D		D		D		D		D		D		D		D		D		D		D		D		D
DL	M		M		M		M		M		M		M		M		M		M		M		M		M		M	
DM		C		DQo47		DQo45		WDQSo1_c		DQo43		DQo41		ECCo3		C		DQk47		DQk45		WDQSk1_c		DQk43		DQk41		ECCk3
DN	C		DBIo5		DQo46		DQo44		WDQSo1_t		DQo42		DQo40		C		DBIk5		DQk46		DQk44		WDQSk1_t		DQk42		DQk40	
DP		N		N		N		N		N		N		N		N		N		N		N		N		N		N
DR	C		DBIo4		DQo38		DQo36		DPARo1		DQo34		DQo32		C		DBIk4		DQk38		DQk36		DPARK1		DQk34		DQk32	
DT		C		DQo39		DQo37		RDs2		DQo35		DQo33		ECCo2		C		DQk39		DQk37		RDk2		DQk35		DQk33		ECCk2
DU	D		D		D		D		D		D		D		D		D		D		D		D		D		D	
DV		M		M		M		M		M		M		M		M		M		M		M		M		M		M
DW	C		AERRo		Ro8		Ro6		Ro5		Ro0		Ro2		C		AERRk		Rk8		Rk6		Rk5		Rk0		Rk2	
DY		C		Ro9		Ro7		CKo_c		Ro4		Ro3		Ro1		C		Rk9		Rk7		CKk_c		Rk4		Rk3		Rk1
EA	C		RAo		APARo		Co6		CKo_t		Ca3		Ca1		C		RAK		APARK		CKk6		CKK_t		CKk3		CKk1	
EB		C		ARFUo		Co7		Co5		Co4		Co2		Co0		C		ARFUK		CKk7		CKk5		CKk4		CKk2		CKk0
EC	D		D		D		D		D		D		D		D		D		D		D		D		D		D	
ED		M		M		M		M		M		M		M		M		M		M		M		M		M		M
EE	C		DBIo3		DQo30		DQo28		DERRo0		DQo26		DQo24		C		DBIk3		DQk30		DQk28		DERRk0		DQk26		DQk24	
EF		C		DQo31		DQo29		RDs1		DQo27		DQo25		SEVo1		C		DQk31		DQk29		RDk1		DQk27		DQk25		SEVk1
EG	N		N		N		N		N		N		N		N		N		N		N		N		N		N	
EH		C		DQo23		DQo21		RDQSo0_c		DQo19		DQo17		SEVo0		C		DQk23		DQk21		RDQSk0_c		DQk19		DQk17		SEVk0
EJ	C		DBIo2		DQo22		DQo20		RDQSo0_t		DQo18		DQo16		C		DBIk2		DQk22		DQk20		RDQSk0_t		DQk18		DQk16	
EK		D		D		D		D		D		D		D		D		D		D		D		D		D		D
EL	M		M		M		M		M		M		M		M		M		M		M		M		M		M	
EM		C		DQo15		DQo13		WDQSo0_c		DQo11		DQo9		ECCo1		C		DQk15		DQk13		WDQSk0_c		DQk11		DQk9		ECCk1
EN	C		DBIo1		DQo14		DQo12		WDQSo0_t		DQo10		DQo8		C		DBIk1		DQk14		DQk12		WDQSk0_t		DQk10		DQk8	
EP		N		N		N		N		N		N		N		N		N		N		N		N		N		N
ER	C		DBIo0		DQo6		DQo4		DPARo0		DQo2		DQo0		C		DBIk0		DQk6		DQk4		DPARK0		DQk2		DQk0	
ET		C		DQo7		DQo5		RDs0		DQo3		DQo1		ECCo0		C		DQk7		DQk5		RDk0		DQk3		DQk1		ECCk0
EU	D		D		D		D		D		D		D		D		D		D		D		D		D		D	


Table 135 : HBM3 Ballout – Part 4 (Columns 91 to 118) (cont'd)

	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118
EV		M		M		M		M		M		M		M		M		M		M		M		M		M		M
EW	C		DBIp7		DQp62		DQp60		DERRp1		DQp58		DQp56		C		DBI17		DQI62		DQI60		DERR11		DQI58		DQI56	
EY		C		DQp63		DQp61		RDp3		DQp59		DQp57		SEVp3		C		DQI63		DQI61		RD13		DQI59		DQI57		SEV13
FA	N		N		N		N		N		N		N		N		N		N		N		N		N		N	
FB		C		DQp55		DQp53		RDQSp1_c		DQp51		DQp49		SEVp2		C		DQI55		DQI53		RDQSI1_c		DQI51		DQI49		SEV12
FC	C		DBIp6		DQp54		DQp52		RDQSp1_t		DQp50		DQp48		C		DBI16		DQI54		DQI52		RDQSI1_t		DQI50		DQI48	
FD		D		D		D		D		D		D		D		D		D		D		D		D		D		D
FE	M		M		M		M		M		M		M		M		M		M		M		M		M		M	
FF		C		DQp47		DQp45		WDQSp1_c		DQp43		DQp41		ECCp3		C		DQI47		DQI45		WDQSI1_c		DQI43		DQI41		ECC13
FG	C		DBIp5		DQp46		DQp44		WDQSp1_t		DQp42		DQp40		C		DBI15		DQI46		DQI44		WDQSI1_t		DQI42		DQI40	
FH		N		N		N		N		N		N		N		N		N		N		N		N		N		N
FJ	C		DBIp4		DQp38		DQp36		DPARp1		DQp34		DQp32		C		DBI14		DQI38		DQI36		DPAR11		DQI34		DQI32	
FK		C		DQp39		DQp37		RDp2		DQp35		DQp33		ECCp2		C		DQI39		DQI37		RD12		DQI35		DQI33		ECC12
FL	D		D		D		D		D		D		D		D		D		D		D		D		D		D	
FM		M		M		M		M		M		M		M		M		M		M		M		M		M		M
FN	C		AERRp		Rp8		Rp6		Rp5		Rp0		Rp2		C		AERR1		RI8		RI6		RI5		RI0		RI2	
FP		C		Rp9		Rp7		CKp_c		Rp4		Rp3		Rp1		C		RI9		RI7		CKI_c		RI4		RI3		RI1
FR	C		RAp		APARp		Cp6		CKp_t		Cp3		Cp1		C		RA1		APAR1		CI6		CKI_t		CI3		CI1	
FT		C		ARFUp		Cp7		Cp5		Cp4		Cp2		Cp0		C		ARFU1		CI7		CI5		CI4		CI2		CI0
FU	D		D		D		D		D		D		D		D		D		D		D		D		D		D	
FV		M		M		M		M		M		M		M		M		M		M		M		M		M		M
FW	C		DBIp3		DQp30		DQp28		DERRp0		DQp26		DQp24		C		DBI13		DQI30		DQI28		DERR10		DQI26		DQI24	
FY		C		DQp31		DQp29		RDp1		DQp27		DQp25		SEVp1		C		DQI31		DQI29		RD11		DQI27		DQI25		SEV11
GA	N		N		N		N		N		N		N		N		N		N		N		N		N		N	
GB		C		DQp23		DQp21		RDQSp0_c		DQp19		DQp17		SEVp0		C		DQI23		DQI21		RDQSI0_c		DQI19		DQI17		SEV10
GC	C		DBIp2		DQp22		DQp20		RDQSp0_t		DQp18		DQp16		C		DBI12		DQI22		DQI20		RDQSI0_t		DQI18		DQI16	
GD		D		D		D		D		D		D		D		D		D		D		D		D		D		D
GE	M		M		M		M		M		M		M		M		M		M		M		M		M		M	
GF		C		DQp15		DQp13		WDQSp0_c		DQp11		DQp9		ECCp1		C		DQI15		DQI13		WDQSI0_c		DQI11		DQI9		ECC11
GG	C		DBIp1		DQp14		DQp12		WDQSp0_t		DQp10		DQp8		C		DBI11		DQI14		DQI12		WDQSI0_t		DQI10		DQI8	
GH		N		N		N		N		N		N		N		N		N		N		N		N		N		N
GJ	C		DBIp0		DQp6		DQp4		DPARp0		DQp2		DQp0		C		DBI10		DQI6		DQI4		DPAR10		DQI2		DQI0	
GK		C		DQp7		DQp5		RDp0		DQp3		DQp1		ECCp0		C		DQI7		DQI5		RD10		DQI3		DQI1		ECC10
GL	D		D		D		D		D		D		D		D		D		D		D		D		D		D	


Table 135 : HBM3 Ballout – Part 4 (Columns 91 to 118) (cont'd)

	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118
GM		M		M		M		M		M		M		M		M		M		M		M		M		M		M
GN	C		C		C		C		C		C		C		C		C		C		C		C		C		C	
GP		N		N		N		N		N		N		N		N		N		N		N		N		N		N
GR	C		C		C		C		C		C		C		C		C		C		C		C		C		C	
GT		D		E		C		D		E		C		D		E		C		D		E		C		D		E
GU	D		C		E		D		C		E		D		C		E		D		C		E		D		C	
GV		D		E		C		D		E		C		D		E		C		D		E		C		D		E
GW	D		C		E		D		C		E		D		C		E		D		C		E		D		C	
GY		D		E		C		D		E		C		D		E		C		D		E		C		D		E
HA	D		C		E		D		C		E		D		C		E		D		C		E		D		C	



Table 136: HBM3 Ballout – Part 5 (Columns 119 to 148)

	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148
A	E		D		C		E		D		C		E		D		C		E		D		C		E		D		B	
B		C		D		E		C		D		E		C		D		E		C		D		E		C		D		B
C	E		D		C		E		D		C		E		D		C		E		D		C		E		D		B	
D		C		D		E		C		D		E		C		D		E		C		D		E		C		D		B
E	E		D		C		E		D		C		E		D		C		E		D		C		E		D		B	
F		C		D		E		C		D		E		C		D		E		C		D		E		C		D		B
G	C		C		C		C		C		C		C		C		C		C		C		C		C		C		B	
H		N		N		N		N		N		N		N		N		N		N		N		N		N		N		B
J	C		C		C		C		C		C		C		C		C		C		C		C		C		C		B	
K		M		M		M		M		M		M		M		M		M		M		M		M		M		M		B


Table 136: HBM3 Ballout – Part 5 (Columns 119 to 148) (cont'd)

	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148
L	D		D		D		D		D		D		D		D		D		D		D		D		D		D		B	
M		C		DQe7		DQe5		RDe0		DQe3		DQe1		ECCe0		C		DQa7		DQa5		RDa0		DQa3		DQa1		ECCa0		B
N	C		DBIe0		DQe6		DQe4		DPARe0		DQe2		DQe0		C		DBIa0		DQa6		DQa4		DPARa0		DQa2		DQa0		N	B
P		N		N		N		N		N		N		N		N		N		N		N		N		N		N		B
R	C		DBIe1		DQe14		DQe12		WDQSe0_t		DQe10		DQe8		C		DBIa1		DQa14		DQa12		WDQSa0_t		DQa10		DQa8		B	
T		C		DQe15		DQe13		WDQSe0_c		DQe11		DQe9		ECCe1		C		DQa15		DQa13		WDQSa0_c		DQa11		DQa9		ECCa1		B
U	M		M		M		M		M		M		M		M		M		M		M		M		M		M		B	
V		D		D		D		D		D		D		D		D		D		D		D		D		D		D		B
W	C		DBIe2		DQe22		DQe20		RDQSe0_c		DQe18		DQe16		C		DBIa2		DQa22		DQa20		RDQSa0_c		DQa18		DQa16		B	
Y		C		DQe23		DQe21		RDQSe0_c		DQe19		DQe17		SEVe0		C		DQa23		DQa21		RDQSa0_c		DQa19		DQa17		SEVa0		B
AA	N		N		N		N		N		N		N		N		N		N		N		N		N		N		B	
AB		C		DQe31		DQe29		RDe1		DQe27		DQe25		SEVe1		C		DQa31		DQa29		RDa1		DQa27		DQa25		SEVa1		B
AC	C		DBIe3		DQe30		DQe28		DERRe0		DQe26		DQe24		C		DBIa3		DQa30		DQa28		DERRa0		DQa26		DQa24		B	
AD		M		M		M		M		M		M		M		M		M		M		M		M		M		M		B
AE	D		D		D		D		D		D		D		D		D		D		D		D		D		D		B	
AF		C		ARFUe		Ce7		Ce5		Ce4		Ce2		Ce0		C		ARFUa		Ca7		Ca5		Ca4		Ca2		Ca0		B
AG	C		RAe		APARe		Ce6		CKe_t		Ce3		Ce1		C		RAa		APARa		Ca6		CKa_t		Ca3		Ca1		B	
AH		C		Re9		Re7		CKe_c		Re4		Re3		Re1		C		Ra9		Ra7		CKa_c		Ra4		Ra3		Ra1		B
AJ	C		AERRe		Re8		Re6		Re5		Re0		Re2		C		AERRa		Ra8		Ra6		Ra5		Ra0		Ra2		B	
AK		M		M		M		M		M		M		M		M		M		M		M		M		M		M		B
AL	D		D		D		D		D		D		D		D		D		D		D		D		D		D		B	
AM		C		DQe39		DQe37		RDe2		DQe35		DQe33		ECCe2		C		DQa39		DQa37		RDa2		DQa35		DQa33		ECCa2		B
AN	C		DBIe4		DQe38		DQe36		DPARe1		DQe34		DQe32		C		DBIa4		DQa38		DQa36		DPARa1		DQa34		DQa32		B	
AP		N		N		N		N		N		N		N		N		N		N		N		N		N		N		B
AR	C		DBIe5		DQe46		DQe44		WDQSe1_t		DQe42		DQe40		C		DBIa5		DQa46		DQa44		WDQSa1_t		DQa42		DQa40		B	
AT		C		DQe47		DQe45		WDQSe1_c		DQe43		DQe41		ECCe3		C		DQa47		DQa45		WDQSa1_c		DQa43		DQa41		ECCa3		B
AU	M		M		M		M		M		M		M		M		M		M		M		M		M		M		B	
AV		D		D		D		D		D		D		D		D		D		D		D		D		D		D		B
AW	C		DBIe6		DQe54		DQe52		RDQSe1_t		DQe50		DQe48		C		DBIa6		DQa54		DQa52		RDQSa1_t		DQa50		DQa48		B	
AY		C		DQe55		DQe53		RDQSe1_c		DQe51		DQe49		SEVe2		C		DQa55		DQa53		RDQSa1_c		DQa51		DQa49		SEVa2		B
BA	N		N		N		N		N		N		N		N		N		N		N		N		N		N		B	
BB		C		DQe63		DQe61		RDe3		DQe59		DQe57		SEVe3		C		DQa63		DQa61		RDa3		DQa59		DQa57		SEVa3		B
BC	C		DBIe7		DQe62		DQe60		DERRe1		DQe58		DQe56		C		DBIa7		DQa62		DQa60		DERRa1		DQa58		DQa56		B	
BD		M		M		M		M		M		M		M		M		M		M		M		M		M		M		B


Table 136: HBM3 Ballout – Part 5 (Columns 119 to 148) (cont'd)

	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	
BE	D		D		D		D		D		D		D		D		D		D		D		D		D		D		B		
BF		C		DQf7		DQf5		RDf0		DQf3		DQf1		ECCf0		C		DQb7		DQb5		RDb0		DQb3		DQb1		ECCb0		B	
BG	C		DBf0		DQf6		DQf4		DPARf0		DQf2		DQf0		C		DBf0		DQb6		DQb4		DPARb0		DQb2		DQb0		B		
BH		N		N		N		N		N		N		N		N		N		N		N		N		N		N		B	
BJ	C		DBf1		DQf14		DQf12		WDQSB0_t		DQf10		DQf8		C		DBf1		DQb14		DQb12		WDQSB0_t		DQb10		DQb8		B		
BK		C		DQf15		DQf13		WDQSB0_c		DQf11		DQf9		ECCf1		C		DQb15		DQb13		WDQSB0_c		DQb11		DQb9		ECCb1		B	
BL	M		M		M		M		M		M		M		M		M		M		M		M		M		M		B		
BM		D		D		D		D		D		D		D		D		D		D		D		D		D		D		B	
BN	C		DBf2		DQf22		DQf20		RDQSB0_t		DQf18		DQf16		C		DBf2		DQb22		DQb20		RDQSB0_t		DQb18		DQb16		B		
BP		C		DQf23		DQf21		RDQSB0_c		DQf19		DQf17		SEVf0		C		DQb23		DQb21		RDQSB0_c		DQb19		DQb17		SEVb0		B	
BR	N		N		N		N		N		N		N		N		N		N		N		N		N		N		B		
BT		C		DQf31		DQf29		RDf1		DQf27		DQf25		SEVf1		C		DQb31		DQb29		RDb1		DQb27		DQb25		SEVb1		B	
BU	C		DBf3		DQf30		DQf28		DERRf0		DQf26		DQf24		C		DBf3		DQb30		DQb28		DERRb0		DQb26		DQb24		B		
BV		M		M		M		M		M		M		M		M		M		M		M		M		M		M		B	
BW	D		D		D		D		D		D		D		D		D		D		D		D		D		D		B		
BY		C		ARFUb		Cl7		Cl5		Cl4		Cl2		Cl0		C		ARFUb		Cb7		Cb5		Cb4		Cb2		Cb0		B	
CA	C		RAf		APARf		Cl6		CKf_t		Cl3		Cl1		C		RAf		APARb		Cb6		CKb_t		Cb3		Cb1		B		
CB		C		Rf9		Rf7		CKf_c		Rf4		Rf3		Rf1		C		Rb9		Rb7		CKb_c		Rb4		Rb3		Rb1		B	
CC	C		AERRf		Rf8		Rf6		Rf5		Rf0		Rf2		C		AERRb		Rb8		Rb6		Rb5		Rb0		Rb2		B		
CD		M		M		M		M		M		M		M		M		M		M		M		M		M		M		B	
CE	D		D		D		D		D		D		D		D		D		D		D		D		D		D		B		
CF		C		DQf39		DQf37		RDf2		DQf35		DQf33		ECCf2		C		DQb39		DQb37		RDb2		DQb35		DQb33		ECCb2		B	
CG	C		DBf4		DQf38		DQf36		DPARf1		DQf34		DQf32		C		DBf4		DQb38		DQb36		DPARb1		DQb34		DQb32		B		
CH		N		N		N		N		N		N		N		N		N		N		N		N		N		N		B	
CJ	C		DBf5		DQf46		DQf44		WDQSB1_t		DQf42		DQf40		C		DBf5		DQb46		DQb44		WDQSB1_t		DQb42		DQb40		B		
CK		C		DQf47		DQf45		WDQSB1_c		DQf43		DQf41		ECCb1		C		DQb47		DQb45		WDQSB1_c		DQb43		DQb41		ECCb3		B	
CL	M		M		M		M		M		M		M		M		M		M		M		M		M		M		B		
CM		D		D		D		D		D		D		D		D		D		D		D		D		D		D		B	
CN	C		DBf6		DQf54		DQf52		RDQSB1_t		DQf50		DQf48		C		DBf6		DQb54		DQb52		RDQSB1_t		DQb50		DQb48		B		
CP		C		DQf55		DQf53		RDQSB1_c		DQf51		DQf49		SEVf2		C		DQb55		DQb53		RDQSB1_c		DQb51		DQb49		SEVb2		B	
CR	N		N		N		N		N		N		N		N		N		N		N		N		N		N		B		
CT		C		DQf63		DQf61		RDf3		DQf59		DQf57		SEVb1		C		DQb63		DQb61		RDb3		DQb59		DQb57		SEVb3		B	
CU	C		DBf7		DQf62		DQf60		DERRf1		DQf58		DQf56		C		DBf7		DQb62		DQb60		DERRb1		DQb58		DQb56		B		
CV		M		M		M		M		M		M		M		M		M		M		M		M		M		M		B	



Table 136: HBM3 Ballout – Part 5 (Columns 119 to 148) (cont’d)

	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148
CW	D		D		D		D		D		D		D		D		D		D		D		D		D		D		B	
CY		WSOe		WSOe		WSOc		WSOc		WSOa		WSOa		WSI		WSI		CAPTURE WR		CAPTURE WR		WRCK		WRCK		RESET_n		RESET_n		B
DA	M		M		M		M		M		M		M		M		M		M		M		M		M		M		B	
DB		WSOf		WSOf		WSOd		WSOd		WSOb		WSOb		SELECT WIR		SELECT WIR		UPDATE WR		UPDATE WR		SHIFTWR		SHIFT WR		WRST_n		WRST_n		B
DC	D		D		D		D		D		D		D		D		D		D		D		D		D		D		B	


Table 136: HBM3 Ballout – Part 5 (Columns 119 to 148) (cont'd)

	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148
DD		M		M		M		M		M		M		M		M		M		M		M		M		M		M		B
DE	C		DBlg7		DQg62		DQg60		DERRg1		DQg58		DQg56		C		DBlc7		DQc62		DQc60		DERRc1		DQc58		DQc56		B	
DF		C		DQg63		DQg61		RDg3		DQg59		DQg57		SEVg3		C		DQc63		DQc61		RDc3		DQc59		DQc57		SEVc3		B
DG	N		N		N		N		N		N		N		N		N		N		N		N		N		N		N	
DH		C		DQg55		DQg53		RDQsg1_c		DQg51		DQg49		SEVg2		C		DQc55		DQc53		RDQSc1_c		DQc51		DQc49		SEVc2		B
DJ	C		DBlg6		DQg54		DQg52		RDQsg1_t		DQg50		DQg48		C		DBlc6		DQc54		DQc52		RDQSc1_t		DQc50		DQc48		B	
DK		D		D		D		D		D		D		D		D		D		D		D		D		D		D		B
DL	M		M		M		M		M		M		M		M		M		M		M		M		M		M		M	
DM		C		DQg47		DQg45		WDQsg1_c		DQg43		DQg41		ECCg3		C		DQc47		DQc45		WDQSc1_c		DQc43		DQc41		ECCc3		B
DN	C		DBlg5		DQg46		DQg44		WDQsg1_t		DQg42		DQg40		C		DBlc5		DQc46		DQc44		WDQSc1_t		DQc42		DQc40		B	
DP		N		N		N		N		N		N		N		N		N		N		N		N		N		N		B
DR	C		DBlg4		DQg38		DQg36		DPARG1		DQg34		DQg32		C		DBlc4		DQc38		DQc36		DPARC1		DQc34		DQc32		B	
DT		C		DQg39		DQg37		RDg2		DQg35		DQg33		ECCg2		C		DQc39		DQc37		RDc2		DQc35		DQc33		ECCc2		B
DU	D		D		D		D		D		D		D		D		D		D		D		D		D		D		D	
DV		M		M		M		M		M		M		M		M		M		M		M		M		M		M		B
DW	C		AERRg		Rg8		Rg6		Rg5		Rg0		Rg2		C		AERRc		Rc8		Rc6		Rc5		Rc0		Rc2		B	
DY		C		Rg9		Rg7		CKg_c		Rg4		Rg3		Rg1		C		Rc9		Rc7		CKc_c		Rc4		Rc3		Rc1		B
EA	C		RAg		APARG		Cg6		CKg_t		Cg3		Cg1		C		RAc		APARC		Cc6		CKc_t		Cc3		Cc1		B	
EB		C		ARFug		Cg7		Cg5		Cg4		Cg2		Cg0		C		ARFuc		Cc7		Cc5		Cc4		Cc2		Cc0		B
EC	D		D		D		D		D		D		D		D		D		D		D		D		D		D		D	
ED		M		M		M		M		M		M		M		M		M		M		M		M		M		M		B
EE	C		DBlg3		DQg30		DQg28		DERRg0		DQg26		DQg24		C		DBlc3		DQc30		DQc28		DERRc0		DQc26		DQc24		B	
EF		C		DQg31		DQg29		RDg1		DQg27		DQg25		SEVg1		C		DQc31		DQc29		RDc1		DQc27		DQc25		SEVc1		B
EG	N		N		N		N		N		N		N		N		N		N		N		N		N		N		N	
EH		C		DQg23		DQg21		RDQsg0_c		DQg19		DQg17		SEVg0		C		DQc23		DQc21		RDQSc0_c		DQc19		DQc17		SEVc0		B
EJ	C		DBlg2		DQg22		DQg20		RDQsg0_t		DQg18		DQg16		C		DBlc2		DQc22		DQc20		RDQSc0_t		DQc18		DQc16		B	
EK		D		D		D		D		D		D		D		D		D		D		D		D		D		D		B
EL	M		M		M		M		M		M		M		M		M		M		M		M		M		M		M	
EM		C		DQg15		DQg13		WDQsg0_c		DQg11		DQg9		ECCg1		C		DQc15		DQc13		WDQSc0_c		DQc11		DQc9		ECCc1		B
EN	C		DBlg1		DQg14		DQg12		WDQsg0_t		DQg10		DQg8		C		DBlc1		DQc14		DQc12		WDQSc0_t		DQc10		DQc8		B	
EP		N		N		N		N		N		N		N		N		N		N		N		N		N		N		B
ER	C		DBlg0		DQg6		DQg4		DPARG0		DQg2		DQg0		C		DBlc0		DQc6		DQc4		DPARC0		DQc2		DQc0		B	
ET		C		DQg7		DQg5		RDg0		DQg3		DQg1		ECCg0		C		DQc7		DQc5		RDc0		DQc3		DQc1		ECCc0		B
EU	D		D		D		D		D		D		D		D		D		D		D		D		D		D		D	


Table 136: HBM3 Ballout – Part 5 (Columns 119 to 148) (cont'd)

	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148
EV		M		M		M		M		M		M		M		M		M		M		M		M		M		M		B
EW	C		DBb7		DQh62		DQh60		DERRh1		DQh58		DQh56		C		DBId7		DQd62		DQd60		DERRd1		DQd58		DQd56		B	
EY		C		DQh63		DQh61		RDh3		DQh59		DQh57		SEVh3		C		DQd63		DQd61		RDd3		DQd59		DQd57		SEVd3		B
FA	N		N		N		N		N		N		N		N		N		N		N		N		N		N		N	
FB		C		DQh55		DQh53		RDQSh1_c		DQh51		DQh49		SEVh2		C		DQd55		DQd53		RDQSD1_c		DQd51		DQd49		SEVd2		B
FC	C		DBb6		DQh54		DQh52		RDQSh1_t		DQh50		DQh48		C		DBId6		DQd54		DQd52		RDQSD1_t		DQd50		DQd48		B	
FD		D		D		D		D		D		D		D		D		D		D		D		D		D		D		B
FE	M		M		M		M		M		M		M		M		M		M		M		M		M		M		M	
FF		C		DQh47		DQh45		WDQSh1_c		DQh43		DQh41		ECCh3		C		DQd47		DQd45		WDQSD1_c		DQd43		DQd41		ECCd3		B
FG	C		DBb5		DQh46		DQh44		WDQSh1_t		DQh42		DQh40		C		DBId5		DQd46		DQd44		WDQSD1_t		DQd42		DQd40		B	
FH		N		N		N		N		N		N		N		N		N		N		N		N		N		N		B
FJ	C		DBb4		DQh38		DQh36		DPARh1		DQh34		DQh32		C		DBId4		DQd38		DQd36		DPARD1		DQd34		DQd32		B	
FK		C		DQh39		DQh37		RDh2		DQh35		DQh33		ECCh2		C		DQd39		DQd37		RDd2		DQd35		DQd33		ECCd2		B
FL	D		D		D		D		D		D		D		D		D		D		D		D		D		D		D	
FM		M		M		M		M		M		M		M		M		M		M		M		M		M		M		B
FN	C		AERRh		Rh8		Rh6		Rh5		Rh0		Rh2		C		AERRd		Rd8		Rd6		Rd5		Rd0		Rd2		B	
FP		C		Rh9		Rh7		CKh_c		Rh4		Rh3		Rh1		C		Rd9		Rd7		CKd_c		Rd4		Rd3		Rd1		B
FR	C		RAh		APARh		Ch6		CKh_t		Ch3		Ch1		C		RAd		APARd		Cd6		CKd_t		Cd3		Cd1		B	
FT		C		ARFUh		Ch7		Ch5		Ch4		Ch2		Ch0		C		ARFUd		Cd7		Cd5		Cd4		Cd2		Cd0		B
FU	D		D		D		D		D		D		D		D		D		D		D		D		D		D		D	
FV		M		M		M		M		M		M		M		M		M		M		M		M		M		M		B
FW	C		DBb3		DQh30		DQh28		DERRh0		DQh26		DQh24		C		DBId3		DQd30		DQd28		DERRd0		DQd26		DQd24		B	
FY		C		DQh31		DQh29		RDh1		DQh27		DQh25		SEVh1		C		DQd31		DQd29		RDd1		DQd27		DQd25		SEVd1		B
GA	N		N		N		N		N		N		N		N		N		N		N		N		N		N		N	
GB		C		DQh23		DQh21		RDQSh0_c		DQh19		DQh17		SEVh0		C		DQd23		DQd21		RDQSD0_c		DQd19		DQd17		SEVd0		B
GC	C		DBb2		DQh22		DQh20		RDQSh0_t		DQh18		DQh16		C		DBId2		DQd22		DQd20		RDQSD0_t		DQd18		DQd16		B	
GD		D		D		D		D		D		D		D		D		D		D		D		D		D		D		B
GE	M		M		M		M		M		M		M		M		M		M		M		M		M		M		M	
GF		C		DQh15		DQh13		WDQSh0_c		DQh11		DQh9		ECCh1		C		DQd15		DQd13		WDQSD0_c		DQd11		DQd9		ECCd1		B
GG	C		DBb1		DQh14		DQh12		WDQSh0_t		DQh10		DQh8		C		DBId1		DQd14		DQd12		WDQSD0_t		DQd10		DQd8		B	
GH		N		N		N		N		N		N		N		N		N		N		N		N		N		N		B
GJ	C		DBb0		DQh6		DQh4		DPARh0		DQh2		DQh0		C		DBId0		DQd6		DQd4		DPARD0		DQd2		DQd0		B	
GK		C		DQh7		DQh5		RDh0		DQh3		DQh1		ECCh0		C		DQd7		DQd5		RDd0		DQd3		DQd1		ECCd0		B
GL	D		D		D		D		D		D		D		D		D		D		D		D		D		D		B	



Table 136: HBM3 Ballout – Part 5 (Columns119 to 148) (cont’d)

	119	120	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148
GM		M		M		M		M		M		M		M		M		M		M		M		M		M		M		B
GN	C		C		C		C		C		C		C		C		C		C		C		C		C		C		B	
GP		N		N		N		N		N		N		N		N		N		N		N		N		N		N		B
GR	C		C		C		C		C		C		C		C		C		C		C		C		C		C		B	
GT		C		D		E		C		D		E		C		D		E		C		D		E		C		D		B
GU	E		D		C		E		D		C		E		D		C		E		D		C		E		D		B	
GV		C		D		E		C		D		E		C		D		E		C		D		E		C		D		B
GW	E		D		C		E		D		C		E		D		C		E		D		C		E		D		B	
GY		C		D		E		C		D		E		C		D		E		C		D		E		C		D		B
HA	E		D		C		E		D		C		E		D		C		E		D		C		E		D		B	



Revision History

Rev. L – 06/2025

- AC Timings Part 2: CK clock frequency for read latency updated.

Rev. K – 01/2025

- Figure for Mechanical Outline Drawing updated
- 8-High IDD5P values updated
- 12-High IDD values added

Rev. J – 07/2024

- Updated legal status to Production.
- IDD Specification: IDD DC Operating values added
- AC Timings: ^tWDQS2DQ_I value changed

Rev. I – 11/2023

- Mode Register: Duty Cycle Training Mode enable assigned to MR12 OP3
- Duty Cycle Monitor (DCM): Updated sequence for WDQS duty cycle adjustment
- DWORD Register Mode: Clarified the latched UI bits during a write command

Rev. H – 09/2023

- Product Numbering: Added 9.2 Gb/s data rate
- IDD Specification: Added speed bin -92 (9.2 Gb/s) for 8H and 12H
- AC Timings Part 1: Added speed bin -92 (9.2 Gb/s)
- AC Timings Part 2: CK clock frequency for read and write latency updated

Rev. G – 04/2023

- Updated legal status to Preliminary
- Mode Register: WL, RL and WR ranges in MR1, MR2 and MR3 updated
- IEEE1500 Test Port: note added regarding differential inputs in EXTEXT_RX
- AC Timings: ^tCKRLnn and ^tCKWLnn values added

Rev. F – 03/2023

- Part number changed

Rev. E – 01/2023

- IDD Specification: timings used for IDD measurement loop pattern updated
- AC Timings: ^tWDQS2DQ_O, ^tPARAC and ^tPARDQ values changed

Rev. D – 07/2022

- Error Correction Code (ECC): description of write and read data in MD off case updated
- Self Repair: number of repairs per auto repair step updated
- IEEE1500 Test Port: RFM related bits in DEVICE_ID WDR updated; options for “all channels” added to CHANNEL_DISABLE instruction
- AC Timings: ^tRFCab value changed

**Rev. C – 04/2022**

- Mode Register: WL range in MR1 and RL range in MR2 updated
- Operation: ERRTH value added in Error Check and Scrub section; temperature trip points changed in Temperature Compensated Refresh Reporting section
- IEEE1500 Test Port: row address range in SOFT_REPAIR and HARD_REPAIR WDR tables updated (typo)
- Operating Conditions: Operating temperature range changed
- AC Timings: $t_{WDQS2DQ_I}$, $t_{WDQS2DQ_I_VOLT}$, $t_{WDQS2DQ_I_TEMP}$, $t_{WDQS2DQ_O}$, $t_{WDQS2DQ_O_TEMP}$, t_{RRDL} , t_{RRDS} , t_{CCDR} , t_{WRPDE} , t_{WRAPDE} , t_{WRMRS} and t_{PARDQ} values changed

Rev. B – 02/2022

- Mode Registers: parity latency (PL) in MR1 updated
- Operation: Timing diagrams in Read and Write sections updated to PL = 3

Rev. A – 01/2022

- Initial release



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This data sheet contains minimum and maximum limits specified over the power supply and temperature range set forth herein. Although considered final, these specifications are subject to change, as further product development and data characterization sometimes occur.